Project: **Mansourah Massarah Gold Project**

Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**

3120 FOUNDATION DESIGN OF CONVEYOR 3120-CNV-001

DESIGN CALCULATION

AS BUILT

MD-216-3120-EG-CV-CAL-1001

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Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

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REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria
SBC 301 :2007	Saudi Structural - Loading & Forces Code
SBC 304 :2007	Saudi Code for Concrete Structures

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
RL	:	Reduced Level
C.L.	:	Centre Line
CJ	:	Construction Joint
FDN	:	Foundation
TOC	:	Top of concrete
FGL	:	Finished Ground Level
FFL	:	Finished Floor Level
BOF	:	Bottom of footing
TYP	:	Typical
TOC	:	Top of concrete
IP	:	Insert Plate
RCC	:	Reinforced Cement Concrete
MKD.	:	Marked
THK.	:	Thick

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1. INTENT

The document is prepared for Concrete design calculation of Conveyor Gallery 3120-CNV-001.
This conveyor is between TP-2 and Secondary Crusher House and belt width of the conveyor is 1400 mm.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.
MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.
The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

This document is prepared for foundation design of Conveyor 3120-CNV-001. All the loads applied to the structure for analysis & design followed Civil -Structural Design Criteria and SBC 301:2007
The column and beam reinforcement design have been performed in Staad-Pro Software. The foundation of Ground Portal GR-1 to GR-4; TR-4 & VGTU have been performed in MS Excel.
Foundation of TR-1, TR-2 & TR-3 has been designed by FEM analysis in Staad-Pro Software.

4. CONCRETE STRUCTURE

4.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304 and ACI 318M.

4.2 Materials

Concrete Strength of 28 days
· Foundation and Structural Members: 35 MPa
· Grade slab and pavement: 30 MPa
· Blinding Concrete below foundation bases: 15 Mpa

Reinforcement Grade
· Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
· Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

4.3 Foundations for Structures

Foundation shall be Isolated and combined Shallow Concrete foundation. The design calculations are prepared as per recommendation of standard SBC-304 Clause 13.3.

4.4 Minimum Concrete Cover

Items	Minimum Cover(mm)
Foundation (Side & Bottom)	75
Foundation (Top)	50
Column & Pedestal	50
Beam	50
Slab	40

4.5 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 500 kN/m2 at 1m & 3m depth below FGL (ie. RL 911.500m & RL 909.500m respectively) refer Table 4 - BH-17,18,19,25,29 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

4.6 Reference

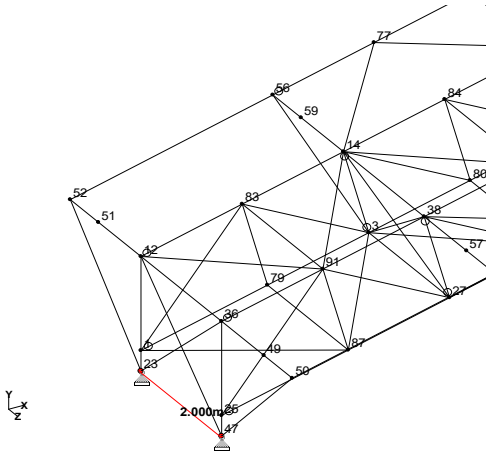
Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

5 INPUT LOADS FROM SUPERSTRUCTURE

For load calculation of superstructure refer Document No: MD-216-3120-EG-ST-CAL-1001

5.1 Support Reaction from TR-1

Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN
23	1 DL	-0.4	31.8	-2.4
	2 LL	0.0	34.3	-2.8
	3 ML - CONVEYOR MATERIAL L	0.0	27.0	1.6
	4 WL (+Z)	-0.8	-15.2	-12.1
	5 WL (-Z)	0.8	16.6	12.2
	6 TL1	0.1	0.1	7.3
	7 TL2	-0.1	-0.1	-7.3
	8 EARTHQUAKE LOAD (+X)	-1.1	-0.3	0.0
	9 EARTHQUAKE LOAD (+Z)	0.0	-0.8	-0.5
47	1 DL	0.4	74.8	-1.7
	2 LL	0.0	34.1	2.8
	3 ML - CONVEYOR MATERIAL L	0.0	26.7	-1.6
	4 WL (+Z)	0.8	15.2	-8.7
	5 WL (-Z)	-0.8	-16.6	8.7
	6 TL1	-0.1	-0.1	-7.2
	7 TL2	0.1	0.1	7.2
	8 EARTHQUAKE LOAD (+X)	-1.1	-0.3	0.0
	9 EARTHQUAKE LOAD (+Z)	0.0	0.8	-0.4



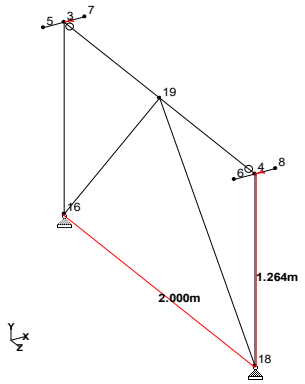
Staad model of 12m Gallery End portal which is acting as TR1

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5.2 Support Reaction from TR-2

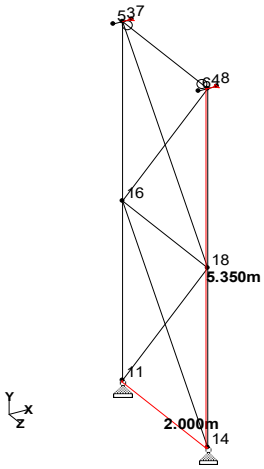
Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN
16	1 DEAD LOAD	1.3	175.9	-5.8
	2 LIVE LOAD	0.7	78.0	0.0
	3 MATERIAL LOAD	0.5	70.3	0.0
	4 WL (+Z)	-0.8	-111.6	-35.5
	5 WL (-Z)	0.8	111.2	35.4
	6 WL (+X)	-0.8	0.0	0.0
	7 WL (-X)	0.8	0.0	0.0
	8 TL1	0.0	0.0	0.0
	9 TL2	0.0	0.0	0.0
	10 EL (+X)	-0.1	0.0	0.0
	11 EL (+Z)	0.0	-7.7	-4.1
18	1 DEAD LOAD	1.0	117.3	-6.1
	2 LIVE LOAD	0.5	78.0	0.0
	3 MATERIAL LOAD	0.5	71.1	0.0
	4 WL (+Z)	0.8	111.6	-35.5
	5 WL (-Z)	-0.8	-111.2	35.4
	6 WL (+X)	-0.8	0.0	0.0
	7 WL (-X)	0.8	0.0	0.0
	8 TL1	0.0	0.0	0.0
	9 TL2	0.0	0.0	0.0
	10 EL (+X)	-0.1	-0.1	0.0
	11 EL (+Z)	0.0	7.7	-4.1



Staad model of TR2 superstructure (along with dimension)

5.3 Support Reaction from TR-3

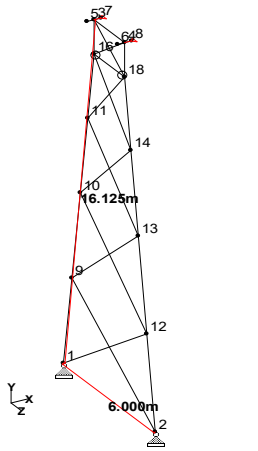
Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN
11	1 DEAD LOAD	0.3	155.8	1.8
	2 LIVE LOAD	0.2	78.0	4.1
	3 MATERIAL LOAD	0.1	70.3	3.7
	4 WL (+Z)	-0.2	-265.0	-39.6
	5 WL (-Z)	0.2	264.0	39.5
	6 WL (+X)	-3.2	0.0	0.0
	7 WL (-X)	3.2	0.0	0.0
	8 TL1	0.0	0.0	15.9
	9 TL2	0.0	0.0	-15.9
	10 EL (+X)	0.0	0.0	0.0
	11 EL (+Z)	0.0	-15.2	-2.4
14	1 DEAD LOAD	0.2	145.8	-13.7
	2 LIVE LOAD	0.1	78.0	-4.1
	3 MATERIAL LOAD	0.1	71.1	-3.7
	4 WL (+Z)	0.2	265.0	-39.6
	5 WL (-Z)	-0.2	-264.0	39.4
	6 WL (+X)	-3.2	0.0	0.0
	7 WL (-X)	3.2	0.0	0.0
	8 TL1	0.0	0.0	-15.9
	9 TL2	0.0	0.0	15.9
	10 EL (+X)	0.0	-0.1	0.0
	11 EL (+Z)	0.0	15.2	-2.4



Staad model of TR3 superstructure (along with dimension)

C01 **5.4 Support Reaction from TR-4**

Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN
1	1 DEAD LOAD	0.1	98.5	21.3
	2 LIVE LOAD	0.0	72.0	13.0
	3 MATERIAL LOAD	0.0	57.5	10.3
	4 WL (+Z)	0.0	-226.5	-47.1
	5 WL (-Z)	0.0	226.5	47.1
	6 WL (+X)	-9.7	0.0	0.0
	7 WL (-X)	9.7	0.0	0.0
	8 TL1	0.0	0.0	55.2
	9 TL2	0.0	0.0	-55.2
	10 EL (+X)	0.0	0.0	0.0
	11 EL (+Z)	0.0	-12.0	-2.1
2	1 DEAD LOAD	-0.1	203.4	-32.5
	2 LIVE LOAD	0.0	72.0	-13.0
	3 MATERIAL LOAD	0.0	57.5	-10.3
	4 WL (+Z)	0.0	226.5	-47.1
	5 WL (-Z)	0.0	-226.5	47.1
	6 WL (+X)	-9.7	0.0	0.0
	7 WL (-X)	9.7	0.0	0.0
	8 TL1	0.0	0.0	-55.2
	9 TL2	0.0	0.0	55.2
	10 EL (+X)	0.0	0.0	0.0
	11 EL (+Z)	0.0	12.0	-2.1



Staad model of TR4 superstructure (along with dimension)

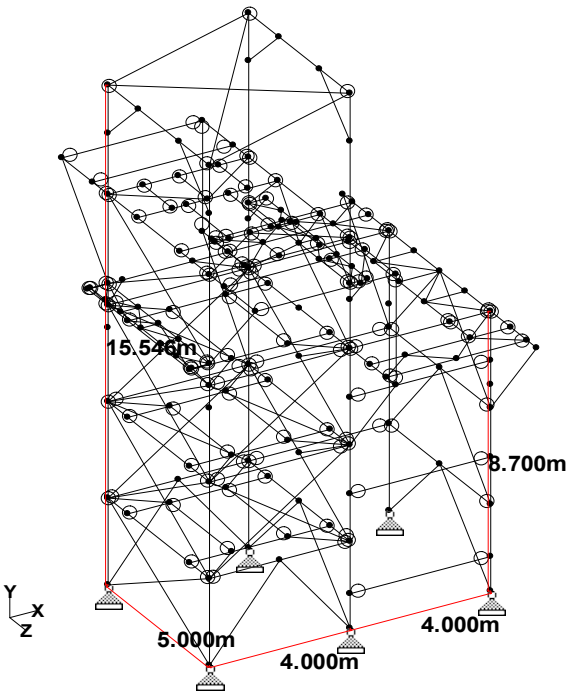
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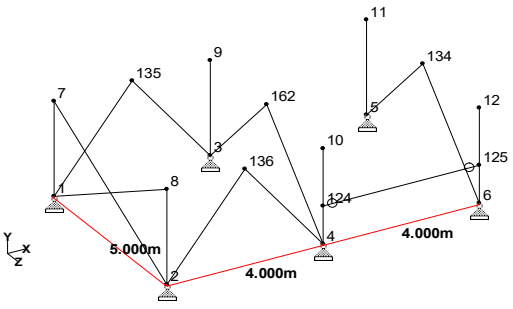
C01

5.5 Support Reaction from VGTU

Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN
1	1 DEAD LOAD	26.4	249.3	11.1
	2 LIVE LOAD	2.2	243.8	12.9
	3 MATERIAL LOAD	0.0	29.1	1.7
	4 WL (+Z)	1.2	-95.8	-27.5
	5 WL (-Z)	-1.2	95.8	27.5
	6 WL (+X)	-39.9	-161.9	-6.2
	7 WL (-X)	39.9	161.9	6.2
	8 TL1	-3.0	0.1	42.5
	9 TL2	3.0	-0.1	-42.5
	10 EL (+X)	-1.5	-7.4	-0.3
	11 EL (+Z)	0.0	-4.9	-1.4
2	1 DEAD LOAD	25.8	312.6	-17.0
	2 LIVE LOAD	3.1	204.1	-12.9
	3 MATERIAL LOAD	0.0	28.4	-1.6
	4 WL (+Z)	-1.2	95.8	-27.7
	5 WL (-Z)	1.2	-95.8	27.7
	6 WL (+X)	-39.8	-163.1	6.4
	7 WL (-X)	39.8	163.1	-6.4
	8 TL1	1.4	-19.3	-32.3
	9 TL2	-1.4	19.3	32.3
	10 EL (+X)	-1.6	-7.6	0.3
	11 EL (+Z)	0.0	4.9	-1.4
3	1 DEAD LOAD	24.3	19.9	-0.2
	2 LIVE LOAD	-3.2	271.9	8.1
	3 MATERIAL LOAD	0.0	28.4	-0.1
	4 WL (+Z)	1.6	-98.9	-23.8
	5 WL (-Z)	-1.6	98.9	23.8
	6 WL (+X)	-39.9	161.3	-0.1
	7 WL (-X)	39.9	-161.3	0.1
	8 TL1	-5.5	14.7	-0.6
	9 TL2	5.5	-14.7	0.6
	10 EL (+X)	-1.5	7.4	0.0
	11 EL (+Z)	0.1	-4.2	-0.9
4	1 DEAD LOAD	23.6	91.2	-4.2
	2 LIVE LOAD	-2.1	281.2	-9.2
	3 MATERIAL LOAD	0.0	29.1	-0.2
	4 WL (+Z)	-1.6	98.9	-23.8
	5 WL (-Z)	1.6	-98.9	23.8
	6 WL (+X)	-41.2	163.6	-0.5
	7 WL (-X)	41.2	-163.6	0.5
	8 TL1	-0.5	4.5	-1.7
	9 TL2	0.5	-4.5	1.7
	10 EL (+X)	-1.6	7.7	0.0
	11 EL (+Z)	-0.1	4.2	-0.9
5	1 DEAD LOAD	0.0	17.4	-0.1
	2 LIVE LOAD	0.0	62.1	0.6
	3 MATERIAL LOAD	0.0	0.2	0.1
	4 WL (+Z)	0.0	-21.2	-10.2
	5 WL (-Z)	0.0	21.2	10.2
	6 WL (+X)	-1.2	0.6	0.2
	7 WL (-X)	1.2	-0.6	-0.2
	8 TL1	-0.1	0.6	0.6
	9 TL2	0.1	-0.6	-0.6
	10 EL (+X)	0.0	0.0	0.0
	11 EL (+Z)	0.0	-1.8	-0.8
6	1 DEAD LOAD	-0.1	20.8	-1.0
	2 LIVE LOAD	0.0	51.9	0.6
	3 MATERIAL LOAD	0.0	-0.2	0.1
	4 WL (+Z)	0.0	21.2	-10.2
	5 WL (-Z)	0.0	-21.2	10.2
	6 WL (+X)	0.1	-0.6	0.2
	7 WL (-X)	-0.1	0.6	-0.2
	8 TL1	-0.5	-0.6	-0.3
	9 TL2	0.5	0.6	0.3
	10 EL (+X)	0.0	0.0	0.0
	11 EL (+Z)	0.0	1.8	-0.7



Staad model of VGTU superstructure (along witgh dimension)



Node number and orientation of VGTU superstructure

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5.6 LOAD COMBINATIONS

5.6.1 Load Combination for Base pressure calculation

(As per Section 2.4 of SBC 301 -2007)

Load Cases	Reference
LOAD COMB 301 1.0 DL	(Eq. 2.4.1-1 of SBC 301)
LOAD COMB 302 1.0 (DL + TL1) + 1.0 (LL+ML)	(Eq. 2.4.1-2 of SBC 301)
LOAD COMB 303 1.0 (DL + TL1) + 1.0 (LL+ML)	
LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75 (LL+ML)	(Eq. 2.4.1-6 of SBC 301)
LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75 (LL+ML)	
LOAD COMB 306 1.0 DL+ 0.75 WL(+X) + 0.75 (LL+ML)	
LOAD COMB 307 1.0 DL+ 0.75 WL(-X) + 0.75 (LL+ML)	
LOAD COMB 308 1.0 DL + 0.525 EL(+X) + 0.75 (LL+ML)	
LOAD COMB 309 1.0 DL + 0.525 EL(-X) + 0.75 (LL+ML)	
LOAD COMB 310 1.0 DL + 0.525 EL(+Z) + 0.75 (LL+ML)	
LOAD COMB 311 1.0 DL + 0.525 EL(-Z) + 0.75 (LL+ML)	
LOAD COMB 312 0.6 DL + 1.0 WL(+Z)	(Eq. 2.4.1-7 of SBC 301)
LOAD COMB 313 0.6 DL + 1.0 WL(-Z)	
LOAD COMB 314 0.6 DL + 1.0 WL(+X)	
LOAD COMB 315 0.6 DL + 1.0 WL(-X)	
LOAD COMB 316 0.6 DL + 0.7 EL(+X)	(Eq. 2.4.1-8 of SBC 301)
LOAD COMB 317 0.6 DL + 0.7 EL(-X)	
LOAD COMB 318 0.6 DL + 0.7 EL(+Z)	
LOAD COMB 319 0.6 DL + 0.7 EL(-Z)	

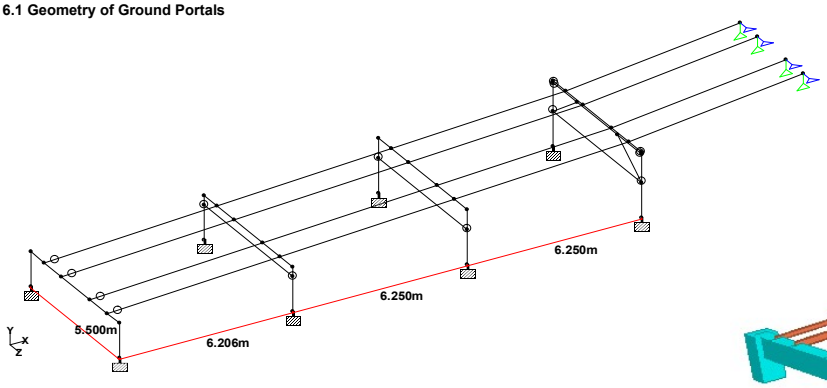
5.6.2 Load Combination for Concrete design

Load Case	Reference
LOAD COMB 401 1.4 DL	(Eq. 2.3.2-1 of SBC 301)
LOAD COMB 402 1.4 (DL + TL1) + 1.7 (LL+ML)	(Eq. 2.3.2-2 of SBC 301)
LOAD COMB 403 1.4 (DL + TL1) + 1.7 (LL+ML)	
LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5 (LL+ML)	(Eq. 2.3.2-4 of SBC 301)
LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5 (LL+ML)	
LOAD COMB 406 1.2 DL+ 1.6 WL(+X) + 0.5 (LL+ML)	
LOAD COMB 407 1.2 DL+ 1.6 WL(-X) + 0.5 (LL+ML)	
LOAD COMB 408 1.2 DL + 1.0 EL(+X) + 0.5 (LL+ML)	(Eq. 2.3.2-5 of SBC 301)
LOAD COMB 409 1.2 DL + 1.0 EL(-X) + 0.5 (LL+ML)	
LOAD COMB 410 1.2 DL + 1.0 EL(+Z) + 0.5 (LL+ML)	
LOAD COMB 411 1.2 DL + 1.0 EL(-Z) + 0.5 (LL+ML)	
LOAD COMB 412 0.9 DL + 1.6 WL(+Z)	(Eq. 2.3.2-6 of SBC 301)
LOAD COMB 413 0.9 DL + 1.6 WL(-Z)	
LOAD COMB 414 0.9 DL + 1.6 WL(+X)	
LOAD COMB 415 0.9 DL + 1.6 WL(-X)	
LOAD COMB 416 0.9 DL + 1.0 EL(+X)	(Eq. 2.3.2-7 of SBC 301)
LOAD COMB 417 0.9 DL + 1.0 EL(-X)	
LOAD COMB 418 0.9 DL + 1.0 EL(+Z)	
LOAD COMB 419 0.9 DL + 1.0 EL(-Z)	

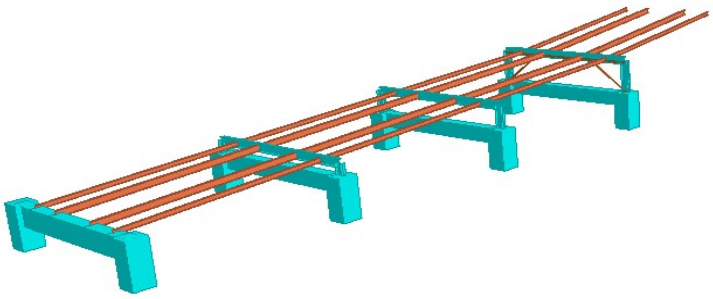
Eq. 2.3.2-3 of SBC 301 - 2007 has not been considered as it would not produce worst load combination in the absence of Roof Live Load & Rain Load

6 GEOMETRY

6.1 Geometry of Ground Portals



Staad model of Ground Portals (along witgh dimension)

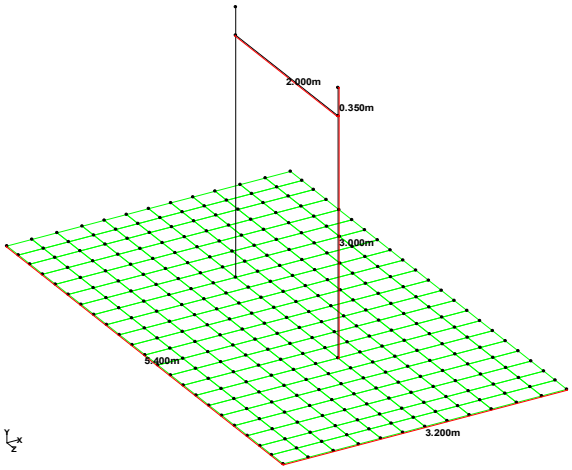


Isometric view of Ground Portals (From Staad)

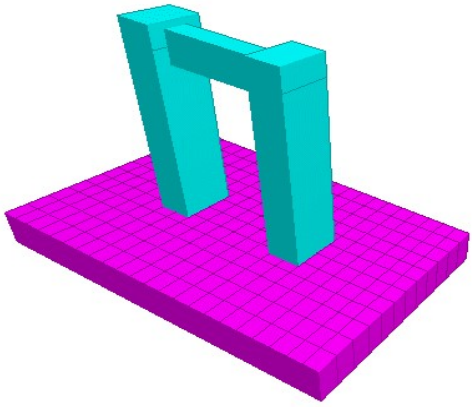
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Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
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6.2 Geometry of TR-3 Foundation



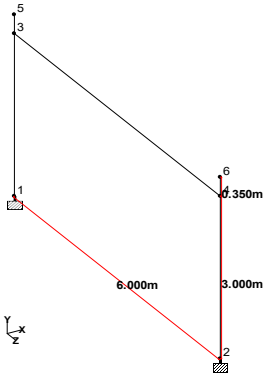
Staad Model of TR3 Foundation mkd F2 (Along with dimension)



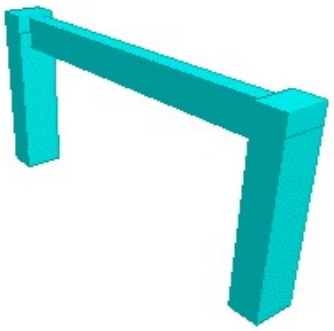
Isometric view of TR3 Foundation mkd F2

Same size as TR-3 footing has been provided for TR-1 & TR-2 as height of these trestles are less than TR-3. Moreover wind load are also less for those trestles.

6.3 Geometry of TR-4 Foundation

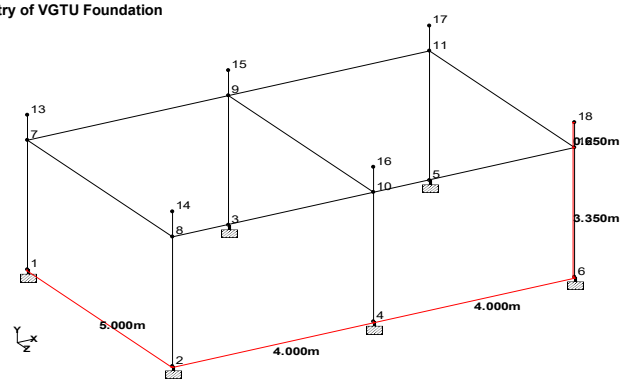


Staad Model of TR4 Foundation mkd F5 (Along with dimension)

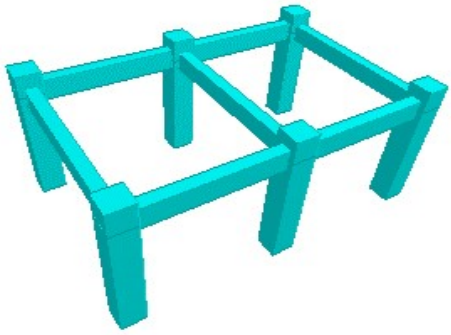


Isometric view of of TR4 Foundation mkd F5

6.4 Geometry of VGTU Foundation



Staad Model of VGTU Foundation (Along with dimension)



Isometric view of VGTU foundation (From Staad)

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7. DESIGN OF CONCRETE COMPONENTS

All foundations have been designed using worksheet in accordance to SBC-304.
All beams & columns have been designed in Staad-Pro by ACI 318M.
For Design Results, Refer Attachemnt 3 to 6 - Staad Output file.

7.1 Design Summary of Column

7.1.1 Column Reinforcement Summary

Column mark	Width (Along conv)	Depth (Across conv)	Reinforcement	Stirrup	Remarks	Numbers
P1	600	600	12 No.- D20	3 Nos D10 @ 200c/c	GP1	2
P2	600	600	12 No.- D20	3 Nos D10 @ 200c/c	GP2 & GP3	4
P2A	600	600	12 No.- D20	3 Nos D10 @ 200c/c	GP4	2
P3	600	600	12 No.- D20	3 Nos D10 @ 200c/c	TR1 & TR2	4
P4	700	500	12 No.- D20	3 Nos D10 @ 200c/c	TR3 & TR4	4
P5	600	600	12 No.- D20	3 Nos D10 @ 200c/c	VGTU Grid A-1, A-2, B-1 & B-2	4
P6	600	600	12 No.- D20	3 Nos D10 @ 200c/c	VGTU Grid C-1 & C-2	2
P7	450	450	8 No.- D20	2 Nos D10 @ 200c/c	VGTU Post	2

7.2 Design Summary of Beam

7.2.1 Beam Reinforcement Summary

Beam mark	Width (mm)	Depth (mm)	Top Reinforcement	Bottom Reinforcement	Stirrup	Numbers
TB1	350	600	3 Nos D16	3 Nos D16	2L D10 STRPS @ 200c/c	1
TB2	300	500	3 Nos D16	3 Nos D16	2L D10 STRPS @ 200c/c	3
TB3	300	500	3 Nos D16	3 Nos D16	2L D10 STRPS @ 200c/c	3
TB4	300	500	3 Nos D16	3 Nos D16	2L D10 STRPS @ 200c/c	3
TB5	300	600	3 Nos D16	3 Nos D16	2L D10 STRPS @ 200c/c	4
TB6	300	600	3 Nos D16	3 Nos D16	2L D10 STRPS @ 200c/c	1

7.3 Design of Foundation

C01

7.3.1 Foundation marking, Max/min Base pressure and Reinforcement Summary

Structure Name	Foundation mark	Length (Along conv) (mm)	Width (Across conv) (mm)	Thickness (mm)	Max pressure (kN/m^2)	Top reinforcement	Bottom reinforcement	F.O.S.	Numbers
GP1	F1A	1500	2000	300	118	D12 @ 200c/c B/W	D16 @ 200c/c B/W	0.23	2
GP2, GP3 & GP4	F1	1500	2000	300	118	D12 @ 200c/c B/W	D16 @ 200c/c B/W	0.23	6
TR1, TR2 & TR3 *(Combined footing)*	F2	3200	5400	550	121	D12 @ 200c/c B/W	D16 @ 200c/c B/W	0.22	3
VGTU Grid A-1, A-2, C-1 & C-2	F3	2500	2500	450	196	D12 @ 200c/c B/W	D16 @ 200c/c B/W	0.35	4
VGTU Grid B-1 & B-2	F4	3000	2500	450	145	D12 @ 200c/c B/W	D16 @ 200c/c B/W	0.26	2
TR4	F5	2500	3000	450	149	D12 @ 200c/c B/W	D16 @ 200c/c B/W	0.27	2

C01

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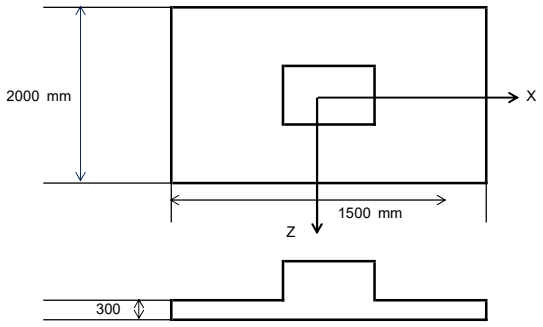
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7.3.2 Design of foundation mkd. F1A & F1 for supporting trestle mkd. GP1; GP2, GP3 & GP4.

Finished Ground level	F.G.L	=	-0.500	m
Bottom of foundation	B.O.F	=	-1.500	m
Staad support reaction level		=	-1.500	m
Ground Water Level	G.W.L	=	-10.000	m
Surcharge	Ws	=	0.000	KN/m²
Length of footing	Lz	=	2.00	m
Width of footing	Lx	=	1.50	m
Thickness of footing	D	=	0.30	m
Depth of pedestal	Bx	=	0.60	m
Width of pedestal	Bz	=	0.60	m
Height of Load point from bottom of footing	H	=	0.000	m
Section Modulus of Pedestal	Zxx	=	1.00	m³
Section Modulus of Pedestal	Zzz	=	0.75	m³
Allowable Net Bearing Capacity		=	500	KN/m²
Gross Bearing Capacity		=	518	KN/m²
Grade of concrete		=	35	N/mm²
Grade of steel		=	420	N/mm²
Clear cover		=	75	mm
Top Reinfocement		=	12	mm
Bottom Reinforcement		=	16	mm
Co-efficient of friction between soil & concrete		=	0.3	
Unit wt of soil		=	18	KN/m³
Unit wt of concrete		=	24	KN/m³

Staad Node no	=	44	
Weight of soil	=	33.26	KN
Weight of foundation	=	21.60	KN
Uplift	=	0.00	KN
Surcharge	=	0	KN
Pressure on top for tension	=	18.3	KN/m²

*inputs



$$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7}$$ β_1 = 0.8 *Table 22.2.2.4.3 SBC 304*

CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
301 1.0 DL	0.1	51.794	13.4	18.83	0.0	-0.1	106.7	54.48	16.62	1	0.11
302 1.0 (DL + TL1) +	2.1	97.835	26.6	62.24	-0.001	-3.3	152.7	117.58	-15.78	1	0.23
303 1.0 (DL + TL1) +	-1.6	97.707	22.1	42.76	0	2.6	152.6	97.11	4.61	1	0.19

Max P_{max} = 117.58 KN/m²
Max P_{min} = -15.78 KN/m²
Resultant F.O.S = 0.23

BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
401 1.4 DL	0.1	72.5	18.7	26.37	0.0	-0.10	50.68	-2.33
402 1.4 (DL + TL1) +	3.0	150.8	40.6	97.23	-0.002	-4.75	153.82	-53.32
403 1.4 (DL + TL1) +	-2.1	150.6	34.2	69.96	0	3.58	124.93	-24.54

Max Net factored B.P = 153.82 KN/m²

DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 0.45 m
Moment about Z Muz = 15.57 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$ = 321.51 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section
Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$ = 190.8 mm²

Effective Depth Along X = 217 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement $A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*
greater of
or A_{stmin} = 0.0014bd

A_{stmin} = 390.6 mm²
A_{req} = 190.8 mm²
Required Spacing = **514.8** mm

Length of cantilever along Z = 0.7 m
Moment about X Mux = 37.69 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$ = 275.84 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section
Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$ = 504.3 mm²

Effective Depth Along X = 201 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement $A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*
greater of
or A_{stmin} = 0.0014bd

A_{stmin} = 361.8 mm²
A_{req} = 504.3 mm²
Required Spacing = **398.7** mm

Provide Bottom Reinforcement = ***16 mm dia @ 200 c/c*** Ast prov = 1005.3 mm²

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Top Reinforcement

Length of cantilever along X = 0.45 m

Moment about Z: Muz = 1.85 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 327.46 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 22.4 mm^2

Effective Depth Along X = 219 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of

A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% o

or

A_{stmin} = 50% of 0.0014bd more prov ir bottom

Table 24.4.3.2 SBC 304

A_{stmin} = 197.1 mm^2

A_{req} = 22.4 mm^2

Required Spacing = **573.8** mm

Length of cantilever along Z = 0.7 m

Moment about X: Mux = 4.48 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 292.56 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 57.3 mm^2

Effective Depth Along Z = 207 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of

A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% o

or

A_{stmin} = 50% of 0.0014bd more prov ir bottom

Table 24.4.3.2 SBC 304

A_{stmin} = 186.3 mm^2

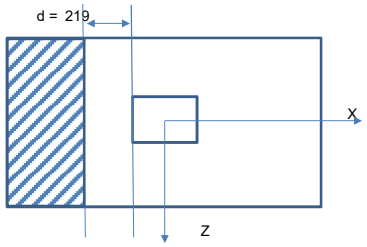
A_{req} = 57.3 mm^2

Required Spacing = **607.1** mm

Provide top Reinforcement = ***12 mm dia @ 200 c/c***

Ast prov = 565.5 mm^2

ONE WAY SHEAR CKECK



Along X Direction Cantilever: L = 0.231 m

Reinforcement ratio: ρ_w = 0.0046

Effective Depth Along X: d = 219 mm

Shear Force at d distance: V_u = 35.53 KN

Allowable shear strength: φVc = 168.29 KN O.K

Utility ratio = 0.21

Least of

i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ *Table 22.5.5.1 SBC 304*

λ = 1 Normal concrete

Mu = 4.10 KNm

φ = 0.75 for shear

φVc = 179.8 KN

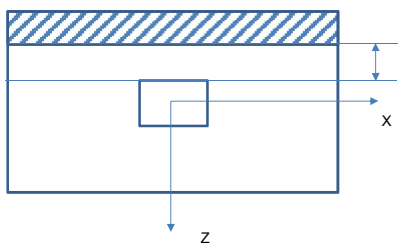
ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$

φVc = 168.3 KN *Table 22.5.5.1 SBC 304*

iii) $V_c = 0.29\lambda\sqrt{f_c'} b_w d$

= 281.8 KN

ONE WAY SHEAR CKECK



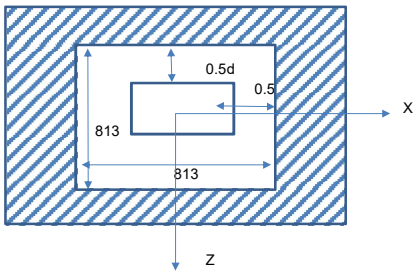
Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T	0	1	2
Date	17-03-2020	30-04-2020	13/10/2023	Date	17-03-2020	30-04-2020	13/10/2023
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Checked	SBY	SBY	SBY	Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

Along Z Direction	Cantilever	L	=	0.493	m		Least of	i)	V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$
Reinforcement ratio		ρ_w	=	0.0049					Table 22.5.5.1 SBC 304
Effective Depth Along Z		d	=	207	mm				λ = 1 Normal concrete
Shear Force at d distance		V_u	=	75.83	KN				Mu = 18.69 KNm
Allowable shear strength		φVc	=	157.72	KN	O.K			φ = 0.75 for shear
Utility ratio			=	0.48					φVc = 157.7 KN
								ii)	V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$
									φVc = 159.8 KN
								iii)	V_c = $0.29\lambda\sqrt{f_c'}b_w d$
									φVc = 266.4 KN

TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



Total area		=	3	m2		
Punching area		=	0.66	m2		
	λ	=	1			
	β	=	1			
	αs	=	40			
	b0	=	3252.0	mm		

Allowable shear stress — least of — Table 22.6.5.2 SBC 304

v_c	$0.33\lambda\sqrt{f_c'}$	=	1.95 N/mm²
v_c	$0.17\left(1+\frac{2}{\beta}\right)\lambda\sqrt{f_c'}$	=	3.02 N/mm²
v_c	$0.083\left(2+\frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$	=	2.27 N/mm²

Design vc = 1.95 N/mm²

Check at 0.5d from column face

Perimeter at 0.5d from column face	u	=	3.25	m		Vc	=	$v_c b_0 d$
Depth at section	d	=	213	mm			=	1352.32 KN
Shear Force at section	Vu	=	359.8	KN		φ	=	0.75 for shear
Allowable shear strength	φVc	=	1014.24	KN	O.K	φV_n	=	φVc
Utility ratio			0.35			φVc	=	1014.2 KN

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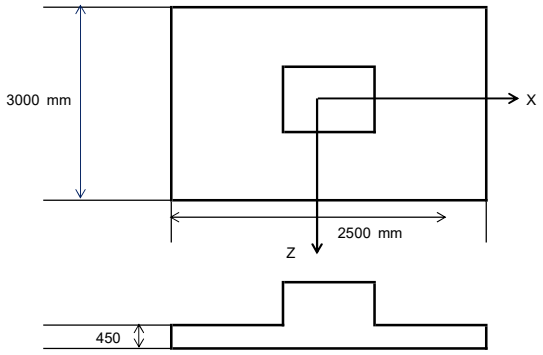
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7.3.3 Design of foundation F5 for supporting trestle mkd. TR4

Finished Ground level	F.G.L	=	-0.500	m	Staad Node no	=	2
Bottom of foundation	B.O.F	=	-3.500	m	Weight of soil	=	328.19 KN
Staad support reaction level		=	-3.500	m	Weight of foundation	=	81.00 KN
Ground Water Level	G.W.L	=	-10.000	m	Uplift	=	0.00 KN
Surcharge	Ws	=	0.000	KN/m^2	Surcharge	=	0 KN
Length of footing	Lz	=	3.00	m	Pressure on top for tension	=	54.6 KN/m^2
Width of footing	Lx	=	2.50	m			
Thickness of footing	D	=	0.45	m			
Depth of pedestal	Bx	=	0.70	m			
Width of pedestal	Bz	=	0.50	m			
Height of Load point from bottom of footing	H	=	0.000	m			
Section Modulus of Pedestal	Zxx	=	3.75	m^3			
Section Modulus of Pedestal	Zzz	=	3.13	m^3			
Allowable Net Bearing Capacity		=	500	KN/m^2			
Gross Bearing Capacity		=	554	KN/m^2			
Grade of concrete		=	35	N/mm^2			
Grade of steel		=	420	N/mm^2			
Clear cover		=	75	mm			
Top Reinfocement		=	12	mm			
Bottom Reinforcement		=	16	mm			
Co-efficient of friction between soil & concrete		=	0.3				
Unit wt of soil		=	18	KN/m^3			
Unit wt of concrete		=	24	KN/m^3			

*inputs



$$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7}$$

β_1 = 0.8 *Table 22.2.2.4.3 SBC 304*

CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
301 1.0 DL	0.1	246.057	7.4	14.38	0.0	-0.2	655.2	91.26	83.48	1	0.16
302 1.0 (DL + TL1) +	0.1	375.623	-2.1	7.55	0.016	-0.2	784.8	106.73	102.55	1	0.19
303 1.0 (DL + TL1) +	0.1	375.623	11.3	17.16	0.016	-0.2	784.8	109.29	99.99	1	0.20
304 1.0 DL+ 0.75 WL	0.0	527.7	40.6	87.35	0.009	-0.1	936.9	148.25	101.58	1	0.27
305 1.0 DL+ 0.75 WL	0.1	158.762	-30.0	-61.63	0.021	-0.3	567.9	92.27	59.19	1	0.17
306 1.0 DL+ 0.75 WL	7.3	343.231	5.3	12.86	0.0	-24.5	752.4	111.60	89.04	1	0.20
307 1.0 DL+ 0.75 WL	-7.2	343.231	5.3	12.86	0.015	24.1	752.4	111.46	89.19	1	0.20
308 1.0 DL + 0.525 L	0.1	343.214	5.3	12.86	0.015	-0.3	752.4	103.84	96.80	1	0.19
309 1.0 DL + 0.525 L	0.1	343.249	5.3	12.86	0.015	-0.2	752.4	103.81	96.84	1	0.19
310 1.0 DL + 0.525 L	0.1	350.006	6.4	15.23	0.016	-0.2	759.2	105.36	97.09	1	0.19
311 1.0 DL + 0.525 L	0.1	336.457	4.2	10.49	0.0	-0.2	745.6	102.29	96.55	1	0.18
312 0.6 DL + 1.0 WL	0.0	393.593	51.5	107.95	-0.002	0.0	802.8	135.83	78.24	1	0.25
313 0.6 DL + 1.0 WL	0.1	-98.325	-42.6	-90.69	0.015	-0.2	310.9	65.71	17.19	1	0.12
314 0.6 DL + 1.0 WL	9.7	147.634	4.5	8.63	0.0	-32.5	556.8	86.95	61.54	1	0.16
315 0.6 DL + 1.0 WL	-9.6	147.634	4.5	8.63	0.007	32.3	556.8	86.88	61.60	1	0.16
316 0.6 DL + 0.7 EL	0.1	147.611	4.5	8.63	0.007	-0.2	556.8	76.60	71.88	1	0.14
317 0.6 DL + 0.7 EL	0.0	147.657	4.5	8.63	0.007	0.0	556.8	76.56	71.93	1	0.14
318 0.6 DL + 0.7 EL	0.0	156.667	6.0	11.79	0.008	-0.1	565.9	78.63	72.27	1	0.14
319 0.6 DL + 0.7 EL	0.0	138.601	3.0	5.47	0.0	-0.1	547.8	74.53	71.55	1	0.13

Max P_{max}	=	148.25	KN/m^2
Max P_{min}	=	17.19	KN/m^2
Resultant F.O.S	=	0.27	

BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
401 1.4 DL	0.1	344.5	10.4	20.13	0.0	-0.24	51.38	40.49
402 1.4 (DL + TL1) +	0.1	564.7	-3.8	9.96	0.024	-0.37	78.07	72.52
403 1.4 (DL + TL1) +	0.1	564.7	15.0	23.41	0.024	-0.37	81.66	68.94
404 1.2 DL+ 1.6 WL	0.0	753.6	82.8	175.16	0.002	-0.03	147.20	53.76
405 1.2 DL+ 1.6 WL	0.1	-33.5	-67.8	-142.67	0.03	-0.46	33.73	-42.66
406 1.2 DL+ 1.6 WL	15.6	360.1	7.5	16.24	0.016	-52.10	69.01	27.00
407 1.2 DL+ 1.6 WL	-15.4	360.1	7.5	16.24	0.016	51.61	68.85	27.16
408 1.2 DL + 1.0 EL	0.1	360.0	7.5	16.24	0.016	-0.34	52.44	43.56
409 1.2 DL + 1.0 EL	0.0	360.1	7.5	16.24	0.016	-0.14	52.39	43.63
410 1.2 DL + 1.0 EL	0.1	373.0	9.6	20.76	0.017	-0.27	55.35	44.11
411 1.2 DL + 1.0 EL	0.1	347.1	5.4	11.73	0.015	-0.22	49.49	43.09
412 0.9 DL + 1.6 WL	0.0	615.0	82.0	171.85	-0.003	0.06	127.84	36.15
413 0.9 DL + 1.6 WL	0.1	-172.1	-68.6	-145.97	0.024	-0.37	16.10	-61.99
414 0.9 DL + 1.6 WL	15.5	221.5	6.7	12.94	0.01	-52.01	49.62	9.43
415 0.9 DL + 1.6 WL	-15.4	221.5	6.7	12.94	0.01	51.70	49.52	9.53
416 0.9 DL + 1.0 EL	0.1	221.4	6.7	12.94	0.011	-0.26	33.06	25.99
417 0.9 DL + 1.0 EL	0.0	221.5	6.7	12.94	0.01	-0.06	33.00	26.06
418 0.9 DL + 1.0 EL	0.1	234.4	8.8	17.46	0.012	-0.18	35.96	26.54
419 0.9 DL + 1.0 EL	0.0	208.5	4.6	8.43	0.009	-0.13	30.10	25.52

Max Net factored B.P = 147.20 KN/m^2

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DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 0.9 m

Moment about Z: Muz = 59.61 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 919.61 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 432.8 mm^2

Length of cantilever along Z = 1.25 m

Moment about X: Mux = 115.00 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 841.17 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 881.3 mm^2

Provide Bottom Reinforcement = ***16 mm dia @ 200 c/c***

Effective Depth Along X = 367 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement $A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

or greater of

A_{stmin} = 0.0014bd

A_{stmin} = 660.6 mm^2

A_{req} = 432.8 mm^2

Required Spacing = **304.4** mm

Effective Depth Along X = 351 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement $A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

or greater of

A_{stmin} = 0.0014bd

A_{stmin} = 631.8 mm^2

A_{req} = 881.3 mm^2

Required Spacing = **228.2** mm

Ast prov = 1005.3 mm^2

Top Reinforcement

Length of cantilever along X = 0.9 m

Moment about Z: Muz = 0.00 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 929.66 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 0.0 mm^2

Length of cantilever along Z = 1.25 m

Moment about X: Mux = 0.00 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 870.17 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 0.0 mm^2

Provide top Reinforcement = ***12 mm dia @ 200 c/c***

Effective Depth Along X = 369 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% o

or greater of A_{stmin} = 50% of 0.0014bd more prov ir bottom

Table 24.4.3.2 SBC 304

A_{stmin} = 332.1 mm^2

A_{req} = 0.0 mm^2

Required Spacing = **340.6** mm

Effective Depth Along Z = 357 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% o

or greater of A_{stmin} = 50% of 0.0014bd more prov ir bottom

Table 24.4.3.2 SBC 304

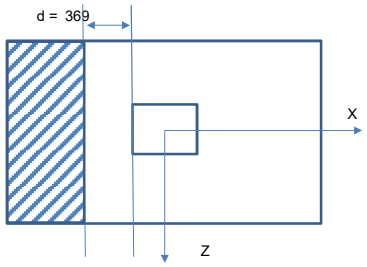
A_{stmin} = 321.3 mm^2

A_{req} = 0.0 mm^2

Required Spacing = **352.0** mm

Ast prov = 565.5 mm^2

ONE WAY SHEAR CKECK



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Along X Direction Cantilever

L	=	0.531	m		
Reinforcement ratio	ρ_w	=	0.0027		
Effective Depth Along X	d	=	369	mm	
Shear Force at d distance	V_u	=	78.16	KN	
Allowable shear strength	φVc	=	274.78	KN	O.K
Utility ratio		=	0.28		

Least of

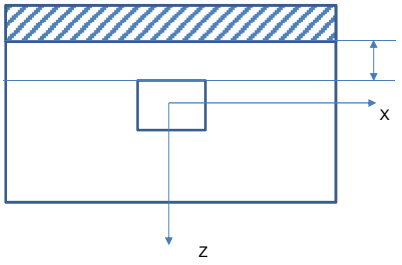
i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ — Table 22.5.5.1 SBC 304

λ = 1 Normal concrete
Mu = 20.75 KNm
φ = 0.75 for shear
φVc = 279.8 KN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right)b_w d$
φVc = 274.8 KN — Table 22.5.5.1 SBC 304

iii) $V_c = 0.29\lambda\sqrt{f_c'}b_w d$
= 474.8 KN

ONE WAY SHEAR CKECK



Along Z Direction Cantilever

L	=	0.893	m		
Reinforcement ratio	ρ_w	=	0.0028		
Effective Depth Along Z	d	=	357	mm	
Shear Force at d distance	V_u	=	131.45	KN	
Allowable shear strength	φVc	=	263.69	KN	O.K
Utility ratio		=	0.50		

Least of

i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ — Table 22.5.5.1 SBC 304

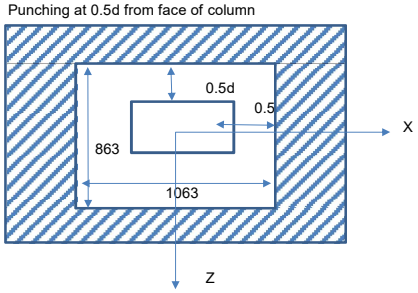
λ = 1 Normal concrete
Mu = 58.69 KNm
φ = 0.75 for shear
φVc = 263.7 KN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right)b_w d$
φVc = 266.3 KN

iii) $V_c = 0.29\lambda\sqrt{f_c'}b_w d$
φVc = 459.4 KN

TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



Total area = 7.5 m2
Punching area = 0.92 m2
λ = 1
β = 1.4
αs = 40
b0 = 3852.0 mm

Allowable shear stress — least of — Table 22.6.5.2 SBC 304

$v_c = 0.33\lambda\sqrt{f_c'}$ = 1.95 N/mm²

$v_c = 0.17\left(1+\frac{2}{\beta}\right)\lambda\sqrt{f_c'}$ = 2.44 N/mm²

$v_c = 0.083\left(2+\frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$ = 2.83 N/mm²

Design vc = 1.95 N/mm²

Check at 0.5d from column face

Perimeter at 0.5d from column face	u	=	3.85	m	
Depth at section	d	=	363	mm	
Shear Force at section	Vu	=	968.9	KN	
Allowable shear strength	φVc	=	2047.40	KN	O.K
Utility ratio			0.47		

Vc = $v_c b_0 d$
= 2729.86 KN
φ = 0.75 for shear
ϕV_n = φVc
φVc = 2047.4 KN

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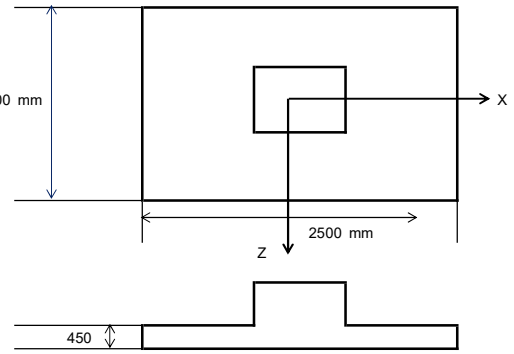
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Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

C01

7.3.4 Design of foundation F3 for supporting VGTU pedestals along grid A-1 & A-2.

Finished Ground level	F.G.L	=	-0.500	m	Staad Node no	=	2
Bottom of foundation	B.O.F	=	-3.500	m	Weight of soil	=	270.35 KN
Staad support reaction level		=	-3.500	m	Weight of foundation	=	67.50 KN
Ground Water Level	G.W.L	=	-10.000	m	Uplift	=	0.00 KN
Surcharge	Ws	=	0.000	KN/m^2	Surcharge	=	0 KN
Length of footing	Lz	=	2.50	m	Pressure on top for tension	=	54.1 KN/m^2
Width of footing	Lx	=	2.50	m			
Thickness of footing	D	=	0.45	m			
Depth of pedestal	Bx	=	0.60	m			
Width of pedestal	Bz	=	0.60	m			
Height of Load point from bottom of footing	H	=	0.000	m			
Section Modulus of Pedestal	Zxx	=	2.60	m^3			
Section Modulus of Pedestal	Zzz	=	2.60	m^3			
Allowable Net Bearing Capacity		=	500	KN/m^2			
Gross Bearing Capacity		=	554	KN/m^2			
Grade of concrete		=	35	N/mm^2			
Grade of steel		=	420	N/mm^2			
Clear cover		=	75	mm			
Top Reinfocement		=	12	mm			
Bottom Reinforcement		=	16	mm			
Co-efficient of friction between soil & concrete		=	0.3				
Unit wt of soil		=	18	KN/m^3			
Unit wt of concrete		=	24	KN/m^3			

*inputs



$$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7}$$

β_1 = 0.8 *Table 22.2.2.4.3 SBC 304*

CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
301 1.0 DL	-14.1	375.571	2.5	8.25	0.2	36.0	713.4	131.14	97.15	1	0.24
302 1.0 (DL + TL1) +	-14.7	588.261	-13.8	-14.60	-0.792	37.8	926.1	168.29	128.07	1	0.30
303 1.0 (DL + TL1) +	-14.8	630.643	12.2	25.15	0.983	37.7	968.5	179.08	130.84	1	0.32
304 1.0 DL+ 0.75 W	-14.0	632.225	20.5	62.96	0.706	35.7	970.1	193.11	117.31	1	0.35
305 1.0 DL+ 0.75 W	-15.2	469.739	-20.4	-50.93	-0.459	38.9	807.6	163.69	94.74	1	0.30
306 1.0 DL+ 0.75 W	0.5	415.538	1.1	6.81	0.1	-4.0	753.4	124.68	116.40	1	0.23
307 1.0 DL+ 0.75 W	-29.7	686.425	-1.0	5.23	0.193	78.6	1024.3	196.06	131.71	1	0.35
308 1.0 DL + 0.525	-14.2	546.622	0.1	6.07	0.127	36.2	884.5	157.74	125.29	1	0.28
309 1.0 DL + 0.525	-15.0	555.341	0.0	5.96	0.121	38.4	893.2	159.95	125.87	1	0.29
310 1.0 DL + 0.525	-14.6	553.896	0.7	7.95	0.159	37.2	891.7	160.03	125.33	1	0.29
311 1.0 DL + 0.525	-14.6	548.067	-0.7	4.09	0.1	37.3	885.9	157.65	125.84	1	0.28
312 0.6 DL + 1.0 WL	-7.7	333.666	28.8	80.88	0.902	19.5	671.5	146.00	68.89	1	0.26
313 0.6 DL + 1.0 WL	-9.3	117.019	-25.7	-70.98	-0.652	23.7	454.9	109.13	36.43	1	0.20
314 0.6 DL + 1.0 WL	11.7	44.751	2.9	6.00	0.0	-33.4	382.6	76.36	46.08	1	0.14
315 0.6 DL + 1.0 WL	-28.6	405.934	0.2	3.90	0.217	76.6	743.8	149.93	88.08	1	0.27
316 0.6 DL + 0.7 EL	-7.9	219.53	1.6	5.02	0.129	20.1	557.4	98.83	79.53	1	0.18
317 0.6 DL + 0.7 EL	-9.0	231.156	1.5	4.88	0.121	23.1	569.0	101.78	80.30	1	0.18
318 0.6 DL + 0.7 EL	-8.4	229.229	2.4	7.53	0.172	21.5	567.1	101.90	79.57	1	0.18
319 0.6 DL + 0.7 EL	-8.5	221.456	0.6	2.38	0.1	21.7	559.3	98.72	80.26	1	0.18

C01

Max P_{max}	=	196.06	KN/m^2
Max P_{min}	=	36.43	KN/m^2
Resultant F.O.S	=	0.35	

BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
401 1.4 DL	-19.8	525.8	3.5	11.55	0.3	50.41	107.92	60.33
402 1.4 (DL + TL1) +	-20.8	893.7	-20.3	-21.34	-1.143	53.38	171.69	114.31
403 1.4 (DL + TL1) +	-21.0	953.1	16.1	34.31	1.343	53.26	186.12	118.86
404 1.2 DL+ 1.6 WL	-16.0	740.9	45.0	129.89	1.437	40.74	184.07	53.03
405 1.2 DL+ 1.6 WL	-18.6	394.3	-42.2	-113.07	-1.05	47.39	124.71	1.47
406 1.2 DL+ 1.6 WL	14.9	278.7	3.5	10.09	0.045	-43.98	65.35	23.83
407 1.2 DL+ 1.6 WL	-49.5	856.6	-0.8	6.73	0.342	132.11	190.36	83.74
408 1.2 DL + 1.0 EL	-16.5	559.3	1.5	8.51	0.199	41.94	108.87	70.12
409 1.2 DL + 1.0 EL	-18.0	575.9	1.3	8.31	0.187	46.19	113.08	71.22
410 1.2 DL + 1.0 EL	-17.2	573.2	2.7	12.09	0.261	43.98	113.24	70.18
411 1.2 DL + 1.0 EL	-17.3	562.1	0.1	4.73	0.125	44.15	108.70	71.16
412 0.9 DL + 1.6 WL	-11.4	511.3	45.9	128.91	1.43	29.08	142.48	21.14
413 0.9 DL + 1.6 WL	-14.0	164.7	-41.3	-114.06	-1.056	35.73	83.87	-31.17
414 0.9 DL + 1.6 WL	19.5	49.1	4.4	9.11	0.039	-55.64	32.71	-17.01
415 0.9 DL + 1.6 WL	-44.9	627.0	0.1	5.75	0.335	120.45	148.77	51.86
416 0.9 DL + 1.0 EL	-11.9	329.7	2.4	7.53	0.193	30.28	67.27	38.23
417 0.9 DL + 1.0 EL	-13.5	346.3	2.2	7.33	0.181	34.53	71.48	39.34
418 0.9 DL + 1.0 EL	-12.7	343.6	3.6	11.11	0.255	32.32	71.65	38.29
419 0.9 DL + 1.0 EL	-12.7	332.5	1.0	3.75	0.119	32.49	67.11	39.28

Max Net factored B.P = 190.36 KN/m^2

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	17-03-2020	30-04-2020	13/10/2023
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Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

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DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 0.95 m

Moment about Z: Muz = 85.90 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 919.61 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 626.0 mm^2

Length of cantilever along Z = 0.95 m

Moment about X: Mux = 85.90 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 841.17 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 655.3 mm^2

Provide Bottom Reinforcement = ***16 mm dia @ 200 c/c***

Effective Depth Along X = 367 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement greater of

$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

or

A_{stmin} = 0.0014bd

A_{stmin} = 660.6 mm^2

A_{req} = 626.0 mm^2

Required Spacing = **304.4** mm

Effective Depth Along X = 351 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement greater of

$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

or

A_{stmin} = 0.0014bd

A_{stmin} = 631.8 mm^2

A_{req} = 655.3 mm^2

Required Spacing = **306.8** mm

Ast prov = 1005.3 mm^2

Top Reinforcement

Length of cantilever along X = 0.95 m

Moment about Z: Muz = 0.00 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 929.66 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 0.0 mm^2

Length of cantilever along Z = 0.95 m

Moment about X: Mux = 0.00 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 870.17 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 0.0 mm^2

Provide top Reinforcement = ***12 mm dia @ 200 c/c***

Effective Depth Along X = 369 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement greater of

A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% o

or

A_{stmin} = 50% of 0.0014bd more prov ir bottom

Table 24.4.3.2 SBC 304

A_{stmin} = 332.1 mm^2

A_{req} = 0.0 mm^2

Required Spacing = **340.6** mm

Effective Depth Along Z = 357 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement greater of

A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% o

or

A_{stmin} = 50% of 0.0014bd more prov ir bottom

Table 24.4.3.2 SBC 304

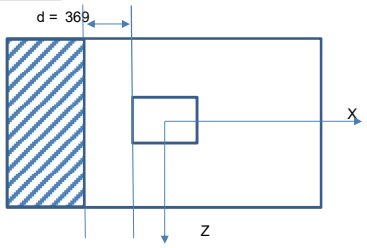
A_{stmin} = 321.3 mm^2

A_{req} = 0.0 mm^2

Required Spacing = **352.0** mm

Ast prov = 565.5 mm^2

ONE WAY SHEAR CKECK



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Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T	0	1	2
Date	17-03-2020	30-04-2020	13/10/2023	Date	17-03-2020	30-04-2020	13/10/2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED C01	AS BUILT	Revision Tex	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED C01	AS BUILT
Prepared By	APL	APL	APL	Prepared By	APL	APL	APL
Checked	SBY	SBY	SBY	Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

Along X Direction Cantilever

L	=	0.581	m		
Reinforcement ratio ρ_w	=	0.0027			
Effective Depth Along X d	=	369	mm		
Shear Force at d distance V_u	=	110.60	KN		
Allowable shear strength φVc	=	274.78	KN	O.K	
Utility ratio	=	0.40			

Least of

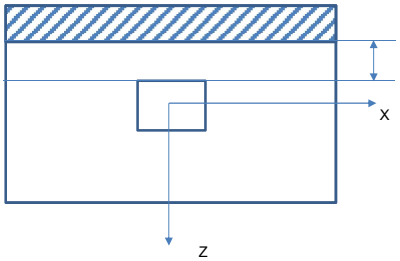
i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ — *Table 22.5.5.1 SBC 304*

λ = 1 Normal concrete
Mu = 32.13 KNm
φ = 0.75 for shear
φVc = 278.2 KN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right)b_w d$
φVc = 274.8 KN — *Table 22.5.5.1 SBC 304*

iii) $V_c = 0.29\lambda\sqrt{f_c'}b_w d$
= 474.8 KN

ONE WAY SHEAR CKECK



Along Z Direction Cantilever

L	=	0.593	m		
Reinforcement ratio ρ_w	=	0.0028			
Effective Depth Along Z d	=	357	mm		
Shear Force at d distance V_u	=	112.89	KN		
Allowable shear strength φVc	=	266.26	KN	O.K	
Utility ratio	=	0.42			

Least of

i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ — *Table 22.5.5.1 SBC 304*

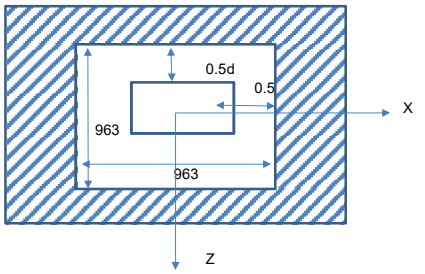
λ = 1 Normal concrete
Mu = 33.47 KNm
φ = 0.75 for shear
φVc = 268.9 KN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right)b_w d$
φVc = 266.3 KN

iii) $V_c = 0.29\lambda\sqrt{f_c'}b_w d$
φVc = 459.4 KN

TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



Total area = 6.25 m2
Punching area = 0.93 m2
λ = 1
β = 1
αs = 40
b0 = 3852.0 mm

Allowable shear stress — least of — *Table 22.6.5.2 SBC 304*

$v_c = 0.33\lambda\sqrt{f_c'}$ = 1.95 N/mm²

$v_c = 0.17\left(1+\frac{2}{\beta}\right)\lambda\sqrt{f_c'}$ = 3.02 N/mm²

$v_c = 0.083\left(2+\frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$ = 2.83 N/mm²

Design vc = 1.95 N/mm²

Check at 0.5d from column face

Perimeter at 0.5d from column face u	=	3.85	m		
Depth at section d	=	363	mm		
Shear Force at section Vu	=	1013.2	KN		
Allowable shear strength φVc	=	2047.40	KN	O.K	
Utility ratio		0.49			

Vc = $v_c b_0 d$ = 2729.86 KN
φ = 0.75 for shear
φV_n = φVc
φVc = 2047.4 KN

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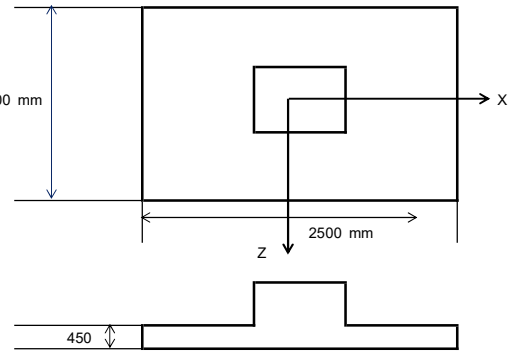
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7.3.5 Design of foundation F3 for supporting VGTU pedestals along grid C-1 & C-2.

Finished Ground level	F.G.L	=	-0.500	m	Staad Node no	=	6	
Bottom of foundation	B.O.F	=	-3.500	m	Weight of soil	=	270.35	KN
Staad support reaction level		=	-3.500	m	Weight of foundation	=	67.50	KN
Ground Water Level	G.W.L	=	-10.000	m	Uplift	=	0.00	KN
Surcharge	Ws	=	0.000	KN/m^2	Surcharge	=	0	KN
Length of footing	Lz	=	2.50	m	Pressure on top for tension	=	54.1	KN/m^2
Width of footing	Lx	=	2.50	m				
Thickness of footing	D	=	0.45	m				
Depth of pedestal	Bx	=	0.60	m				
Width of pedestal	Bz	=	0.60	m				
Height of Load point from bottom of footing	H	=	0.000	m				
Section Modulus of Pedestal	Zxx	=	2.60	m^3				
Section Modulus of Pedestal	Zzz	=	2.60	m^3				
Allowable Net Bearing Capacity		=	500	KN/m^2				
Gross Bearing Capacity		=	554	KN/m^2				
Grade of concrete		=	35	N/mm^2				
Grade of steel		=	420	N/mm^2				
Clear cover		=	75	mm				
Top Reinfocement		=	12	mm				
Bottom Reinforcement		=	16	mm				
Co-efficient of friction between soil & concrete		=	0.3					
Unit wt of soil		=	18	KN/m^3				
Unit wt of concrete		=	24	KN/m^3				

*inputs



$$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7}$$

β_1 = 0.8 *Table 22.2.2.4.3 SBC 304*

CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
301 1.0 DL	-15.6	62.073	3.5	4.96	0.3	36.4	399.9	79.88	48.10	1	0.14
302 1.0 (DL + TL1) +	-15.1	114.367	2.9	3.48	0.386	36.8	452.2	87.83	56.88	1	0.16
303 1.0 (DL + TL1) +	-15.0	115.301	3.0	3.54	0.544	36.7	453.2	87.94	57.07	1	0.16
304 1.0 DL+ 0.75 W	-14.1	122.687	11.8	28.39	2.005	34.7	460.5	97.90	49.48	1	0.18
305 1.0 DL+ 0.75 W	-16.3	80.601	-5.7	-20.65	-1.167	38.7	418.5	89.73	44.17	1	0.16
306 1.0 DL+ 0.75 W	6.6	114.243	3.0	3.65	0.5	-10.2	452.1	77.66	67.01	1	0.14
307 1.0 DL+ 0.75 W	-37.0	89.045	3.2	4.09	0.375	83.5	426.9	101.96	34.65	1	0.18
308 1.0 DL + 0.525	-14.6	101.993	3.1	3.86	0.421	35.4	439.8	85.44	55.31	1	0.15
309 1.0 DL + 0.525	-15.8	101.295	3.1	3.88	0.417	37.9	439.1	86.32	54.20	1	0.16
310 1.0 DL + 0.525	-15.2	102.816	3.5	4.96	0.44	36.6	440.7	86.47	54.54	1	0.16
311 1.0 DL + 0.525	-15.2	100.472	2.7	2.79	0.4	36.7	438.3	85.30	54.96	1	0.15
312 0.6 DL + 1.0 WL	-7.9	65.301	13.8	35.67	2.282	19.2	403.2	85.57	43.44	1	0.15
313 0.6 DL + 1.0 WL	-10.8	9.187	-9.6	-29.72	-1.947	24.5	347.0	76.36	34.69	1	0.14
314 0.6 DL + 1.0 WL	19.7	54.042	2.0	2.68	0.2	-40.7	391.9	79.34	46.06	1	0.14
315 0.6 DL + 1.0 WL	-38.5	20.445	2.2	3.27	0.109	84.4	358.3	90.98	23.67	1	0.16
316 0.6 DL + 0.7 EL	-8.6	37.709	2.1	2.96	0.17	20.1	375.6	68.96	51.22	1	0.12
317 0.6 DL + 0.7 EL	-10.2	36.779	2.1	2.99	0.166	23.6	374.6	70.14	49.74	1	0.13
318 0.6 DL + 0.7 EL	-9.3	38.807	2.6	4.42	0.196	21.8	376.7	70.33	50.20	1	0.13
319 0.6 DL + 0.7 EL	-9.4	35.681	1.6	1.53	0.1	21.9	373.5	68.77	50.76	1	0.12

Max P_{max}	=	101.96	KN/m^2
Max P_{min}	=	23.67	KN/m^2
Resultant F.O.S	=	0.18	

BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
401 1.4 DL	-21.9	86.9	4.9	6.94	0.4	51.00	36.15	-8.34
402 1.4 (DL + TL1) +	-21.0	175.9	3.9	4.44	0.596	51.63	49.68	6.62
403 1.4 (DL + TL1) +	-20.8	177.3	4.0	4.52	0.818	51.42	49.84	6.88
404 1.2 DL+ 1.6 WL	-16.1	145.8	22.6	57.54	3.812	39.59	60.62	-13.97
405 1.2 DL+ 1.6 WL	-20.7	56.0	-14.8	-47.09	-2.955	48.15	45.53	-27.61
406 1.2 DL+ 1.6 WL	28.1	127.7	3.7	4.75	0.523	-56.15	43.83	-2.95
407 1.2 DL+ 1.6 WL	-65.0	74.0	4.1	5.70	0.334	143.89	69.28	-45.60
408 1.2 DL + 1.0 EL	-17.3	101.5	3.9	5.21	0.432	41.43	34.15	-1.66
409 1.2 DL + 1.0 EL	-19.6	100.2	3.9	5.24	0.425	46.31	35.83	-3.76
410 1.2 DL + 1.0 EL	-18.4	103.1	4.7	7.29	0.469	43.77	36.10	-3.11
411 1.2 DL + 1.0 EL	-18.5	98.6	3.2	3.16	0.388	43.97	33.88	-2.31
412 0.9 DL + 1.6 WL	-11.7	100.8	21.8	56.77	3.635	28.50	48.87	-16.62
413 0.9 DL + 1.6 WL	-16.3	11.0	-15.5	-47.85	-3.132	37.07	34.36	-30.85
414 0.9 DL + 1.6 WL	32.5	82.7	3.0	3.99	0.346	-67.24	40.59	-14.11
415 0.9 DL + 1.6 WL	-60.6	29.0	3.3	4.94	0.157	132.80	57.53	-48.25
416 0.9 DL + 1.0 EL	-12.9	56.5	3.1	4.44	0.255	30.34	22.40	-4.31
417 0.9 DL + 1.0 EL	-15.2	55.2	3.2	4.48	0.249	35.23	24.08	-6.41
418 0.9 DL + 1.0 EL	-14.0	58.1	3.9	6.53	0.293	32.69	24.35	-5.76
419 0.9 DL + 1.0 EL	-14.1	53.6	2.4	2.40	0.211	32.88	22.13	-4.97

Max Net factored B.P = 69.28 KN/m^2

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
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DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 0.95 m

Moment about Z: Muz = 31.26 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 919.61 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 226.1 mm^2

Effective Depth Along X = 367 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement greater of

$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

or

A_{stmin} = 0.0014bd

A_{stmin} = 660.6 mm^2

A_{req} = 226.1 mm^2

Required Spacing = **304.4** mm

Length of cantilever along Z = 0.95 m

Moment about X: Mux = 31.26 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 841.17 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 236.5 mm^2

Effective Depth Along X = 351 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement greater of

$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

or

A_{stmin} = 0.0014bd

A_{stmin} = 631.8 mm^2

A_{req} = 236.5 mm^2

Required Spacing = **318.2** mm

Provide Bottom Reinforcement = ***16 mm dia @ 200 c/c***

Ast prov = 1005.3 mm^2

Top Reinforcement

Length of cantilever along X = 0.95 m

Moment about Z: Muz = 0.00 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 929.66 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 0.0 mm^2

Effective Depth Along X = 369 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement greater of

A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% o

or more prov ir

A_{stmin} = 50% of 0.0014bd bottom

Table 24.4.3.2 SBC 304

A_{stmin} = 332.1 mm^2

A_{req} = 0.0 mm^2

Required Spacing = **340.6** mm

Length of cantilever along Z = 0.95 m

Moment about X: Mux = 0.00 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 870.17 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 0.0 mm^2

Effective Depth Along Z = 357 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement greater of

A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% o

or more prov ir

A_{stmin} = 50% of 0.0014bd bottom

Table 24.4.3.2 SBC 304

A_{stmin} = 321.3 mm^2

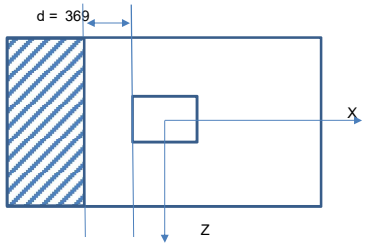
A_{req} = 0.0 mm^2

Required Spacing = **352.0** mm

Provide top Reinforcement = ***12 mm dia @ 200 c/c***

Ast prov = 565.5 mm^2

ONE WAY SHEAR CKECK



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Project Country	KINGDOM OF SAUDI ARABIA

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Along X Direction Cantilever — L = 0.581 m

Reinforcement ratio — ρ_w = 0.0027

Effective Depth Along X — d = 369 mm

Shear Force at d distance — V_u = 40.25 KN

Allowable shear strength — φVc = 274.78 KN — O.K

Utility ratio = 0.15

Least of

i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ — *Table 22.5.5.1 SBC 304*

λ = 1 Normal concrete

Mu = 11.69 KNm

φ = 0.75 for shear

φVc = 278.2 KN

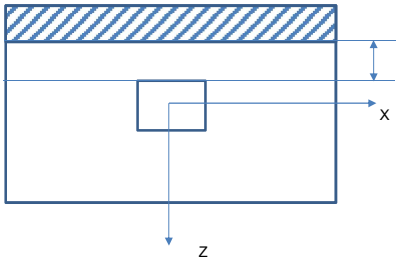
ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right)b_w d$

φVc = 274.8 KN — *Table 22.5.5.1 SBC 304*

iii) $V_c = 0.29\lambda\sqrt{f_c'}b_w d$

= 474.8 KN

ONE WAY SHEAR CKECK



Along Z Direction Cantilever — L = 0.593 m

Reinforcement ratio — ρ_w = 0.0028

Effective Depth Along Z — d = 357 mm

Shear Force at d distance — V_u = 41.08 KN

Allowable shear strength — φVc = 266.26 KN — O.K

Utility ratio = 0.15

Least of

i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ — *Table 22.5.5.1 SBC 304*

λ = 1 Normal concrete

Mu = 12.18 KNm

φ = 0.75 for shear

φVc = 268.9 KN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right)b_w d$

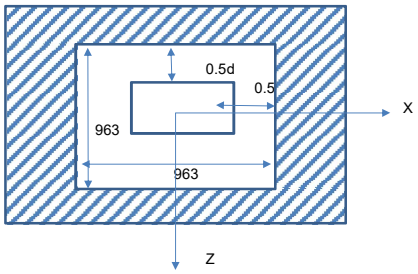
φVc = 266.3 KN

iii) $V_c = 0.29\lambda\sqrt{f_c'}b_w d$

φVc = 459.4 KN

TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



Total area = 6.25 m2

Punching area = 0.93 m2

λ = 1

β = 1

αs = 40

b0 = 3852.0 mm

Allowable shear stress — least of — *Table 22.6.5.2 SBC 304*

$v_c = 0.33\lambda\sqrt{f_c'}$ = 1.95 N/mm²

$v_c = 0.17\left(1 + \frac{2}{\beta}\right)\lambda\sqrt{f_c'}$ = 3.02 N/mm²

$v_c = 0.083\left(2 + \frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$ = 2.83 N/mm²

Design vc = 1.95 N/mm²

Check at 0.5d from column face

Perimeter at 0.5d from column face — u = 3.85 m

Depth at section — d = 363 mm

Shear Force at section — Vu = 368.7 KN

Allowable shear strength — φVc = 2047.40 KN — O.K

Utility ratio = 0.18

Vc = $v_c b_0 d$ = 2729.86 KN

φ = 0.75 for shear

φV_n = φVc

φVc = 2047.4 KN

Global Project ID	900707
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Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	17-03-2020	30-04-2020	13/10/2023
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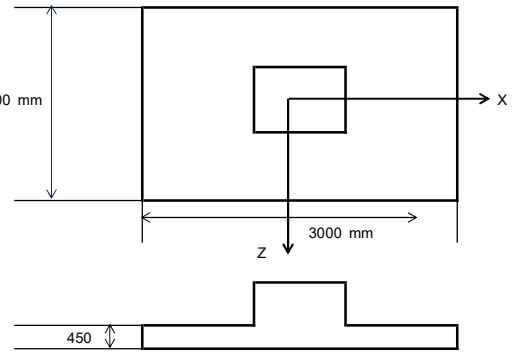
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C01

7.3.6 Design of foundation F4 for supporting VGTU pedestals along grid B-1 & B-2.

Finished Ground level	F.G.L	=	-0.500	m	Staad Node no	=	4
Bottom of foundation	B.O.F	=	-3.500	m	Weight of soil	=	327.73 KN
Staad support reaction level		=	-3.500	m	Weight of foundation	=	81.00 KN
Ground Water Level	G.W.L	=	-10.000	m	Uplift	=	0.00 KN
Surcharge	Ws	=	0.000	KN/m^2	Surcharge	=	0 KN
Length of footing	Lz	=	2.50	m	Pressure on top for tension	=	54.5 KN/m^2
Width of footing	Lx	=	3.00	m			
Thickness of footing	D	=	0.45	m			
Depth of pedestal	Bx	=	0.60	m			
Width of pedestal	Bz	=	0.60	m			
Height of Load point from bottom of footing	H	=	0.000	m			
Section Modulus of Pedestal	Zxx	=	3.13	m^3			
Section Modulus of Pedestal	Zzz	=	3.75	m^3			
Allowable Net Bearing Capacity		=	500	KN/m^2			
Gross Bearing Capacity		=	554	KN/m^2			
Grade of concrete		=	35	N/mm^2			
Grade of steel		=	420	N/mm^2			
Clear cover		=	75	mm			
Top Reinfocement		=	12	mm			
Bottom Reinforcement		=	16	mm			
Co-efficient of friction between soil & concrete		=	0.3				
Unit wt of soil		=	18	KN/m^3			
Unit wt of concrete		=	24	KN/m^3			

*inputs



$$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7}$$

β_1 = 0.8 *Table 22.2.2.4.3 SBC 304*

CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
301 1.0 DL	-19.5	151.852	4.4	8.43	0.4	41.2	560.6	88.43	61.05	1	0.16
302 1.0 (DL + TL1) +	-20.5	464.134	2.7	8.09	-0.136	43.2	872.9	130.50	102.27	1	0.24
303 1.0 (DL + TL1) +	-20.0	455.544	2.9	7.49	1.123	42.7	864.3	129.02	101.46	1	0.23
304 1.0 DL+ 0.75 W	-19.0	464.838	20.2	55.43	2.232	40.5	873.6	145.01	87.95	1	0.26
305 1.0 DL+ 0.75 W	-21.2	300.846	-13.8	-39.53	-1.297	44.6	709.6	119.15	70.07	1	0.22
306 1.0 DL+ 0.75 W	3.7	505.687	3.3	8.36	0.5	-7.4	914.4	126.56	117.28	1	0.23
307 1.0 DL+ 0.75 W	-43.8	259.997	3.0	7.54	0.484	92.4	668.7	116.22	62.11	1	0.21
308 1.0 DL + 0.525	-19.4	386.874	3.2	7.94	0.47	41.2	795.6	119.60	92.56	1	0.22
309 1.0 DL + 0.525	-20.7	378.811	3.2	7.96	0.466	43.9	787.5	119.25	90.76	1	0.22
310 1.0 DL + 0.525	-20.0	385.297	3.7	9.36	0.514	42.5	794.0	120.19	91.55	1	0.22
311 1.0 DL + 0.525	-20.1	380.388	2.7	6.54	0.4	42.6	789.1	118.66	91.77	1	0.21
312 0.6 DL + 1.0 WL	-10.2	200.439	25.3	68.37	2.587	22.0	609.2	108.97	53.48	1	0.20
313 0.6 DL + 1.0 WL	-13.1	-18.217	-20.0	-58.25	-2.118	27.5	390.5	78.03	26.10	1	0.14
314 0.6 DL + 1.0 WL	20.0	254.905	2.8	5.60	0.2	-41.8	663.6	101.42	75.55	1	0.18
315 0.6 DL + 1.0 WL	-43.4	-72.682	2.4	4.52	0.256	91.3	336.0	70.59	19.03	1	0.13
316 0.6 DL + 0.7 EL	-10.8	96.487	2.6	5.05	0.238	22.9	505.2	75.09	59.64	1	0.14
317 0.6 DL + 0.7 EL	-12.5	85.735	2.6	5.07	0.232	26.5	494.5	74.63	57.23	1	0.13
318 0.6 DL + 0.7 EL	-11.6	94.383	3.3	6.94	0.297	24.7	503.1	75.88	58.29	1	0.14
319 0.6 DL + 0.7 EL	-11.7	87.839	2.0	3.18	0.2	24.8	496.6	73.84	58.58	1	0.13

C01

Max P_{max}	=	145.01	KN/m^2
Max P_{min}	=	19.03	KN/m^2
Resultant F.O.S	=	0.26	

BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
401 1.4 DL	-27.3	212.6	6.2	11.81	0.5	57.71	47.51	9.18
402 1.4 (DL + TL1) +	-28.9	742.2	3.2	11.13	-0.16	61.04	118.80	79.12
403 1.4 (DL + TL1) +	-28.3	730.2	3.6	10.29	1.603	60.28	116.72	77.99
404 1.2 DL+ 1.6 WL	-21.4	511.1	40.7	111.10	4.285	45.95	115.96	20.35
405 1.2 DL+ 1.6 WL	-26.1	161.3	-31.8	-91.50	-3.243	54.71	65.38	-22.36
406 1.2 DL+ 1.6 WL	26.9	598.3	4.8	10.67	0.486	-56.10	98.14	61.40
407 1.2 DL+ 1.6 WL	-74.5	74.1	4.2	8.93	0.555	156.76	54.55	-34.78
408 1.2 DL + 1.0 EL	-22.5	343.9	4.5	9.78	0.525	47.74	61.71	29.99
409 1.2 DL + 1.0 EL	-25.0	328.5	4.5	9.82	0.516	52.92	61.06	26.55
410 1.2 DL + 1.0 EL	-23.7	340.9	5.4	12.48	0.609	50.23	62.84	28.07
411 1.2 DL + 1.0 EL	-23.8	331.5	3.5	7.12	0.432	50.44	59.93	28.48
412 0.9 DL + 1.6 WL	-15.2	311.6	40.2	108.89	4.116	32.71	85.11	-2.02
413 0.9 DL + 1.6 WL	-19.8	-38.3	-32.3	-93.71	-3.412	41.48	35.95	-46.15
414 0.9 DL + 1.6 WL	33.2	398.7	4.3	8.46	0.318	-69.34	74.36	31.97
415 0.9 DL + 1.6 WL	-68.2	-125.4	3.6	6.72	0.387	143.53	23.71	-57.15
416 0.9 DL + 1.0 EL	-16.3	144.3	4.0	7.57	0.356	34.51	30.87	7.62
417 0.9 DL + 1.0 EL	-18.8	129.0	4.0	7.61	0.348	39.69	30.22	4.18
418 0.9 DL + 1.0 EL	-17.5	141.3	4.9	10.27	0.441	36.99	32.00	5.69
419 0.9 DL + 1.0 EL	-17.6	132.0	3.0	4.91	0.264	37.20	29.09	6.11

Max Net factored B.P = 118.80 KN/m^2

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Approved	PRI	PRI	PRI

Outotec / L&T	0	1	D01
Date	17-03-2020	30-04-2020	13/10/2023
Revision Tex	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED C01	AS BUILT
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Approved	PRI	PRI	PRI

DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 1.2 m

Moment about Z: Muz = 85.53 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 919.61 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 623.3 mm^2

Length of cantilever along Z = 0.95 m

Moment about X: Mux = 53.61 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 841.17 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 406.9 mm^2

Provide Bottom Reinforcement = ***16 mm dia @ 200 c/c***

Effective Depth Along X = 367 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of

$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

or

A_{stmin} = 0.0014bd

A_{stmin} = 660.6 mm^2

A_{req} = 623.3 mm^2

Required Spacing = **304.4** mm

Effective Depth Along X = 351 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of

$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

or

A_{stmin} = 0.0014bd

A_{stmin} = 631.8 mm^2

A_{req} = 406.9 mm^2

Required Spacing = **318.2** mm

Ast prov = 1005.3 mm^2

Top Reinforcement

Length of cantilever along X = 1.2 m

Moment about Z: Muz = 0.00 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 929.66 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 0.0 mm^2

Length of cantilever along Z = 0.95 m

Moment about X: Mux = 0.00 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 870.17 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 0.0 mm^2

Provide top Reinforcement = ***12 mm dia @ 200 c/c***

Effective Depth Along X = 369 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of

A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% o

or more prov ir

A_{stmin} = 50% of 0.0014bd bottom

Table 24.4.3.2 SBC 304

A_{stmin} = 332.1 mm^2

A_{req} = 0.0 mm^2

Required Spacing = **340.6** mm

Effective Depth Along Z = 357 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of

A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% o

or more prov ir

A_{stmin} = 50% of 0.0014bd bottom

Table 24.4.3.2 SBC 304

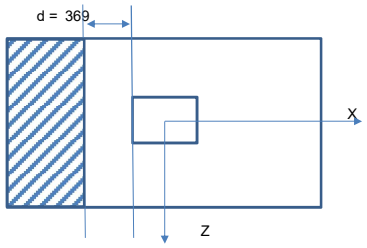
A_{stmin} = 321.3 mm^2

A_{req} = 0.0 mm^2

Required Spacing = **352.0** mm

Ast prov = 565.5 mm^2

ONE WAY SHEAR CKECK



Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T	0	1	2
Date	17-03-2020	30-04-2020	13/10/2023	Date	17-03-2020	30-04-2020	13/10/2023
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Checked	SBY	SBY	SBY	Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

Along X Direction Cantilever

L	=	0.831	m		
Reinforcement ratio ρ_w	=	0.0027			
Effective Depth Along X d	=	369	mm		
Shear Force at d distance V_u	=	98.72	KN		
Allowable shear strength φVc	=	273.35	KN	O.K	
Utility ratio	=	0.36			

Least of

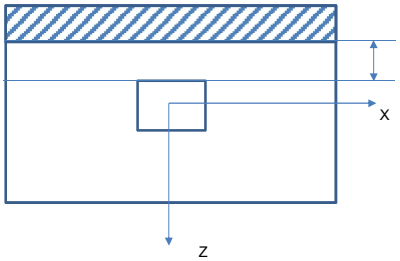
i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ — Table 22.5.5.1 SBC 304

λ = 1 Normal concrete
Mu = 41.02 KNm
φ = 0.75 for shear
φVc = 273.3 KN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right)b_w d$
φVc = 274.8 KN — Table 22.5.5.1 SBC 304

iii) $V_c = 0.29\lambda\sqrt{f_c'}b_w d$
= 474.8 KN

ONE WAY SHEAR CKECK



Along Z Direction Cantilever

L	=	0.593	m		
Reinforcement ratio ρ_w	=	0.0028			
Effective Depth Along Z d	=	357	mm		
Shear Force at d distance V_u	=	70.45	KN		
Allowable shear strength φVc	=	266.26	KN	O.K	
Utility ratio	=	0.26			

Least of

i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ — Table 22.5.5.1 SBC 304

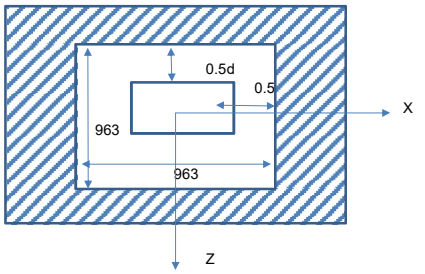
λ = 1 Normal concrete
Mu = 20.89 KNm
φ = 0.75 for shear
φVc = 268.9 KN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right)b_w d$
φVc = 266.3 KN

iii) $V_c = 0.29\lambda\sqrt{f_c'}b_w d$
φVc = 459.4 KN

TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



Total area = 7.5 m2
Punching area = 0.93 m2
λ = 1
β = 1
αs = 40
b0 = 3852.0 mm

Allowable shear stress least of — Table 22.6.5.2 SBC 304

$v_c = 0.33\lambda\sqrt{f_c'}$ = 1.95 N/mm²

$v_c = 0.17\left(1+\frac{2}{\beta}\right)\lambda\sqrt{f_c'}$ = 3.02 N/mm²

$v_c = 0.083\left(2+\frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$ = 2.83 N/mm²

Design vc = 1.95 N/mm²

Check at 0.5d from column face

Perimeter at 0.5d from column face u	=	3.85	m		
Depth at section d	=	363	mm		
Shear Force at section Vu	=	780.8	KN		
Allowable shear strength φVc	=	2047.40	KN	O.K	
Utility ratio		0.38			

Vc = $v_c b_0 d$ = 2729.86 KN
φ = 0.75 for shear
φV_n = φVc
φVc = 2047.4 KN

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
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Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	17-03-2020	30-04-2020	13/10/2023
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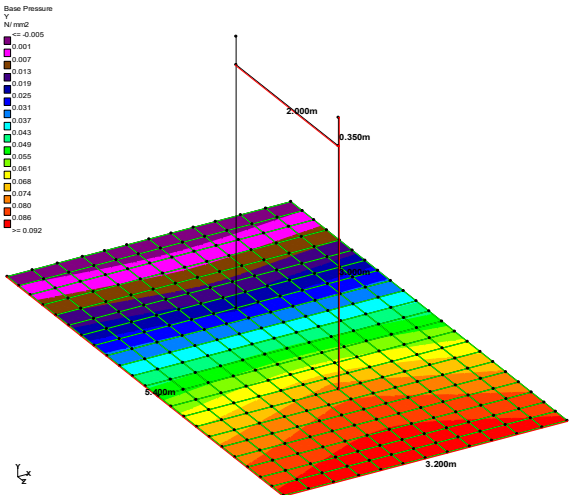
Outotec / L&T Revision No.:	0	1	2
Date	07-03-2020	30-04-2020	13/10/2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED (C01)	AS BUILT
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7.3.7 Design of combined footing for Trestle mkd. TR1, TR2 & TR3 and foundation mkd. as F2

For Calculation of Base pressure and design forces for column & foundation Raft, Refer STAAD File **CF-1 GR-J-5.STD**

Staad Supoort Nodes are 5 & 6

Finished Floor level	F.F.L	=	0.0	m	(Corresponds to RL +913.0m)
Finished Ground level	F.G.L	=	-0.5	m	
Bottom of foundation	B.O.F	=	-2.5	m	
Top of pedestal (staad support)	T.O.F	=	-0.15	m	
Ground Water Level	G.W.L	=	-10	m	
Length of footing	Lz	=	3.200	m	
Width of footing	Lx	=	5.400	m	
Thickness of footing	D	=	0.55	m	
Depth of pedestal	Bx	=	0.60	m	
Width of pedestal	Bz	=	0.60	m	
Unit wt of soil		=	18.0	KN/m^3	
Allowable Net Bearing Capacity		=	500	KN/m^2	
Gross Bearing Capacity		=	536	KN/m^2	
Soil load on footing in staad		=	26.1	KN/m^2	staad input



1) Check for Gross Base Pressure :

Maximum Base Pressure	=	120.9	KN/m^2	Load Case 304
Minimum Base Pressure	=	-5.5	KN/m^2	Load Case 312

2) Design of foundation Raft :

Slab Thickness =	550 mm	Yield Strength of Main Reinforcement , fy =	420 N/mm^2
Width =	1000 mm	Yield Strength of Shear Reinforcement , fyt =	420 N/mm^2
Clear Cover =	75 mm	Strength Reduction Factor of Main Reinforcement , Φ =	0.9
Compressive Strength of Concrete , fc' =	35 N/mm^2	Strength Reduction Factor of Shear Reinforcement , Φt =	0.75

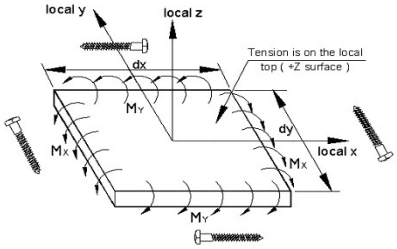
STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

(Staad Group : MX,MY)

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	33.025	kN-m	Corresponding Mxy	2.113	kN-m	*(Plate No. - 115 , L/C No. - 412)*
Min. Mx	-77.613	kN-m	Corresponding Mxy	11.633	kN-m	*(Plate No. - 24 , L/C No. - 404)*
Max. My	83.022	kN-m	Corresponding Mxy	2.705	kN-m	*(Plate No. - 114 , L/C No. - 412)*
Min. My	-131.997	kN-m	Corresponding Mxy	3.757	kN-m	*(Plate No. - 25 , L/C No. - 404)*

FOR TOP FACE REINFORCEMRNT :

Mx1 =	Mx + abs(Mxy) =	33.025 + 2.113 =	35.138	kN-m
My1 =	My + abs(Mxy) =	83.022 + 2.705 =	85.727	kN-m
Mx2 =	Mx+abs(Mxy^2/My) =	33.025 + 0.054 =	33.079	kN-m
My2 =	My+abs(Mxy^2/Mx) =	83.022 + 0.151 =	83.173	kN-m



DESIGN MOMENT CALCULATION FOR TOP FACE REINFORCEMENT :

Mxd =	35.138 kN-m
Myd =	85.727 kN-m

ALONG LOCAL X - DIRECTION :

Mu =	35.138 kN-m						
Φ*Kn =	Mu/bd^2				deff. =	469 mm	
Mu/bd^2 =	35.138X10^6/(1000X469^2) =	0.16	N/mm^2	<	6.9	N/mm^2	
Pt =	0.0004	Reqd. Ast =	0.0004X1000X469/100 =	1.88	cm^2		
Provided	12 mm @	200 C/C	(Provide Ast =	5.65	cm^2)		
			Total Provide Ast =	5.65	cm^2	O.K.	
			Total Provide Pt =	0.0012			

ALONG LOCAL Y - DIRECTION :

Mu =	85.727 kN-m						
Φ*Kn =	Mu/bd^2				deff. =	457 mm	
Mu/bd^2 =	85.727X10^6/(1000X457^2) =	0.39	N/mm^2	<	6.9	N/mm^2	
Pt =	0.0010	Reqd. Ast =	0.001X1000X469/100 =	4.57	cm^2		
Provided	12 mm @	200 C/C	(Provide Ast =	5.65	cm^2)		
			Total Provide Ast =	5.65	cm^2	O.K.	
			Total Provide Pt =	0.0012			

Global Project ID	900707
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Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2		
Date	17-03-2020	30-04-2020	13/10/2023				Date	07-03-2020	30-04-2020	13/10/2023		
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Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

FOR BOTTOM FACE REINFORCEMRNT :

Mx1 = Mx - abs(Mxy) = -77.613 - 11.633 = -89.246 kN-m
My1 = My - abs(Mxy) = -131.997 - 3.757 = -135.754 kN-m
Mx2 = Mx-abs(Mxy^2/My) = -77.613 - 1.025 = -78.638 kN-m
My2 = My-abs(Mxy^2/Mx) = -131.997 - 0.182 = -131.815 kN-m

DESIGN MOMENT CALCULATION FOR BOTTOM FACE REINFORCEMENT :

Mxd = -89.246 kN-m
Myd = -135.754 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 89.246 kN-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 89.246X10^6/(1000X467^2) = 0.409 N/mm^2 deff. = 467 mm < 6.9 N/mm^2
Pt = 0.0011 Reqd. Ast = 0.0011X1000X467/100 = 5.14 cm^2
Provided 16 mm @ 200 C/C (Provide Ast = 10.05)cm^2

Total Provide Ast = 10.05 cm^2 O.K.
Total Provide Pt = 0.0022

ALONG LOCAL Y - DIRECTION :

Mu = 135.754 kN-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 135.754X10^6/(1000X451^2) = 0.667 N/mm^2 deff. = 451 mm < 6.9 N/mm^2
Pt = 0.0018 Reqd. Ast = 0.0018X1000X451/100 = 8.12 cm^2
Provided 16 mm @ 200 C/C (Provide Ast = 10.05)cm^2

Total Provide Ast = 10.05 cm^2 O.K.
Total Provide Pt = 0.0022

SHEAR CHECK

(Staad Group : SHEAR)
(FROM FACTORED LOAD COMBINATION)

ALONG LOCAL X - DIRECTION :

SQx = -0.184 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 288 , L/C No. - 404)*
Vu = 85.93 kN

Provided Pt = 0.0022
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0022X85.928X10^3X467/89.246/10^6)X1000X467/10^3
= 449.903 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0022)X1000X467/10^3
= 459.515 kN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X467/10^3
= 801.215 kN
HENCE, Vc = 449.903 kN
Φt*Vc = 0.75X449.903 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 337.427 kN > 85.928T , Hence O.K.

ALONG LOCAL Y - DIRECTION :

SQy = -0.263 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 169 , L/C No. - 404)*
Vu = 118.613 kN

Pt provided = 0.0022
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0022X118.613X10^3X451/135.754/10^3)X1000X451/10^3
= 433.551 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0022)X1000X451/10^3
= 443.772 KN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X451/10^3
= 773.764 kN
HENCE, Vc = 433.551 kN
Φt*Vc = 0.75X433.551 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 325.163 kN > 118.613T , Hence O.K.

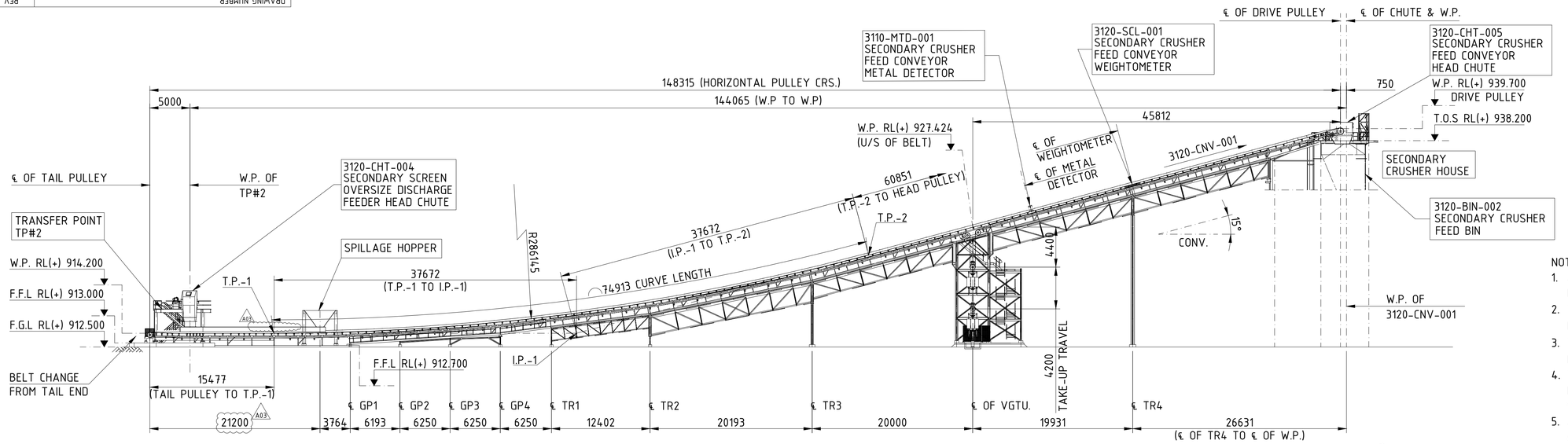
Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2		
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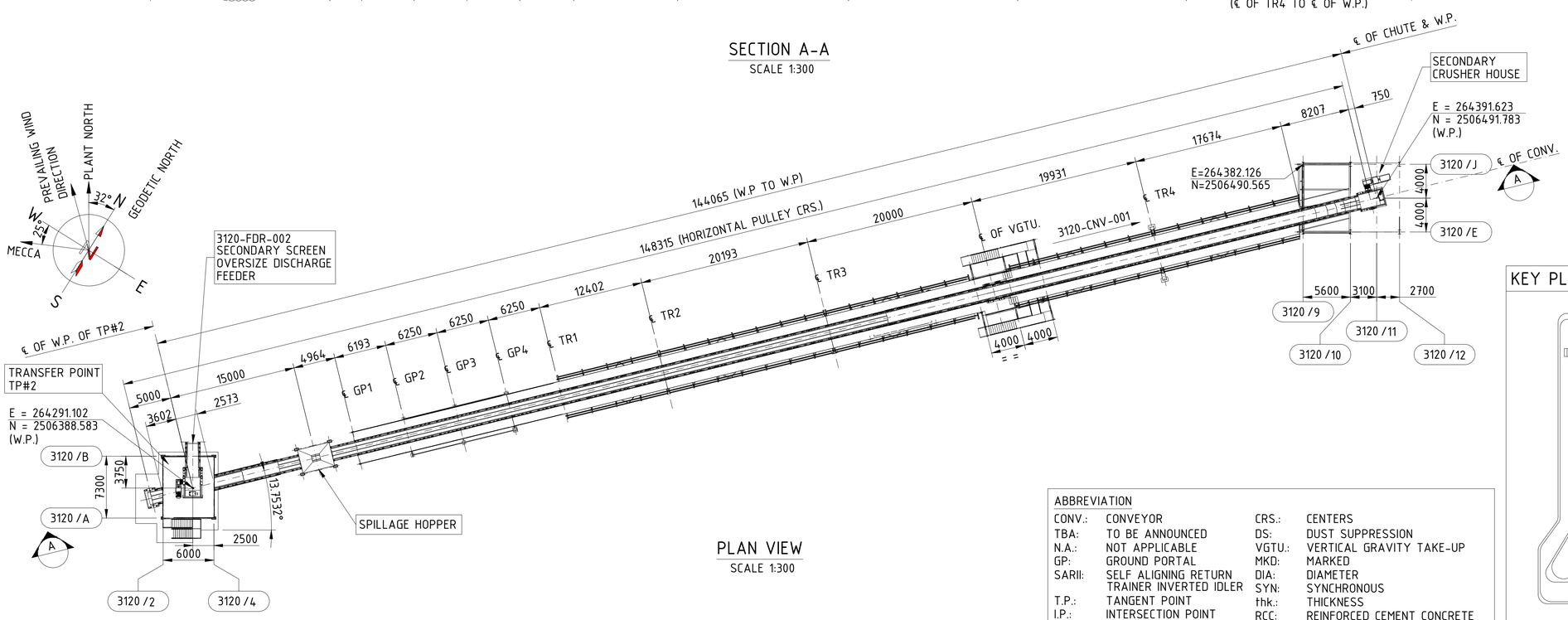
C01

ATTACHMENT 1 MECHANICAL GENERAL ARRANGEMENT DRAWINGS

DRAWING NUMBER MD-216-3120-EG-ME-LPX-1001 REV A03



SECTION A-A
SCALE 1:300

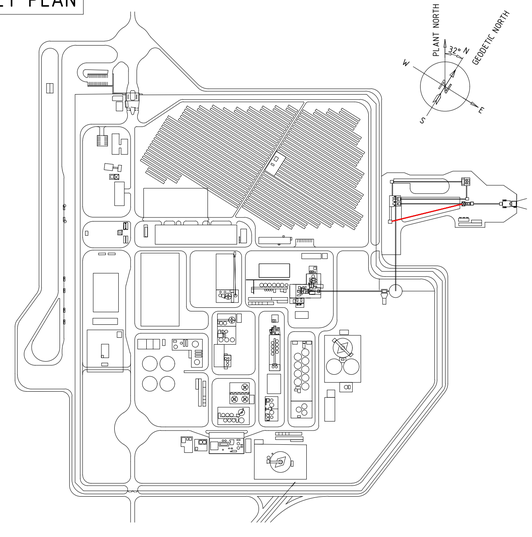


PLAN VIEW
SCALE 1:300

NOTE:
1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
2. FINISHED FLOOR LEVEL (F.F.L) FOR CRUSHING AREA = RL(+) 913.000 m
3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+) 912.500 m
4. F.F.L, F.G.L, T.O.C, = FINISH FLOOR LEVEL, FINISH GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B= TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN TENTATIVE.

CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL

KEY PLAN



ABBREVIATION

CONV.:	CONVEYOR	CRS.:	CENTERS
TBA:	TO BE ANNOUNCED	DS:	DUST SUPPRESSION
N.A.:	NOT APPLICABLE	VGTU.:	VERTICAL GRAVITY TAKE-UP
GP:	GROUND PORTAL	MKD:	MARKED
SARII:	SELF ALIGNING RETURN TRAINER INVERTED IDLER	DIA:	DIAMETER
		SYN:	SYNCHRONOUS
T.P.:	TANGENT POINT	thk.:	THICKNESS
I.P.:	INTERSECTION POINT	RCC:	REINFORCED CEMENT CONCRETE
W.P.:	WORKING POINT	B/W:	BELT WIDTH
STD.:	STANDARD	MAX:	MAXIMUM
STR.:	STRINGER	deg.:	DEGREE
SEC:	SECTION	U/S.:	UNDERSIDE
℄:	CENTER LINE	B.O.M.R:	BOTTOM OF MONORAIL
DN:	DOWNWARD	T.O.W.W.:	TOP OF WALKWAY
UP:	UPWARD	TR:	TRESTLE
TP:	TRANSFER POINT	TYP.:	TYPICAL

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		A03	15/04/2020	ISSUED FOR APPROVAL & REVISED AS MARKED (A03)	HKD	BRD	BD	BGR	
MD-216-3120-EG-ME-ELV-1003	CONV. 3120-CNV-001-ELEVATION	A02	16/03/2020	ISSUED FOR APPROVAL & REVISED AS PER EMPLOYER	AFO	BRD	BD	BGR	
MD-216-3120-EG-ME-ELV-1002	CONV. 3120-CNV-001-ELEVATION			COMMENTS DT. 27.02.20 & REVISED AS MARKED (A02)					
MD-216-3120-EG-ME-ELV-1001	CONV. 3120-CNV-001-ELEVATION	A01	29/01/2020	ISSUED FOR APPROVAL	AFO	BRD	BD	BGR	

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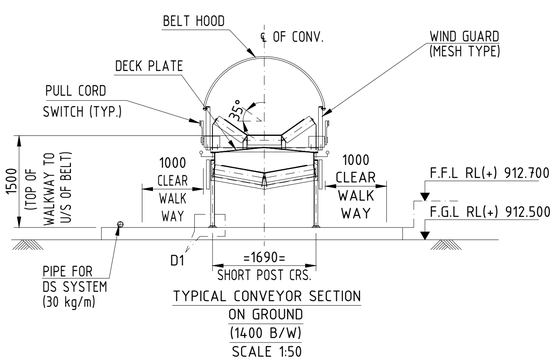
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CHECKED	BRD	28/01/2020
ENGINEER	BD	28/01/2020
APPROVED	BGR	29/01/2020
EMPLOYER		
SCALE	1:300	A1

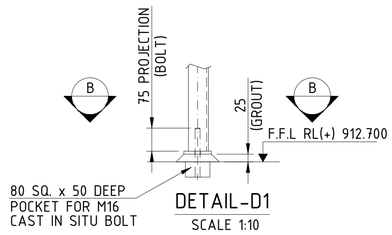
MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
SECONDARY CRUSHING
CONV. 3120-CNV-001 - PLAN VIEW

DRAWING NUMBER MD-216-3120-EG-ME-LPX-1001 REV A03

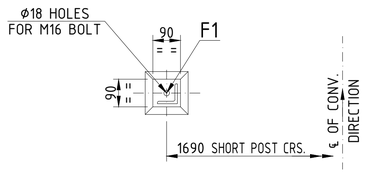
DRAWING NUMBER MD-216-3120-EG-ME-ELV-1001 REV A03



TYPICAL CONVEYOR SECTION ON GROUND (1400 B/W)
SCALE 1:50



DETAIL-D1
SCALE 1:10



VIEW B-B
SCALE 1:10

TABLE-1 : LOAD FOR CONVEYOR SHORT POST

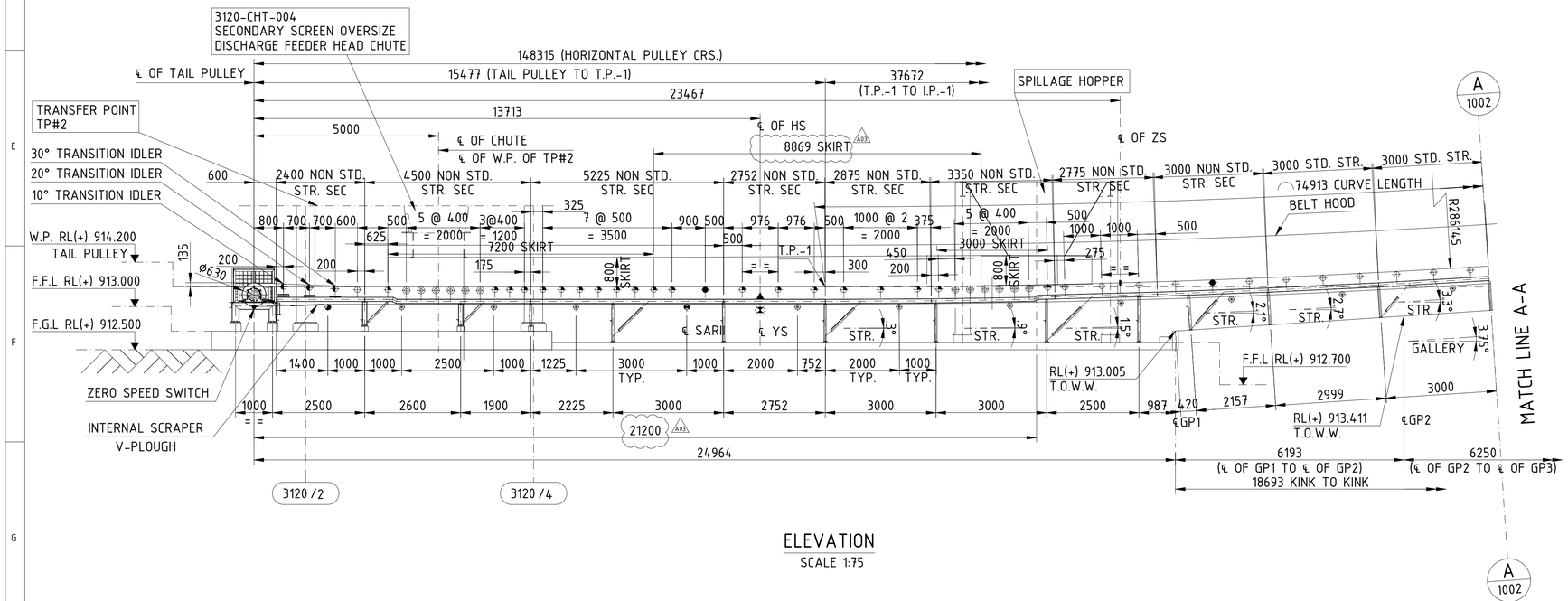
FOR CONV. 3120-CNV-001

POINT	VERTICAL LOAD IN "t"		HORIZONTAL LOAD IN "t"	
	DEAD LOAD	LIVE LOAD	ALONG	ACROSS
F1	(+)0.60	(+)0.65	(±)0.15	(±)0.08

SIGN CONVENTION

UPWARD (-)
DOWNWARD (+)
TOWARDS DRIVE PULLEY=(+)
TOWARDS TAIL PULLEY=(-)
REVERSIBLE LOAD=(±)

Note: Wind load & seismic to be considered during civil & structural design



ELEVATION
SCALE 1:75

CONVEYOR CHARACTERISTICS

	PARTICULARS	SPECIFICATION	QUANTITY
MISCELLANEOUS EQUIPMENT	TAKE-UP / TRAVEL	VGTU. / 4200 mm	
	TAKE-UP LOAD	~4760 kg.(APPROXIMATE) RCC BLOCKS, 40 kg EACH	
	SCRAPER AT DISCHARGE PULLEY	EXTERNAL / MAIN & PRE CLEANER	1 NO. + 1 NO.
	SCRAPER AT TAKE-UP	INTERNAL V-PLOUGH	1 NO.
	SCRAPER AT TAIL PULLEY	INTERNAL V-PLOUGH	1 NO.
	BELT SWAY SWITCH (ZS) ▮	@ 50 m INTERVAL	4 PAIRS
	PULL CORD SWITCH (HS) ▲	@ 30 m INTERVAL	6 PAIRS
	ZERO SPEED SWITCH (SS) ◐	AT TAIL PULLEY	1 NO.
	BELT RIP SWITCH (YS) ⋇	AFTER FEEDING ZONE	1 NO.
	DECK PLATE	3.15 mm thk. CONTINUOUS	1 LOT.
	BELT HOOD	GALVANIZED CORRUGATED SHEET	1 LOT.

ABBREVIATION

CONV.:	CONVEYOR	CRS.:	CENTERS
TBA:	TO BE ANNOUNCED	DS:	DUST SUPPRESSION
N.A.:	NOT APPLICABLE	VGTU.:	VERTICAL GRAVITY TAKE-UP
GP:	GROUND PORTAL	MKD:	MARKED
SARII:	SELF ALIGNING RETURN TRAINER INVERTED IDLER	DIA:	DIAMETER
		SYN:	SYNCHRONOUS
T.P.:	TANGENT POINT	thk.:	THICKNESS
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W.P.:	WORKING POINT	B/W:	BELT WIDTH
STD.:	STANDARD	MAX:	MAXIMUM
STR.:	STRINGER	deg.:	DEGREE
SEC:	SECTION	U/S.:	UNDERSIDE
℄:	CENTER LINE	B.O.M.R:	BOTTOM OF MONORAIL
DN:	DOWNWARD	T.O.W.W.:	TOP OF WALKWAY
UP:	UPWARD	TR:	TRESTLE
TP:	TRANSFER POINT	TYP.:	TYPICAL

CONVEYOR CHARACTERISTICS

		PARTICULARS	SPECIFICATION	QUANTITY
CONVEYOR		EQUIPMENT TAG NO.	3120-CNV-001	1 NO.
		EQUIPMENT DESCRIPTION	SECONDARY CRUSHER FEED CONVEYOR	
		LOCATION	TP-2 TO SECONDARY CRUSHER HOUSE	
		CAPACITY RATED / DESIGN (t/h)	771 / 850	
		BELT WIDTH (mm)	1400	
		BELT SPEED (m/sec)	1.673	
		TRIPPER FIXED / MOVABLE	N.A.	
		CONVEYOR LENGTH / LIFT (mm)	148315 / 25500	
MATERIAL		MATERIAL	GOLD ORE : FRESH ORE / OXIDE ORE	
		BULK DENSITY (t/m³)	1.6	
		MAX. LUMP SIZE (mm)	(-)340	
		MOISTURE CONTENT (%)	2	
		SURCHARGE ANGLE (deg.)	20	
BELTING		MAX. RUNNING TENSION (T1 kN)	84.613	
		SLACK SIDE TENSION (T2 kN)	32.257	
		TAIL TENSION (T0 kN)	32.246	
		BELT DESIGNATION	EP-1000 / 4 (HEAVY DUTY)	
		TOP / BOTTOM COVER THICKNESS (mm)	8 / 3	
		COVER GRADE / FABRIC	DIN-X / EP	
PULLEYS		DISCHARGE WITH DRIVE (MKD. x DIA.)	D-2 x 800 mm	1 NO.
		DISCHARGE WITHOUT DRIVE (MKD. x DIA.)	N.A.	
		SNUB (MKD. x DIA.)	ND-4 x 500 mm	1 NO.
		TAIL (MKD. x DIA.)	ND-3 x 630 mm	1 NO.
		TAKE-UP (MKD. x DIA.)	ND-3 x 630 mm	1 NO.
		DEFLECTOR (MKD. x DIA.)	DR-2 x 273 mm	1 NO.
		TRIPPER DISCHARGE (MKD. x DIA.)	N.A.	
		TRIPPER BEND (MKD. x DIA.)	N.A.	
		BEND (MKD. x DIA.)	ND-3 x 630 mm	2 NOS.
		LAGGING (mm)	DRIVE: 16thk CERAMIC	
			NON-DRIVE: 16thk PLAIN RUBBER	
BEARING/SLEEVE PLUMMER BLOCK		DISCHARGE WITH DRIVE	TBA	2 SETS
		DISCHARGE WITHOUT DRIVE	N.A.	
		SNUB	TBA	2 SETS
		TAIL	TBA	2 SETS
		TAKE-UP	TBA	2 SETS
		DEFLECTOR	TBA	2 SETS
		BEND	TBA	4 SETS
		TRIPPER DISCHARGE	N.A.	
		TRIPPER BEND	N.A.	
IDLER		TROUGHING ⊕ (DIA. x deg.)	152.4 x 35°	105 NOS.
		TROUGHING RETRACTABLE ◔ (DIA. x deg.)	152.4 x 45°	20 NOS.
		TROUGHING TRAINER ● (DIA. x deg.)	152.4 x 35°	10 NOS.
		RETURN ⊕ (DIA. x deg.)	152.4 x 10°	39 NOS.
		SELF ALIGNING RETURN TRAINER INVERTED IDLER ⊕ (DIA. x deg.)	152.4 x 10°	6 NOS.
		IMPACT RETRACTABLE ⊕ (DIA. x deg.)	190 x 45°	12 NOS.
		TRANSITION (ADJUSTABLE) ◐ (DIA. x deg.)	152.4 x 10°/20°/30°	(2+2+1) NOS.
		RETURN SELF CLEANING ◔ (DIA. x deg.)	190 x 0°	2 NOS.
		NON MAGNETIC TROUGHING ◔ (DIA. x deg.)	152.4 x 35°	2 NOS.
		WEIGHING IDLER ◔ (DIA. x deg.)	152.4 x 35°	4 NOS.
		SPECIAL MACHINED IDLER ◭ (DIA. x deg.)	152.4 x 35°	8 NOS.
		RETURN FLAT IDLER ◕ (DIA. x deg.)	152.4 x 0°	3 NOS.
DRIVE	MOTOR	RATING kW/rpm (SYN)	132 / 1800	1 NO.
		FRAME SIZE	TBA	
		TYPE	SQUIRREL CAGE	
		SUPPLY	400V, 60Hz	
	REDUCER	TYPE	BEVEL HELICAL	1 NO.
		MODEL NO./COOLING/RATIO	TBA	
		RATIO	45	
		HOLD BACK	YES	
	BRAKE	TYPE	ELECTRO HYDRAULIC THRUSTER	1 NO.
		MODEL NO.	TBA	
		DRAM DIAMETER	TBA	
	COUPLING LOW SPEED	TYPE	FLUID COUPLING	1 NO.
		MODEL NO.	TBA	
	COUPLING HIGH SPEED	TYPE	GEAR	1 NO.
		MODEL NO.	TBA	

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		A03	15/04/2020	ISSUED FOR APPROVAL & REVISED AS PER EMPLOYER	HKD	BRD	BD	BGR	
				COMMENTS DT. 06.04.2020 & REVISED AS MARKED A03					
MD-216-3120-EG-ME-ELV-1003	CONV. 3120-CNV-001-ELEVATION	A02	16/03/2020	ISSUED FOR APPROVAL & REVISED AS PER EMPLOYER	AFO	BRD	BD	BGR	
MD-216-3120-EG-ME-ELV-1002	CONV. 3120-CNV-001-ELEVATION			COMMENTS DT. 27.02.20 & REVISED AS MARKED A02					
MD-216-3120-EG-ME-LPX-1001	CONV. 3120-CNV-001-PLAN VIEW	A01	29/01/2020	ISSUED FOR APPROVAL	AFO	BRD	BD	BGR	

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LARSEN & TOUBRO DRAWING NUMBER		REVISION
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OUTOTEC DRAWING NUMBER	DOCUMENT ID	REVISION
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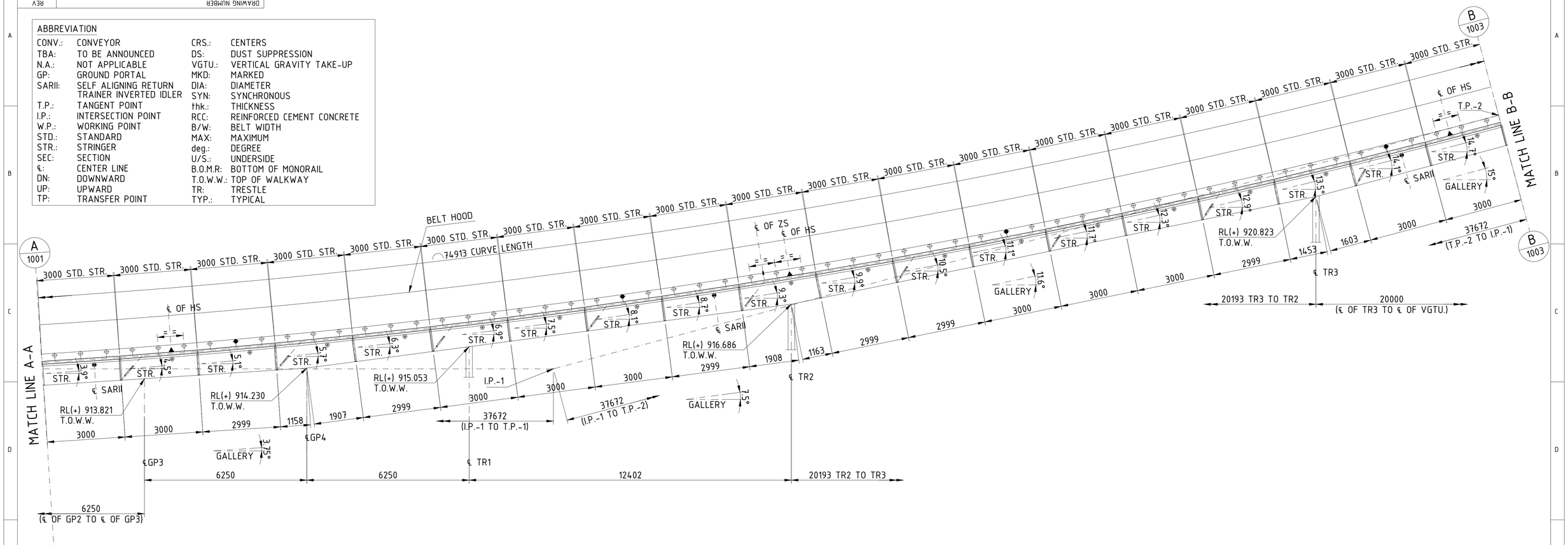
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CHECKED	BRD	28/01/2020
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APPROVED	BGR	29/01/2020
EMPLOYER		
SCALE	1:75, 1:50, 1:10	A1

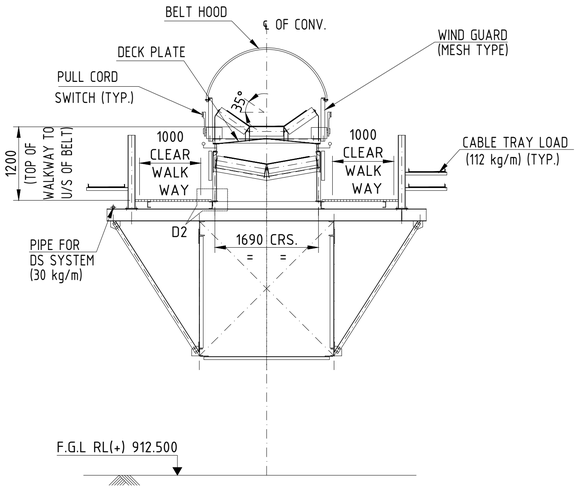
MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
SECONDARY CRUSHING
CONV. 3120-CNV-001 - ELEVATION

DRAWING NUMBER	REV
MD-216-3120-EG-ME-ELV-1001	A03

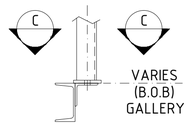
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ABBREVIATION			
CONV.:	CONVEYOR	CRS.:	CENTERS
TBA:	TO BE ANNOUNCED	DS:	DUST SUPPRESSION
N.A.:	NOT APPLICABLE	VGTU.:	VERTICAL GRAVITY TAKE-UP
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℄:	CENTER LINE	B.O.M.R:	BOTTOM OF MONORAIL
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UP:	UPWARD	TR:	TRESTLE
TP:	TRANSFER POINT	TYP.:	TYPICAL

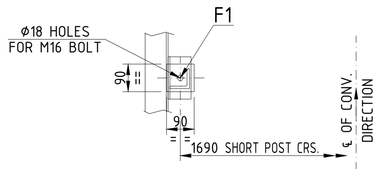




TYPICAL CONVEYOR SECTION ON GALLERY (1400 B/W)
SCALE 1:50



DETAIL-D2
SCALE 1:10



VIEW C-C
SCALE 1:10

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		A03	15/04/2020	ISSUED FOR DETAIL DESIGN	HKD	BRD	BD	BGR	
MD-216-3120-EG-ME-ELV-1003	CONV. 3120-CNV-001-ELEVATION	A02	16/03/2020	ISSUED FOR APPROVAL & REVISED AS PER EMPLOYER	AFO	BRD	BD	BGR	
MD-216-3120-EG-ME-ELV-1001	CONV. 3120-CNV-001-ELEVATION			COMMENTS DT. 27.02.20 & REVISED AS MARKED A02					
MD-216-3120-EG-ME-LPX-1001	CONV. 3120-CNV-001-PLAN VIEW	A01	29/01/2020	ISSUED FOR APPROVAL	AFO	BRD	BD	BGR	

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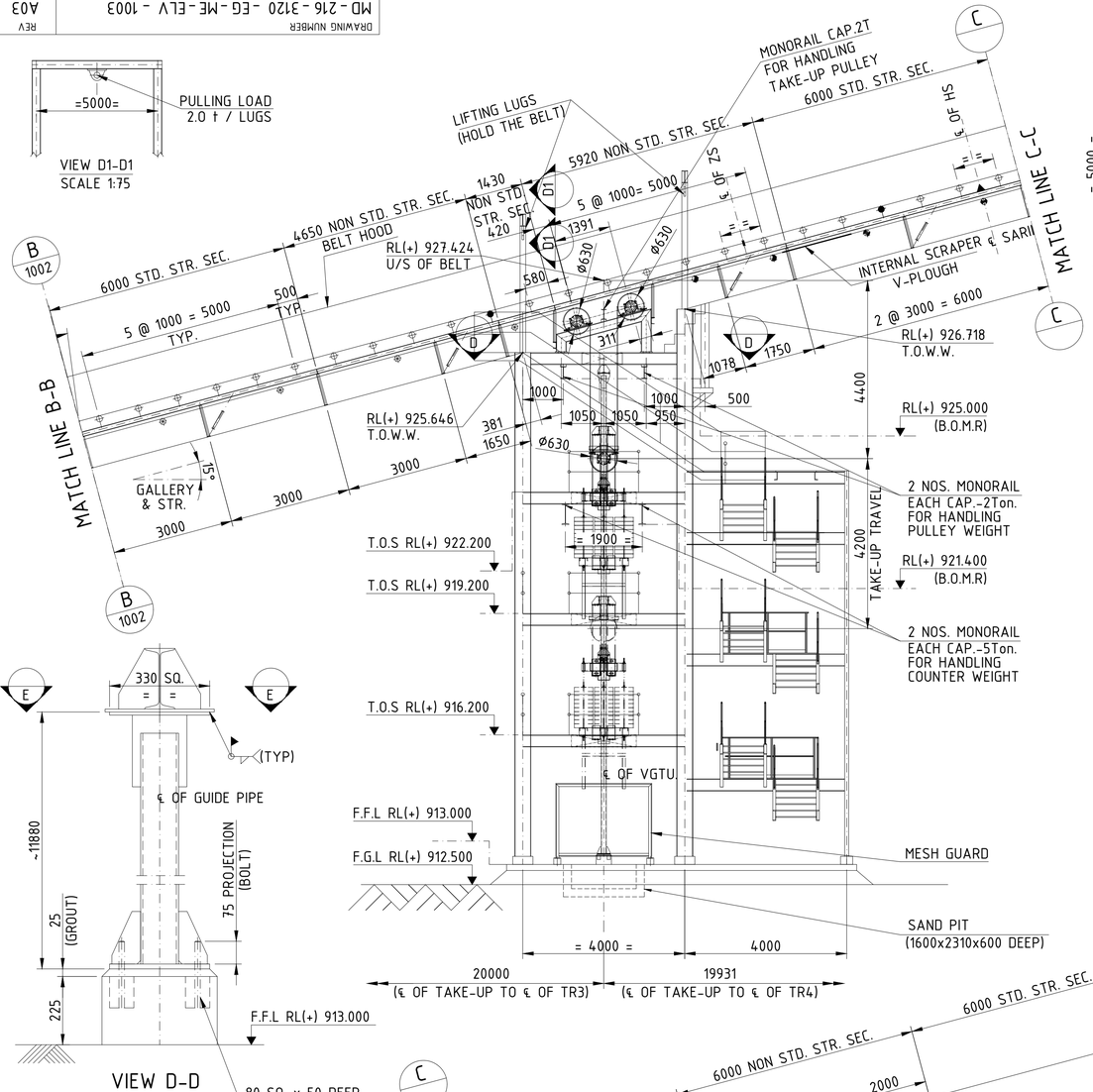
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CHECKED	BRD	28/01/2020
ENGINEER	BD	28/01/2020
APPROVED	BGR	29/01/2020
EMPLOYER		
SCALE	1:75, 1:50, 1:10	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
SECONDARY CRUSHING
CONV. 3120-CNV-001 - ELEVATION

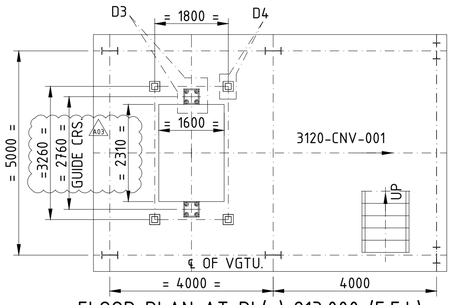
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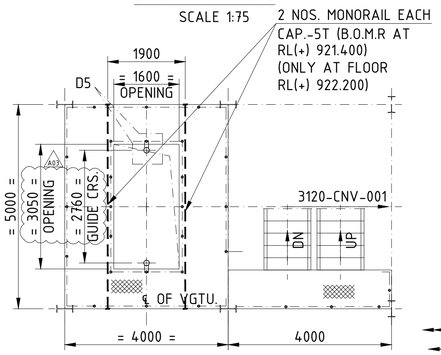
VIEW D1-D1
SCALE 1:75



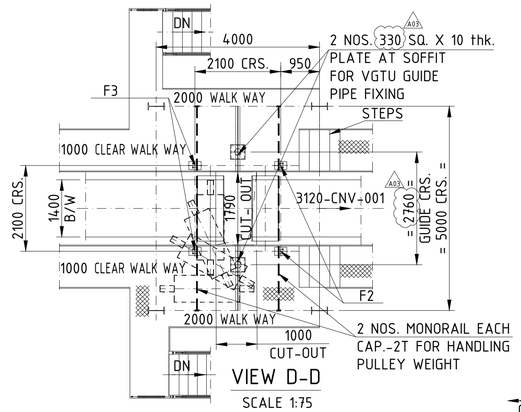
VIEW D-D
SCALE 1:10



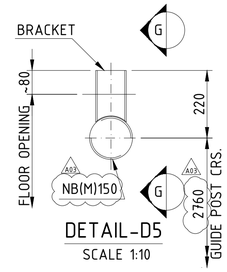
FLOOR PLAN AT RL(+) 913.000 (F.F.L)
(VGTU. TOWER)



FLOOR PLAN AT RL(+) 922.200 T.O.S
FLOOR PLAN AT RL(+) 919.200 T.O.S
FLOOR PLAN AT RL(+) 916.200 T.O.S
(VGTU. TOWER)
SCALE 1:75



VIEW D-D
SCALE 1:75

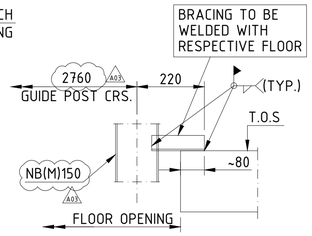


DETAIL-D5
SCALE 1:10

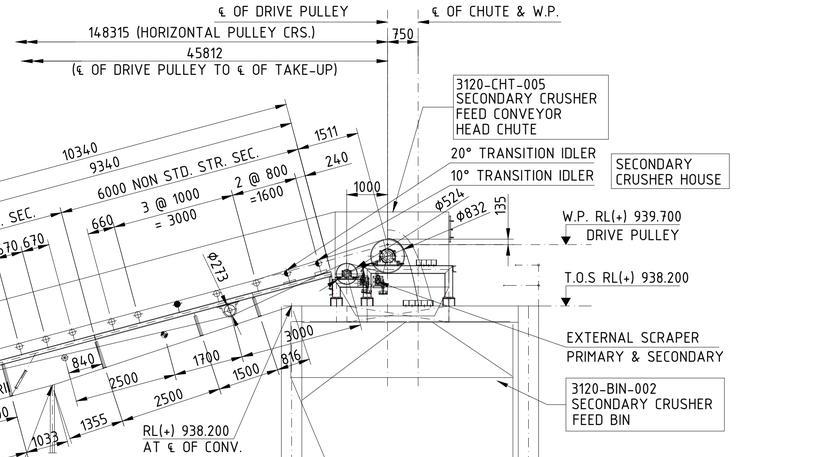
TABLE-2 : LOAD FOR TAKE UP BEND FRAME

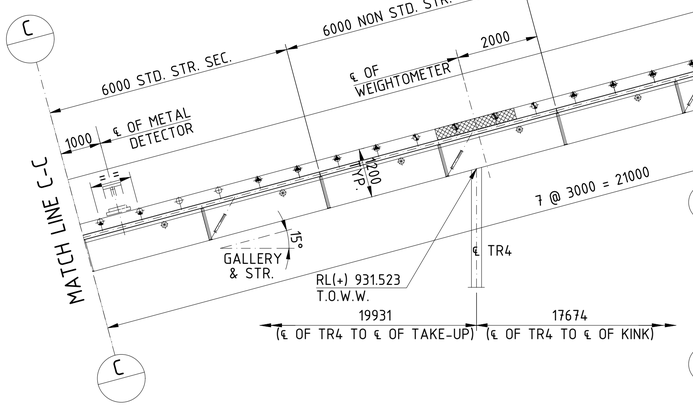
FOR CONV. 3120-CNV-001

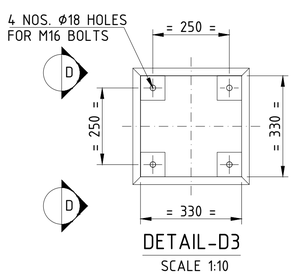
POINTS	VERTICAL LOAD (IN "t")	HORIZONTAL LOAD	
		ALONG (IN "t")	ACROSS (IN "t")
F2	(+) 4.00	(+) 1.50	(±) 0.40
F3	(+) 4.00	(-) 1.50	(±) 0.40



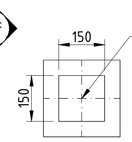
VIEW G-G
SCALE 1:10



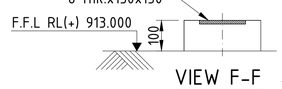




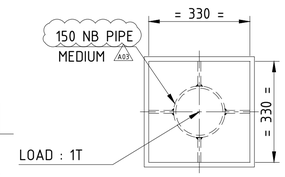
DETAIL-D3
SCALE 1:10



DETAIL-D4
SCALE 1:10



VIEW F-F
SCALE 1:10



VIEW E-E
SCALE 1:10

ABBREVIATION

CONV.:	CONVEYOR	CRS.:	CENTERS
TBA:	TO BE ANNOUNCED	DS:	DUST SUPPRESSION
N.A.:	NOT APPLICABLE	VGTU.:	VERTICAL GRAVITY TAKE-UP
GP:	GROUND PORTAL	MKD:	MARKED
SARII:	SELF ALIGNING RETURN TRAINER INVERTED IDLER	DIA:	DIAMETER
		SYN:	SYNCHRONOUS
T.P.:	TANGENT POINT	thk.:	THICKNESS
I.P.:	INTERSECTION POINT	RCC:	REINFORCED CEMENT CONCRETE
W.P.:	WORKING POINT	B/W:	BELT WIDTH
STD.:	STANDARD	MAX:	MAXIMUM
STR.:	STRINGER	deg.:	DEGREE
SEC:	SECTION	U/S.:	UNDERSIDE
℄:	CENTER LINE	B.O.M.R:	BOTTOM OF MONORAIL
DN:	DOWNWARD	T.O.W.W.:	TOP OF WALKWAY
UP:	UPWARD	TR:	TRESTLE
TP:	TRANSFER POINT	TYP.:	TYPICAL

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		A03	15/04/2020	ISSUED FOR APPROVAL & REVISED AS PER EMPLOYER	HKD	BRD	BD	BGR	
				COMMENTS DT. 06.04.2020 & REVISED AS MARKED A03					
MD-216-3120-EG-ME-ELV-1002	CONV. 3120-CNV-001-ELEVATION	A02	16/03/2020	ISSUED FOR APPROVAL & REVISED AS PER EMPLOYER	AFO	BRD	BD	BGR	
MD-216-3120-EG-ME-ELV-1001	CONV. 3120-CNV-001-ELEVATION			COMMENTS DT. 27.02.20 & REVISED AS MARKED A02					
MD-216-3120-EG-ME-LPX-1001	CONV. 3120-CNV-001-PLAN VIEW	A01	29/01/2020	ISSUED FOR APPROVAL	AFO	BRD	BD	BGR	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO DRAWING NUMBER: O19045_100_H_G_16112_004 | REVISION: 2

OUTOTEC DRAWING NUMBER: 900707__GAU01__ZC01__MTL01__31244 | DOCUMENT ID: OU100669103 | REVISION: 2

LARSEN & TOUBRO

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MA'ADEN معادن Saudi Arabian Mining Company شركة معادن العربية السعودية

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DRAWN	AFO	27/01/2020
CHECKED	BRD	28/01/2020
ENGINEER	BD	28/01/2020
APPROVED	BGR	29/01/2020
EMPLOYER		
SCALE	1:75, 1:10	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
SECONDARY CRUSHING
CONV. 3120-CNV-001 - ELEVATION

DRAWING NUMBER: MD-216-3120-EG-ME-ELV-1003 | REV: A03

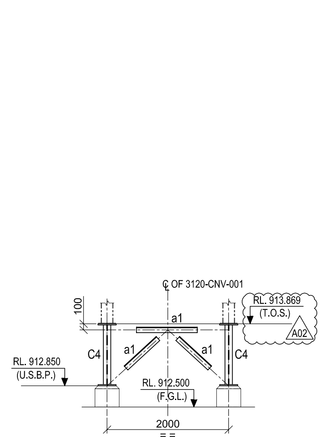
Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01			
Date	19-03-2020	30-04-2020	13/10/2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED C01	AS BUILT			
Prepared By	APL	APL	APL			
Checked	SBY	SBY	SBY			
Approved	PRI	PRI	PRI			

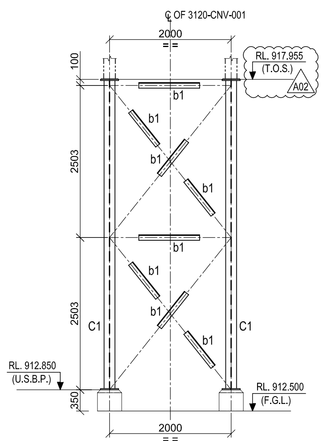
Outotec / L&T Revision No.:	0	1	2			
Date	19-03-2020	30-04-2020	13/10/2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED C01	AS BUILT			
Prepared By	APL	APL	APL			
Checked	SBY	SBY	SBY			
Approved	PRI	PRI	PRI			

C01

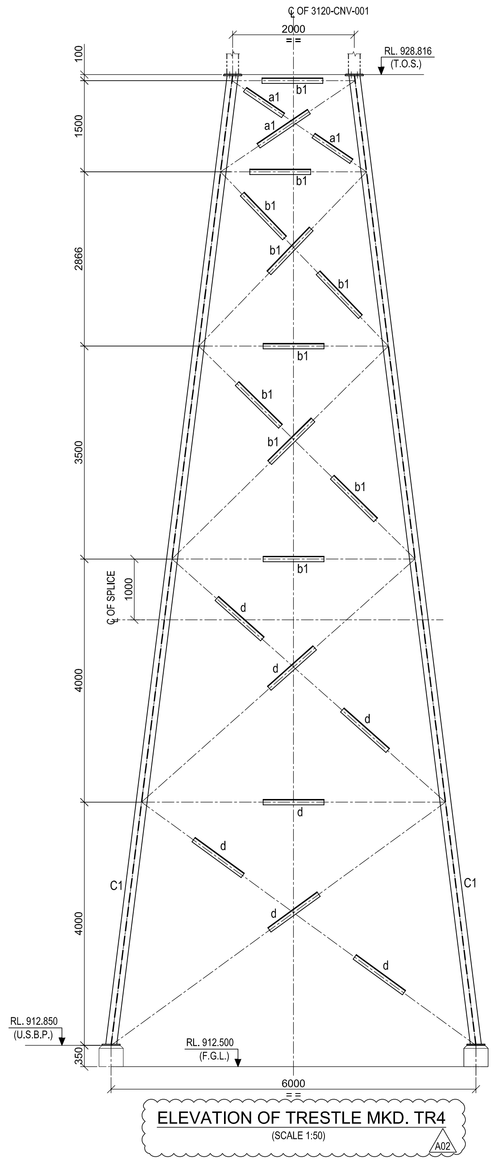
ATTACHMENT 2 STRUCTURAL GENERAL ARRANGEMENT DRAWING CONTAINING BASE PLATE DETAIL



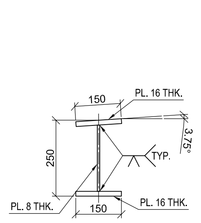
ELEVATION OF TRESTLE MKD. TR2
(SCALE 1:50)



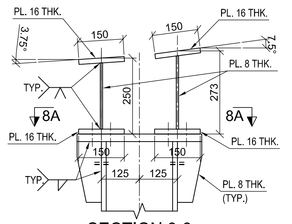
ELEVATION OF TRESTLE MKD. TR3
(SCALE 1:50)



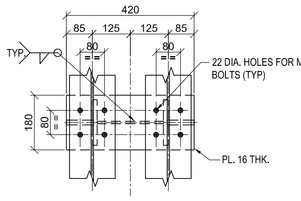
ELEVATION OF TRESTLE MKD. TR4
(SCALE 1:50)



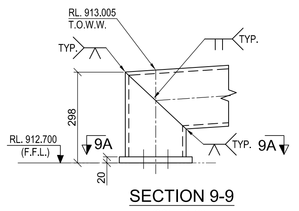
SECTION 7-7
(SCALE 1:10)



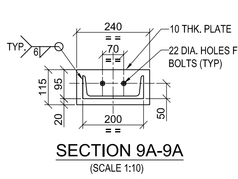
SECTION 8-8
(SCALE 1:10)



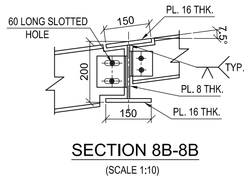
SECTION 8A-8A
(SCALE 1:10)



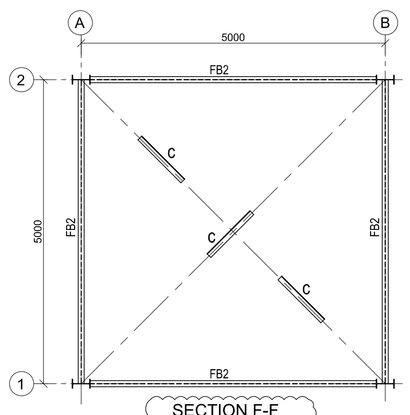
SECTION 9-9
(SCALE 1:10)



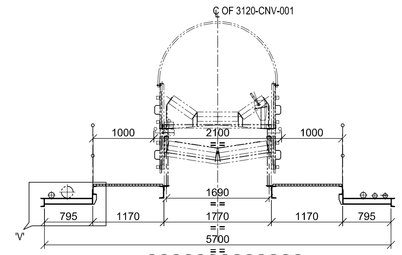
SECTION 9A-9A
(SCALE 1:10)



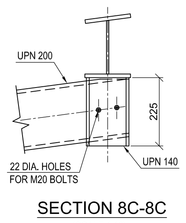
SECTION 8B-8B
(SCALE 1:10)



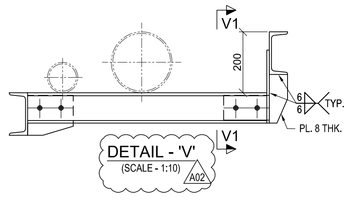
SECTION F-F
(REFER DRAWING NO. 1004)
(SCALE 1:50)
A02



SECTION 1A-1A
(REFER DRAWING NO. 1001)
(SCALE - 1:50)
A02

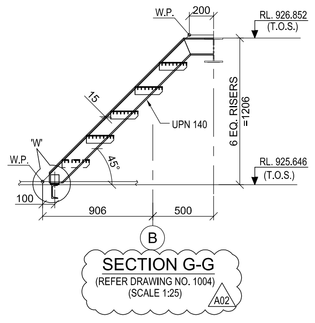


SECTION 8C-8C
(SCALE 1:10)

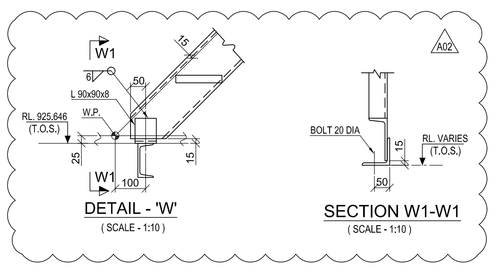


DETAIL - 'V'
(SCALE - 1:10)
A02

SECTION V1-V1
(SCALE 1:10)
A02

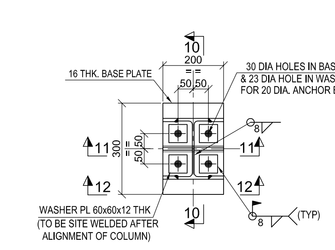


SECTION G-G
(REFER DRAWING NO. 1004)
(SCALE 1:25)
A02

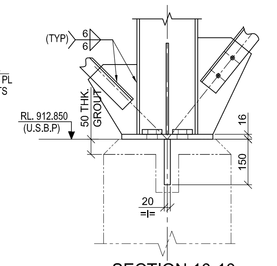


DETAIL - 'W'
(SCALE - 1:10)

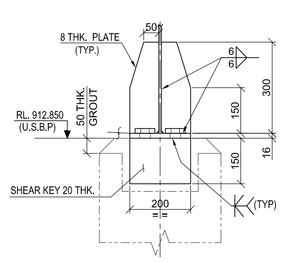
SECTION W1-W1
(SCALE - 1:10)



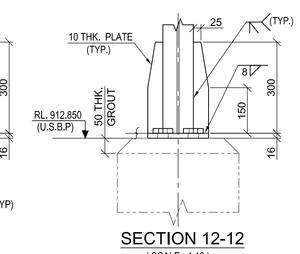
BASE PLATE DETAIL FOR TR1
(SCALE - 1:10)



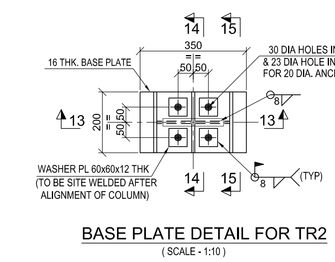
SECTION 10-10
(SCALE - 1:10)



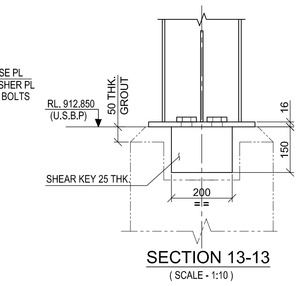
SECTION 11-11
(SCALE - 1:10)



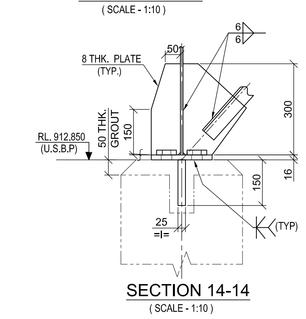
SECTION 12-12
(SCALE - 1:10)



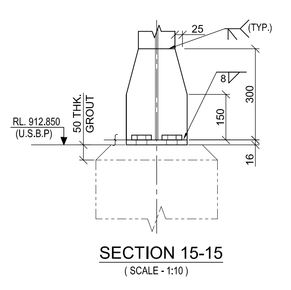
BASE PLATE DETAIL FOR TR2
(SCALE - 1:10)



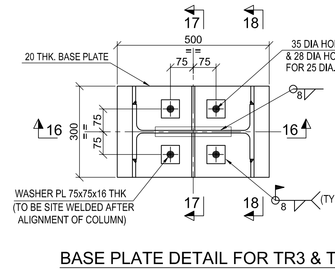
SECTION 13-13
(SCALE - 1:10)



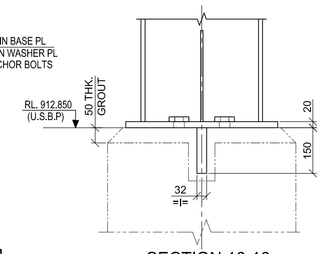
SECTION 14-14
(SCALE - 1:10)



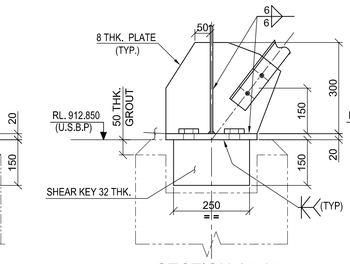
SECTION 15-15
(SCALE - 1:10)



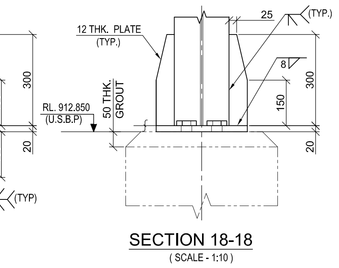
BASE PLATE DETAIL FOR TR3 & TR4
(SCALE - 1:10)



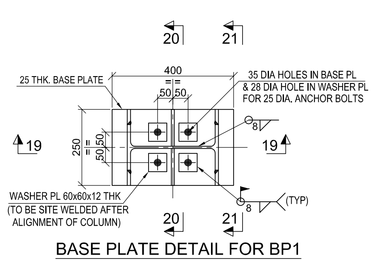
SECTION 16-16
(SCALE - 1:10)



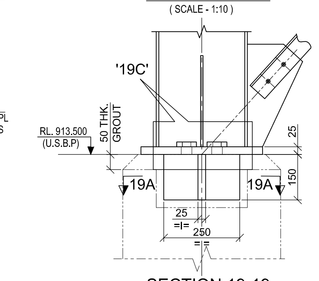
SECTION 17-17
(SCALE - 1:10)



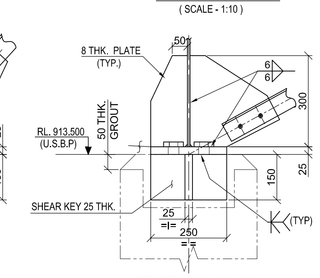
SECTION 18-18
(SCALE - 1:10)



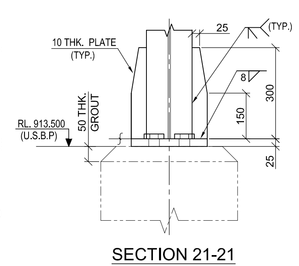
BASE PLATE DETAIL FOR BP1
(SCALE - 1:10)



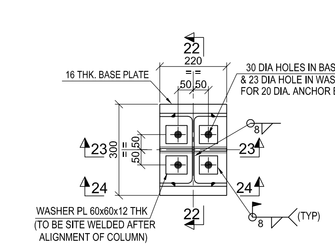
SECTION 19-19
(SCALE - 1:10)



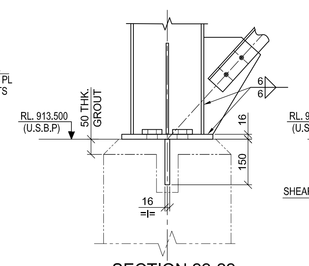
SECTION 20-20
(SCALE - 1:10)



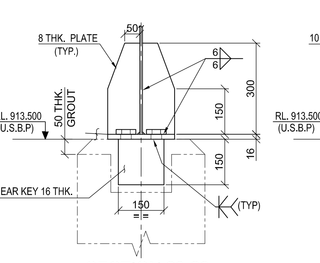
SECTION 21-21
(SCALE - 1:10)



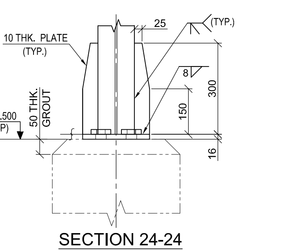
BASE PLATE DETAIL FOR BP2
(SCALE - 1:10)



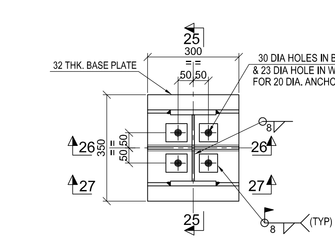
SECTION 22-22
(SCALE - 1:10)



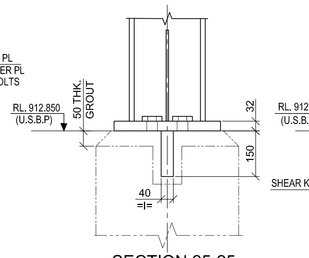
SECTION 23-23
(SCALE - 1:10)



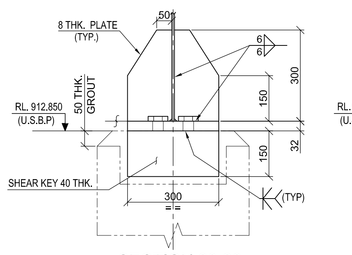
SECTION 24-24
(SCALE - 1:10)



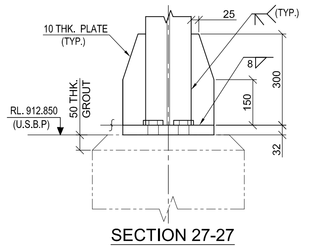
BASE PLATE DETAIL FOR GP2 & GP3
(SCALE - 1:10)



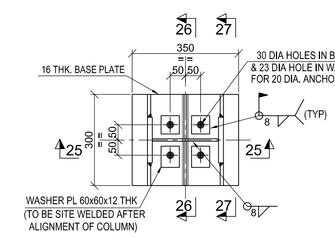
SECTION 25-25
(SCALE - 1:10)



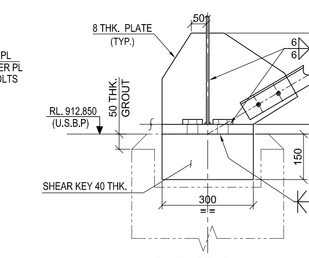
SECTION 26-26
(SCALE - 1:10)



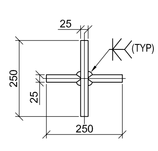
SECTION 27-27
(SCALE - 1:10)



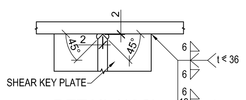
BASE PLATE DETAIL FOR GP4
(SCALE - 1:10)



SECTION 26-26
(SCALE - 1:10)



VIEW 19A-19A
(SCALE - 1:10)



DETAIL '19C'
(SCALE - 1:10)

SCHEDULE OF MEMBERS FOR VGTU, TR2 TO TR4					
MEMBER MARKED	MEMBER	SECTON	FACTORED AXIAL FORCE(KN)	FACTORED SHEAR FORCE(KN)	REMARKS
C1	IPE 400	I	650	-	MATERIAL GRADE:S275 JR
C2	IPE 300	I	720	-	
C3	IPE 240	I	120	-	
C4	1. PL. 120X10 2. PL. 240X8	I-1 2	410 A02	-	
FB1	IPE 200	I	-	-	
FB2	IPE 240	I	-	-	
FB3	IPE 300	I	-	-	
FB4	IPE 450	I	-	-	
B1	UPN 120	[	-	-	
B2	UPN 100	[	-	-	
B3	IPE 200	I	-	-	
B4	IPE 240	I	-	-	
B5	UPN 200	[	-	-	
B6	IPE 300	I	-	-	
a	L 50X50X6	L	-	-	
a1	2L 50X50X6	T	90	-	
b	L 65X65X6	L	-	-	
b1	2L 65X65X6	T	140	-	
c	2L 75X75X8	T	-	-	
c1	L 75X75X8	L	-	-	
d	2L 90X90X8	T	50	-	
P1	IPE 200	I	-	-	
MR1	1. IPE 240 2. PL. 90X10	I-1 2	-	-	
MR2	1. IPE 450 2. PL. 140X12	I-1 2	-	-	

ALL GUSSET PLATE ARE 8mm THK.
FOR DEVELOPING CONNECTIONS MEMBER END FORCES MENTIONED IN SCHEDULE OF MEMBERS SHALL BE FOLLOWED.
MINIMUM 6mm FILLET WELD OF 150 mm LENGTH SHALL BE PROVIDED FOR WELDED CONNECTION, FOR BOLTED CONNECTION, MINIMUM 2 Nos. M20 BOLTS SHALL BE PROVIDED.

LEGEND :-
R.L. - REDUCED LEVEL
℄ - CENTRE LINE
T.O.S. - TOP OF STEEL.
F.G.L. - FINISHED GROUND LEVEL.
F.F.L. - FINISHED FLOOR LEVEL.
TYP. - TYPICAL.
B.O.M.R. - BOTTOM OF MONORAIL.
MKD. - MARKED.
THK. - THICK
F/S - FAR SIDE
N/S - NEAR SIDE
W.P.-WORKING POINT
U.S.B.P. - UNDER SIDE OF BASE PLATE
U.N.O. - UNLESS NOTED OTHERWISE
T.O.W.W. - TOP OF WALKWAY

NOTES :-
1. ALL DIMENSIONS ARE IN MILLIMETER AND LEVELS & COORDINATES ARE IN METER UNO.
2. GRADE OF STEEL USED IS S275R AS PER EN10025.
3. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH DRAWING
MD - 216 - 3120 -EG -ST - GAD - 1001 TO MD - 216 - 3120 -EG -ST - GAD - 1002,
MD - 216 - 3120 -EG -ST - GAD - 1004 TO MD - 216 - 3120 -EG -ST - GAD - 1005.
4. ALL STRUCTURAL WORK SHALL CONFORM TO CIVIL STRUCTURAL DESIGN CRITARIYA NO MD - 216 - 1000 -EG -ST - CRT - 1001

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3120-EG-ME-LPX-1001	CONV. 3120-CNV-001-PLAN VIEW								
MD-216-3120-EG-ME-ELV-1003	CONV. 3120-CNV-001-ELEVATION								
MD-216-3120-EG-ME-ELV-1002	CONV. 3120-CNV-001-ELEVATION	A02	30/04/2020	REVISED AS PER EMPLOYER COMMENT ON 24.04.2020	SUS	SBY	APR	PRI	
MD-216-3120-EG-ME-ELV-1001	CONV. 3120-CNV-001-ELEVATION	A01	18/03/2020	ISSUED FOR REFERENCE	SUS	SBY	APR	PRI	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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LARSEN & TOUBRO DRAWING NUMBER		REVISION
019045_100_S_G_26013_010		1
OUTOTEC DRAWING NUMBER	DOCUMENT ID	REVISION
900707 __ GAU01 __ ZC01 __ MTL01 __ 31249	OU100670426	1

DRAWN	SUS	09/03/2020
CHECKED	SBY	16/03/2020
ENGINEER	APR	15/03/2020
APPROVED	PRI	18/03/2020
EMPLOYER		
SCALE	1:100,50,20,10	A0

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
STRUCTURAL GA FOR 3120-CNV-001, SH3
PLAN, ELEVATIONS & DETAILS

DRAWING NUMBER	REV
MD - 216 - 3120 - EG - ST - GAD - 1003	A02

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2			
Date	19-03-2020	30-04-2020	13/10/2023				Date	19-03-2020	30-04-2020	13/10/2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED C01	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED C01	AS BUILT			
Prepared By	APL	APL	APL				Prepared By	APL	APL	APL			
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

ATTACHMENT 3 **STAAD INPUT & OUTPUT FILE OF FOUNDATION FOR GP**

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 17-Mar-20
JOB NAME MMGP-SAUDI ARABIA
JOB CLIENT MA'ADEN
JOB NO O19045/900707
JOB REV 0
JOB PART GROUND PORTAL FOUNDATION
JOB REF FOUNDATIONS FOR CONVEYOR 3120-CNV-001
ENGINEER NAME APL
CHECKER NAME SBY
APPROVED NAME PRI
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 0 0; 2 6.25 0 0; 3 12.5 0 0; 4 0 0.361 0; 5 6.25 0.77 0; 6 12.5 1.18 0;
7 0 0 5.5; 8 6.25 0 5.5; 9 12.5 0 5.5; 10 0 0.361 5.5; 11 6.25 0.77 5.5;
12 12.5 1.18 5.5; 13 0 0.361 0.795; 14 0 0.361 1.865; 15 0 0.361 3.635;
16 0 0.361 4.705; 17 6.25 0.77 0.795; 18 12.5 1.18 0.795; 23 18.75 2 0.795;
24 6.25 0.77 1.865; 25 12.5 1.18 1.865; 26 18.75 2 1.865; 27 6.25 0.77 3.635;
28 12.5 1.18 3.635; 29 18.75 2 3.635; 30 6.25 0.77 4.705; 31 12.5 1.18 4.705;
32 18.75 2 4.705; 33 -6.193 -0.0449997 0.795; 34 -6.193 -0.0449997 1.865;
35 -6.193 -0.0449997 3.635; 36 -6.193 -0.0449997 4.705; 37 12.5 1.1 0;
38 12.5 1.1 5.5; 39 12.5 1.1 1.5; 40 12.5 1.1 4; 41 0 -1.35 0; 42 6.25 -1.35 0;
43 12.5 -1.35 0; 44 0 -1.35 5.5; 45 6.25 -1.35 5.5; 46 12.5 -1.35 5.5;
47 -6.206 -1.35 0; 48 -6.206 0 0; 49 -6.206 -1.35 5.5; 50 -6.206 0 5.5;
MEMBER INCIDENCES
1 1 4; 2 2 5; 3 3 37; 4 7 10; 5 8 11; 6 9 38; 7 4 13; 8 5 17; 9 6 18; 10 13 14;
11 14 15; 12 15 16; 13 16 10; 14 17 24; 15 13 17; 16 18 25; 17 17 18; 22 18 23;
23 24 27; 24 14 24; 25 25 28; 26 24 25; 28 25 26; 29 27 30; 30 15 27; 31 28 31;
32 27 28; 34 28 29; 35 30 11; 36 16 30; 37 31 12; 38 30 31; 40 31 32; 41 33 13;
42 34 14; 43 35 15; 44 36 16; 45 3 39; 46 9 40; 47 37 6; 48 38 12; 49 37 39;
50 39 40; 51 40 38; 52 41 1; 53 44 7; 54 42 2; 55 45 8; 56 43 3; 57 46 9;
58 1 7; 59 2 8; 60 3 9; 61 47 48; 62 49 50; 63 48 33; 64 33 34; 65 34 35;
66 35 36; 67 36 50;
DEFINE MATERIAL START
ISOTROPIC STEEL
E 2.05e+008
POISSON 0.3
DENSITY 76.8195
ALPHA 1.2e-005
DAMP 0.03
TYPE STEEL
STRENGTH FY 253200 FU 407800 RY 1.5 RT 1.2
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
```

```
MEMBER PROPERTY EUROPEAN
1 TO 8 10 TO 14 23 29 35 47 48 TAPERED 0.25 0.008 0.25 0.15 0.016 0.15 0.016
9 16 25 31 37 TAPERED 0.25 0.008 0.25 0.2 0.016 0.2 0.016
15 17 22 36 38 40 41 44 TABLE ST UPN140
24 26 28 30 32 34 42 43 TABLE ST UPN200
45 46 TABLE ST L50X50X6
49 TO 51 TABLE LD L75X75X8 SP 0.01
MEMBER PROPERTY EUROPEAN
52 TO 57 61 62 PRIS YD 0.6 ZD 0.6
MEMBER PROPERTY EUROPEAN
58 TO 60 PRIS YD 0.5 ZD 0.3
MEMBER PROPERTY EUROPEAN
63 TO 67 PRIS YD 0.6 ZD 0.35
CONSTANTS
BETA 90 MEMB 1 TO 6 47 48
MATERIAL STEEL MEMB 1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 51
MATERIAL CONCRETE MEMB 52 TO 67
SUPPORTS
23 26 29 32 FIXED BUT FX MX MY MZ
41 TO 47 49 FIXED
MEMBER TRUSS
45 46
MEMBER RELEASE
1 TO 6 9 41 TO 44 49 START MP 0.99
37 51 END MP 0.99
MEMBER OFFSET
63 TO 67 START 0 -0.3 0
63 TO 67 END 0 -0.3 0
58 TO 60 START 0 -0.25 0
58 TO 60 END 0 -0.25 0
LOAD 1 DL
SELFWEIGHT Y -1.1
*WALKWAY LOAD @50KG/SQ. M Including Seal Plate & H/R
MEMBER LOAD
15 17 22 36 38 40 41 44 UNI GY -0.32
24 26 28 30 32 34 42 43 UNI GY -0.5
*PIPE DEAD LOAD
MEMBER LOAD
7 TO 9 CON GY -2.03
13 35 37 CON GY -3.57
*CONVEYOR DEAD LOAD
MEMBER LOAD
42 43 CON GY -6 0.4
42 43 CON GY -6 2.55
42 43 CON GY -6 5.55
24 30 CON GY -6 2.37
24 30 CON GY -6 5.37
26 32 CON GY -6 2.1
26 32 CON GY -6 5.1
28 34 CON GY -6 1.9
28 34 CON GY -6 4.9
*
MEMBER LOAD
42 43 CON GX 1.5 0.4
```

```
42 43 CON GX 1.5 2.55
42 43 CON GX 1.5 5.55
24 30 CON GX 1.5 2.37
24 30 CON GX 1.5 5.37
26 32 CON GX 1.5 2.1
26 32 CON GX 1.5 5.1
28 34 CON GX 1.5 1.9
28 34 CON GX 1.5 4.9
LOAD 2 LL
*WALKWAY LOAD @400KG/SQ. MTER(FOR GALLERY)
MEMBER LOAD
15 17 22 24 26 28 30 32 34 36 38 40 TO 44 UNI GY -2
LOAD 3 ML
*CONVEYOR MATERIAL LOAD
MEMBER LOAD
42 43 CON GY -6.5 0.4
42 43 CON GY -6.5 2.55
42 43 CON GY -6.5 5.55
24 30 CON GY -6.5 2.37
24 30 CON GY -6.5 5.37
26 32 CON GY -6.5 2.1
26 32 CON GY -6.5 5.1
28 34 CON GY -6.5 1.9
28 34 CON GY -6.5 4.9
** Spillage Load @75kg/ Sq. m on seal plate
24 26 28 30 32 34 42 43 UNI GY -0.375
LOAD 4 TL1
TEMPERATURE LOAD
1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 46 TEMP 25
LOAD 5 TL2
TEMPERATURE LOAD
1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 46 TEMP -25
LOAD 11 G
REPEAT LOAD
1 1.0
LOAD 12 G + QN + MN
REPEAT LOAD
1 1.0 2 1.0 3 1.0
****LOAD COMBINATION for structural design
LOAD COMB 201 1.4 DL
1 1.4
LOAD COMB 202 1.2 (DL + TL1) + 1.6  (LL+ML)
1 1.2 4 1.2 2 1.6 3 1.6
LOAD COMB 203 1.2 (DL + TL1) + 1.6  (LL+ML)
1 1.2 5 1.2 2 1.6 3 1.6
****LOAD COMBINATION FOR BEARING PRESSURE CHECK
LOAD COMB 301 1.0 DL
1 1.0
LOAD COMB 302 1.0 (DL + TL1) + 1.0  (LL+ML)
1 1.0 4 1.0 2 1.0 3 1.0
LOAD COMB 303 1.0 (DL + TL1) + 1.0  (LL+ML)
1 1.0 5 1.0 2 1.0 3 1.0
****LOAD COMBINATION FOR FOUNDATION DESIGN
LOAD COMB 401 1.4 DL
```

```
1 1.4
LOAD COMB 402 1.4 (DL + TL1) + 1.7  (LL+ML)
1 1.4 4 1.4 2 1.7 3 1.7
LOAD COMB 403 1.4 (DL + TL1) + 1.7  (LL+ML)
1 1.4 5 1.4 2 1.7 3 1.7
**********************************************************************
PERFORM ANALYSIS
STEEL TAKE OFF LIST 1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 48
LOAD LIST 201 TO 203
PARAMETER 1
CODE LRFD
FYLD 275000 MEMB 1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 51
FU 410000 MEMB 1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 51
TRACK 2 MEMB 1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 51
BEAM 1 MEMB 1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 51
TMAIN 300 MEMB 1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 51
MAIN 200 MEMB 1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 51
NSF 0.8 MEMB 1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 51
UNT 1.9 MEMB 7 TO 14 16 23 25 29 31 35 37
LY 1.9 MEMB 7 TO 14 16 23 25 29 31 35 37
LX 1.9 MEMB 7 TO 14 16 23 25 29 31 35 37
LZ 5.5 MEMB 7 TO 14 16 23 25 29 31 35 37
LY 3 MEMB 15 17 22 24 26 28 30 32 34 36 38 40 TO 44
LX 3 MEMB 15 17 22 24 26 28 30 32 34 36 38 40 TO 44
UNT 3 MEMB 15 17 22 24 26 28 30 32 34 36 38 40 TO 44
UNB 3 MEMB 15 17 22 24 26 28 30 32 34 36 38 40 TO 44
LY 5.5 MEMB 49 TO 51
CHECK CODE MEMB 1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 51
**********************************************************************
LOAD LIST 401 TO 403
START CONCRETE DESIGN
CODE ACI 2008
FC 35000 MEMB 52 TO 67
FYMAIN 420000 MEMB 52 TO 67
TRACK 1 MEMB 52 TO 67
CLB 0.05 MEMB 52 TO 67
CLT 0.05 MEMB 52 TO 67
CLS 0.05 MEMB 52 TO 67
MAXMAIN 25 MEMB 52 TO 67
MINMAIN 16 MEMB 52 TO 67
MINSEC 8 MEMB 52 TO 67
REINF 0 MEMB 52 TO 57 61 62
DESIGN COLUMN 52 TO 57 61 62
DESIGN BEAM 58 TO 60 63 TO 67
END CONCRETE DESIGN
FINISH
```

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6           *
*           Version  20.07.11.70                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    APR 30, 2020                *
*           Time=    18:55: 5                    *
*                                                *
*      USER ID: lnt                              *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: D:\APL\Assignments\13-Maaden-Crushing_n_Screening\Conv-3120-CNV-001\Staad\Civil\3120-CNV-0.. .STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 17-MAR-20
    4. JOB NAME MMGP-SAUDI ARABIA
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045/900707
    7. JOB REV 0
    8. JOB PART GROUND PORTAL FOUNDATION
    9. JOB REF FOUNDATIONS FOR CONVEYOR 3120-CNV-001
   10. ENGINEER NAME APL
   11. CHECKER NAME SBY
   12. APPROVED NAME PRI
   13. END JOB INFORMATION
   14. INPUT WIDTH 79
   15. UNIT METER KN
   16. JOINT COORDINATES
   17. 1 0 0 0; 2 6.25 0 0; 3 12.5 0 0; 4 0 0.361 0; 5 6.25 0.77 0; 6 12.5 1.18 0
   18. 7 0 0 5.5; 8 6.25 0 5.5; 9 12.5 0 5.5; 10 0 0.361 5.5; 11 6.25 0.77 5.5
   19. 12 12.5 1.18 5.5; 13 0 0.361 0.795; 14 0 0.361 1.865; 15 0 0.361 3.635
   20. 16 0 0.361 4.705; 17 6.25 0.77 0.795; 18 12.5 1.18 0.795; 23 18.75 2 0.795
   21. 24 6.25 0.77 1.865; 25 12.5 1.18 1.865; 26 18.75 2 1.865; 27 6.25 0.77 3.635
   22. 28 12.5 1.18 3.635; 29 18.75 2 3.635; 30 6.25 0.77 4.705; 31 12.5 1.18 4.705
   23. 32 18.75 2 4.705; 33 -6.193 -0.0449997 0.795; 34 -6.193 -0.0449997 1.865
   24. 35 -6.193 -0.0449997 3.635; 36 -6.193 -0.0449997 4.705; 37 12.5 1.1 0
   25. 38 12.5 1.1 5.5; 39 12.5 1.1 1.5; 40 12.5 1.1 4; 41 0 -1.35 0; 42 6.25 -1.35 0
   26. 43 12.5 -1.35 0; 44 0 -1.35 5.5; 45 6.25 -1.35 5.5; 46 12.5 -1.35 5.5
   27. 47 -6.206 -1.35 0; 48 -6.206 0 0; 49 -6.206 -1.35 5.5; 50 -6.206 0 5.5
   28. MEMBER INCIDENCES
   29. 1 1 4; 2 2 5; 3 3 37; 4 7 10; 5 8 11; 6 9 38; 7 4 13; 8 5 17; 9 6 18; 10 13 14
   30. 11 14 15; 12 15 16; 13 16 10; 14 17 24; 15 13 17; 16 18 25; 17 17 18; 22 18 23
   31. 23 24 27; 24 14 24; 25 25 28; 26 24 25; 28 25 26; 29 27 30; 30 15 27; 31 28 31
   32. 32 27 28; 34 28 29; 35 30 11; 36 16 30; 37 31 12; 38 30 31; 40 31 32; 41 33 13
   33. 42 34 14; 43 35 15; 44 36 16; 45 3 39; 46 9 40; 47 37 6; 48 38 12; 49 37 39
   34. 50 39 40; 51 40 38; 52 41 1; 53 44 7; 54 42 2; 55 45 8; 56 43 3; 57 46 9
   35. 58 1 7; 59 2 8; 60 3 9; 61 47 48; 62 49 50; 63 48 33; 64 33 34; 65 34 35
   36. 66 35 36; 67 36 50
   37. DEFINE MATERIAL START
   38. ISOTROPIC STEEL
```

```
39. E 2.05E+008
40. POISSON 0.3
41. DENSITY 76.8195
42. ALPHA 1.2E-005
43. DAMP 0.03
44. TYPE STEEL
45. STRENGTH FY 253200 FU 407800 RY 1.5 RT 1.2
46. ISOTROPIC CONCRETE
47. E 2.17185E+007
48. POISSON 0.17
49. DENSITY 23.5616
50. ALPHA 1E-005
51. DAMP 0.05
52. TYPE CONCRETE
53. STRENGTH FCU 27579
54. END DEFINE MATERIAL
55. MEMBER PROPERTY EUROPEAN
56. 1 TO 8 10 TO 14 23 29 35 47 48 TAPERED 0.25 0.008 0.25 0.15 0.016 0.15 0.016
57. 9 16 25 31 37 TAPERED 0.25 0.008 0.25 0.2 0.016 0.2 0.016
58. 15 17 22 36 38 40 41 44 TABLE ST UPN140
59. 24 26 28 30 32 34 42 43 TABLE ST UPN200
60. 45 46 TABLE ST L50X50X6
61. 49 TO 51 TABLE LD L75X75X8 SP 0.01
62. MEMBER PROPERTY EUROPEAN
63. 52 TO 57 61 62 PRIS YD 0.6 ZD 0.6
64. MEMBER PROPERTY EUROPEAN
65. 58 TO 60 PRIS YD 0.5 ZD 0.3
66. MEMBER PROPERTY EUROPEAN
67. 63 TO 67 PRIS YD 0.6 ZD 0.35
68. CONSTANTS
69. BETA 90 MEMB 1 TO 6 47 48
70. MATERIAL STEEL MEMB 1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 51
71. MATERIAL CONCRETE MEMB 52 TO 67
72. SUPPORTS
73. 23 26 29 32 FIXED BUT FX MX MY MZ
74. 41 TO 47 49 FIXED
75. MEMBER TRUSS
76. 45 46
77. MEMBER RELEASE
78. 1 TO 6 9 41 TO 44 49 START MP 0.99
79. 37 51 END MP 0.99
80. MEMBER OFFSET
81. 63 TO 67 START 0 -0.3 0
82. 63 TO 67 END 0 -0.3 0
83. 58 TO 60 START 0 -0.25 0
84. 58 TO 60 END 0 -0.25 0
85. LOAD 1 DL
86. SELFWEIGHT Y -1.1
87. *WALKWAY LOAD @50KG/SQ. M INCLUDING SEAL PLATE _H/R
88. MEMBER LOAD
89. 15 17 22 36 38 40 41 44 UNI GY -0.32
90. 24 26 28 30 32 34 42 43 UNI GY -0.5
91. *PIPE DEAD LOAD
92. MEMBER LOAD
93. 7 TO 9 CON GY -2.03
94. 13 35 37 CON GY -3.57
```

```
 95. *CONVEYOR DEAD LOAD
 96. MEMBER LOAD
 97. 42 43 CON GY -6 0.4
 98. 42 43 CON GY -6 2.55
 99. 42 43 CON GY -6 5.55
100. 24 30 CON GY -6 2.37
101. 24 30 CON GY -6 5.37
102. 26 32 CON GY -6 2.1
103. 26 32 CON GY -6 5.1
104. 28 34 CON GY -6 1.9
105. 28 34 CON GY -6 4.9
106. *
107. MEMBER LOAD
108. 42 43 CON GX 1.5 0.4
109. 42 43 CON GX 1.5 2.55
110. 42 43 CON GX 1.5 5.55
111. 24 30 CON GX 1.5 2.37
112. 24 30 CON GX 1.5 5.37
113. 26 32 CON GX 1.5 2.1
114. 26 32 CON GX 1.5 5.1
115. 28 34 CON GX 1.5 1.9
116. 28 34 CON GX 1.5 4.9
117. LOAD 2 LL
118. *WALKWAY LOAD @400KG/SQ. MTER(FOR GALLERY)
119. MEMBER LOAD
120. 15 17 22 24 26 28 30 32 34 36 38 40 TO 44 UNI GY -2
121. LOAD 3 ML
122. *CONVEYOR MATERIAL LOAD
123. MEMBER LOAD
124. 42 43 CON GY -6.5 0.4
125. 42 43 CON GY -6.5 2.55
126. 42 43 CON GY -6.5 5.55
127. 24 30 CON GY -6.5 2.37
128. 24 30 CON GY -6.5 5.37
129. 26 32 CON GY -6.5 2.1
130. 26 32 CON GY -6.5 5.1
131. 28 34 CON GY -6.5 1.9
132. 28 34 CON GY -6.5 4.9
133. ** SPILLAGE LOAD @75KG/ SQ. M ON SEAL PLATE
134. 24 26 28 30 32 34 42 43 UNI GY -0.375
135. LOAD 4 TL1
136. TEMPERATURE LOAD
137. 1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 46 TEMP 25
138. LOAD 5 TL2
139. TEMPERATURE LOAD
140. 1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 46 TEMP -25
141. LOAD 11 G
142. REPEAT LOAD
143. 1 1.0
144. LOAD 12 G + QN + MN
145. REPEAT LOAD
146. 1 1.0 2 1.0 3 1.0
147. ****LOAD COMBINATION FOR STRUCTURAL DESIGN
148. LOAD COMB 201 1.4 DL
149. 1 1.4
150. LOAD COMB 202 1.2 (DL + TL1) + 1.6  (LL+ML)
```

```
151. 1 1.2 4 1.2 2 1.6 3 1.6
152. LOAD COMB 203 1.2 (DL + TL1) + 1.6  (LL+ML)
153. 1 1.2 5 1.2 2 1.6 3 1.6
154. ****LOAD COMBINATION FOR BEARING PRESSURE CHECK
155. LOAD COMB 301 1.0 DL
156. 1 1.0
157. LOAD COMB 302 1.0 (DL + TL1) + 1.0  (LL+ML)
158. 1 1.0 4 1.0 2 1.0 3 1.0
159. LOAD COMB 303 1.0 (DL + TL1) + 1.0  (LL+ML)
160. 1 1.0 5 1.0 2 1.0 3 1.0
161. ****LOAD COMBINATION FOR FOUNDATION DESIGN
162. LOAD COMB 401 1.4 DL
163. 1 1.4
164. LOAD COMB 402 1.4 (DL + TL1) + 1.7  (LL+ML)
165. 1 1.4 4 1.4 2 1.7 3 1.7
166. LOAD COMB 403 1.4 (DL + TL1) + 1.7  (LL+ML)
167. 1 1.4 5 1.4 2 1.7 3 1.7
168. ************************************************************
169. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**46**	**NUMBER OF MEMBERS**	**60**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**12**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 36/ 8/ 52 DOF
TOTAL PRIMARY LOAD CASES = 7, TOTAL DEGREES OF FREEDOM = 220
TOTAL LOAD COMBINATION CASES = 9 SO FAR.
SIZE OF STIFFNESS MATRIX = 12 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.2/ 49575.1 MB

```
170. STEEL TAKE OFF LIST 1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 48
```

STEEL TAKE-OFF

PROFILE	**LENGTH(METE)**	**WEIGHT(KN)**
Tapered MembNo: 1	15.62	7.853
Tapered MembNo: 9	5.50	3.441
ST UPN140	50.07	7.847
ST UPN200	50.07	12.386
ST L50X50X6	3.72	0.163
	TOTAL =	31.690

********** END OF DATA FROM INTERNAL STORAGE **********

171. LOAD LIST 201 TO 203
172. PARAMETER 1
173. CODE LRFD
174. FYLD 275000 MEMB 1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 51
175. FU 410000 MEMB 1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 51
176. TRACK 2 MEMB 1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 51
177. BEAM 1 MEMB 1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 51
178. TMAIN 300 MEMB 1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 51
179. MAIN 200 MEMB 1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 51
180. NSF 0.8 MEMB 1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 51
181. UNT 1.9 MEMB 7 TO 14 16 23 25 29 31 35 37
182. LY 1.9 MEMB 7 TO 14 16 23 25 29 31 35 37
183. LX 1.9 MEMB 7 TO 14 16 23 25 29 31 35 37
184. LZ 5.5 MEMB 7 TO 14 16 23 25 29 31 35 37
185. LY 3 MEMB 15 17 22 24 26 28 30 32 34 36 38 40 TO 44
186. LX 3 MEMB 15 17 22 24 26 28 30 32 34 36 38 40 TO 44
187. UNT 3 MEMB 15 17 22 24 26 28 30 32 34 36 38 40 TO 44
188. UNB 3 MEMB 15 17 22 24 26 28 30 32 34 36 38 40 TO 44
189. LY 5.5 MEMB 49 TO 51
190. CHECK CODE MEMB 1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 51

STAAD.PRO CODE CHECKING - (LRFD 3RD EDITION) v1.0

****

```
|-------------------------------------------------------------------------|
|                                                 Y        PROPERTIES     |
|*************                                    |        IN  CMS UNIT   |
|            *  |============================|   ==| |==     ------------   |
|MEMBER   1  *  |  EUROPEAN SECTIONS         |     | |       AX=0.6544E+2   |
|            *  |     TAP ERED               |     | |  --Z  AY=0.2000E+2   |
|DESIGN CODE *  |                            |     | |       AZ=0.3200E+2   |
| LRFD 2001  *  ================================   ==| |==     PY=0.1835E+3   |
|            *                                                 PZ=0.6566E+3   |
|            *  |<---LENGTH (M)=     0.36 --->|                RY=0.3710E+1   |
|*************                                                 RZ=0.1054E+2   |
|                                                                         |
|                101.4 (KNS-METRE)                                        |
|PARAMETER        |                                      L202CAPACITIES     |
|IN KNS  CMS      |                                   L202    IN KNS METRE   |
|---------------  +                             L202L202        ------------|
| KL/R-Y=   9.73  |                         L202                PNC=0.1521E+4|
| KL/R-Z=   3.42  +                                             pnc=0.1063E+3|
| UNL   =  36.10  |               L202L202                      PNT=0.1610E+4|
| CB    =   1.00  +                                             pnt=0.0000E+0|
| PHIC  =   0.85  |           L202                              MNZ=0.1625E+3|
| PHIB  =   0.90  +   L203L202                                  mnz=0.7444E+0|
| FYLD  =  27.50  |L203                                         MNY=0.4460E+2|
| NSF   =   0.80  +---+---+---+---+---+---+---+---+---+---+   mny=0.5838E+0|
| DFF =     0.00  -2.3                                        VN =0.2970E+3|
| dff =     0.00               ABSOLUTE MZ ENVELOPE           vn =0.2788E+3|
|                                 (WITH LOAD NO.)                         |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY  (KNS-METRE)              |
|                     -------------------------                           |
|                                                                         |
|                AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z   |
|                                                                         |
|      VALUE     106.3       278.8        2.6         0.6       101.4     |
|   LOCATION       0.0         0.0        0.0         0.0         0.4     |
|    LOADING       202         202        202         203         202     |
|                                                                         |
|*************************************************************************|
|*                                                                       *|
|*                        DESIGN SUMMARY  (KNS-METRE)                    *|
|*                        --------------                                 *|
|*                                                                       *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/         *|
|         FX              MY             MZ           LOCATION           |
|        ==================================================             |
|          PASS        SHEAR-Y          0.939            202              |
|        106.26 C          0.58        -0.74            0.00              |
|*                                                                       *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
```

STAAD.PRO CODE CHECKING - (LRFD 3RD EDITION) v1.0
****

```
|-------------------------------------------------------------------------|
|                                                 Y        PROPERTIES     |
|*************                                    |        IN  CMS UNIT  |
|            *  |==============================|   ==| |==     ------------ |
|MEMBER   2  *  |  EUROPEAN SECTIONS           |     | |       AX=0.6544E+2 |
|            *  |     TAP ERED                 |     | |  --Z  AY=0.2000E+2 |
|DESIGN CODE *  |                              |     | |       AZ=0.3200E+2 |
| LRFD 2001  *  ================================   ==| |==     PY=0.1835E+3 |
|            *                                                 PZ=0.6566E+3 |
|            *  |<---LENGTH (M)=     0.77 --->|                RY=0.3710E+1 |
|*************                                                 RZ=0.1054E+2 |
|                                                                         |
|                 90.1 (KNS-METRE)                                        |
|PARAMETER        |                                        L202CAPACITIES   |
|IN KNS  CMS      |                                    L202    IN KNS METRE |
|---------------  +                            L202L202        ------------|
| KL/R-Y=  20.75  |                        L202                PNC=0.1493E+4|
| KL/R-Z=   7.30  +                                            pnc=0.9177E+2|
| UNL   =  77.00  |               L202L202                     PNT=0.1610E+4|
| CB    =   1.00  +                                            pnt=0.0000E+0|
| PHIC  =   0.85  |           L202                             MNZ=0.1625E+3|
| PHIB  =   0.90  +   L202L202                                 mnz=0.9013E+2|
| FYLD  =  27.50  |L203                                        MNY=0.4460E+2|
| NSF   =   0.80  +---+---+---+---+---+---+---+---+---+---+   mny=0.1963E+0|
| DFF =     0.00  -4.3                                         VN =0.2970E+3|
| dff =     0.00               ABSOLUTE MZ ENVELOPE            vn =0.1174E+3|
|                                 (WITH LOAD NO.)                         |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY  (KNS-METRE)              |
|                     -------------------------                           |
|                                                                         |
|                AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z   |
|                                                                         |
|      VALUE      92.6       117.4        0.9         0.5        90.1    |
|   LOCATION       0.0         0.0        0.0         0.0         0.8    |
|    LOADING       203         202        203         202         202    |
|                                                                         |
|*************************************************************************|
|*                                                                       *|
|*                        DESIGN SUMMARY  (KNS-METRE)                    *|
|*                        --------------                                 *|
|*                                                                       *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/         *|
|         FX              MY             MZ           LOCATION           |
|        ======================================================           |
|          PASS        LRFD-H1-1B-C     0.590            202              |
|         91.77 C         -0.20       -90.13            0.77             |
|*                                                                       *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
```

STAAD.PRO CODE CHECKING - (LRFD 3RD EDITION) v1.0

****

```
|-------------------------------------------------------------------------|
|                                                 Y        PROPERTIES     |
|*************                                    |        IN  CMS UNIT   |
|            *  |============================|   ==| |==     ------------   |
|MEMBER   3  *  |  EUROPEAN SECTIONS         |     | |       AX=0.6544E+2   |
|            *  |     TAP ERED               |     | |  --Z  AY=0.2000E+2   |
|DESIGN CODE *  |                            |     | |       AZ=0.3200E+2   |
| LRFD 2001  *  ================================   ==| |==     PY=0.1835E+3   |
|            *                                                 PZ=0.6566E+3   |
|            *  |<---LENGTH (M)=     1.10 --->|                RY=0.3710E+1   |
|*************                                                 RZ=0.1054E+2   |
|                                                                         |
|                 14.7 (KNS-METRE)                                        |
|PARAMETER        |                                        L203CAPACITIES   |
|IN KNS  CMS      |                                    L203    IN KNS METRE |
|---------------  +                            L203L203        ------------|
| KL/R-Y=  29.65  |                        L203                PNC=0.1455E+4|
| KL/R-Z=  10.44  +                                            pnc=0.1051E+3|
| UNL    =110.00  |               L203L203                     PNT=0.1610E+4|
| CB     =  1.00  +                                            pnt=0.0000E+0|
| PHIC   =  0.85  |           L203                             MNZ=0.1625E+3|
| PHIB   =  0.90  +   L203L203                                 mnz=0.1471E+2|
| FYLD   = 27.50  |L202                                        MNY=0.4460E+2|
| NSF    =  0.80  +---+---+---+---+---+---+---+---+---+---|   mny=0.1702E-1|
| DFF =    0.00   -0.6                                       VN =0.2970E+3|
| dff =    0.00               ABSOLUTE MZ ENVELOPE           vn =0.1321E+2|
|                                (WITH LOAD NO.)                          |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY  (KNS-METRE)              |
|                     -------------------------                           |
|                                                                         |
|                AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z   |
|                                                                         |
|      VALUE     105.9        13.2        0.3         0.4        14.7     |
|   LOCATION       0.0         0.0        0.0         0.0         1.1     |
|    LOADING       203         203        202         202         203     |
|                                                                         |
|*************************************************************************|
|*                                                                       *|
|*                         DESIGN SUMMARY  (KNS-METRE)                   *|
|*                         --------------                                *|
|*                                                                       *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/         *|
|         FX              MY             MZ           LOCATION           |
|        ==================================================             |
|          PASS        LRFD-H1-1B-C     0.127            203              |
|        105.13 C         -0.02       -14.71            1.10             |
|*                                                                       *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
```

STAAD.PRO CODE CHECKING - (LRFD 3RD EDITION) v1.0

****

```
|-------------------------------------------------------------------------|
|                                                 Y        PROPERTIES     |
|*************                                    |        IN  CMS UNIT   |
|            *  |============================|   ==| |==     ------------   |
|MEMBER   4  *  |  EUROPEAN SECTIONS         |     | |       AX=0.6544E+2   |
|            *  |     TAP ERED               |     | |  --Z  AY=0.2000E+2   |
|DESIGN CODE *  |                            |     | |       AZ=0.3200E+2   |
| LRFD 2001  *  ================================   ==| |==     PY=0.1835E+3   |
|            *                                                 PZ=0.6566E+3   |
|            *  |<---LENGTH (M)=     0.36 --->|                RY=0.3710E+1   |
|*************                                                 RZ=0.1054E+2   |
|                                                                         |
|                101.4 (KNS-METRE)                                        |
|PARAMETER        |                                      L202CAPACITIES     |
|IN KNS  CMS      |                                   L202    IN KNS METRE   |
|---------------  +                             L202L202        ------------|
| KL/R-Y=   9.73  |                         L202                PNC=0.1521E+4|
| KL/R-Z=   3.42  +                                             pnc=0.1079E+3|
| UNL   =  36.10  |                L202L202                     PNT=0.1610E+4|
| CB    =   1.00  +                                             pnt=0.0000E+0|
| PHIC  =   0.85  |           L202                              MNZ=0.1625E+3|
| PHIB  =   0.90  +   L203L202                                  mnz=0.7629E+0|
| FYLD  =  27.50  |L203                                         MNY=0.4460E+2|
| NSF   =   0.80  +---+---+---+---+---+---+---+---+---+---+---|   mny=0.5843E+0|
| DFF =     0.00  -2.3                                          VN =0.2970E+3|
| dff =     0.00                ABSOLUTE MZ ENVELOPE            vn =0.2788E+3|
|                                  (WITH LOAD NO.)                        |
|                                                                         |
|                      MAX FORCE/ MOMENT SUMMARY  (KNS-METRE)             |
|                      -------------------------                          |
|                                                                         |
|                AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z    |
|                                                                         |
|      VALUE     107.9       278.8        2.6         0.6       101.4     |
|   LOCATION       0.0         0.0        0.0         0.0         0.4     |
|    LOADING       202         202        202         203         202     |
|                                                                         |
|*************************************************************************|
|*                                                                       *|
|*                         DESIGN SUMMARY  (KNS-METRE)                   *|
|*                         --------------                                *|
|*                                                                       *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/         *|
|         FX              MY             MZ           LOCATION           |
|        ======================================================           |
|          PASS        SHEAR-Y          0.939            202              |
|        107.85 C          0.58         0.76            0.00              |
|*                                                                       *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
```

STAAD.PRO CODE CHECKING - (LRFD 3RD EDITION) v1.0
****

```
|-------------------------------------------------------------------------|
|                                                 Y        PROPERTIES     |
|*************                                    |        IN  CMS UNIT  |
|            *  |==============================|   ==| |==     ------------ |
|MEMBER   5  *  |  EUROPEAN SECTIONS           |     | |       AX=0.6544E+2 |
|            *  |     TAP ERED                 |     | |  --Z  AY=0.2000E+2 |
|DESIGN CODE *  |                              |     | |       AZ=0.3200E+2 |
| LRFD 2001  *  ================================   ==| |==     PY=0.1835E+3 |
|            *                                                 PZ=0.6566E+3 |
|            *  |<---LENGTH (M)=     0.77 --->|                RY=0.3710E+1 |
|*************                                                 RZ=0.1054E+2 |
|                                                                         |
|                 90.1 (KNS-METRE)                                        |
|PARAMETER        |                                        L202CAPACITIES   |
|IN KNS  CMS      |                                    L202    IN KNS METRE |
|---------------  +                             L202L202        ------------|
| KL/R-Y=  20.75  |                         L202                PNC=0.1493E+4|
| KL/R-Z=   7.30  +                                             pnc=0.9336E+2|
| UNL   =  77.00  |               L202L202                      PNT=0.1610E+4|
| CB    =   1.00  +                                             pnt=0.0000E+0|
| PHIC  =   0.85  |           L202                              MNZ=0.1625E+3|
| PHIB  =   0.90  +   L202L202                                  mnz=0.9014E+2|
| FYLD  =  27.50  |L203                                         MNY=0.4460E+2|
| NSF   =   0.80  +---+---+---+---+---+---+---+---+---+---+---|   mny=0.1964E+0|
| DFF =     0.00  -4.3                                        VN =0.2970E+3|
| dff =     0.00               ABSOLUTE MZ ENVELOPE           vn =0.1174E+3|
|                                 (WITH LOAD NO.)                         |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY  (KNS-METRE)              |
|                     -------------------------                           |
|                                                                         |
|                AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z   |
|                                                                         |
|      VALUE      94.2       117.4        0.9         0.5        90.1    |
|   LOCATION       0.0         0.0        0.0         0.0         0.8    |
|    LOADING       203         202        203         202         202    |
|                                                                         |
|*************************************************************************|
|*                                                                       *|
|*                        DESIGN SUMMARY  (KNS-METRE)                    *|
|*                        --------------                                 *|
|*                                                                       *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/         *|
|         FX              MY             MZ           LOCATION           |
|        ========================================================         |
|          PASS        LRFD-H1-1B-C     0.590            202              |
|         93.36 C         -0.20        90.14            0.77             |
|*                                                                       *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
```

STAAD.PRO CODE CHECKING - (LRFD 3RD EDITION) v1.0

****

```
|-------------------------------------------------------------------------|
|                                                 Y        PROPERTIES     |
|*************                                    |        IN  CMS UNIT   |
|            *  |==============================|   ==| |==     ------------ |
|MEMBER   6  *  |  EUROPEAN SECTIONS           |     | |       AX=0.6544E+2 |
|            *  |     TAP ERED                 |     | |  --Z  AY=0.2000E+2 |
|DESIGN CODE *  |                              |     | |       AZ=0.3200E+2 |
| LRFD 2001  *  ================================   ==| |==     PY=0.1835E+3 |
|            *                                                 PZ=0.6566E+3 |
|            *  |<---LENGTH (M)=     1.10 --->|                RY=0.3710E+1 |
|*************                                                 RZ=0.1054E+2 |
|                                                                         |
|                 14.7 (KNS-METRE)                                        |
|PARAMETER        |                                        L203CAPACITIES   |
|IN KNS  CMS      |                                    L203    IN KNS METRE |
|---------------  +                            L203L203        ------------|
| KL/R-Y=  29.65  |                         L203               PNC=0.1455E+4|
| KL/R-Z=  10.44  +                                            pnc=0.1067E+3|
| UNL    =110.00  |               L203L203                     PNT=0.1610E+4|
| CB     =  1.00  +                                            pnt=0.0000E+0|
| PHIC   =  0.85  |           L203                             MNZ=0.1625E+3|
| PHIB   =  0.90  +   L203L203                                 mnz=0.1471E+2|
| FYLD   = 27.50  |L202                                        MNY=0.4460E+2|
| NSF    =  0.80  +---+---+---+---+---+---+---+---+---+---|   mny=0.1702E-1|
| DFF =    0.00   -0.6                                       VN =0.2970E+3|
| dff =    0.00               ABSOLUTE MZ ENVELOPE           vn =0.1322E+2|
|                                (WITH LOAD NO.)                          |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY  (KNS-METRE)              |
|                     -------------------------                           |
|                                                                         |
|                AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z   |
|                                                                         |
|      VALUE     107.4        13.2        0.3         0.4        14.7     |
|   LOCATION       0.0         0.0        0.0         0.0         1.1     |
|    LOADING       203         203        202         202         203     |
|                                                                         |
|*************************************************************************|
|*                                                                       *|
|*                         DESIGN SUMMARY  (KNS-METRE)                   *|
|*                         --------------                                *|
|*                                                                       *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/         *|
|         FX              MY             MZ           LOCATION           |
|        ==================================================             |
|          PASS        LRFD-H1-1B-C     0.128            203              |
|        106.71 C         -0.02        14.71            1.10             |
|*                                                                       *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
```

STAAD.PRO CODE CHECKING - (LRFD 3RD EDITION) v1.0
****

```
|-------------------------------------------------------------------------|
|                                                 Y        PROPERTIES     |
|*************                                    |        IN  CMS UNIT  |
|            *  |==============================|   ==| |==     ------------ |
|MEMBER   7  *  |  EUROPEAN SECTIONS           |     | |       AX=0.6544E+2 |
|            *  |     TAP ERED                 |     | |  --Z  AY=0.2000E+2 |
|DESIGN CODE *  |                              |     | |       AZ=0.3200E+2 |
| LRFD 2001  *  ================================   ==| |==     PY=0.1835E+3 |
|            *                                                 PZ=0.6566E+3 |
|            *  |<---LENGTH (M)=     0.80 --->|                RY=0.3710E+1 |
|*************                                                 RZ=0.1054E+2 |
|                                                                         |
|                101.4 (KNS-METRE)                                        |
|PARAMETER        |L202                                      CAPACITIES   |
|IN KNS  CMS      |   L202                                   IN KNS METRE |
|---------------  +       L202L202                           -------------|
| KL/R-Y=  51.21  |               L202                       PNC=0.1310E+4|
| KL/R-Z=  52.18  +                                          pnc=0.2788E+3|
| UNL   =  79.50  |                   L202L202               PNT=0.1610E+4|
| CB    =   1.00  +                                          pnt=0.0000E+0|
| PHIC  =   0.85  |                           L202           MNZ=0.1625E+3|
| PHIB  =   0.90  +                               L202L202   mnz=0.1014E+3|
| FYLD  =  27.50  |                                       L202MNY=0.4460E+2|
| NSF   =   0.80  +---+---+---+---+---+---+---+---+---+---|   mny=0.1609E-2|
| DFF =     0.00  13.7                                     VN =0.2970E+3|
| dff =     0.00              ABSOLUTE MZ ENVELOPE         vn =0.1060E+3|
|                                (WITH LOAD NO.)                          |
|                                                                         |
|                    MAX FORCE/ MOMENT SUMMARY  (KNS-METRE)               |
|                    -------------------------                            |
|                                                                         |
|               AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z    |
|                                                                         |
|      VALUE     278.8       106.0        2.6         2.1       101.4     |
|   LOCATION       0.0         0.0        0.0         0.8         0.0     |
|    LOADING       202         202        202         202         202     |
|                                                                         |
|*************************************************************************|
|*                                                                       *|
|*                        DESIGN SUMMARY  (KNS-METRE)                    *|
|*                        --------------                                 *|
|*                                                                       *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/         *|
|         FX              MY             MZ           LOCATION           |
|        ======================================================           |
|          PASS        LRFD-H1-1A-C     0.767            202              |
|        278.82 C         -0.00       101.40            0.00              |
|*                                                                       *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
```

STAAD.PRO CODE CHECKING - (LRFD 3RD EDITION) v1.0

****

```
|-------------------------------------------------------------------------|
|                                                 Y        PROPERTIES     |
|*************                                    |        IN  CMS UNIT   |
|            *  |==============================|   ==| |==     ------------ |
|MEMBER   8  *  |  EUROPEAN SECTIONS           |     | |       AX=0.6544E+2 |
|            *  |     TAP ERED                 |     | |  --Z  AY=0.2000E+2 |
|DESIGN CODE *  |                              |     | |       AZ=0.3200E+2 |
| LRFD 2001  *  ================================   ==| |==     PY=0.1835E+3 |
|            *                                          |       PZ=0.6566E+3 |
|            *  |<---LENGTH (M)=     0.80 --->|         |       RY=0.3710E+1 |
|*************                                          |       RZ=0.1054E+2 |
|                                                                         |
|                 90.1 (KNS-METRE)                                        |
|PARAMETER        |L202                                   CAPACITIES      |
|IN KNS  CMS      |   L202                                IN KNS METRE    |
|---------------  +       L202L202                        -------------|
| KL/R-Y=  51.21  |               L202                    PNC=0.1310E+4|
| KL/R-Z=  52.18  +                                       pnc=0.1174E+3|
| UNL   =  79.50  |                   L202L202            PNT=0.1610E+4|
| CB    =   1.00  +                                       pnt=0.0000E+0|
| PHIC  =   0.85  |                           L202        MNZ=0.1625E+3|
| PHIB  =   0.90  +                               L202L202  mnz=0.9013E+2|
| FYLD  =  27.50  |                                       L202MNY=0.4460E+2|
| NSF   =   0.80  +---+---+---+---+---+---+---+---+---+---|   mny=0.4734E-3|
| DFF =     0.00  14.4                                    VN =0.2970E+3|
| dff =     0.00              ABSOLUTE MZ ENVELOPE        vn =0.9177E+2|
|                                (WITH LOAD NO.)                          |
|                                                                         |
|                    MAX FORCE/ MOMENT SUMMARY  (KNS-METRE)               |
|                    -------------------------                            |
|                                                                         |
|               AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z    |
|                                                                         |
|      VALUE     117.4        92.1        0.9         0.7        90.1     |
|   LOCATION       0.0         0.0        0.0         0.8         0.0     |
|    LOADING       202         203        203         203         202     |
|                                                                         |
|*************************************************************************|
|*                                                                       *|
|*                        DESIGN SUMMARY  (KNS-METRE)                    *|
|*                        --------------                                 *|
|*                                                                       *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/         *|
|         FX              MY             MZ           LOCATION           |
|        ======================================================           |
|          PASS        LRFD-H1-1B-C     0.599            202              |
|        117.43 C         -0.00        90.13            0.00              |
|*                                                                       *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
```

STAAD.PRO CODE CHECKING - (LRFD 3RD EDITION) v1.0

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```
|-------------------------------------------------------------------------|
|                                                 Y        PROPERTIES     |
|*************                                    |        IN  CMS UNIT   |
|            *  |==============================|   ==| |==     ------------ |
|MEMBER   9  *  |  EUROPEAN SECTIONS           |     | |       AX=0.8144E+2 |
|            *  |     TAP ERED                 |     | |  --Z  AY=0.2000E+2 |
|DESIGN CODE *  |                              |     | |       AZ=0.4267E+2 |
| LRFD 2001  *  ================================   ==| |==     PY=0.3235E+3 |
|            *                                                 PZ=0.8438E+3 |
|            *  |<---LENGTH (M)=     0.80 --->|                RY=0.5119E+1 |
|*************                                                 RZ=0.1078E+2 |
|                                                                         |
|                 75.7 (KNS-METRE)                                        |
|PARAMETER        |                                        L203CAPACITIES   |
|IN KNS  CMS      |                                     L203    IN KNS METRE |
|---------------  +                              L203L203        ------------|
| KL/R-Y=  37.11  |                         L203                 PNC=0.1642E+4|
| KL/R-Z=  51.02  +                                              pnc=0.1161E+3|
| UNL   = 190.00  |                    L203                      PNT=0.2003E+4|
| CB    =   1.00  +                 L203                         pnt=0.0000E+0|
| PHIC  =   0.85  |            L203                              MNZ=0.2089E+3|
| PHIB  =   0.90  +L202   L203                                   mnz=0.7493E+2|
| FYLD  =  27.50  |   L203                                       MNY=0.7923E+2|
| NSF   =   0.80  +---+---+---+---+---+---+---+---+---+---+      mny=0.2307E+0|
| DFF =     0.00  -3.7                                           VN =0.2970E+3|
| dff =     0.00                ABSOLUTE MZ ENVELOPE             vn =0.1018E+3|
|                                  (WITH LOAD NO.)                        |
|                                                                         |
|                      MAX FORCE/ MOMENT SUMMARY  (KNS-METRE)             |
|                      -------------------------                          |
|                                                                         |
|                AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z    |
|                                                                         |
|      VALUE     116.1       104.9        0.3         0.2        75.7     |
|   LOCATION       0.0         0.0        0.0         0.8         0.8     |
|    LOADING       202         202        202         203         203     |
|                                                                         |
|*************************************************************************|
|*                                                                       *|
|*                         DESIGN SUMMARY  (KNS-METRE)                   *|
|*                         --------------                                *|
|*                                                                       *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/         *|
|         FX              MY             MZ           LOCATION           |
|        ======================================================           |
|          PASS        LRFD-H1-1B-C     0.397            202              |
|        116.07 C          0.23       -74.93            0.80              |
|*                                                                       *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
```

STAAD.PRO CODE CHECKING - (LRFD 3RD EDITION) v1.0
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```
|-------------------------------------------------------------------------|
|                                                 Y        PROPERTIES     |
|*************                                    |        IN  CMS UNIT  |
|            *  |==============================|   ==| |==     ------------ |
|MEMBER  10  *  |  EUROPEAN SECTIONS           |     | |       AX=0.6544E+2 |
|            *  |     TAP ERED                 |     | |  --Z  AY=0.2000E+2 |
|DESIGN CODE *  |                              |     | |       AZ=0.3200E+2 |
| LRFD 2001  *  ================================   ==| |==     PY=0.1835E+3 |
|            *                                                 PZ=0.6566E+3 |
|            *  |<---LENGTH (M)=     1.07 --->|                RY=0.3710E+1 |
|*************                                                 RZ=0.1054E+2 |
|                                                                         |
|                 85.9 (KNS-METRE)                                        |
|PARAMETER        |                                        L203CAPACITIES   |
|IN KNS  CMS      |                                    L203    IN KNS METRE |
|---------------  +                             L203L203        ------------|
| KL/R-Y=  51.21  |                         L203                PNC=0.1310E+4|
| KL/R-Z=  52.18  +                                             pnc=0.2076E+3|
| UNL    = 190.00 |                    L203                     PNT=0.1610E+4|
| CB     =   1.00 +                 L203                        pnt=0.0000E+0|
| PHIC   =   0.85 |            L203                             MNZ=0.1613E+3|
| PHIB   =   0.90 +L202   L203                                  mnz=0.8588E+2|
| FYLD   =  27.50 |   L203                                      MNY=0.4460E+2|
| NSF    =   0.80 +---+---+---+---+---+---+---+---+---+---+---|   mny=0.1250E+1|
| DFF =      0.00   7.6                                         VN =0.2970E+3|
| dff =      0.00                ABSOLUTE MZ ENVELOPE           vn =0.7530E+2|
|                                   (WITH LOAD NO.)                       |
|                                                                         |
|                      MAX FORCE/ MOMENT SUMMARY  (KNS-METRE)             |
|                      -------------------------                          |
|                                                                         |
|                AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z   |
|                                                                         |
|      VALUE     278.8        76.7        2.5         1.9        85.9     |
|   LOCATION       0.0         0.0        0.0         0.0         1.1     |
|    LOADING       202         202        203         202         203     |
|                                                                         |
|*************************************************************************|
|*                                                                       *|
|*                         DESIGN SUMMARY  (KNS-METRE)                   *|
|*                         --------------                                *|
|*                                                                       *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/         *|
|         FX              MY             MZ           LOCATION           |
|        ======================================================           |
|          PASS        LRFD-H1-1B-C     0.640            203              |
|        207.58 C          1.25       -85.88            1.07             |
|*                                                                       *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
```

STAAD.PRO CODE CHECKING - (LRFD 3RD EDITION) v1.0

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|-------------------------------------------------------------------------|
|                                                 Y        PROPERTIES     |
|*************                                    |        IN  CMS UNIT   |
|            *  |============================|   ==| |==     ------------   |
|MEMBER  11  *  |  EUROPEAN SECTIONS         |     | |       AX=0.6544E+2   |
|            *  |     TAP ERED               |     | |  --Z  AY=0.2000E+2   |
|DESIGN CODE *  |                            |     | |       AZ=0.3200E+2   |
| LRFD 2001  *  ================================   ==| |==     PY=0.1835E+3   |
|            *                                                 PZ=0.6566E+3   |
|            *  |<---LENGTH (M)=     1.77 --->|                RY=0.3710E+1   |
|*************                                                 RZ=0.1054E+2   |
|                                                                             |
|                 86.3 (KNS-METRE)                                            |
|PARAMETER        |                   L203L203L203            CAPACITIES    |
|IN KNS  CMS      |               L203            L203        IN KNS METRE  |
|---------------  +                                   L203    -------------|
| KL/R-Y=  51.21  |           L203                              PNC=0.1310E+4|
| KL/R-Z=  52.18  +                                       L203pnc=0.2076E+3|
| UNL    =190.00  |       L203                                  PNT=0.1610E+4|
| CB     =  1.00  +                                           pnt=0.0000E+0|
| PHIC   =  0.85  |   L203                                      MNZ=0.1613E+3|
| PHIB   =  0.90  +                                           mnz=0.8627E+2|
| FYLD   = 27.50  |L203                                       MNY=0.4460E+2|
| NSF    =  0.80  +---+---+---+---+---+---+---+---+---+---|   mny=0.1157E+1|
| DFF =     0.00  85.9                                        VN =0.2970E+3|
| dff =     0.00              ABSOLUTE MZ ENVELOPE            vn =0.3185E-1|
|                                (WITH LOAD NO.)                          |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY  (KNS-METRE)              |
|                     -------------------------                           |
|                                                                         |
|               AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z    |
|                                                                         |
|      VALUE     278.9         0.8        0.0         1.2        86.3     |
|   LOCATION       0.0         0.0        0.0         1.8         1.0     |
|    LOADING       202         201        201         203         203     |
|                                                                         |
|*************************************************************************|
|*                                                                       *|
|*                        DESIGN SUMMARY  (KNS-METRE)                    *|
|*                        --------------                                 *|
|*                                                                       *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/         *|
|         FX              MY             MZ           LOCATION           |
|        ======================================================           |
|          PASS        LRFD-H1-1B-C     0.640            203              |
|        207.59 C          1.16       -86.27            1.03              |
|*                                                                       *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
```

STAAD.PRO CODE CHECKING - (LRFD 3RD EDITION) v1.0

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```
|-------------------------------------------------------------------------|
|                                                 Y        PROPERTIES     |
|*************                                    |        IN  CMS UNIT   |
|            *  |==============================|   ==| |==     ------------ |
|MEMBER  12  *  |  EUROPEAN SECTIONS           |     | |       AX=0.6544E+2 |
|            *  |     TAP ERED                 |     | |  --Z  AY=0.2000E+2 |
|DESIGN CODE *  |                              |     | |       AZ=0.3200E+2 |
| LRFD 2001  *  ================================   ==| |==     PY=0.1835E+3 |
|            *                                                 PZ=0.6566E+3 |
|            *  |<---LENGTH (M)=     1.07 --->|                RY=0.3710E+1 |
|*************                                                 RZ=0.1054E+2 |
|                                                                         |
|                 86.1 (KNS-METRE)                                        |
|PARAMETER        |L203                                        CAPACITIES   |
|IN KNS  CMS      |   L203                                     IN KNS METRE |
|---------------  +       L203L203                             ------------|
| KL/R-Y=  51.21  |               L203                         PNC=0.1310E+4|
| KL/R-Z=  52.18  +                                            pnc=0.2076E+3|
| UNL   = 190.00  |                   L203                     PNT=0.1610E+4|
| CB    =   1.00  +                       L203                 pnt=0.0000E+0|
| PHIC  =   0.85  |                           L203             MNZ=0.1613E+3|
| PHIB  =   0.90  +                               L203    L202mnz=0.8611E+2|
| FYLD  =  27.50  |                                   L203     MNY=0.4460E+2|
| NSF   =   0.80  +---+---+---+---+---+---+---+---+---+---|   mny=0.1252E+1|
| DFF =     0.00   8.1                                         VN =0.2970E+3|
| dff =     0.00              ABSOLUTE MZ ENVELOPE             vn =0.7504E+2|
|                                (WITH LOAD NO.)                          |
|                                                                         |
|                    MAX FORCE/ MOMENT SUMMARY  (KNS-METRE)               |
|                    -------------------------                            |
|                                                                         |
|               AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z     |
|                                                                         |
|      VALUE     278.8        76.4        2.5         1.9        86.1     |
|   LOCATION       0.0         1.1        0.0         1.1         0.0     |
|    LOADING       202         202        203         202         203     |
|                                                                         |
|*************************************************************************|
|*                                                                       *|
|*                        DESIGN SUMMARY  (KNS-METRE)                    *|
|*                        --------------                                 *|
|*                                                                       *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/         *|
|         FX              MY             MZ           LOCATION            |
|        ======================================================           |
|          PASS        LRFD-H1-1B-C     0.641            203               |
|        207.58 C          1.25       -86.11            0.00              |
|*                                                                       *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
```

STAAD.PRO CODE CHECKING - (LRFD 3RD EDITION) v1.0
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```
|-------------------------------------------------------------------------|
|                                                 Y        PROPERTIES     |
|*************                                    |        IN  CMS UNIT  |
|            *  |==============================|   ==| |==     ------------ |
|MEMBER  13  *  |  EUROPEAN SECTIONS           |     | |       AX=0.6544E+2 |
|            *  |     TAP ERED                 |     | |  --Z  AY=0.2000E+2 |
|DESIGN CODE *  |                              |     | |       AZ=0.3200E+2 |
| LRFD 2001  *  ================================   ==| |==     PY=0.1835E+3 |
|            *                                                 PZ=0.6566E+3 |
|            *  |<---LENGTH (M)=     0.79 --->|                RY=0.3710E+1 |
|*************                                                 RZ=0.1054E+2 |
|                                                                         |
|                101.4 (KNS-METRE)                                        |
|PARAMETER        |                                      L202CAPACITIES     |
|IN KNS  CMS      |                                   L202    IN KNS METRE  |
|---------------  +                             L202L202        ------------|
| KL/R-Y=  51.21  |                         L202                PNC=0.1310E+4|
| KL/R-Z=  52.18  +                                             pnc=0.2788E+3|
| UNL    =  79.50 |                L202L202                     PNT=0.1610E+4|
| CB     =   1.00 +                                             pnt=0.0000E+0|
| PHIC   =   0.85 |           L202                              MNZ=0.1625E+3|
| PHIB   =   0.90 +   L202L202                                  mnz=0.1014E+3|
| FYLD   =  27.50 |L202                                         MNY=0.4460E+2|
| NSF    =   0.80 +---+---+---+---+---+---+---+---+---+---+---|  mny=0.1609E-2|
| DFF =      0.00 |13.1                                         VN =0.2970E+3|
| dff =      0.00               ABSOLUTE MZ ENVELOPE            vn =0.1076E+3|
|                                  (WITH LOAD NO.)                        |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY  (KNS-METRE)              |
|                     -------------------------                           |
|                                                                         |
|                AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z   |
|                                                                         |
|      VALUE     278.8       107.6        2.6         2.1       101.4     |
|   LOCATION       0.0         0.8        0.0         0.0         0.8     |
|    LOADING       202         202        202         202         202     |
|                                                                         |
|*************************************************************************|
|*                                                                       *|
|*                        DESIGN SUMMARY  (KNS-METRE)                    *|
|*                        --------------                                 *|
|*                                                                       *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/         *|
|         FX              MY             MZ           LOCATION           |
|        ======================================================           |
|          PASS        LRFD-H1-1A-C     0.767            202              |
|        278.82 C         -0.00       101.42            0.79              |
|*                                                                       *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
```

STAAD.PRO CODE CHECKING - (LRFD 3RD EDITION) v1.0
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```
|-------------------------------------------------------------------------|
|                                                 Y        PROPERTIES     |
|*************                                    |        IN  CMS UNIT  |
|            *  |==============================|   ==| |==     ------------ |
|MEMBER  14  *  |  EUROPEAN SECTIONS           |     | |       AX=0.6544E+2 |
|            *  |     TAP ERED                 |     | |  --Z  AY=0.2000E+2 |
|DESIGN CODE *  |                              |     | |       AZ=0.3200E+2 |
| LRFD 2001  *  ================================   ==| |==     PY=0.1835E+3 |
|            *                                                 PZ=0.6566E+3 |
|            *  |<---LENGTH (M)=     1.07 --->|                RY=0.3710E+1 |
|*************                                                 RZ=0.1054E+2 |
|                                                                         |
|                 64.2 (KNS-METRE)                                        |
|PARAMETER        |                                        L203CAPACITIES   |
|IN KNS  CMS      |                                   L203    IN KNS METRE |
|---------------  +                                 L203        ------------|
| KL/R-Y=  51.21  |                             L203            PNC=0.1310E+4|
| KL/R-Z=  52.18  +                         L203                pnc=0.1011E+3|
| UNL    = 190.00 |                     L203                    PNT=0.1610E+4|
| CB     =   1.00 +                 L203                        pnt=0.0000E+0|
| PHIC   =   0.85 |L202                                         MNZ=0.1613E+3|
| PHIB   =   0.90 +   L202    L203                               mnz=0.6416E+2|
| FYLD   =  27.50 |       L202                                  MNY=0.4460E+2|
| NSF    =   0.80 +---+---+---+---+---+---+---+---+---+---+---|  mny=0.8982E+0|
| DFF =      0.00   3.3                                        VN =0.2970E+3|
| dff =      0.00               ABSOLUTE MZ ENVELOPE           vn =0.6562E+2|
|                                  (WITH LOAD NO.)                        |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY  (KNS-METRE)              |
|                     -------------------------                           |
|                                                                         |
|                AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z   |
|                                                                         |
|      VALUE     117.4        66.3        1.5         1.0        64.2     |
|   LOCATION       0.0         0.0        0.0         1.1         1.1     |
|    LOADING       202         203        203         202         203     |
|                                                                         |
|*************************************************************************|
|*                                                                       *|
|*                         DESIGN SUMMARY  (KNS-METRE)                   *|
|*                         --------------                                *|
|*                                                                       *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/         *|
|         FX              MY             MZ           LOCATION           |
|        ======================================================           |
|          PASS        LRFD-H1-1B-C     0.456            203              |
|        101.08 C          0.90       -64.16            1.07              |
|*                                                                       *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
```

STAAD.PRO CODE CHECKING - (LRFD 3RD EDITION) v1.0

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|-------------------------------------------------------------------------|
|                                                 Y        PROPERTIES     |
|*************                                    |        IN  CMS UNIT   |
|            *  |============================|    |===     ------------   |
|MEMBER  15  *  |  EUROPEAN SECTIONS         |    |          AX=0.2040E+2  |
|            *  | ST  UPN140                 |    |     --Z  AY=0.9800E+1  |
|DESIGN CODE *  |                            |    |          AZ=0.8000E+1  |
| LRFD 2001  *  ==============================    |===       PY=0.2620E+2  |
|            *                                                PZ=0.1032E+3  |
|            *  |<---LENGTH (M)=     6.26 --->|               RY=0.1753E+1  |
|*************                                               RZ=0.5446E+1  |
|                                                                         |
|                 15.1 (KNS-METRE)                                        |
|PARAMETER        |L203                                      CAPACITIES     |
|IN KNS  CMS      |                                          IN KNS METRE   |
|---------------  +                                      L202-------------|
| KL/R-Y= 171.12  |                                          PNC=0.1006E+3|
| KL/R-Z= 115.01  +   L203                                   pnc=0.0000E+0|
| UNL   = 300.00  |                                    L202  PNT=0.5018E+3|
| CB    =   1.00  +               L203L203L203               pnt=0.1867E+1|
| PHIC  =   0.85  |       L202                               MNZ=0.1965E+2|
| PHIB  =   0.90  +                           L203L202       mnz=0.1502E+2|
| FYLD  =  27.50  |           L202                           MNY=0.5542E+1|
| NSF   =   0.80  +---+---+---+---+---+---+---+---+---+---|  mny=0.8944E-1|
| DFF =     0.00  -0.5                                     VN =0.1455E+3|
| dff =     0.00              ABSOLUTE MZ ENVELOPE         vn =0.1238E+2|
|                                (WITH LOAD NO.)                          |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY  (KNS-METRE)              |
|                     -------------------------                           |
|                                                                         |
|               AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z    |
|                                                                         |
|      VALUE      -6.3        12.4        0.0         0.1        15.1     |
|   LOCATION       6.3         0.0        0.0         0.0         0.0     |
|    LOADING       203         203        202         202         203     |
|                                                                         |
|*************************************************************************|
|*                                                                       *|
|*                        DESIGN SUMMARY  (KNS-METRE)                    *|
|*                        --------------                                 *|
|*                                                                       *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/         *|
|         FX              MY             MZ           LOCATION           |
|        ======================================================           |
|          PASS        LRFD-H1-1B-T     0.782            202              |
|          1.87 T          0.09        15.02            0.00             |
|*                                                                       *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
```

STAAD.PRO CODE CHECKING - (LRFD 3RD EDITION) v1.0

****

```
|-------------------------------------------------------------------------|
|                                                 Y        PROPERTIES     |
|*************                                    |        IN  CMS UNIT   |
|            *  |============================|   ==| |==     ------------   |
|MEMBER  16  *  |  EUROPEAN SECTIONS         |     | |       AX=0.8144E+2   |
|            *  |     TAP ERED               |     | |  --Z  AY=0.2000E+2   |
|DESIGN CODE *  |                            |     | |       AZ=0.4267E+2   |
| LRFD 2001  *  ================================   ==| |==     PY=0.3235E+3   |
|            *                                                 PZ=0.8438E+3   |
|            *  |<---LENGTH (M)=     1.07 --->|                RY=0.5119E+1   |
|*************                                                 RZ=0.1078E+2   |
|                                                                         |
|                155.4 (KNS-METRE)                                        |
|PARAMETER        |                                      L203CAPACITIES     |
|IN KNS  CMS      |                                 L203L203    IN KNS METRE |
|---------------  +                            L203              ------------|
| KL/R-Y=  37.11  |                        L203                  PNC=0.1642E+4|
| KL/R-Z=  51.02  +                    L203                      pnc=0.1160E+3|
| UNL    = 190.00 |                L203                          PNT=0.2003E+4|
| CB     =   1.00 +                                              pnt=0.0000E+0|
| PHIC   =   0.85 |       L203L203                               MNZ=0.2089E+3|
| PHIB   =   0.90 +   L203                                      mnz=0.1546E+3|
| FYLD   =  27.50 |L203                                          MNY=0.7923E+2|
| NSF    =   0.80 +---+---+---+---+---+---+---+---+---+---+---|  mny=0.3821E+1|
| DFF =      0.00  71.3                                         VN =0.2970E+3|
| dff =      0.00               ABSOLUTE MZ ENVELOPE            vn =0.7403E+2|
|                                  (WITH LOAD NO.)                        |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY  (KNS-METRE)              |
|                     -------------------------                           |
|                                                                         |
|                AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z   |
|                                                                         |
|      VALUE     116.0        74.9        3.4         3.8       155.4     |
|   LOCATION       0.0         0.0        0.0         1.1         1.1     |
|    LOADING       202         203        202         202         203     |
|                                                                         |
|*************************************************************************|
|*                                                                       *|
|*                         DESIGN SUMMARY  (KNS-METRE)                   *|
|*                         --------------                                *|
|*                                                                       *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/         *|
|         FX              MY             MZ           LOCATION           |
|        ==================================================             |
|          PASS        LRFD-H1-1B-C     0.824            202              |
|        116.05 C          3.82      -154.60            1.07              |
|*                                                                       *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
```

STAAD.PRO CODE CHECKING - (LRFD 3RD EDITION) v1.0

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```
|-------------------------------------------------------------------------|
|                                                 Y        PROPERTIES     |
|*************                                    |        IN  CMS UNIT   |
|            *  |============================|    |===     ------------   |
|MEMBER  17  *  |  EUROPEAN SECTIONS         |    |          AX=0.2040E+2  |
|            *  | ST  UPN140                 |    |     --Z  AY=0.9800E+1  |
|DESIGN CODE *  |                            |    |          AZ=0.8000E+1  |
| LRFD 2001  *  ==============================    |===       PY=0.2620E+2  |
|            *                                                PZ=0.1032E+3  |
|            *  |<---LENGTH (M)=     6.26 --->|               RY=0.1753E+1  |
|*************                                               RZ=0.5446E+1  |
|                                                                         |
|                 14.8 (KNS-METRE)                                        |
|PARAMETER        |                                        L202CAPACITIES   |
|IN KNS  CMS      |                                            IN KNS METRE |
|---------------  +L203                                       -------------|
| KL/R-Y= 171.12  |                                            PNC=0.1006E+3|
| KL/R-Z= 115.01  +                                   L202     pnc=0.0000E+0|
| UNL   = 300.00  |   L203                                     PNT=0.5018E+3|
| CB    =   1.00  +               L202L202L202                 pnt=0.3891E+1|
| PHIC  =   0.85  |                                 L202       MNZ=0.1965E+2|
| PHIB  =   0.90  +       L203L202                             mnz=0.1476E+2|
| FYLD  =  27.50  |                            L201            MNY=0.5542E+1|
| NSF   =   0.80  +---+---+---+---+---+---+---+---+---+---+   mny=0.5477E-1|
| DFF =     0.00  -0.7                                         VN =0.1455E+3|
| dff =     0.00              ABSOLUTE MZ ENVELOPE             vn =0.1237E+2|
|                                (WITH LOAD NO.)                          |
|                                                                         |
|                    MAX FORCE/ MOMENT SUMMARY  (KNS-METRE)               |
|                    -------------------------                            |
|                                                                         |
|               AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z    |
|                                                                         |
|      VALUE      -4.0        12.4        0.0         0.1        14.8     |
|   LOCATION       6.3         6.3        0.0         0.0         6.3     |
|    LOADING       203         202        202         202         202     |
|                                                                         |
|*************************************************************************|
|*                                                                       *|
|*                        DESIGN SUMMARY  (KNS-METRE)                    *|
|*                        --------------                                 *|
|*                                                                       *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/         *|
|         FX              MY             MZ           LOCATION           |
|        ======================================================           |
|          PASS        LRFD-H1-1B-T     0.765            202              |
|          3.89 T         -0.05        14.76            6.26             |
|*                                                                       *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
```

STAAD.PRO CODE CHECKING - (LRFD 3RD EDITION) v1.0

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|-------------------------------------------------------------------------|
|                                                 Y        PROPERTIES     |
|*************                                    |        IN  CMS UNIT   |
|            *  |============================|    |===     ------------   |
|MEMBER  22  *  |  EUROPEAN SECTIONS         |    |          AX=0.2040E+2  |
|            *  | ST  UPN140                 |    |     --Z  AY=0.9800E+1  |
|DESIGN CODE *  |                            |    |          AZ=0.8000E+1  |
| LRFD 2001  *  ==============================    |===       PY=0.2620E+2  |
|            *                                                PZ=0.1032E+3  |
|            *  |<---LENGTH (M)=     6.30 --->|               RY=0.1753E+1  |
|*************                                               RZ=0.5446E+1  |
|                                                                         |
|                 14.8 (KNS-METRE)                                        |
|PARAMETER        |L202                                      CAPACITIES     |
|IN KNS  CMS      |                                          IN KNS METRE   |
|---------------  +                  L203L203                -------------|
| KL/R-Y= 171.12  |               L203        L203           PNC=0.1006E+3|
| KL/R-Z= 115.75  +   L202                        L203       pnc=0.1861E+1|
| UNL   = 300.00  |                                          PNT=0.5018E+3|
| CB    =   1.00  +                                 L203     pnt=0.0000E+0|
| PHIC  =   0.85  |           L203                           MNZ=0.1965E+2|
| PHIB  =   0.90  +       L202                               mnz=0.1474E+2|
| FYLD  =  27.50  |                                      L0  MNY=0.5542E+1|
| NSF   =   0.80  +---+---+---+---+---+---+---+---+---+---|   mny=0.5620E-1|
| DFF =     0.00  -0.8                                     VN =0.1455E+3|
| dff =     0.00              ABSOLUTE MZ ENVELOPE           vn =0.1418E+2|
|                                (WITH LOAD NO.)                          |
|                                                                         |
|                    MAX FORCE/ MOMENT SUMMARY  (KNS-METRE)               |
|                    -------------------------                            |
|                                                                         |
|               AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z    |
|                                                                         |
|      VALUE       1.9        14.2        0.0         0.1        14.8     |
|   LOCATION       0.0         0.0        0.0         0.0         0.0     |
|    LOADING       202         202        203         203         202     |
|                                                                         |
|*************************************************************************|
|*                                                                       *|
|*                        DESIGN SUMMARY  (KNS-METRE)                    *|
|*                        --------------                                 *|
|*                                                                       *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/         *|
|         FX              MY             MZ           LOCATION           |
|        ======================================================           |
|          PASS        LRFD-H1-1B-C     0.769            203              |
|          1.86 C         -0.06        14.74            0.00             |
|*                                                                       *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
```

STAAD.PRO CODE CHECKING - (LRFD 3RD EDITION) v1.0
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|-------------------------------------------------------------------------|
|                                                 Y        PROPERTIES     |
|*************                                    |        IN  CMS UNIT  |
|            *  |==============================|   ==| |==     ------------ |
|MEMBER  23  *  |  EUROPEAN SECTIONS           |     | |       AX=0.6544E+2 |
|            *  |     TAP ERED                 |     | |  --Z  AY=0.2000E+2 |
|DESIGN CODE *  |                              |     | |       AZ=0.3200E+2 |
| LRFD 2001  *  ================================   ==| |==     PY=0.1835E+3 |
|            *                                                 PZ=0.6566E+3 |
|            *  |<---LENGTH (M)=     1.77 --->|                RY=0.3710E+1 |
|*************                                                 RZ=0.1054E+2 |
|                                                                         |
|                 64.6 (KNS-METRE)                                        |
|PARAMETER        |                   L203L203L203            CAPACITIES    |
|IN KNS  CMS      |               L203            L203        IN KNS METRE  |
|---------------  +                                   L203    -------------|
| KL/R-Y=  51.21  |           L203                              PNC=0.1310E+4|
| KL/R-Z=  52.18  +                                       L203pnc=0.1011E+3|
| UNL    =190.00  |       L203                                  PNT=0.1610E+4|
| CB     =  1.00  +                                           pnt=0.0000E+0|
| PHIC   =  0.85  |   L203                                      MNZ=0.1613E+3|
| PHIB   =  0.90  +                                           mnz=0.6456E+2|
| FYLD   = 27.50  |L203                                       MNY=0.4460E+2|
| NSF    =  0.80  +---+---+---+---+---+---+---+---+---+---|   mny=0.8198E+0|
| DFF =    0.00   64.2                                        VN =0.2970E+3|
| dff =    0.00              ABSOLUTE MZ ENVELOPE             vn =0.3346E-1|
|                               (WITH LOAD NO.)                           |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY  (KNS-METRE)              |
|                     -------------------------                           |
|                                                                         |
|               AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z    |
|                                                                         |
|      VALUE     117.4         0.8        0.0         0.9        64.6     |
|   LOCATION       0.0         0.0        0.0         1.8         1.0     |
|    LOADING       202         201        201         202         203     |
|                                                                         |
|*************************************************************************|
|*                                                                       *|
|*                        DESIGN SUMMARY  (KNS-METRE)                    *|
|*                        --------------                                 *|
|*                                                                       *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/         *|
|         FX              MY             MZ           LOCATION           |
|        ======================================================           |
|          PASS        LRFD-H1-1B-C     0.457            203              |
|        101.07 C          0.82       -64.56            1.03              |
|*                                                                       *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
```

STAAD.PRO CODE CHECKING - (LRFD 3RD EDITION) v1.0

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|-------------------------------------------------------------------------|
|                                                 Y        PROPERTIES     |
|*************                                    |        IN  CMS UNIT   |
|            *  |============================|    |===     ------------   |
|MEMBER  24  *  |  EUROPEAN SECTIONS         |    |          AX=0.3220E+2  |
|            *  | ST  UPN200                 |    |     --Z  AY=0.1700E+2  |
|DESIGN CODE *  |                            |    |          AZ=0.1150E+2  |
| LRFD 2001  *  ==============================    |===       PY=0.5340E+2  |
|            *                                                PZ=0.2292E+3  |
|            *  |<---LENGTH (M)=     6.26 --->|               RY=0.2144E+1  |
|*************                                               RZ=0.7702E+1  |
|                                                                         |
|                 42.6 (KNS-METRE)                                        |
|PARAMETER        |L202                                      CAPACITIES     |
|IN KNS  CMS      |                                       L203IN KNS METRE  |
|---------------  +                                          -------------|
| KL/R-Y= 139.93  |   L202                                   PNC=0.2329E+3|
| KL/R-Z=  81.32  +                                          pnc=0.0000E+0|
| UNL   = 300.00  |                                          PNT=0.7921E+3|
| CB    =   1.00  +       L202    L203               L203    pnt=0.5054E+1|
| PHIC  =   0.85  |                   L203L203               MNZ=0.4467E+2|
| PHIB  =   0.90  +                                          mnz=0.4256E+2|
| FYLD  =  27.50  |           L202            L203L202       MNY=0.9990E+1|
| NSF   =   0.80  +---+---+---+---+---+---+---+---+---+---|   mny=0.4422E-1|
| DFF =     0.00   0.1                                       VN =0.2524E+3|
| dff =     0.00              ABSOLUTE MZ ENVELOPE           vn =0.2938E+2|
|                                (WITH LOAD NO.)                          |
|                                                                         |
|                    MAX FORCE/ MOMENT SUMMARY  (KNS-METRE)               |
|                    -------------------------                            |
|                                                                         |
|               AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z    |
|                                                                         |
|      VALUE      -8.7        35.7        0.0         0.1        42.6     |
|   LOCATION       2.1         6.3        0.0         6.3         0.0     |
|    LOADING       201         203        201         201         202     |
|                                                                         |
|*************************************************************************|
|*                                                                       *|
|*                        DESIGN SUMMARY  (KNS-METRE)                    *|
|*                        --------------                                 *|
|*                                                                       *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/         *|
|         FX              MY             MZ           LOCATION            |
|        ======================================================           |
|          PASS        LRFD-H1-1B-T     0.960            202               |
|          5.05 T          0.04        42.56            0.00              |
|*                                                                       *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
```

STAAD.PRO CODE CHECKING - (LRFD 3RD EDITION) v1.0

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|-------------------------------------------------------------------------|
|                                                 Y        PROPERTIES     |
|*************                                    |        IN  CMS UNIT   |
|            *  |==============================|   ==| |==     ------------ |
|MEMBER  25  *  |  EUROPEAN SECTIONS           |     | |       AX=0.8144E+2 |
|            *  |     TAP ERED                 |     | |  --Z  AY=0.2000E+2 |
|DESIGN CODE *  |                              |     | |       AZ=0.4267E+2 |
| LRFD 2001  *  ================================   ==| |==     PY=0.3235E+3 |
|            *                                                 PZ=0.8438E+3 |
|            *  |<---LENGTH (M)=     1.77 --->|                RY=0.5119E+1 |
|*************                                                 RZ=0.1078E+2 |
|                                                                         |
|                155.8 (KNS-METRE)                                        |
|PARAMETER        |                   L203L203                CAPACITIES   |
|IN KNS  CMS      |               L203        L203             IN KNS METRE |
|---------------  +                                 L203        ------------|
| KL/R-Y=  37.11  |           L203                    L203     PNC=0.1642E+4|
| KL/R-Z=  51.02  +                                      L203pnc=0.1160E+3|
| UNL    = 190.00 |       L203                                 PNT=0.2003E+4|
| CB     =   1.00 +                                           pnt=0.0000E+0|
| PHIC   =   0.85 |   L203                                     MNZ=0.2089E+3|
| PHIB   =   0.90 +                                           mnz=0.1550E+3|
| FYLD   =  27.50 |L203                                        MNY=0.7923E+2|
| NSF    =   0.80 +---+---+---+---+---+---+---+---+---+---+   mny=0.3761E+1|
| DFF =     0.00 155.4                                        VN =0.2970E+3|
| dff =     0.00              ABSOLUTE MZ ENVELOPE            vn =0.9696E-2|
|                                (WITH LOAD NO.)                          |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY  (KNS-METRE)              |
|                     -------------------------                           |
|                                                                         |
|               AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z    |
|                                                                         |
|      VALUE     116.0         1.0        0.0         3.8       155.8     |
|   LOCATION       0.0         0.0        0.0         1.8         1.0     |
|    LOADING       202         201        201         202         203     |
|                                                                         |
|*************************************************************************|
|*                                                                       *|
|*                        DESIGN SUMMARY  (KNS-METRE)                    *|
|*                        --------------                                 *|
|*                                                                       *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/         *|
|         FX              MY             MZ           LOCATION           |
|        ======================================================           |
|          PASS        LRFD-H1-1B-C     0.825            202              |
|        116.02 C          3.76      -155.05            1.03              |
|*                                                                       *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
```

STAAD.PRO CODE CHECKING - (LRFD 3RD EDITION) v1.0

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|-------------------------------------------------------------------------|
|                                                 Y        PROPERTIES     |
|*************                                    |        IN  CMS UNIT   |
|            *  |============================|    |===     ------------   |
|MEMBER  26  *  |  EUROPEAN SECTIONS         |    |          AX=0.3220E+2  |
|            *  | ST  UPN200                 |    |     --Z  AY=0.1700E+2  |
|DESIGN CODE *  |                            |    |          AZ=0.1150E+2  |
| LRFD 2001  *  ==============================    |===       PY=0.5340E+2  |
|            *                                                PZ=0.2292E+3  |
|            *  |<---LENGTH (M)=     6.26 --->|               RY=0.2144E+1  |
|*************                                               RZ=0.7702E+1  |
|                                                                         |
|                 40.2 (KNS-METRE)                                        |
|PARAMETER        |                                        L203CAPACITIES   |
|IN KNS  CMS      |L203                                        IN KNS METRE |
|---------------  +                                           -------------|
| KL/R-Y= 139.93  |                                           PNC=0.2329E+3|
| KL/R-Z=  81.32  +                                    L203   pnc=0.0000E+0|
| UNL   = 300.00  |   L203                                    PNT=0.7921E+3|
| CB    =   1.00  +               L202L202                    pnt=0.2567E+1|
| PHIC  =   0.85  |                       L202                MNZ=0.4467E+2|
| PHIB  =   0.90  +       L203                    L203        mnz=0.4011E+2|
| FYLD  =  27.50  |           L202            L202            MNY=0.9990E+1|
| NSF   =   0.80  +---+---+---+---+---+---+---+---+---+---|   mny=0.7923E-1|
| DFF =     0.00   1.0                                       VN =0.2524E+3|
| dff =     0.00              ABSOLUTE MZ ENVELOPE           vn =0.3580E+2|
|                                (WITH LOAD NO.)                          |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY  (KNS-METRE)              |
|                     -------------------------                           |
|                                                                         |
|               AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z    |
|                                                                         |
|      VALUE      -5.5        35.8        0.0         0.1        40.2     |
|   LOCATION       2.1         6.3        0.0         6.3         6.3     |
|    LOADING       201         202        201         201         203     |
|                                                                         |
|*************************************************************************|
|*                                                                       *|
|*                        DESIGN SUMMARY  (KNS-METRE)                    *|
|*                        --------------                                 *|
|*                                                                       *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/         *|
|         FX              MY             MZ           LOCATION           |
|        ========================================================         |
|          PASS        LRFD-H1-1B-T     0.908            202              |
|          2.57 T         -0.08        40.11            6.26             |
|*                                                                       *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
```

STAAD.PRO CODE CHECKING - (LRFD 3RD EDITION) v1.0

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```
|-------------------------------------------------------------------------|
|                                                 Y        PROPERTIES     |
|*************                                    |        IN  CMS UNIT   |
|            *  |============================|    |===     ------------   |
|MEMBER  28  *  |  EUROPEAN SECTIONS         |    |          AX=0.3220E+2  |
|            *  | ST  UPN200                 |    |     --Z  AY=0.1700E+2  |
|DESIGN CODE *  |                            |    |          AZ=0.1150E+2  |
| LRFD 2001  *  ==============================    |===       PY=0.5340E+2  |
|            *                                                PZ=0.2292E+3  |
|            *  |<---LENGTH (M)=     6.30 --->|               RY=0.2144E+1  |
|*************                                               RZ=0.7702E+1  |
|                                                                         |
|                 40.2 (KNS-METRE)                                        |
|PARAMETER        |L203                                      CAPACITIES     |
|IN KNS  CMS      |                       L202L202           IN KNS METRE  |
|---------------  +               L202L202                    -------------|
| KL/R-Y= 139.93  |                                 L202     PNC=0.2329E+3|
| KL/R-Z=  81.85  +   L203                                    pnc=0.1281E+1|
| UNL   = 300.00  |                                          PNT=0.7921E+3|
| CB    =   1.00  +           L202                   L202    pnt=0.0000E+0|
| PHIC  =   0.85  |                                          MNZ=0.4467E+2|
| PHIB  =   0.90  +       L203                               mnz=0.4020E+2|
| FYLD  =  27.50  |                                      L0  MNY=0.9990E+1|
| NSF   =   0.80  +---+---+---+---+---+---+---+---+---+---|   mny=0.5379E-1|
| DFF =     0.00  -2.2                                       VN =0.2524E+3|
| dff =     0.00              ABSOLUTE MZ ENVELOPE           vn =0.3744E+2|
|                                (WITH LOAD NO.)                          |
|                                                                         |
|                    MAX FORCE/ MOMENT SUMMARY  (KNS-METRE)               |
|                    -------------------------                            |
|                                                                         |
|               AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z     |
|                                                                         |
|      VALUE      -3.6        37.4        0.0         0.1        40.2     |
|   LOCATION       6.3         0.0        0.0         0.0         0.0     |
|    LOADING       202         203        203         203         203     |
|                                                                         |
|*************************************************************************|
|*                                                                       *|
|*                        DESIGN SUMMARY  (KNS-METRE)                    *|
|*                        --------------                                 *|
|*                                                                       *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/         *|
|         FX              MY             MZ           LOCATION            |
|        ======================================================           |
|          PASS        LRFD-H1-1B-C     0.908            203               |
|          1.28 C         -0.05        40.20            0.00              |
|*                                                                       *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
```

STAAD.PRO CODE CHECKING - (LRFD 3RD EDITION) v1.0

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```
|-------------------------------------------------------------------------|
|                                                 Y        PROPERTIES     |
|*************                                    |        IN  CMS UNIT   |
|            *  |==============================|   ==| |==     ------------ |
|MEMBER  29  *  |  EUROPEAN SECTIONS           |     | |       AX=0.6544E+2 |
|            *  |     TAP ERED                 |     | |  --Z  AY=0.2000E+2 |
|DESIGN CODE *  |                              |     | |       AZ=0.3200E+2 |
| LRFD 2001  *  ================================   ==| |==     PY=0.1835E+3 |
|            *                                                 PZ=0.6566E+3 |
|            *  |<---LENGTH (M)=     1.07 --->|                RY=0.3710E+1 |
|*************                                                 RZ=0.1054E+2 |
|                                                                         |
|                 64.4 (KNS-METRE)                                        |
|PARAMETER        |L203                                        CAPACITIES   |
|IN KNS  CMS      |   L203                                     IN KNS METRE |
|---------------  +       L203                                 ------------|
| KL/R-Y=  51.21  |           L203                             PNC=0.1310E+4|
| KL/R-Z=  52.18  +               L203                       pnc=0.1011E+3|
| UNL    = 190.00 |                   L203                     PNT=0.1610E+4|
| CB     =   1.00 +                       L203                 pnt=0.0000E+0|
| PHIC   =   0.85 |                                        L202MNZ=0.1613E+3|
| PHIB   =   0.90 +                           L203    L202    mnz=0.6439E+2|
| FYLD   =  27.50 |                               L202         MNY=0.4460E+2|
| NSF    =   0.80 +---+---+---+---+---+---+---+---+---+---|   mny=0.8997E+0|
| DFF =     0.00   2.8                                       VN =0.2970E+3|
| dff =     0.00               ABSOLUTE MZ ENVELOPE          vn =0.6536E+2|
|                                 (WITH LOAD NO.)                         |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY  (KNS-METRE)              |
|                     -------------------------                           |
|                                                                         |
|                AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z   |
|                                                                         |
|      VALUE     117.4        66.1        1.5         1.0        64.4     |
|   LOCATION       0.0         1.1        0.0         0.0         0.0     |
|    LOADING       202         203        203         202         203     |
|                                                                         |
|*************************************************************************|
|*                                                                       *|
|*                        DESIGN SUMMARY  (KNS-METRE)                    *|
|*                        --------------                                 *|
|*                                                                       *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/         *|
|         FX              MY             MZ           LOCATION           |
|        ========================================================         |
|          PASS        LRFD-H1-1B-C     0.458            203              |
|        101.08 C          0.90       -64.39            0.00              |
|*                                                                       *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
```

STAAD.PRO CODE CHECKING - (LRFD 3RD EDITION) v1.0

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|-------------------------------------------------------------------------|
|                                                 Y        PROPERTIES     |
|*************                                    |        IN  CMS UNIT   |
|            *  |============================|    |===     ------------   |
|MEMBER  30  *  |  EUROPEAN SECTIONS         |    |          AX=0.3220E+2  |
|            *  | ST  UPN200                 |    |     --Z  AY=0.1700E+2  |
|DESIGN CODE *  |                            |    |          AZ=0.1150E+2  |
| LRFD 2001  *  ==============================    |===       PY=0.5340E+2  |
|            *                                                PZ=0.2292E+3  |
|            *  |<---LENGTH (M)=     6.26 --->|               RY=0.2144E+1  |
|*************                                               RZ=0.7702E+1  |
|                                                                         |
|                 42.6 (KNS-METRE)                                        |
|PARAMETER        |L202                                      CAPACITIES   |
|IN KNS  CMS      |                                       L203IN KNS METRE |
|---------------  +                                          -------------|
| KL/R-Y= 139.93  |   L202                                   PNC=0.2329E+3|
| KL/R-Z=  81.32  +                                          pnc=0.0000E+0|
| UNL   = 300.00  |                                          PNT=0.7921E+3|
| CB    =   1.00  +       L202    L203               L203    pnt=0.5047E+1|
| PHIC  =   0.85  |                   L203L203               MNZ=0.4467E+2|
| PHIB  =   0.90  +                                          mnz=0.4255E+2|
| FYLD  =  27.50  |           L202            L203L202       MNY=0.9990E+1|
| NSF   =   0.80  +---+---+---+---+---+---+---+---+---+---|   mny=0.4487E-1|
| DFF =     0.00   0.1                                       VN =0.2524E+3|
| dff =     0.00              ABSOLUTE MZ ENVELOPE           vn =0.2938E+2|
|                                (WITH LOAD NO.)                          |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY  (KNS-METRE)              |
|                     -------------------------                           |
|                                                                         |
|               AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z    |
|                                                                         |
|      VALUE      -8.7        35.7        0.0         0.1        42.6     |
|   LOCATION       2.1         6.3        0.0         6.3         0.0     |
|    LOADING       201         203        201         201         202     |
|                                                                         |
|*************************************************************************|
|*                                                                       *|
|*                        DESIGN SUMMARY  (KNS-METRE)                    *|
|*                        --------------                                 *|
|*                                                                       *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/         *|
|         FX              MY             MZ           LOCATION           |
|        ======================================================           |
|          PASS        LRFD-H1-1B-T     0.960            202              |
|          5.05 T         -0.04        42.55            0.00             |
|*                                                                       *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
```

STAAD.PRO CODE CHECKING - (LRFD 3RD EDITION) v1.0

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|-------------------------------------------------------------------------|
|                                                 Y        PROPERTIES     |
|*************                                    |        IN  CMS UNIT   |
|            *  |==============================|   ==| |==     ------------ |
|MEMBER  31  *  |  EUROPEAN SECTIONS           |     | |       AX=0.8144E+2 |
|            *  |     TAP ERED                 |     | |  --Z  AY=0.2000E+2 |
|DESIGN CODE *  |                              |     | |       AZ=0.4267E+2 |
| LRFD 2001  *  ================================   ==| |==     PY=0.3235E+3 |
|            *                                                  PZ=0.8438E+3 |
|            *  |<---LENGTH (M)=     1.07 --->|                 RY=0.5119E+1 |
|*************                                                  RZ=0.1078E+2 |
|                                                                         |
|                155.6 (KNS-METRE)                                        |
|PARAMETER        |L203                                        CAPACITIES   |
|IN KNS  CMS      |   L203L203                                 IN KNS METRE |
|---------------  +           L203                             ------------|
| KL/R-Y=  37.11  |               L203                         PNC=0.1642E+4|
| KL/R-Z=  51.02  +                   L203                   pnc=0.1160E+3|
| UNL    =190.00  |                       L203               PNT=0.2003E+4|
| CB     =  1.00  +                                          pnt=0.0000E+0|
| PHIC   =  0.85  |                           L203L203       MNZ=0.2089E+3|
| PHIB   =  0.90  +                                  L203    mnz=0.1548E+3|
| FYLD   = 27.50  |                                      L203MNY=0.7923E+2|
| NSF    =  0.80  +---+---+---+---+---+---+---+---+---+---|   mny=0.3823E+1|
| DFF =    0.00  71.8                                        VN =0.2970E+3|
| dff =    0.00              ABSOLUTE MZ ENVELOPE            vn =0.7376E+2|
|                               (WITH LOAD NO.)                           |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY  (KNS-METRE)              |
|                     -------------------------                           |
|                                                                         |
|               AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z    |
|                                                                         |
|      VALUE     116.0        74.7        3.4         3.8       155.6     |
|   LOCATION       0.0         1.1        0.0         0.0         0.0     |
|    LOADING       202         203        202         202         203     |
|                                                                         |
|*************************************************************************|
|*                                                                       *|
|*                        DESIGN SUMMARY  (KNS-METRE)                    *|
|*                        --------------                                 *|
|*                                                                       *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/         *|
|         FX              MY             MZ           LOCATION           |
|        ======================================================           |
|          PASS        LRFD-H1-1B-C     0.825            202              |
|        116.05 C          3.82      -154.84            0.00              |
|*                                                                       *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
```

STAAD.PRO CODE CHECKING - (LRFD 3RD EDITION) v1.0

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|-------------------------------------------------------------------------|
|                                                 Y        PROPERTIES     |
|*************                                    |        IN  CMS UNIT   |
|            *  |============================|    |===     ------------   |
|MEMBER  32  *  |  EUROPEAN SECTIONS         |    |          AX=0.3220E+2  |
|            *  | ST  UPN200                 |    |     --Z  AY=0.1700E+2  |
|DESIGN CODE *  |                            |    |          AZ=0.1150E+2  |
| LRFD 2001  *  ==============================    |===       PY=0.5340E+2  |
|            *                                                PZ=0.2292E+3  |
|            *  |<---LENGTH (M)=     6.26 --->|               RY=0.2144E+1  |
|*************                                               RZ=0.7702E+1  |
|                                                                         |
|                 40.2 (KNS-METRE)                                        |
|PARAMETER        |                                        L203CAPACITIES   |
|IN KNS  CMS      |L203                                        IN KNS METRE |
|---------------  +                                            ------------|
| KL/R-Y= 139.93  |                                            PNC=0.2329E+3|
| KL/R-Z=  81.32  +                                  L203      pnc=0.0000E+0|
| UNL   = 300.00  |   L203                                     PNT=0.7921E+3|
| CB    =   1.00  +               L202L202                     pnt=0.2563E+1|
| PHIC  =   0.85  |                       L202                 MNZ=0.4467E+2|
| PHIB  =   0.90  +       L203                    L203         mnz=0.4011E+2|
| FYLD  =  27.50  |           L202            L202             MNY=0.9990E+1|
| NSF   =   0.80  +---+---+---+---+---+---+---+---+---+---|   mny=0.7838E-1|
| DFF =     0.00   1.0                                       VN =0.2524E+3|
| dff =     0.00              ABSOLUTE MZ ENVELOPE           vn =0.3580E+2|
|                                (WITH LOAD NO.)                          |
|                                                                         |
|                    MAX FORCE/ MOMENT SUMMARY  (KNS-METRE)               |
|                    -------------------------                            |
|                                                                         |
|               AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z    |
|                                                                         |
|      VALUE      -5.5        35.8        0.0         0.1        40.2     |
|   LOCATION       2.1         6.3        0.0         6.3         6.3     |
|    LOADING       201         202        201         201         203     |
|                                                                         |
|*************************************************************************|
|*                                                                       *|
|*                        DESIGN SUMMARY  (KNS-METRE)                    *|
|*                        --------------                                 *|
|*                                                                       *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/         *|
|         FX              MY             MZ           LOCATION           |
|        ======================================================           |
|          PASS        LRFD-H1-1B-T     0.907            202              |
|          2.56 T          0.08        40.11            6.26             |
|*                                                                       *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
```

STAAD.PRO CODE CHECKING - (LRFD 3RD EDITION) v1.0

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|-------------------------------------------------------------------------|
|                                                 Y        PROPERTIES     |
|*************                                    |        IN  CMS UNIT   |
|            *  |============================|    |===     ------------   |
|MEMBER  34  *  |  EUROPEAN SECTIONS         |    |          AX=0.3220E+2  |
|            *  | ST  UPN200                 |    |     --Z  AY=0.1700E+2  |
|DESIGN CODE *  |                            |    |          AZ=0.1150E+2  |
| LRFD 2001  *  ==============================    |===       PY=0.5340E+2  |
|            *                                                PZ=0.2292E+3  |
|            *  |<---LENGTH (M)=     6.30 --->|               RY=0.2144E+1  |
|*************                                               RZ=0.7702E+1  |
|                                                                         |
|                 40.2 (KNS-METRE)                                        |
|PARAMETER        |L203                                      CAPACITIES     |
|IN KNS  CMS      |                       L202L202           IN KNS METRE  |
|---------------  +               L202L202                    -------------|
| KL/R-Y= 139.93  |                                 L202     PNC=0.2329E+3|
| KL/R-Z=  81.85  +   L203                                    pnc=0.1281E+1|
| UNL   = 300.00  |                                          PNT=0.7921E+3|
| CB    =   1.00  +           L202                  L202     pnt=0.0000E+0|
| PHIC  =   0.85  |                                          MNZ=0.4467E+2|
| PHIB  =   0.90  +       L203                               mnz=0.4020E+2|
| FYLD  =  27.50  |                                      L0  MNY=0.9990E+1|
| NSF   =   0.80  +---+---+---+---+---+---+---+---+---+---|   mny=0.5383E-1|
| DFF =     0.00  -2.2                                       VN =0.2524E+3|
| dff =     0.00              ABSOLUTE MZ ENVELOPE           vn =0.3744E+2|
|                                (WITH LOAD NO.)                          |
|                                                                         |
|                    MAX FORCE/ MOMENT SUMMARY  (KNS-METRE)               |
|                    -------------------------                            |
|                                                                         |
|               AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z     |
|                                                                         |
|      VALUE      -3.6        37.4        0.0         0.1        40.2     |
|   LOCATION       6.3         0.0        0.0         0.0         0.0     |
|    LOADING       202         203        203         203         203     |
|                                                                         |
|*************************************************************************|
|*                                                                        *|
|*                        DESIGN SUMMARY  (KNS-METRE)                     *|
|*                        --------------                                  *|
|*                                                                        *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/          *|
|         FX              MY             MZ           LOCATION            |
|        ==================================================              |
|          PASS        LRFD-H1-1B-C     0.908            203               |
|          1.28 C          0.05        40.20            0.00              |
|*                                                                        *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
```

STAAD.PRO CODE CHECKING - (LRFD 3RD EDITION) v1.0

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|-------------------------------------------------------------------------|
|                                                 Y        PROPERTIES     |
|*************                                    |        IN  CMS UNIT   |
|            *  |============================|   ==| |==     ------------   |
|MEMBER  35  *  |  EUROPEAN SECTIONS         |     | |       AX=0.6544E+2   |
|            *  |     TAP ERED               |     | |  --Z  AY=0.2000E+2   |
|DESIGN CODE *  |                            |     | |       AZ=0.3200E+2   |
| LRFD 2001  *  ================================   ==| |==     PY=0.1835E+3   |
|            *                                                 PZ=0.6566E+3   |
|            *  |<---LENGTH (M)=     0.79 --->|                RY=0.3710E+1   |
|*************                                                 RZ=0.1054E+2   |
|                                                                         |
|                 90.1 (KNS-METRE)                                        |
|PARAMETER        |                                        L202CAPACITIES   |
|IN KNS  CMS      |                                    L202    IN KNS METRE |
|---------------  +                             L202L202        ------------|
| KL/R-Y=  51.21  |                         L202                PNC=0.1310E+4|
| KL/R-Z=  52.18  +                                             pnc=0.1174E+3|
| UNL   =  79.50  |               L202L202                      PNT=0.1610E+4|
| CB    =   1.00  +                                             pnt=0.0000E+0|
| PHIC  =   0.85  |           L202                              MNZ=0.1625E+3|
| PHIB  =   0.90  +   L202L202                                  mnz=0.9014E+2|
| FYLD  =  27.50  |L202                                         MNY=0.4460E+2|
| NSF   =   0.80  +---+---+---+---+---+---+---+---+---+---+---|   mny=0.4733E-3|
| DFF =     0.00  13.8                                        VN =0.2970E+3|
| dff =     0.00               ABSOLUTE MZ ENVELOPE           vn =0.9336E+2|
|                                 (WITH LOAD NO.)                         |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY  (KNS-METRE)              |
|                     -------------------------                           |
|                                                                         |
|               AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z    |
|                                                                         |
|      VALUE     117.4        93.7        0.9         0.7        90.1     |
|   LOCATION       0.0         0.8        0.0         0.0         0.8     |
|    LOADING       202         203        203         203         202     |
|                                                                         |
|*************************************************************************|
|*                                                                       *|
|*                        DESIGN SUMMARY  (KNS-METRE)                    *|
|*                        --------------                                 *|
|*                                                                       *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/         *|
|         FX              MY             MZ           LOCATION           |
|        ==================================================             |
|          PASS        LRFD-H1-1B-C     0.599            202              |
|        117.43 C         -0.00        90.14            0.79             |
|*                                                                       *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
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STAAD.PRO CODE CHECKING - (LRFD 3RD EDITION) v1.0

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|-------------------------------------------------------------------------|
|                                                 Y        PROPERTIES     |
|*************                                    |        IN  CMS UNIT   |
|            *  |============================|    |===     ------------   |
|MEMBER  36  *  |  EUROPEAN SECTIONS         |    |          AX=0.2040E+2  |
|            *  | ST  UPN140                 |    |     --Z  AY=0.9800E+1  |
|DESIGN CODE *  |                            |    |          AZ=0.8000E+1  |
| LRFD 2001  *  ==============================    |===       PY=0.2620E+2  |
|            *                                                PZ=0.1032E+3  |
|            *  |<---LENGTH (M)=     6.26 --->|               RY=0.1753E+1  |
|*************                                               RZ=0.5446E+1  |
|                                                                         |
|                 15.1 (KNS-METRE)                                        |
|PARAMETER        |L203                                      CAPACITIES     |
|IN KNS  CMS      |                                          IN KNS METRE   |
|---------------  +                                      L202-------------|
| KL/R-Y= 171.12  |                                          PNC=0.1006E+3|
| KL/R-Z= 115.01  +   L203                                   pnc=0.0000E+0|
| UNL   = 300.00  |                                    L202  PNT=0.5018E+3|
| CB    =   1.00  +               L203L203L203               pnt=0.1868E+1|
| PHIC  =   0.85  |       L202                               MNZ=0.1965E+2|
| PHIB  =   0.90  +                           L203L202       mnz=0.1502E+2|
| FYLD  =  27.50  |           L202                           MNY=0.5542E+1|
| NSF   =   0.80  +---+---+---+---+---+---+---+---+---+---|  mny=0.8961E-1|
| DFF =     0.00  -0.5                                     VN =0.1455E+3|
| dff =     0.00              ABSOLUTE MZ ENVELOPE         vn =0.1238E+2|
|                                (WITH LOAD NO.)                          |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY  (KNS-METRE)              |
|                     -------------------------                           |
|                                                                         |
|               AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z    |
|                                                                         |
|      VALUE      -6.3        12.4        0.0         0.1        15.1     |
|   LOCATION       6.3         0.0        0.0         0.0         0.0     |
|    LOADING       203         203        202         202         203     |
|                                                                         |
|*************************************************************************|
|*                                                                       *|
|*                        DESIGN SUMMARY  (KNS-METRE)                    *|
|*                        --------------                                 *|
|*                                                                       *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/         *|
|         FX              MY             MZ           LOCATION           |
|        ======================================================           |
|          PASS        LRFD-H1-1B-T     0.782            202              |
|          1.87 T         -0.09        15.02            0.00             |
|*                                                                       *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
```

STAAD.PRO CODE CHECKING - (LRFD 3RD EDITION) v1.0

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|-------------------------------------------------------------------------|
|                                                 Y        PROPERTIES     |
|*************                                    |        IN  CMS UNIT   |
|            *  |============================|   ==| |==     ------------   |
|MEMBER  37  *  |  EUROPEAN SECTIONS         |     | |       AX=0.8144E+2   |
|            *  |     TAP ERED               |     | |  --Z  AY=0.2000E+2   |
|DESIGN CODE *  |                            |     | |       AZ=0.4267E+2   |
| LRFD 2001  *  ================================   ==| |==     PY=0.3235E+3   |
|            *                                                 PZ=0.8438E+3   |
|            *  |<---LENGTH (M)=     0.79 --->|                RY=0.5119E+1   |
|*************                                                 RZ=0.1078E+2   |
|                                                                             |
|                 76.2 (KNS-METRE)                                            |
|PARAMETER        |L203                                        CAPACITIES     |
|IN KNS  CMS      |   L203                                     IN KNS METRE   |
|---------------  +       L203L203                             -------------- |
| KL/R-Y=  37.11  |               L203                         PNC=0.1642E+4  |
| KL/R-Z=  51.02  +                                            pnc=0.1161E+3  |
| UNL   = 190.00  |                   L203                     PNT=0.2003E+4  |
| CB    =   1.00  +                       L203                 pnt=0.0000E+0  |
| PHIC  =   0.85  |                           L203             MNZ=0.2089E+3  |
| PHIB  =   0.90  +                               L203   L202  mnz=0.7544E+2  |
| FYLD  =  27.50  |                                   L203     MNY=0.7923E+2  |
| NSF   =   0.80  +---+---+---+---+---+---+---+---+---+---+    mny=0.2307E+0  |
| DFF =     0.00  -3.6                                         VN =0.2970E+3  |
| dff =     0.00              ABSOLUTE MZ ENVELOPE             vn =0.1016E+3  |
|                                (WITH LOAD NO.)                              |
|                                                                             |
|                    MAX FORCE/ MOMENT SUMMARY  (KNS-METRE)                   |
|                    -------------------------                                |
|                                                                             |
|               AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z       |
|                                                                             |
|      VALUE     116.1       106.5        0.3         0.2        76.2        |
|   LOCATION       0.0         0.8        0.0         0.0         0.0        |
|    LOADING       202         202        202         203         203        |
|                                                                             |
|*************************************************************************|
|*                                                                        *|
|*                        DESIGN SUMMARY  (KNS-METRE)                     *|
|*                        --------------                                  *|
|*                                                                        *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/          *|
|         FX              MY             MZ           LOCATION             |
|        ======================================================            |
|          PASS        LRFD-H1-1B-C     0.399            202               |
|        116.07 C          0.23       -75.44            0.00               |
|*                                                                        *|
|*************************************************************************|
|                                                                             |
|-----------------------------------------------------------------------------|
```

STAAD.PRO CODE CHECKING - (LRFD 3RD EDITION) v1.0

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|-------------------------------------------------------------------------|
|                                                 Y        PROPERTIES     |
|*************                                    |        IN  CMS UNIT   |
|            *  |============================|    |===     ------------   |
|MEMBER  38  *  |  EUROPEAN SECTIONS         |    |          AX=0.2040E+2  |
|            *  | ST  UPN140                 |    |     --Z  AY=0.9800E+1  |
|DESIGN CODE *  |                            |    |          AZ=0.8000E+1  |
| LRFD 2001  *  ==============================    |===       PY=0.2620E+2  |
|            *                                                PZ=0.1032E+3  |
|            *  |<---LENGTH (M)=     6.26 --->|               RY=0.1753E+1  |
|*************                                               RZ=0.5446E+1  |
|                                                                         |
|                 14.8 (KNS-METRE)                                        |
|PARAMETER        |                                        L202CAPACITIES   |
|IN KNS  CMS      |                                            IN KNS METRE |
|---------------  +L203                                       -------------|
| KL/R-Y= 171.12  |                                            PNC=0.1006E+3|
| KL/R-Z= 115.01  +                                    L202    pnc=0.0000E+0|
| UNL   = 300.00  |   L203                                     PNT=0.5018E+3|
| CB    =   1.00  +               L202L202L202                 pnt=0.3892E+1|
| PHIC  =   0.85  |                                 L202       MNZ=0.1965E+2|
| PHIB  =   0.90  +       L203L202                             mnz=0.1476E+2|
| FYLD  =  27.50  |                           L201             MNY=0.5542E+1|
| NSF   =   0.80  +---+---+---+---+---+---+---+---+---+---+   mny=0.5428E-1|
| DFF =     0.00  -0.7                                         VN =0.1455E+3|
| dff =     0.00              ABSOLUTE MZ ENVELOPE             vn =0.1237E+2|
|                                (WITH LOAD NO.)                          |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY  (KNS-METRE)              |
|                     -------------------------                           |
|                                                                         |
|               AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z    |
|                                                                         |
|      VALUE      -4.0        12.4        0.0         0.1        14.8     |
|   LOCATION       6.3         6.3        0.0         0.0         6.3     |
|    LOADING       203         202        202         202         202     |
|                                                                         |
|*************************************************************************|
|*                                                                        *|
|*                        DESIGN SUMMARY  (KNS-METRE)                     *|
|*                        --------------                                  *|
|*                                                                        *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/          *|
|         FX              MY             MZ           LOCATION            |
|        ========================================================          |
|          PASS        LRFD-H1-1B-T     0.765            202               |
|          3.89 T          0.05        14.76            6.26              |
|*                                                                        *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
```

STAAD.PRO CODE CHECKING - (LRFD 3RD EDITION) v1.0

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|-------------------------------------------------------------------------|
|                                                 Y        PROPERTIES     |
|*************                                    |        IN  CMS UNIT   |
|            *  |============================|    |===     ------------   |
|MEMBER  40  *  |  EUROPEAN SECTIONS         |    |          AX=0.2040E+2  |
|            *  | ST  UPN140                 |    |     --Z  AY=0.9800E+1  |
|DESIGN CODE *  |                            |    |          AZ=0.8000E+1  |
| LRFD 2001  *  ==============================    |===       PY=0.2620E+2  |
|            *                                                PZ=0.1032E+3  |
|            *  |<---LENGTH (M)=     6.30 --->|               RY=0.1753E+1  |
|*************                                               RZ=0.5446E+1  |
|                                                                         |
|                 14.8 (KNS-METRE)                                        |
|PARAMETER        |L202                                      CAPACITIES     |
|IN KNS  CMS      |                                          IN KNS METRE   |
|---------------  +                  L203L203                -------------|
| KL/R-Y= 171.12  |               L203        L203           PNC=0.1006E+3|
| KL/R-Z= 115.75  +   L202                        L203       pnc=0.1861E+1|
| UNL   = 300.00  |                                          PNT=0.5018E+3|
| CB    =   1.00  +                                 L203     pnt=0.0000E+0|
| PHIC  =   0.85  |           L203                           MNZ=0.1965E+2|
| PHIB  =   0.90  +       L202                               mnz=0.1474E+2|
| FYLD  =  27.50  |                                      L0  MNY=0.5542E+1|
| NSF   =   0.80  +---+---+---+---+---+---+---+---+---+---|   mny=0.5637E-1|
| DFF =     0.00  -0.8                                     VN =0.1455E+3|
| dff =     0.00              ABSOLUTE MZ ENVELOPE           vn =0.1418E+2|
|                                (WITH LOAD NO.)                          |
|                                                                         |
|                    MAX FORCE/ MOMENT SUMMARY  (KNS-METRE)               |
|                    -------------------------                            |
|                                                                         |
|               AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z     |
|                                                                         |
|      VALUE       1.9        14.2        0.0         0.1        14.8     |
|   LOCATION       0.0         0.0        0.0         0.0         0.0     |
|    LOADING       202         202        203         203         202     |
|                                                                         |
|*************************************************************************|
|*                                                                        *|
|*                        DESIGN SUMMARY  (KNS-METRE)                     *|
|*                        --------------                                  *|
|*                                                                        *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/          *|
|         FX              MY             MZ           LOCATION            |
|        ======================================================           |
|          PASS        LRFD-H1-1B-C     0.769            203               |
|          1.86 C          0.06        14.74            0.00              |
|*                                                                        *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
```

STAAD.PRO CODE CHECKING - (LRFD 3RD EDITION) v1.0

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|-------------------------------------------------------------------------|
|                                                 Y        PROPERTIES     |
|*************                                    |        IN  CMS UNIT   |
|            *  |============================|    |===     ------------   |
|MEMBER  41  *  |  EUROPEAN SECTIONS         |    |          AX=0.2040E+2  |
|            *  | ST  UPN140                 |    |     --Z  AY=0.9800E+1  |
|DESIGN CODE *  |                            |    |          AZ=0.8000E+1  |
| LRFD 2001  *  ==============================    |===       PY=0.2620E+2  |
|            *                                                PZ=0.1032E+3  |
|            *  |<---LENGTH (M)=     6.21 --->|               RY=0.1753E+1  |
|*************                                               RZ=0.5446E+1  |
|                                                                         |
|                 15.4 (KNS-METRE)                                        |
|PARAMETER        |                                        L202CAPACITIES   |
|IN KNS  CMS      |                                            IN KNS METRE |
|---------------  +               L203                        -------------|
| KL/R-Y= 171.12  |           L203    L203                    PNC=0.1006E+3|
| KL/R-Z= 113.96  +                       L203        L202    pnc=0.1256E-1|
| UNL   = 300.00  |       L203                                 PNT=0.5018E+3|
| CB    =   1.00  +                                            pnt=0.0000E+0|
| PHIC  =   0.85  |   L203                        L202         MNZ=0.1965E+2|
| PHIB  =   0.90  +                           L203             mnz=0.1536E+2|
| FYLD  =  27.50  |L203                                        MNY=0.5542E+1|
| NSF   =   0.80  +---+---+---+---+---+---+---+---+---+---|   mny=0.4831E-1|
| DFF =     0.00  -0.7                                       VN =0.1455E+3|
| dff =     0.00              ABSOLUTE MZ ENVELOPE           vn =0.1419E+2|
|                                (WITH LOAD NO.)                          |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY  (KNS-METRE)              |
|                     -------------------------                           |
|                                                                         |
|               AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z    |
|                                                                         |
|      VALUE     -10.8        14.2        0.0         0.0        15.4     |
|   LOCATION       6.2         6.2        0.0         6.2         6.2     |
|    LOADING       203         202        202         202         202     |
|                                                                         |
|*************************************************************************|
|*                                                                       *|
|*                        DESIGN SUMMARY  (KNS-METRE)                    *|
|*                        --------------                                 *|
|*                                                                       *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/         *|
|         FX              MY             MZ           LOCATION           |
|        ======================================================           |
|          PASS        LRFD-H1-1B-C     0.790            202              |
|          0.01 C         -0.05        15.36            6.21             |
|*                                                                       *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
```

STAAD.PRO CODE CHECKING - (LRFD 3RD EDITION) v1.0

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|-------------------------------------------------------------------------|
|                                                 Y        PROPERTIES     |
|*************                                    |        IN  CMS UNIT   |
|            *  |============================|    |===     ------------   |
|MEMBER  42  *  |  EUROPEAN SECTIONS         |    |          AX=0.3220E+2  |
|            *  | ST  UPN200                 |    |     --Z  AY=0.1700E+2  |
|DESIGN CODE *  |                            |    |          AZ=0.1150E+2  |
| LRFD 2001  *  ==============================    |===       PY=0.5340E+2  |
|            *                                                PZ=0.2292E+3  |
|            *  |<---LENGTH (M)=     6.21 --->|               RY=0.2144E+1  |
|*************                                                RZ=0.7702E+1  |
|                                                                         |
|                 42.6 (KNS-METRE)                                        |
|PARAMETER        |                                        L202CAPACITIES   |
|IN KNS  CMS      |               L203                         IN KNS METRE |
|---------------  +                   L203                     ------------|
| KL/R-Y= 139.93  |           L203        L203                 PNC=0.2329E+3|
| KL/R-Z=  80.58  +       L203                                 pnc=0.0000E+0|
| UNL   = 300.00  |                                   L202    PNT=0.7921E+3|
| CB    =   1.00  +   L203                                     pnt=0.1097E+2|
| PHIC  =   0.85  |                           L203           MNZ=0.4467E+2|
| PHIB  =   0.90  +                               L202        mnz=0.4258E+2|
| FYLD  =  27.50  |L203                                      MNY=0.9990E+1|
| NSF   =   0.80  +---+---+---+---+---+---+---+---+---+---|   mny=0.1529E-1|
| DFF =     0.00  -1.8                                     VN =0.2524E+3|
| dff =     0.00              ABSOLUTE MZ ENVELOPE           vn =0.4563E+2|
|                                (WITH LOAD NO.)                          |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY  (KNS-METRE)              |
|                     -------------------------                           |
|                                                                         |
|               AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z    |
|                                                                         |
|      VALUE     -13.8        45.6        0.0         0.0        42.6     |
|   LOCATION       0.0         6.2        0.0         6.2         6.2     |
|    LOADING       201         202        203         203         202     |
|                                                                         |
|*************************************************************************|
|*                                                                       *|
|*                        DESIGN SUMMARY  (KNS-METRE)                    *|
|*                        --------------                                 *|
|*                                                                       *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/         *|
|         FX              MY             MZ           LOCATION           |
|        ========================================================         |
|          PASS        LRFD-H1-1B-T     0.962            202              |
|         10.97 T         -0.02        42.58            6.21             |
|*                                                                       *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
```

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|-------------------------------------------------------------------------|
|                                                 Y        PROPERTIES     |
|*************                                    |        IN  CMS UNIT   |
|            *  |============================|    |===     ------------   |
|MEMBER  43  *  |  EUROPEAN SECTIONS         |    |          AX=0.3220E+2  |
|            *  | ST  UPN200                 |    |     --Z  AY=0.1700E+2  |
|DESIGN CODE *  |                            |    |          AZ=0.1150E+2  |
| LRFD 2001  *  ==============================    |===       PY=0.5340E+2  |
|            *                                                PZ=0.2292E+3  |
|            *  |<---LENGTH (M)=     6.21 --->|               RY=0.2144E+1  |
|*************                                               RZ=0.7702E+1  |
|                                                                         |
|                 42.6 (KNS-METRE)                                        |
|PARAMETER        |                                        L202CAPACITIES   |
|IN KNS  CMS      |               L203                         IN KNS METRE |
|---------------  +                   L203                     ------------|
| KL/R-Y= 139.93  |           L203        L203                 PNC=0.2329E+3|
| KL/R-Z=  80.58  +       L203                                 pnc=0.0000E+0|
| UNL   = 300.00  |                                   L202    PNT=0.7921E+3|
| CB    =   1.00  +   L203                                     pnt=0.1096E+2|
| PHIC  =   0.85  |                           L203           MNZ=0.4467E+2|
| PHIB  =   0.90  +                               L202        mnz=0.4258E+2|
| FYLD  =  27.50  |L203                                      MNY=0.9990E+1|
| NSF   =   0.80  +---+---+---+---+---+---+---+---+---+---|   mny=0.1559E-1|
| DFF =     0.00  -1.8                                     VN =0.2524E+3|
| dff =     0.00              ABSOLUTE MZ ENVELOPE           vn =0.4563E+2|
|                                (WITH LOAD NO.)                          |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY  (KNS-METRE)              |
|                     -------------------------                           |
|                                                                         |
|               AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z    |
|                                                                         |
|      VALUE     -13.8        45.6        0.0         0.0        42.6     |
|   LOCATION       0.0         6.2        0.0         6.2         6.2     |
|    LOADING       201         202        203         203         202     |
|                                                                         |
|*************************************************************************|
|*                                                                       *|
|*                        DESIGN SUMMARY  (KNS-METRE)                    *|
|*                        --------------                                 *|
|*                                                                       *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/         *|
|         FX              MY             MZ           LOCATION           |
|        ======================================================           |
|          PASS        LRFD-H1-1B-T     0.962            202              |
|         10.96 T          0.02        42.58            6.21             |
|*                                                                       *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
```

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```
|-------------------------------------------------------------------------|
|                                                 Y        PROPERTIES     |
|*************                                    |        IN  CMS UNIT   |
|            *  |============================|    |===     ------------   |
|MEMBER  44  *  |  EUROPEAN SECTIONS         |    |          AX=0.2040E+2  |
|            *  | ST  UPN140                 |    |     --Z  AY=0.9800E+1  |
|DESIGN CODE *  |                            |    |          AZ=0.8000E+1  |
| LRFD 2001  *  ==============================    |===       PY=0.2620E+2  |
|            *                                                PZ=0.1032E+3  |
|            *  |<---LENGTH (M)=     6.21 --->|               RY=0.1753E+1  |
|*************                                               RZ=0.5446E+1  |
|                                                                         |
|                 15.4 (KNS-METRE)                                        |
|PARAMETER        |                                        L202CAPACITIES   |
|IN KNS  CMS      |                                            IN KNS METRE |
|---------------  +               L203                        -------------|
| KL/R-Y= 171.12  |           L203    L203                    PNC=0.1006E+3|
| KL/R-Z= 113.96  +                       L203        L202    pnc=0.1206E-1|
| UNL   = 300.00  |       L203                                 PNT=0.5018E+3|
| CB    =   1.00  +                                            pnt=0.0000E+0|
| PHIC  =   0.85  |   L203                        L202         MNZ=0.1965E+2|
| PHIB  =   0.90  +                           L203             mnz=0.1536E+2|
| FYLD  =  27.50  |L203                                        MNY=0.5542E+1|
| NSF   =   0.80  +---+---+---+---+---+---+---+---+---+---|   mny=0.4839E-1|
| DFF =     0.00  -0.7                                       VN =0.1455E+3|
| dff =     0.00              ABSOLUTE MZ ENVELOPE           vn =0.1419E+2|
|                                (WITH LOAD NO.)                          |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY  (KNS-METRE)              |
|                     -------------------------                           |
|                                                                         |
|               AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z    |
|                                                                         |
|      VALUE     -10.8        14.2        0.0         0.0        15.4     |
|   LOCATION       6.2         6.2        0.0         6.2         6.2     |
|    LOADING       203         202        202         202         202     |
|                                                                         |
|*************************************************************************|
|*                                                                       *|
|*                        DESIGN SUMMARY  (KNS-METRE)                    *|
|*                        --------------                                 *|
|*                                                                       *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/         *|
|         FX              MY             MZ           LOCATION           |
|        ======================================================           |
|          PASS        LRFD-H1-1B-C     0.790            202              |
|          0.01 C          0.05        15.36            6.21             |
|*                                                                       *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
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|-------------------------------------------------------------------------|
|                                                 Y        PROPERTIES     |
|*************                                    |        IN  CMS UNIT  |
|            *  |==============================|   ==| |==     ------------ |
|MEMBER  47  *  |  EUROPEAN SECTIONS           |     | |       AX=0.6544E+2 |
|            *  |     TAP ERED                 |     | |  --Z  AY=0.2000E+2 |
|DESIGN CODE *  |                              |     | |       AZ=0.3200E+2 |
| LRFD 2001  *  ================================   ==| |==     PY=0.1835E+3 |
|            *                                                 PZ=0.6566E+3 |
|            *  |<---LENGTH (M)=     0.08 --->|                RY=0.3710E+1 |
|*************                                                 RZ=0.1054E+2 |
|                                                                         |
|                 14.7 (KNS-METRE)                                        |
|PARAMETER        |L203                                        CAPACITIES   |
|IN KNS  CMS      |   L203                                     IN KNS METRE |
|---------------  +       L203L203                             ------------|
| KL/R-Y=   2.16  |               L203                         PNC=0.1529E+4|
| KL/R-Z=   0.76  +                                            pnc=0.1050E+3|
| UNL   =   8.00  |                   L203                     PNT=0.1610E+4|
| CB    =   1.00  +                       L203                 pnt=0.0000E+0|
| PHIC  =   0.85  |                           L203             MNZ=0.1625E+3|
| PHIB  =   0.90  +                               L203         mnz=0.2031E+1|
| FYLD  =  27.50  |                                   L203L202MNY=0.4460E+2|
| NSF   =   0.80  +---+---+---+---+---+---+---+---+---+---|   mny=0.2424E-1|
| DFF =     0.00   6.7                                       VN =0.2970E+3|
| dff =     0.00              ABSOLUTE MZ ENVELOPE           vn =0.1161E+3|
|                                (WITH LOAD NO.)                          |
|                                                                         |
|                    MAX FORCE/ MOMENT SUMMARY  (KNS-METRE)               |
|                    -------------------------                            |
|                                                                         |
|               AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z     |
|                                                                         |
|      VALUE     105.0       116.1        0.3         0.0        14.7     |
|   LOCATION       0.0         0.0        0.0         0.0         0.0     |
|    LOADING       202         202        202         202         203     |
|                                                                         |
|*************************************************************************|
|*                                                                       *|
|*                        DESIGN SUMMARY  (KNS-METRE)                    *|
|*                        --------------                                 *|
|*                                                                       *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/         *|
|         FX              MY             MZ           LOCATION            |
|        ======================================================           |
|          PASS        SHEAR-Y          0.391            202              |
|        104.97 C          0.02         2.03            0.00              |
|*                                                                       *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
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|-------------------------------------------------------------------------|
|                                                 Y        PROPERTIES     |
|*************                                    |        IN  CMS UNIT   |
|            *  |============================|   ==| |==     ------------   |
|MEMBER  48  *  |  EUROPEAN SECTIONS         |     | |       AX=0.6544E+2   |
|            *  |     TAP ERED               |     | |  --Z  AY=0.2000E+2   |
|DESIGN CODE *  |                            |     | |       AZ=0.3200E+2   |
| LRFD 2001  *  ================================   ==| |==     PY=0.1835E+3   |
|            *                                                 PZ=0.6566E+3   |
|            *  |<---LENGTH (M)=     0.08 --->|                RY=0.3710E+1   |
|*************                                                 RZ=0.1054E+2   |
|                                                                             |
|                 14.7 (KNS-METRE)                                            |
|PARAMETER        |L203                                        CAPACITIES     |
|IN KNS  CMS      |   L203                                     IN KNS METRE   |
|---------------  +       L203L203                             -------------- |
| KL/R-Y=   2.16  |               L203                         PNC=0.1529E+4  |
| KL/R-Z=   0.76  +                                            pnc=0.1066E+3  |
| UNL   =   8.00  |                   L203                     PNT=0.1610E+4  |
| CB    =   1.00  +                       L203                 pnt=0.0000E+0  |
| PHIC  =   0.85  |                           L203             MNZ=0.1625E+3  |
| PHIB  =   0.90  +                               L203         mnz=0.2022E+1  |
| FYLD  =  27.50  |                                   L203L202MNY=0.4460E+2  |
| NSF   =   0.80  +---+---+---+---+---+---+---+---+---+---|   mny=0.2425E-1  |
| DFF =     0.00   6.7                                       VN =0.2970E+3  |
| dff =     0.00              ABSOLUTE MZ ENVELOPE           vn =0.1161E+3  |
|                                (WITH LOAD NO.)                              |
|                                                                             |
|                    MAX FORCE/ MOMENT SUMMARY  (KNS-METRE)                   |
|                    -------------------------                                |
|                                                                             |
|               AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z      |
|                                                                             |
|      VALUE     106.6       116.1        0.3         0.0        14.7       |
|   LOCATION       0.0         0.0        0.0         0.0         0.0       |
|    LOADING       202         202        202         202         203       |
|                                                                             |
|*************************************************************************|
|*                                                                        *|
|*                        DESIGN SUMMARY  (KNS-METRE)                     *|
|*                        --------------                                  *|
|*                                                                        *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/          *|
|         FX              MY             MZ           LOCATION             |
|        ======================================================            |
|          PASS        SHEAR-Y          0.391            202               |
|        106.55 C          0.02        -2.02            0.00               |
|*                                                                        *|
|*************************************************************************|
|                                                                             |
|-----------------------------------------------------------------------------|
```

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|-------------------------------------------------------------------------|
|                                                 Y        PROPERTIES     |
|*************                                    |        IN  CMS UNIT  |
|            *  |==============================|   ==||==      ------------ |
|MEMBER  49  *  |  EUROPEAN SECTIONS           |     ||        AX=0.2300E+2 |
|            *  | LD  L75X75X8                 |     ||   --Z  AY=0.8000E+1 |
|DESIGN CODE *  |                              |     ||        AZ=0.8000E+1 |
| LRFD 2001  *  |------------------------------|     ||        PY=0.6323E+2 |
|            *                                                 PZ=0.4045E+2 |
|            *  |<---LENGTH (M)=     1.50 --->|                RY=0.3496E+1 |
|*************                                                 RZ=0.2288E+1 |
|                                                                         |
|                  0.2 (KNS-METRE)                                        |
|PARAMETER         |                                        L202CAPACITIES  |
|IN KNS  CMS       |                                            IN KNS METRE|
|----------------  +                                    L202    ------------|
| KL/R-Y= 158.60   |                                  L202      PNC=0.1401E+3|
| KL/R-Z=  65.57   +                              L202           pnc=0.1163E+3|
| UNL   = 150.00   |                          L202               PNT=0.5658E+3|
| CB    =   1.00   +                     L203                   pnt=0.0000E+0|
| PHIC  =   0.85   |            L203L203                         MNZ=0.8342E+1|
| PHIB  =   0.90   +   L203L203                                  mnz=0.6396E-1|
| FYLD  =  27.50   |L202                                         MNY=0.8695E+1|
| NSF   =   0.80   +---+---+---+---+---+---+---+---+---+---+---|  mny=0.1151E-1|
| DFF =     0.00   -0.0                                         VN =0.1782E+3|
| dff =     0.00                ABSOLUTE MZ ENVELOPE            vn =0.3960E-2|
|                                  (WITH LOAD NO.)                        |
|                                                                         |
|                      MAX FORCE/ MOMENT SUMMARY  (KNS-METRE)             |
|                      -------------------------                          |
|                                                                         |
|                AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z   |
|                                                                         |
|      VALUE    -117.7         0.3        0.0         0.0         0.2     |
|   LOCATION       0.0         1.5        0.0         1.5         1.5     |
|    LOADING       202         202          0         202         202     |
|                                                                         |
|*************************************************************************|
|*                                                                       *|
|*                         DESIGN SUMMARY  (KNS-METRE)                   *|
|*                         --------------                                *|
|*                                                                       *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/         *|
|         FX              MY             MZ           LOCATION           |
|        ======================================================           |
|          PASS        LRFD-H1-1A-C     0.838            203              |
|        116.29 C          0.01        -0.06            0.75              |
|*                                                                       *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
```

STAAD.PRO CODE CHECKING - (LRFD 3RD EDITION) v1.0

****

```
|-------------------------------------------------------------------------|
|                                                 Y        PROPERTIES     |
|*************                                    |        IN  CMS UNIT   |
|            *  |============================|   ==||==      ------------   |
|MEMBER  50  *  |  EUROPEAN SECTIONS         |     ||        AX=0.2300E+2   |
|            *  | LD  L75X75X8               |     ||   --Z  AY=0.8000E+1   |
|DESIGN CODE *  |                            |     ||        AZ=0.8000E+1   |
| LRFD 2001  *  |----------------------------|     ||        PY=0.6323E+2   |
|            *                                                PZ=0.4045E+2   |
|            *  |<---LENGTH (M)=     2.50 --->|               RY=0.3496E+1   |
|*************                                                RZ=0.2288E+1   |
|                                                                           |
|                  0.2 (KNS-METRE)                                          |
|PARAMETER         |L202                                  L202CAPACITIES   |
|IN KNS  CMS       |                                          IN KNS METRE |
|----------------  +                                          -------------|
| KL/R-Y= 158.60   |                                          PNC=0.1401E+3|
| KL/R-Z= 109.28   +               L203L203L203               pnc=0.1170E+3|
| UNL   = 250.00   |   L202                            L202   PNT=0.5658E+3|
| CB    =   1.00   +                                          pnt=0.0000E+0|
| PHIC  =   0.85   |                                          MNZ=0.8342E+1|
| PHIB  =   0.90   +           L203            L203            mnz=0.1793E+0|
| FYLD  =  27.50   |       L202                    L202       MNY=0.8695E+1|
| NSF   =   0.80   +---+---+---+---+---+---+---+---+---+---|  mny=0.1151E-1|
| DFF =     0.00    0.1                                     VN =0.1782E+3|
| dff =     0.00               ABSOLUTE MZ ENVELOPE          vn =0.1377E-2|
|                                 (WITH LOAD NO.)                           |
|                                                                           |
|                     MAX FORCE/ MOMENT SUMMARY  (KNS-METRE)                |
|                     -------------------------                             |
|                                                                           |
|                AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z     |
|                                                                           |
|      VALUE     117.0         0.3        0.0         0.0         0.2      |
|   LOCATION       0.0         0.0        0.0         2.5         0.0      |
|    LOADING       203         201          0         202         202      |
|                                                                           |
|*************************************************************************|
|*                                                                        *|
|*                         DESIGN SUMMARY  (KNS-METRE)                    *|
|*                         --------------                                 *|
|*                                                                        *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/          *|
|         FX              MY             MZ           LOCATION             |
|        ======================================================            |
|          PASS        LRFD-H1-1A-C     0.855            203               |
|        117.01 C          0.01        -0.18            1.25               |
|*                                                                        *|
|*************************************************************************|
|                                                                           |
|-------------------------------------------------------------------------|
```

STAAD.PRO CODE CHECKING - (LRFD 3RD EDITION) v1.0
****

```
|-------------------------------------------------------------------------|
|                                                 Y        PROPERTIES     |
|*************                                    |        IN  CMS UNIT  |
|            *  |==============================|   ==||==      ------------ |
|MEMBER  51  *  |  EUROPEAN SECTIONS           |     ||        AX=0.2300E+2 |
|            *  | LD  L75X75X8                 |     ||   --Z  AY=0.8000E+1 |
|DESIGN CODE *  |                              |     ||        AZ=0.8000E+1 |
| LRFD 2001  *  |------------------------------|     ||        PY=0.6323E+2 |
|            *                                          PZ=0.4045E+2 |
|            *  |<---LENGTH (M)=     1.50 --->|               RY=0.3496E+1 |
|*************                                          RZ=0.2288E+1 |
|                                                                      |
|                  0.2 (KNS-METRE)                                     |
|PARAMETER         |L202                                 CAPACITIES     |
|IN KNS  CMS       |                                     IN KNS METRE   |
|----------------  +   L202                              -------------|
| KL/R-Y= 158.60   |       L202                          PNC=0.1401E+3|
| KL/R-Z=  65.57   +           L202                       pnc=0.1163E+3|
| UNL   = 150.00   |               L202                  PNT=0.5658E+3|
| CB    =   1.00   +                   L203L203         pnt=0.0000E+0|
| PHIC  =   0.85   |                           L203      MNZ=0.8342E+1|
| PHIB  =   0.90   +                               L203L203 mnz=0.6567E-1|
| FYLD  =  27.50   |                                     L202MNY=0.8695E+1|
| NSF   =   0.80   +---+---+---+---+---+---+---+---+---+---|  mny=0.1151E-1|
| DFF =     0.00   -0.0                                    VN =0.1782E+3|
| dff =     0.00              ABSOLUTE MZ ENVELOPE          vn =0.1657E-2|
|                                (WITH LOAD NO.)                       |
|                                                                      |
|                    MAX FORCE/ MOMENT SUMMARY  (KNS-METRE)            |
|                    -------------------------                         |
|                                                                      |
|               AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z  |
|                                                                      |
|      VALUE    -117.7         0.3        0.0         0.0         0.2   |
|   LOCATION       0.0         0.0        0.0         1.5         0.0   |
|    LOADING       202         202          0         202         202   |
|                                                                      |
|**********************************************************************|
|*                                                                     *|
|*                        DESIGN SUMMARY  (KNS-METRE)                  *|
|*                        --------------                               *|
|*                                                                     *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/       *|
|         FX              MY             MZ           LOCATION         |
|        ==========================================================     |
|          PASS        LRFD-H1-1A-C     0.838            203            |
|        116.30 C          0.01        -0.07            0.75            |
|*                                                                     *|
|**********************************************************************|
|                                                                      |
|----------------------------------------------------------------------|
```

********** END OF TABULATED RESULT OF DESIGN **********

```
191. ************************************************************
192. LOAD LIST 401 TO 403
193. START CONCRETE DESIGN
194. CODE ACI 2008
195. FC 35000 MEMB 52 TO 67
196. FYMAIN 420000 MEMB 52 TO 67
197. TRACK 1 MEMB 52 TO 67
198. CLB 0.05 MEMB 52 TO 67
199. CLT 0.05 MEMB 52 TO 67
200. CLS 0.05 MEMB 52 TO 67
```

==========

COLUMN NO. 52 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	81.59	0.01	0.00007

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	81.59	1.69	0.01533

==========

COLUMN NO. 53 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

```
 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
   M0        P-tens.   Des.Pn    Des.Mn     e/h
 411.95  -1583.37     84.43      0.01    0.00006
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
   M0        P-tens.   Des.Pn    Des.Mn     e/h
 411.95  -1583.37     84.43      1.67    0.01469
 ---------------------------------------------------------
```

==========

COLUMN NO. 54 DESIGN PER ACI 318-08 - AXIAL + BENDING

```
FY - 420.0  FC -  35.0 MPA,  SQRE SIZE - 600.0 X 600.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3600.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
 ---------------------------------------------------------

 12 - 20 MM               1.047      401      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER    8 SPACING 320.00 MM

    COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
   M0        P-tens.   Des.Pn    Des.Mn     e/h
 411.95  -1583.37     75.02      0.01    0.00006
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
   M0        P-tens.   Des.Pn    Des.Mn     e/h
 411.95  -1583.37     75.02     11.43    0.11287
 ---------------------------------------------------------
```

==========

COLUMN NO. 55 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	77.86	0.01	0.00006

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	77.86	11.44	0.10883

==========

COLUMN NO. 56 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	83.46	0.01	0.00009

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	83.46	21.97	0.19499

==========

COLUMN NO. 57 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	86.29	0.01	0.00009

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	86.29	21.96	0.18853

==========

COLUMN NO. 61 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

12 - 20 MM 1.047 401 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	55.70	0.31	0.00414

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	55.70	57.00	0.75801

==========

COLUMN NO. 62 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

12 - 20 MM 1.047 401 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	55.70	0.31	0.00412

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	55.70	57.00	0.75800

**********END OF COLUMN DESIGN RESULTS**********

206. DESIGN BEAM 58 TO 60 63 TO 67

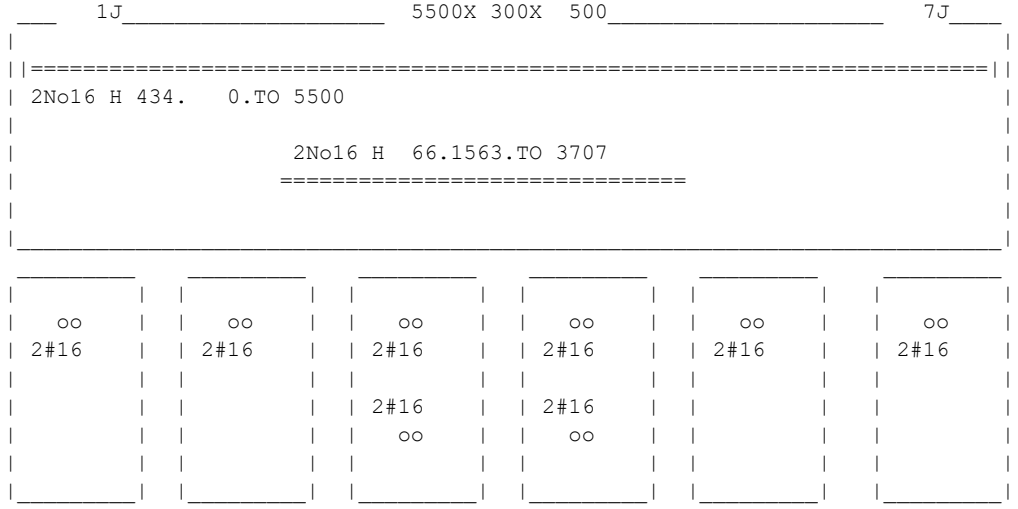
BEAM NO. 58 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 16MM	1563.	3707.	NO	NO
	CRITICAL POS MOMENT= 4.91 KN-MET AT 2750.MM, LOAD 401					
	REQD STEEL= 40.MM2, RHO=0.0003, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 168. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 41100.8 cm^4					
2	434.	2 - 16MM	0.	5500.	YES	YES
	CRITICAL NEG MOMENT= 23.65 KN-MET AT 0.MM, LOAD 402					
	REQD STEEL= 193.MM2, RHO=0.0015, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 168. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 41100.8 cm^4					

B E A M N O. 58 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 12.61 KNS Vc= 110.67 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 4.5 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 12.60 KNS Vc= 110.67 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 4.5 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 59 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

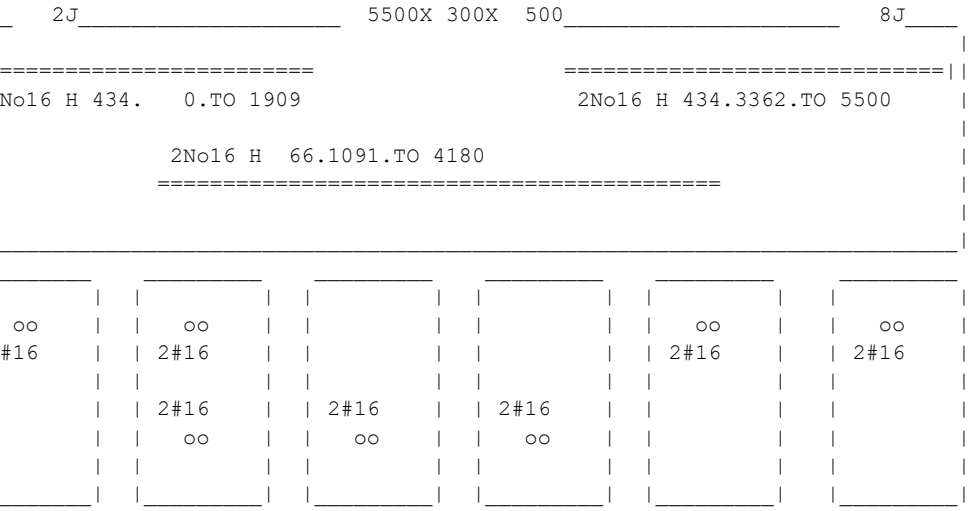
LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 16MM	1091.	4180.	NO	NO
	CRITICAL POS MOMENT= 6.63 KN-MET AT 2750.MM, LOAD 401					
	REQD STEEL= 54.MM2, RHO=0.0004, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 168. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 41100.8 cm^4					
2	434.	2 - 16MM	0.	1909.	YES	NO
	CRITICAL NEG MOMENT= 17.25 KN-MET AT 0.MM, LOAD 402					
	REQD STEEL= 141.MM2, RHO=0.0011, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 168. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 41100.8 cm^4					
3	434.	2 - 16MM	3362.	5500.	NO	YES

```
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     17.24 KN-MET  AT  5500.MM, LOAD  402|
|   REQD STEEL=   141.MM2, RHO=0.0011, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./  168. MMS         |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                       |
|-------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    41100.8 cm^4
```

B E A M N O. 59 D E S I G N R E S U L T S - SHEAR

```
 AT START SUPPORT - Vu=   12.61 KNS  Vc=  121.41 KNS  Vs=    0.00 KNS
 Tu=    0.00 KN-MET  Tc=    4.9 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
 AT END   SUPPORT - Vu=   12.60 KNS  Vc=  121.41 KNS  Vs=    0.00 KNS
 Tu=    0.00 KN-MET  Tc=    4.9 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
```



BEAM NO. 60 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 16MM	1079.	4192.	NO	NO

```
|--------------------------------------------------------------|
|   CRITICAL POS MOMENT=      8.09 KN-MET  AT  2750.MM, LOAD  402|
|   REQD STEEL=    66.MM2, RHO=0.0005, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./  168. MMS         |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                       |
|--------------------------------------------------------------|
```

Cracked Moment of Inertia Iz at above location = 41100.8 cm^4

```
  2        434.     2 - 16MM         0.         1450.        YES  NO
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     13.08 KN-MET  AT     0.MM, LOAD  403|
|   REQD STEEL=   106.MM2, RHO=0.0008, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./  168. MMS         |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                       |
|--------------------------------------------------------------|
```

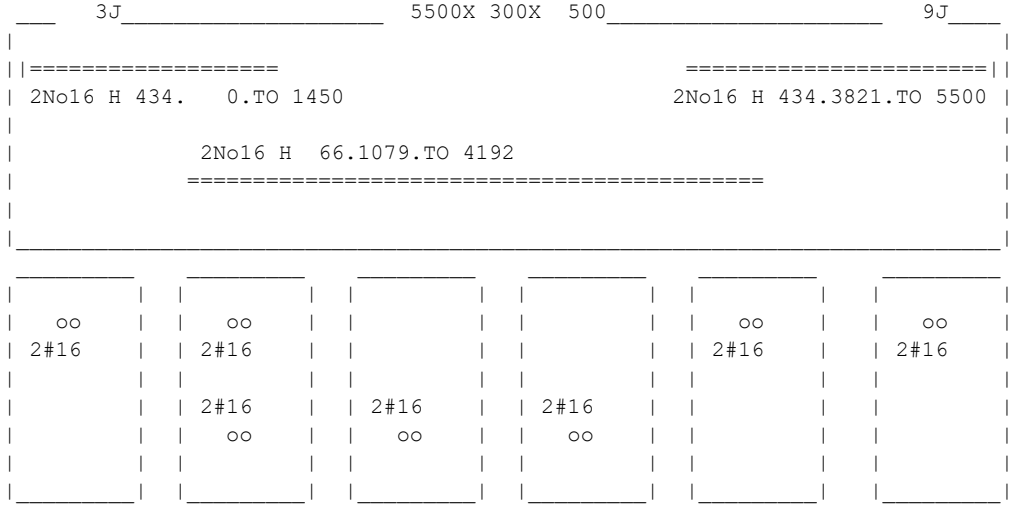
Cracked Moment of Inertia Iz at above location = 41100.8 cm^4

```
  3        434.     2 - 16MM      3821.         5500.        NO   YES
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     13.07 KN-MET  AT  5500.MM, LOAD  403|
|   REQD STEEL=   106.MM2, RHO=0.0008, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./  168. MMS         |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                       |
|--------------------------------------------------------------|
```

Cracked Moment of Inertia Iz at above location = 41100.8 cm^4

B E A M N O. 60 D E S I G N R E S U L T S - SHEAR

```
AT START SUPPORT - Vu=   12.61 KNS  Vc=  122.68 KNS  Vs=    0.00 KNS
Tu=    0.00 KN-MET  Tc=    5.2 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
AT END   SUPPORT - Vu=   12.60 KNS  Vc=  122.68 KNS  Vs=    0.00 KNS
Tu=    0.00 KN-MET  Tc=    5.2 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
```



BEAM NO. 63 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 796. MM FY - 420. FC - 35. MPA, SIZE - 350. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	534.	3 - 16MM	0.	796.	YES	YES

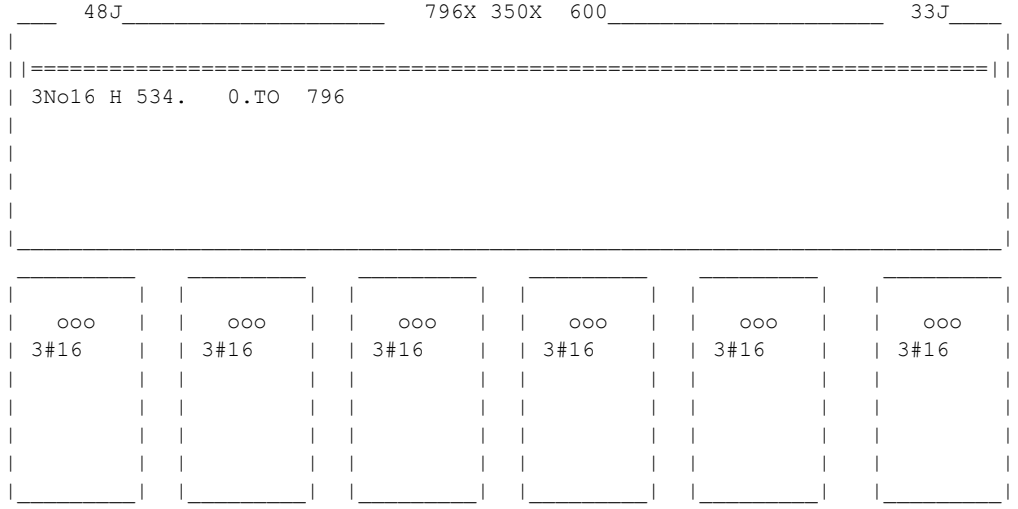
CRITICAL NEG MOMENT= 62.86 KN-MET AT 0.MM, LOAD 403
REQD STEEL= 419.MM2, RHO=0.0022, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 109. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

Cracked Moment of Inertia Iz at above location = 92778.8 cm^4

B E A M N O. 63 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 63 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 63 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



BEAM NO. 64 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 1070. MM FY - 420. FC - 35. MPA, SIZE - 350. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 16MM	0.	1070.	YES	YES

CRITICAL POS MOMENT= 46.73 KN-MET AT 1070.MM, LOAD 403
REQD STEEL= 311.MM2, RHO=0.0017, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 218. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

Cracked Moment of Inertia Iz at above location = 65065.9 cm^4

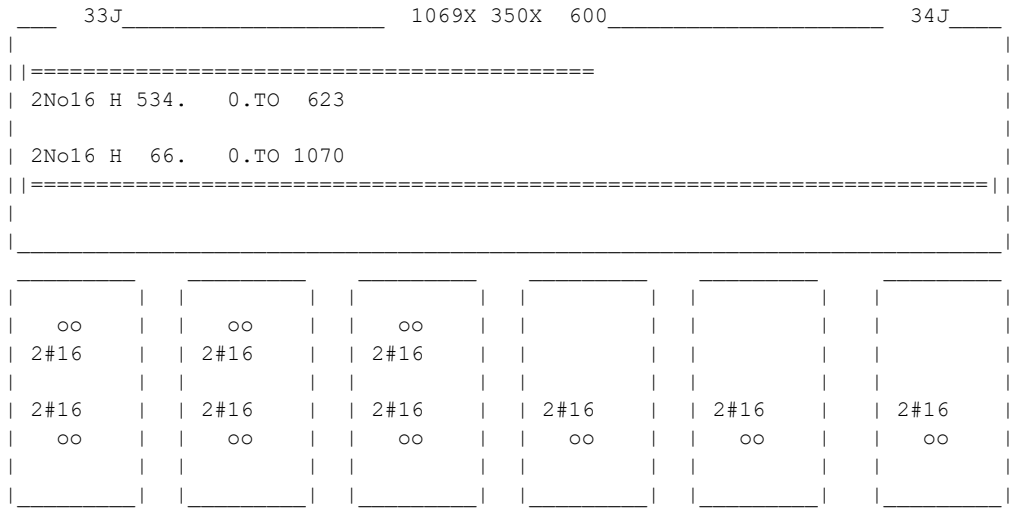
LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
2	534.	2 - 16MM	0.	623.	YES	NO

CRITICAL NEG MOMENT= 7.52 KN-MET AT 0.MM, LOAD 403
REQD STEEL= 50.MM2, RHO=0.0003, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 218. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

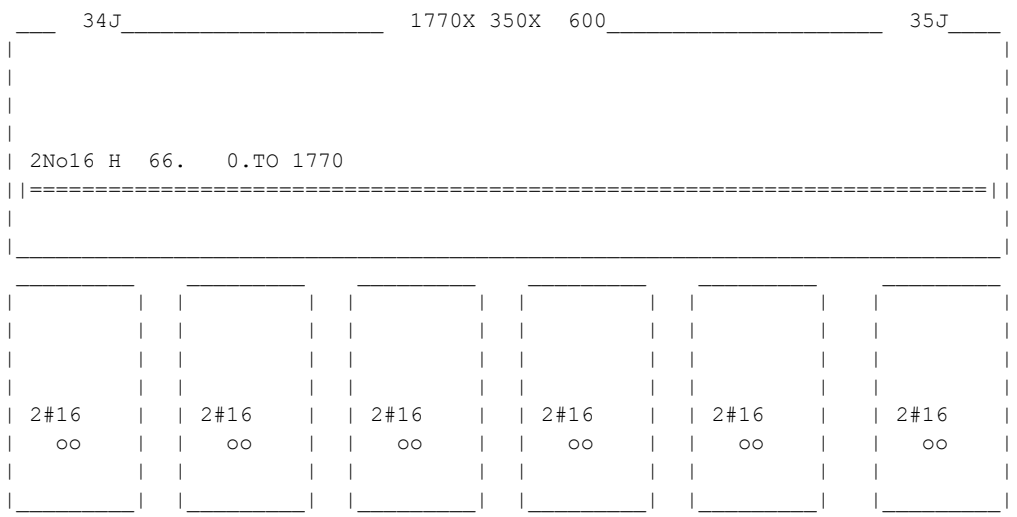
Cracked Moment of Inertia Iz at above location = 65065.9 cm^4

B E A M N O. 64 D E S I G N R E S U L T S - SHEAR

```
   AT START SUPPORT - Vu=   24.62 KNS  Vc=  186.69 KNS  Vs=    0.00 KNS
   Tu=    4.61 KN-MET  Tc=    8.8 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                     STIRRUPS ARE NOT REQUIRED.
   AT END   SUPPORT - Vu=   24.60 KNS  Vc=  186.58 KNS  Vs=    0.00 KNS
   Tu=    4.61 KN-MET  Tc=    8.8 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                     STIRRUPS ARE NOT REQUIRED.
```


33J
1069X 350X 600
34J
2No16 H 534. 0.TO 623
2No16 H 66. 0.TO 1070
oo
2#16
2#16
2#16

AT START SUPPORT - Vu= 2.67 KNS Vc= 174.84 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 8.8 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 2.67 KNS Vc= 174.84 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 8.8 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 66 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 1070. MM FY - 420. FC - 35. MPA, SIZE - 350. X 600. MMS

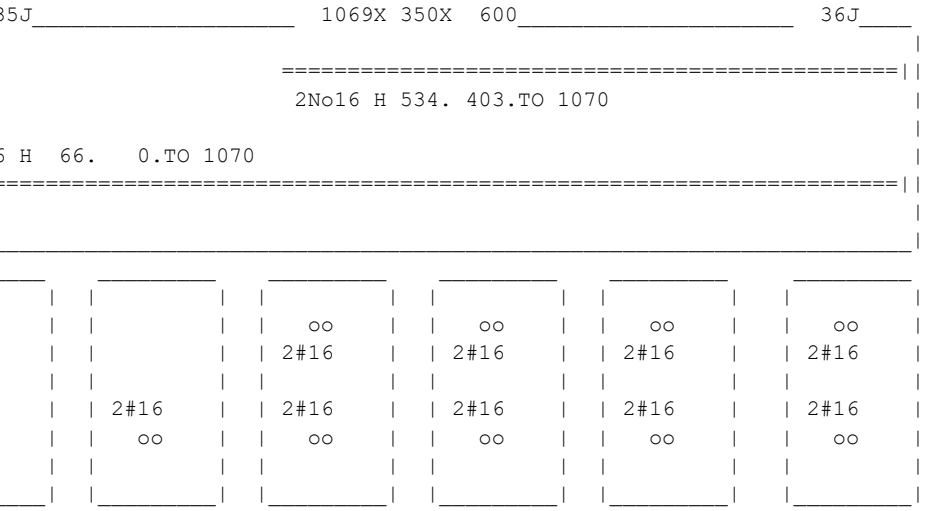
LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 16MM	0.	1070.	YES	YES
	CRITICAL POS MOMENT= 46.73 KN-MET AT 0.MM, LOAD 403					
	REQD STEEL= 311.MM2, RHO=0.0017, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 218. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 65065.9 cm^4					
2	534.	2 - 16MM	403.	1070.	NO	YES

```
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=      7.52 KN-MET  AT  1070.MM, LOAD  403|
|   REQD STEEL=    50.MM2, RHO=0.0003, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./  218. MMS         |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                        |
|--------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    65065.9 cm^4
```

B E A M N O. 66 D E S I G N R E S U L T S - SHEAR

```
 AT START SUPPORT - Vu=   24.61 KNS  Vc=  186.58 KNS  Vs=    0.00 KNS
 Tu=    4.61 KN-MET  Tc=    8.8 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
 AT END   SUPPORT - Vu=   24.62 KNS  Vc=  186.69 KNS  Vs=    0.00 KNS
 Tu=    4.61 KN-MET  Tc=    8.8 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
```



BEAM NO. 67 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 796. MM FY - 420. FC - 35. MPA, SIZE - 350. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	534.	3 - 16MM	0.	796.	YES	YES

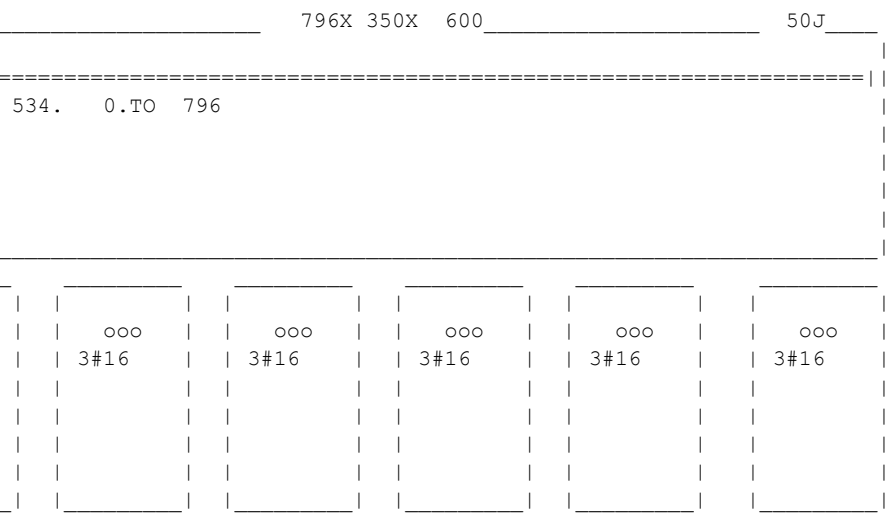
```
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     62.86 KN-MET  AT   796.MM, LOAD  403|
|   REQD STEEL=   419.MM2, RHO=0.0022, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./  109. MMS         |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                       |
|-------------------------------------------------------------|
```

Cracked Moment of Inertia Iz at above location = 92778.8 cm^4

B E A M N O. 67 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 67 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 67 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



**********END OF BEAM DESIGN**********

207. END CONCRETE DESIGN
208. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= APR 30,2020 TIME= 18:55: 7 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2016 Bentley Systems, Inc. *
* http://www.bentley.com *
****

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2			
Date	19-03-2020	30-04-2020	13/10/2023				Date	19-03-2020	30-04-2020	13/10/2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED C01	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED C01	AS BUILT			
Prepared By	APL	APL	APL				Prepared By	APL	APL	APL			
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

ATTACHMENT 4 **STAAD INPUT & OUTPUT FILE OF FOUNDATION FOR TR-3**

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 13-Mar-20
JOB NAME MMGP-SAUDI ARABIA
JOB CLIENT MA'ADEN
JOB NO O19045/900707
JOB REV 0
JOB PART TRESTLE TR3 FOUNDATION
JOB REF FOUNDATIONS FOR CONVEYOR 3120-CNV-001
ENGINEER NAME APL
CHECKER NAME SBY
APPROVED NAME PRI
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 -3 0; 2 0 -3 2; 3 0 1.49012e-008 0; 4 0 1.49012e-008 2; 5 0 0.35 0;
6 0 0.35 2; 7 0 -3 -1.7; 8 0 -3 3.7; 9 1.6 -3 -1.7; 10 1.6 -3 3.7;
11 -1.6 -3 -1.7; 12 -1.6 -3 3.7; 13 -0.75 -3 3.7; 14 -0.75 -3 2.75;
15 -1.6 -3 2.75; 16 -0.5 -3 3.7; 17 -0.5 -3 2.75; 18 -0.25 -3 3.7;
19 -0.25 -3 2.75; 20 0 -3 2.75; 21 0.25 -3 3.7; 22 0.25 -3 2.75; 23 0.5 -3 3.7;
24 0.5 -3 2.75; 25 0.75 -3 3.7; 26 0.75 -3 2.75; 27 1.6 -3 2.75;
28 -0.75 -3 2.5; 29 -1.6 -3 2.5; 30 -0.5 -3 2.5; 31 -0.25 -3 2.5; 32 0 -3 2.5;
33 0.25 -3 2.5; 34 0.5 -3 2.5; 35 0.75 -3 2.5; 36 1.6 -3 2.5; 37 -0.75 -3 2.25;
38 -1.6 -3 2.25; 39 -0.5 -3 2.25; 40 -0.25 -3 2.25; 41 0 -3 2.25;
42 0.25 -3 2.25; 43 0.5 -3 2.25; 44 0.75 -3 2.25; 45 1.6 -3 2.25;
46 -0.75 -3 2; 47 -1.6 -3 2; 48 -0.5 -3 2; 49 -0.25 -3 2; 50 0.25 -3 2;
51 0.5 -3 2; 52 0.75 -3 2; 53 1.6 -3 2; 54 -0.75 -3 1.75; 55 -1.6 -3 1.75;
56 -0.5 -3 1.75; 57 -0.25 -3 1.75; 58 0 -3 1.75; 59 0.25 -3 1.75;
60 0.5 -3 1.75; 61 0.75 -3 1.75; 62 1.6 -3 1.75; 63 -0.75 -3 1.5;
64 -1.6 -3 1.5; 65 -0.5 -3 1.5; 66 -0.25 -3 1.5; 67 0 -3 1.5; 68 0.25 -3 1.5;
69 0.5 -3 1.5; 70 0.75 -3 1.5; 71 1.6 -3 1.5; 72 -0.75 -3 1.25;
73 -1.6 -3 1.25; 74 -0.5 -3 1.25; 75 -0.25 -3 1.25; 76 0 -3 1.25;
77 0.25 -3 1.25; 78 0.5 -3 1.25; 79 0.75 -3 1.25; 80 1.6 -3 1.25;
81 -0.75 -3 1; 82 -1.6 -3 1; 83 -0.5 -3 1; 84 -0.25 -3 1; 85 0 -3 1;
86 0.25 -3 1; 87 0.5 -3 1; 88 0.75 -3 1; 89 1.6 -3 1; 90 -0.75 -3 0.75;
91 -1.6 -3 0.75; 92 -0.5 -3 0.75; 93 -0.25 -3 0.75; 94 0 -3 0.75;
95 0.25 -3 0.75; 96 0.5 -3 0.75; 97 0.75 -3 0.75; 98 1.6 -3 0.75;
99 -0.75 -3 0.5; 100 -1.6 -3 0.5; 101 -0.5 -3 0.5; 102 -0.25 -3 0.5;
103 0 -3 0.5; 104 0.25 -3 0.5; 105 0.5 -3 0.5; 106 0.75 -3 0.5; 107 1.6 -3 0.5;
108 -0.75 -3 0.25; 109 -1.6 -3 0.25; 110 -0.5 -3 0.25; 111 -0.25 -3 0.25;
112 0 -3 0.25; 113 0.25 -3 0.25; 114 0.5 -3 0.25; 115 0.75 -3 0.25;
116 1.6 -3 0.25; 117 -0.75 -3 0; 118 -1.6 -3 0; 119 -0.5 -3 0; 120 -0.25 -3 0;
121 0.25 -3 0; 122 0.5 -3 0; 123 0.75 -3 0; 124 1.6 -3 0; 125 -0.75 -3 -0.25;
126 -1.6 -3 -0.25; 127 -0.5 -3 -0.25; 128 -0.25 -3 -0.25; 129 0 -3 -0.25;
130 0.25 -3 -0.25; 131 0.5 -3 -0.25; 132 0.75 -3 -0.25; 133 1.6 -3 -0.25;
134 -0.75 -3 -0.5; 135 -1.6 -3 -0.5; 136 -0.5 -3 -0.5; 137 -0.25 -3 -0.5;
138 0 -3 -0.5; 139 0.25 -3 -0.5; 140 0.5 -3 -0.5; 141 0.75 -3 -0.5;
142 1.6 -3 -0.5; 143 -0.75 -3 -0.75; 144 -1.6 -3 -0.75; 145 -0.5 -3 -0.75;
146 -0.25 -3 -0.75; 147 0 -3 -0.75; 148 0.25 -3 -0.75; 149 0.5 -3 -0.75;
150 0.75 -3 -0.75; 151 1.6 -3 -0.75; 152 -0.75 -3 -1.7; 153 -0.5 -3 -1.7;
154 -0.25 -3 -1.7; 155 0.25 -3 -1.7; 156 0.5 -3 -1.7; 157 0.75 -3 -1.7;
158 -0.75 -3 -1; 159 -1.6 -3 -1; 160 -0.75 -3 -1.25; 161 -1.6 -3 -1.25;
162 -0.5 -3 -1; 163 -0.5 -3 -1.25; 164 -0.25 -3 -1; 165 -0.25 -3 -1.25;
```

166 0 -3 -1; 167 0 -3 -1.25; 168 0.25 -3 -1; 169 0.25 -3 -1.25; 170 0.5 -3 -1;
171 0.5 -3 -1.25; 172 0.75 -3 -1; 173 0.75 -3 -1.25; 174 1.6 -3 -1;
175 1.6 -3 -1.25; 176 -0.75 -3 3.25; 177 -1.6 -3 3.25; 178 -0.75 -3 3;
179 -1.6 -3 3; 180 -0.5 -3 3.25; 181 -0.5 -3 3; 182 -0.25 -3 3.25;
183 -0.25 -3 3; 184 0 -3 3.25; 185 0 -3 3; 186 0.25 -3 3.25; 187 0.25 -3 3;
188 0.5 -3 3.25; 189 0.5 -3 3; 190 0.75 -3 3.25; 191 0.75 -3 3;
192 1.6 -3 3.25; 193 1.6 -3 3; 194 -0.75 -3 -1.475; 195 -1.6 -3 -1.475;
196 -0.5 -3 -1.475; 197 -0.25 -3 -1.475; 198 0 -3 -1.475; 199 0.25 -3 -1.475;
200 0.5 -3 -1.475; 201 0.75 -3 -1.475; 202 1.6 -3 -1.475; 203 -0.75 -3 3.475;
204 -1.6 -3 3.475; 205 -0.5 -3 3.475; 206 -0.25 -3 3.475; 207 0 -3 3.475;
208 0.25 -3 3.475; 209 0.5 -3 3.475; 210 0.75 -3 3.475; 211 1.6 -3 3.475;
212 1.03333 -3 2.75; 213 1.03333 -3 2.5; 214 1.31667 -3 2.75;
215 1.31667 -3 2.5; 216 1.03333 -3 2.25; 217 1.31667 -3 2.25; 218 1.03333 -3 2;
219 1.31667 -3 2; 220 1.03333 -3 1.75; 221 1.31667 -3 1.75; 222 1.03333 -3 1.5;
223 1.31667 -3 1.5; 224 1.03333 -3 1.25; 225 1.31667 -3 1.25; 226 1.03333 -3 1;
227 1.31667 -3 1; 228 1.03333 -3 0.75; 229 1.31667 -3 0.75; 230 1.03333 -3 0.5;
231 1.31667 -3 0.5; 232 1.03333 -3 0.25; 233 1.31667 -3 0.25; 234 1.03333 -3 0;
235 1.31667 -3 0; 236 1.03333 -3 -0.25; 237 1.31667 -3 -0.25;
238 1.03333 -3 -0.5; 239 1.31667 -3 -0.5; 240 1.03333 -3 -0.75;
241 1.31667 -3 -0.75; 242 1.03333 -3 -1; 243 1.31667 -3 -1;
244 1.03333 -3 -1.25; 245 1.31667 -3 -1.25; 246 1.03333 -3 3.25;
247 1.03333 -3 3; 248 1.31667 -3 3.25; 249 1.31667 -3 3; 250 1.03333 -3 -1.475;
251 1.31667 -3 -1.475; 252 1.03333 -3 -1.7; 253 1.31667 -3 -1.7;
254 1.03333 -3 3.7; 255 1.03333 -3 3.475; 256 1.31667 -3 3.7;
257 1.31667 -3 3.475; 258 -1.31667 -3 2.75; 259 -1.31667 -3 2.5;
260 -1.03333 -3 2.75; 261 -1.03333 -3 2.5; 262 -1.31667 -3 2.25;
263 -1.03333 -3 2.25; 264 -1.31667 -3 2; 265 -1.03333 -3 2;
266 -1.31667 -3 1.75; 267 -1.03333 -3 1.75; 268 -1.31667 -3 1.5;
269 -1.03333 -3 1.5; 270 -1.31667 -3 1.25; 271 -1.03333 -3 1.25;
272 -1.31667 -3 1; 273 -1.03333 -3 1; 274 -1.31667 -3 0.75;
275 -1.03333 -3 0.75; 276 -1.31667 -3 0.5; 277 -1.03333 -3 0.5;
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293 -1.31667 -3 3; 294 -1.03333 -3 3.25; 295 -1.03333 -3 3;
296 -1.31667 -3 -1.475; 297 -1.03333 -3 -1.475; 298 -1.31667 -3 -1.7;
299 -1.03333 -3 -1.7; 300 -1.31667 -3 3.7; 301 -1.31667 -3 3.475;
302 -1.03333 -3 3.7; 303 -1.03333 -3 3.475;

MEMBER INCIDENCES

1 1 3; 2 3 5; 3 2 4; 4 4 6; 5 3 4;

ELEMENT INCIDENCES SHELL

15 14 17 30 28; 16 17 19 31 30; 17 19 20 32 31; 18 20 22 33 32; 19 22 24 34 33;
20 24 26 35 34; 23 28 30 39 37; 24 30 31 40 39; 25 31 32 41 40; 26 32 33 42 41;
27 33 34 43 42; 28 34 35 44 43; 31 37 39 48 46; 32 39 40 49 48; 33 40 41 2 49;
34 41 42 50 2; 35 42 43 51 50; 36 43 44 52 51; 39 46 48 56 54; 40 48 49 57 56;
41 49 2 58 57; 42 2 50 59 58; 43 50 51 60 59; 44 51 52 61 60; 47 54 56 65 63;
48 56 57 66 65; 49 57 58 67 66; 50 58 59 68 67; 51 59 60 69 68; 52 60 61 70 69;
55 63 65 74 72; 56 65 66 75 74; 57 66 67 76 75; 58 67 68 77 76; 59 68 69 78 77;
60 69 70 79 78; 63 72 74 83 81; 64 74 75 84 83; 65 75 76 85 84; 66 76 77 86 85;
67 77 78 87 86; 68 78 79 88 87; 71 81 83 92 90; 72 83 84 93 92; 73 84 85 94 93;
74 85 86 95 94; 75 86 87 96 95; 76 87 88 97 96; 79 90 92 101 99;
80 92 93 102 101; 81 93 94 103 102; 82 94 95 104 103; 83 95 96 105 104;

84 96 97 106 105; 87 99 101 110 108; 88 101 102 111 110; 89 102 103 112 111;
90 103 104 113 112; 91 104 105 114 113; 92 105 106 115 114; 95 108 110 119 117;
96 110 111 120 119; 97 111 112 1 120; 98 112 113 121 1; 99 113 114 122 121;
100 114 115 123 122; 103 117 119 127 125; 104 119 120 128 127;
105 120 1 129 128; 106 1 121 130 129; 107 121 122 131 130; 108 122 123 132 131;
111 125 127 136 134; 112 127 128 137 136; 113 128 129 138 137;
114 129 130 139 138; 115 130 131 140 139; 116 131 132 141 140;
119 134 136 145 143; 120 136 137 146 145; 121 137 138 147 146;
122 138 139 148 147; 123 139 140 149 148; 124 140 141 150 149;
137 143 145 162 158; 138 158 162 163 160; 140 145 146 164 162;
141 162 164 165 163; 143 146 147 166 164; 144 164 166 167 165;
146 147 148 168 166; 147 166 168 169 167; 149 148 149 170 168;
150 168 170 171 169; 152 149 150 172 170; 153 170 172 173 171;
162 176 180 181 178; 163 178 181 17 14; 165 180 182 183 181; 166 181 183 19 17;
168 182 184 185 183; 169 183 185 20 19; 171 184 186 187 185; 172 185 187 22 20;
174 186 188 189 187; 175 187 189 24 22; 177 188 190 191 189; 178 189 191 26 24;
184 160 163 196 194; 185 194 196 153 152; 186 163 165 197 196;
187 196 197 154 153; 188 165 167 198 197; 189 197 198 7 154;
190 167 169 199 198; 191 198 199 155 7; 192 169 171 200 199;
193 199 200 156 155; 194 171 173 201 200; 195 200 201 157 156;
200 13 16 205 203; 201 203 205 180 176; 202 16 18 206 205; 203 205 206 182 180;
204 18 8 207 206; 205 206 207 184 182; 206 8 21 208 207; 207 207 208 186 184;
208 21 23 209 208; 209 208 209 188 186; 210 23 25 210 209; 211 209 210 190 188;
214 26 212 213 35; 215 212 214 215 213; 216 214 27 36 215; 217 35 213 216 44;
218 213 215 217 216; 219 215 36 45 217; 220 44 216 218 52; 221 216 217 219 218;
222 217 45 53 219; 223 52 218 220 61; 224 218 219 221 220; 225 219 53 62 221;
226 61 220 222 70; 227 220 221 223 222; 228 221 62 71 223; 229 70 222 224 79;
230 222 223 225 224; 231 223 71 80 225; 232 79 224 226 88; 233 224 225 227 226;
234 225 80 89 227; 235 88 226 228 97; 236 226 227 229 228; 237 227 89 98 229;
238 97 228 230 106; 239 228 229 231 230; 240 229 98 107 231;
241 106 230 232 115; 242 230 231 233 232; 243 231 107 116 233;
244 115 232 234 123; 245 232 233 235 234; 246 233 116 124 235;
247 123 234 236 132; 248 234 235 237 236; 249 235 124 133 237;
250 132 236 238 141; 251 236 237 239 238; 252 237 133 142 239;
253 141 238 240 150; 254 238 239 241 240; 255 239 142 151 241;
256 150 240 242 172; 257 240 241 243 242; 258 241 151 174 243;
259 172 242 244 173; 260 242 243 245 244; 261 243 174 175 245;
262 190 246 247 191; 263 246 248 249 247; 264 248 192 193 249;
265 191 247 212 26; 266 247 249 214 212; 267 249 193 27 214;
268 173 244 250 201; 269 244 245 251 250; 270 245 175 202 251;
271 201 250 252 157; 272 250 251 253 252; 273 251 202 9 253;
274 25 254 255 210; 275 254 256 257 255; 276 256 10 211 257;
277 210 255 246 190; 278 255 257 248 246; 279 257 211 192 248;
280 15 258 259 29; 281 258 260 261 259; 282 260 14 28 261; 283 29 259 262 38;
284 259 261 263 262; 285 261 28 37 263; 286 38 262 264 47; 287 262 263 265 264;
288 263 37 46 265; 289 47 264 266 55; 290 264 265 267 266; 291 265 46 54 267;
292 55 266 268 64; 293 266 267 269 268; 294 267 54 63 269; 295 64 268 270 73;
296 268 269 271 270; 297 269 63 72 271; 298 73 270 272 82; 299 270 271 273 272;
300 271 72 81 273; 301 82 272 274 91; 302 272 273 275 274; 303 273 81 90 275;
304 91 274 276 100; 305 274 275 277 276; 306 275 90 99 277;
307 100 276 278 109; 308 276 277 279 278; 309 277 99 108 279;
310 109 278 280 118; 311 278 279 281 280; 312 279 108 117 281;
313 118 280 282 126; 314 280 281 283 282; 315 281 117 125 283;
316 126 282 284 135; 317 282 283 285 284; 318 283 125 134 285;

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319 135 284 286 144; 320 284 285 287 286; 321 285 134 143 287;
322 144 286 288 159; 323 286 287 289 288; 324 287 143 158 289;
325 159 288 290 161; 326 288 289 291 290; 327 289 158 160 291;
328 177 292 293 179; 329 292 294 295 293; 330 294 176 178 295;
331 179 293 258 15; 332 293 295 260 258; 333 295 178 14 260;
334 161 290 296 195; 335 290 291 297 296; 336 291 160 194 297;
337 195 296 298 11; 338 296 297 299 298; 339 297 194 152 299;
340 12 300 301 204; 341 300 302 303 301; 342 302 13 203 303;
343 204 301 292 177; 344 301 303 294 292; 345 303 203 176 294;
START GROUP DEFINITION
ELEMENT
_MX 15 16 19 20 23 24 27 28 31 32 35 36 39 40 43 44 47 48 51 52 55 56 59 60 -
63 64 67 68 71 72 75 76 79 80 83 84 87 88 91 92 95 96 99 100 103 104 107 -
108 111 112 115 116 119 120 123 124 137 138 140 141 149 150 152 153 162 163 -
165 166 174 175 177 178 184 TO 187 192 TO 195 200 TO 203 208 TO 211 -
214 TO 345
_MY 15 TO 20 23 TO 28 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 TO 84 87 TO 92 -
111 TO 116 119 TO 124 137 138 140 141 143 144 146 147 149 150 152 153 162 -
163 165 166 168 169 171 172 174 175 177 178 184 TO 195 200 TO 211 -
214 TO 219 226 TO 243 250 TO 285 292 TO 309 316 TO 345
_SHEAR 63 TO 68 71 TO 76 137 138 140 141 143 144 146 147 149 150 152 153 162 -
163 165 166 168 169 171 172 174 175 177 178 184 TO 195 200 TO 211 -
214 TO 345
END GROUP DEFINITION
ELEMENT PROPERTY
15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 138 140 141 -
143 144 146 147 149 150 152 153 162 163 165 166 168 169 171 172 174 175 177 -
178 184 TO 195 200 TO 211 214 TO 345 THICKNESS 0.55
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
1 TO 4 PRIS YD 0.7 ZD 0.5
5 PRIS YD 0.5 ZD 0.3
CONSTANTS
MATERIAL CONCRETE ALL
*****************************************
SUPPORTS
1 2 7 TO 303 ELASTIC MAT DIRECT Y SUBGRADE 72000 PRINT
*****************************************
LOAD 1 DEAD LOAD    DL
SELFWEIGHT Y -1
JOINT LOAD
5 FX -0.297 FY -155.824 FZ -1.798
6 FX -0.238 FY -145.789 FZ 13.698
ELEMENT LOAD
```

```
15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 138 140 141 -
143 144 146 147 149 150 152 153 162 163 165 166 168 169 171 172 174 175 177 -
178 184 TO 195 200 TO 211 214 TO 345 PR GY -44.1
*****************************************
LOAD 2 LIVE LOAD    LL
JOINT LOAD
5 FX -0.164 FY -78 FZ -4.072
6 FX -0.117 FY -78 FZ 4.072
*****************************************
LOAD 3 MATERIAL LOAD    LL
JOINT LOAD
5 FX -0.129 FY -70.3 FZ -3.715
6 FX -0.11 FY -71.1 FZ 3.715
*****************************************
LOAD 4 WIND LOAD ALONG CONVEYOR WL (+Z)
JOINT LOAD
5 FX 0.189 FY 265.049 FZ 39.623
6 FX -0.189 FY -265.049 FZ 39.577
*****************************************
LOAD 5 WIND LOAD ALONG CONVEYOR WL (-Z)
JOINT LOAD
5 FX -0.185 FY -264.046 FZ -39.473
6 FX 0.185 FY 264.046 FZ -39.427
*****************************************
LOAD 6 WIND LOAD ACROSS THE CONV.  WL (+X)
JOINT LOAD
5 FX 3.21 FY 0 FZ 0
6 FX 3.21 FY 0 FZ 0
*****************************************
LOAD 7 WIND LOAD ACROSS THE CONV.   WL (-X)
JOINT LOAD
5 FX -3.21 FY 0 FZ 0
6 FX -3.21 FY 0 FZ 0
*****************************************
LOAD 8 TEMP LOAD    TL1
JOINT LOAD
5 FX 0 FY 0 FZ -15.912
6 FX 0 FY 0 FZ 15.912
*****************************************
LOAD 9 TEMP LOAD    TL2
JOINT LOAD
5 FX 0 FY 0 FZ 15.912
6 FX 0 FY 0 FZ -15.912
*****************************************
LOAD 10 SEISMIC LOAD ALONG THE CONV.    SL (+X)
JOINT LOAD
5 FX 0.018 FY 0 FZ 0.003
6 FX 0.021 FY 0.1 FZ -0.003
*****************************************
LOAD 11 SEISMIC LOAD ACROSS THE CONV.   SL (+Z)
JOINT LOAD
5 FX -0.002 FY 15.194 FZ 2.379
6 FX 0.002 FY -15.194 FZ 2.375
```

```
****************************************
*****Load combination for bearing capacity check
LOAD 301 1.0 DL
REPEAT LOAD
1 1.0
LOAD 302 1.0 (DL + TL1) + 1.0  (LL+ML)
REPEAT LOAD
1 1.0 8 1.0 2 1.0 3 1.0
LOAD 303 1.0 (DL + TL1) + 1.0  (LL+ML)
REPEAT LOAD
1 1.0 9 1.0 2 1.0 3 1.0
LOAD 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 4 0.75 2 0.75 3 0.75
LOAD 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 5 0.75 2 0.75 3 0.75
LOAD 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 6 0.75 2 0.75 3 0.75
LOAD 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 7 0.75 2 0.75 3 0.75
LOAD 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 10 0.525 2 0.75 3 0.75
LOAD 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 10 -0.525 2 0.75 3 0.75
LOAD 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 11 0.525 2 0.75 3 0.75
LOAD 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 11 -0.525 2 0.75 3 0.75
LOAD 312 0.6 DL + 1.0 WL(+Z)
REPEAT LOAD
1 0.6 4 1.0
LOAD 313 0.6 DL + 1.0 WL(-Z)
REPEAT LOAD
1 0.6 5 1.0
LOAD 314 0.6 DL + 1.0 WL(+X)
REPEAT LOAD
1 0.6 6 1.0
LOAD 315 0.6 DL + 1.0 WL(-X)
REPEAT LOAD
1 0.6 7 1.0
LOAD 316 0.6 DL + 0.7 EL(+X)
REPEAT LOAD
1 0.6 10 0.7
LOAD 317 0.6 DL + 0.7 EL(-X)
REPEAT LOAD
1 0.6 10 -0.7
LOAD 318 0.6 DL + 0.7 EL(+Z)
```

```
REPEAT LOAD
1 0.6 11 0.7
LOAD 319 0.6 DL + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 11 -0.7
*****************************************
*****Load combination for foundation design
LOAD 401 1.4 DL
REPEAT LOAD
1 1.4
LOAD 402 1.4 (DL + TL1) + 1.7  (LL+ML)
REPEAT LOAD
1 1.4 8 1.4 2 1.7 3 1.7
LOAD 403 1.4 (DL + TL1) + 1.7  (LL+ML)
REPEAT LOAD
1 1.4 9 1.4 2 1.7 3 1.7
LOAD 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 4 1.6 2 0.5 3 0.5
LOAD 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 5 1.6 2 0.5 3 0.5
LOAD 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 6 1.6 2 0.5 3 0.5
LOAD 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 7 1.6 2 0.5 3 0.5
LOAD 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 10 1.0 2 0.5 3 0.5
LOAD 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 10 -1.0 2 0.5 3 0.5
LOAD 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 11 1.0 2 0.5 3 0.5
LOAD 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 11 -1.0 2 0.5 3 0.5
LOAD 412 0.9 DL + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 4 1.6
LOAD 413 0.9 DL + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 5 1.6
LOAD 414 0.9 DL + 1.6 WL(+X)
REPEAT LOAD
1 0.9 6 1.6
LOAD 415 0.9 DL + 1.6 WL(-X)
REPEAT LOAD
1 0.9 7 1.6
LOAD 416 0.9 DL + 1.0 EL(+X)
REPEAT LOAD
```

```
1 0.9 10 1.0
LOAD 417 0.9 DL + 1.0 EL(-X)
REPEAT LOAD
1 0.9 10 -1.0
LOAD 418 0.9 DL + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 11 1.0
LOAD 419 0.9 DL + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 11 -1.0
*****************************************
PERFORM ANALYSIS
*****************************************
LOAD LIST 401 TO 419
START CONCRETE DESIGN
CODE ACI 2008
FC 35000 MEMB 1 TO 5
FYMAIN 420000 MEMB 1 TO 5
TRACK 1 ALL
CLB 0.05 MEMB 1 TO 5
CLT 0.05 MEMB 1 TO 5
CLS 0.05 ALL
MAXMAIN 25 ALL
MINMAIN 16 ALL
MINSEC 8 ALL
REINF 0 MEMB 1 TO 4
DESIGN COLUMN 1 TO 4
DESIGN BEAM 5
END CONCRETE DESIGN
FINISH
```

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6           *
*           Version  20.07.11.70                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    MAR 19, 2020                *
*           Time=     9:47:47                    *
*                                                *
*      USER ID: lnt                              *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: D:\APL\Assignments\13-Maaden-Crushing_n_Screening\Conv-3120-CNV-001\Staad\3120-CNV-001_TR3.. .STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 13-MAR-20
    4. JOB NAME MMGP-SAUDI ARABIA
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045/900707
    7. JOB REV 0
    8. JOB PART TRESTLE TR3 FOUNDATION
    9. JOB REF FOUNDATIONS FOR CONVEYOR 3120-CNV-001
   10. ENGINEER NAME APL
   11. CHECKER NAME SBY
   12. APPROVED NAME PRI
   13. END JOB INFORMATION
   14. INPUT WIDTH 79
   15. UNIT METER KN
   16. JOINT COORDINATES
   17. 1 0 -3 0; 2 0 -3 2; 3 0 1.49012E-008 0; 4 0 1.49012E-008 2; 5 0 0.35 0
   18. 6 0 0.35 2; 7 0 -3 -1.7; 8 0 -3 3.7; 9 1.6 -3 -1.7; 10 1.6 -3 3.7
   19. 11 -1.6 -3 -1.7; 12 -1.6 -3 3.7; 13 -0.75 -3 3.7; 14 -0.75 -3 2.75
   20. 15 -1.6 -3 2.75; 16 -0.5 -3 3.7; 17 -0.5 -3 2.75; 18 -0.25 -3 3.7
   21. 19 -0.25 -3 2.75; 20 0 -3 2.75; 21 0.25 -3 3.7; 22 0.25 -3 2.75; 23 0.5 -3 3.7
   22. 24 0.5 -3 2.75; 25 0.75 -3 3.7; 26 0.75 -3 2.75; 27 1.6 -3 2.75
   23. 28 -0.75 -3 2.5; 29 -1.6 -3 2.5; 30 -0.5 -3 2.5; 31 -0.25 -3 2.5; 32 0 -3 2.5
   24. 33 0.25 -3 2.5; 34 0.5 -3 2.5; 35 0.75 -3 2.5; 36 1.6 -3 2.5; 37 -0.75 -3 2.25
   25. 38 -1.6 -3 2.25; 39 -0.5 -3 2.25; 40 -0.25 -3 2.25; 41 0 -3 2.25
   26. 42 0.25 -3 2.25; 43 0.5 -3 2.25; 44 0.75 -3 2.25; 45 1.6 -3 2.25
   27. 46 -0.75 -3 2; 47 -1.6 -3 2; 48 -0.5 -3 2; 49 -0.25 -3 2; 50 0.25 -3 2
   28. 51 0.5 -3 2; 52 0.75 -3 2; 53 1.6 -3 2; 54 -0.75 -3 1.75; 55 -1.6 -3 1.75
   29. 56 -0.5 -3 1.75; 57 -0.25 -3 1.75; 58 0 -3 1.75; 59 0.25 -3 1.75
   30. 60 0.5 -3 1.75; 61 0.75 -3 1.75; 62 1.6 -3 1.75; 63 -0.75 -3 1.5
   31. 64 -1.6 -3 1.5; 65 -0.5 -3 1.5; 66 -0.25 -3 1.5; 67 0 -3 1.5; 68 0.25 -3 1.5
   32. 69 0.5 -3 1.5; 70 0.75 -3 1.5; 71 1.6 -3 1.5; 72 -0.75 -3 1.25
   33. 73 -1.6 -3 1.25; 74 -0.5 -3 1.25; 75 -0.25 -3 1.25; 76 0 -3 1.25
   34. 77 0.25 -3 1.25; 78 0.5 -3 1.25; 79 0.75 -3 1.25; 80 1.6 -3 1.25
   35. 81 -0.75 -3 1; 82 -1.6 -3 1; 83 -0.5 -3 1; 84 -0.25 -3 1; 85 0 -3 1
   36. 86 0.25 -3 1; 87 0.5 -3 1; 88 0.75 -3 1; 89 1.6 -3 1; 90 -0.75 -3 0.75
   37. 91 -1.6 -3 0.75; 92 -0.5 -3 0.75; 93 -0.25 -3 0.75; 94 0 -3 0.75
   38. 95 0.25 -3 0.75; 96 0.5 -3 0.75; 97 0.75 -3 0.75; 98 1.6 -3 0.75
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39. 99 -0.75 -3 0.5; 100 -1.6 -3 0.5; 101 -0.5 -3 0.5; 102 -0.25 -3 0.5
40. 103 0 -3 0.5; 104 0.25 -3 0.5; 105 0.5 -3 0.5; 106 0.75 -3 0.5; 107 1.6 -3 0.5
41. 108 -0.75 -3 0.25; 109 -1.6 -3 0.25; 110 -0.5 -3 0.25; 111 -0.25 -3 0.25
42. 112 0 -3 0.25; 113 0.25 -3 0.25; 114 0.5 -3 0.25; 115 0.75 -3 0.25
43. 116 1.6 -3 0.25; 117 -0.75 -3 0; 118 -1.6 -3 0; 119 -0.5 -3 0; 120 -0.25 -3 0
44. 121 0.25 -3 0; 122 0.5 -3 0; 123 0.75 -3 0; 124 1.6 -3 0; 125 -0.75 -3 -0.25
45. 126 -1.6 -3 -0.25; 127 -0.5 -3 -0.25; 128 -0.25 -3 -0.25; 129 0 -3 -0.25
46. 130 0.25 -3 -0.25; 131 0.5 -3 -0.25; 132 0.75 -3 -0.25; 133 1.6 -3 -0.25
47. 134 -0.75 -3 -0.5; 135 -1.6 -3 -0.5; 136 -0.5 -3 -0.5; 137 -0.25 -3 -0.5
48. 138 0 -3 -0.5; 139 0.25 -3 -0.5; 140 0.5 -3 -0.5; 141 0.75 -3 -0.5
49. 142 1.6 -3 -0.5; 143 -0.75 -3 -0.75; 144 -1.6 -3 -0.75; 145 -0.5 -3 -0.75
50. 146 -0.25 -3 -0.75; 147 0 -3 -0.75; 148 0.25 -3 -0.75; 149 0.5 -3 -0.75
51. 150 0.75 -3 -0.75; 151 1.6 -3 -0.75; 152 -0.75 -3 -1.7; 153 -0.5 -3 -1.7
52. 154 -0.25 -3 -1.7; 155 0.25 -3 -1.7; 156 0.5 -3 -1.7; 157 0.75 -3 -1.7
53. 158 -0.75 -3 -1; 159 -1.6 -3 -1; 160 -0.75 -3 -1.25; 161 -1.6 -3 -1.25
54. 162 -0.5 -3 -1; 163 -0.5 -3 -1.25; 164 -0.25 -3 -1; 165 -0.25 -3 -1.25
55. 166 0 -3 -1; 167 0 -3 -1.25; 168 0.25 -3 -1; 169 0.25 -3 -1.25; 170 0.5 -3 -1.
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58. 179 -1.6 -3 3; 180 -0.5 -3 3.25; 181 -0.5 -3 3; 182 -0.25 -3 3.25
59. 183 -0.25 -3 3; 184 0 -3 3.25; 185 0 -3 3; 186 0.25 -3 3.25; 187 0.25 -3 3
60. 188 0.5 -3 3.25; 189 0.5 -3 3; 190 0.75 -3 3.25; 191 0.75 -3 3
61. 192 1.6 -3 3.25; 193 1.6 -3 3; 194 -0.75 -3 -1.475; 195 -1.6 -3 -1.475
62. 196 -0.5 -3 -1.475; 197 -0.25 -3 -1.475; 198 0 -3 -1.475; 199 0.25 -3 -1.475
63. 200 0.5 -3 -1.475; 201 0.75 -3 -1.475; 202 1.6 -3 -1.475; 203 -0.75 -3 3.475
64. 204 -1.6 -3 3.475; 205 -0.5 -3 3.475; 206 -0.25 -3 3.475; 207 0 -3 3.475
65. 208 0.25 -3 3.475; 209 0.5 -3 3.475; 210 0.75 -3 3.475; 211 1.6 -3 3.475
66. 212 1.03333 -3 2.75; 213 1.03333 -3 2.5; 214 1.31667 -3 2.75
67. 215 1.31667 -3 2.5; 216 1.03333 -3 2.25; 217 1.31667 -3 2.25; 218 1.03333 -3 2
68. 219 1.31667 -3 2; 220 1.03333 -3 1.75; 221 1.31667 -3 1.75; 222 1.03333 -3 1.5
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70. 227 1.31667 -3 1; 228 1.03333 -3 0.75; 229 1.31667 -3 0.75; 230 1.03333 -3 0.5
71. 231 1.31667 -3 0.5; 232 1.03333 -3 0.25; 233 1.31667 -3 0.25; 234 1.03333 -3 0
72. 235 1.31667 -3 0; 236 1.03333 -3 -0.25; 237 1.31667 -3 -0.25
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77. 251 1.31667 -3 -1.475; 252 1.03333 -3 -1.7; 253 1.31667 -3 -1.7
78. 254 1.03333 -3 3.7; 255 1.03333 -3 3.475; 256 1.31667 -3 3.7
79. 257 1.31667 -3 3.475; 258 -1.31667 -3 2.75; 259 -1.31667 -3 2.5
80. 260 -1.03333 -3 2.75; 261 -1.03333 -3 2.5; 262 -1.31667 -3 2.25
81. 263 -1.03333 -3 2.25; 264 -1.31667 -3 2; 265 -1.03333 -3 2
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83. 269 -1.03333 -3 1.5; 270 -1.31667 -3 1.25; 271 -1.03333 -3 1.25
84. 272 -1.31667 -3 1; 273 -1.03333 -3 1; 274 -1.31667 -3 0.75
85. 275 -1.03333 -3 0.75; 276 -1.31667 -3 0.5; 277 -1.03333 -3 0.5
86. 278 -1.31667 -3 0.25; 279 -1.03333 -3 0.25; 280 -1.31667 -3 0
87. 281 -1.03333 -3 0; 282 -1.31667 -3 -0.25; 283 -1.03333 -3 -0.25
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89. 287 -1.03333 -3 -0.75; 288 -1.31667 -3 -1; 289 -1.03333 -3 -1.
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93. 299 -1.03333 -3 -1.7; 300 -1.31667 -3 3.7; 301 -1.31667 -3 3.475
94. 302 -1.03333 -3 3.7; 303 -1.03333 -3 3.475

95. MEMBER INCIDENCES
96. 1 1 3; 2 3 5; 3 2 4; 4 4 6; 5 3 4
97. ELEMENT INCIDENCES SHELL
98. 15 14 17 30 28; 16 17 19 31 30; 17 19 20 32 31; 18 20 22 33 32; 19 22 24 34 33
99. 20 24 26 35 34; 23 28 30 39 37; 24 30 31 40 39; 25 31 32 41 40; 26 32 33 42 41
100. 27 33 34 43 42; 28 34 35 44 43; 31 37 39 48 46; 32 39 40 49 48; 33 40 41 2 49
101. 34 41 42 50 2; 35 42 43 51 50; 36 43 44 52 51; 39 46 48 56 54; 40 48 49 57 56
102. 41 49 2 58 57; 42 2 50 59 58; 43 50 51 60 59; 44 51 52 61 60; 47 54 56 65 63
103. 48 56 57 66 65; 49 57 58 67 66; 50 58 59 68 67; 51 59 60 69 68; 52 60 61 70 69
104. 55 63 65 74 72; 56 65 66 75 74; 57 66 67 76 75; 58 67 68 77 76; 59 68 69 78 77
105. 60 69 70 79 78; 63 72 74 83 81; 64 74 75 84 83; 65 75 76 85 84; 66 76 77 86 85
106. 67 77 78 87 86; 68 78 79 88 87; 71 81 83 92 90; 72 83 84 93 92; 73 84 85 94 93
107. 74 85 86 95 94; 75 86 87 96 95; 76 87 88 97 96; 79 90 92 101 99
108. 80 92 93 102 101; 81 93 94 103 102; 82 94 95 104 103; 83 95 96 105 104
109. 84 96 97 106 105; 87 99 101 110 108; 88 101 102 111 110; 89 102 103 112 111
110. 90 103 104 113 112; 91 104 105 114 113; 92 105 106 115 114; 95 108 110 119 117
111. 96 110 111 120 119; 97 111 112 1 120; 98 112 113 121 1; 99 113 114 122 121
112. 100 114 115 123 122; 103 117 119 127 125; 104 119 120 128 127
113. 105 120 1 129 128; 106 1 121 130 129; 107 121 122 131 130; 108 122 123 132 131
114. 111 125 127 136 134; 112 127 128 137 136; 113 128 129 138 137
115. 114 129 130 139 138; 115 130 131 140 139; 116 131 132 141 140
116. 119 134 136 145 143; 120 136 137 146 145; 121 137 138 147 146
117. 122 138 139 148 147; 123 139 140 149 148; 124 140 141 150 149
118. 137 143 145 162 158; 138 158 162 163 160; 140 145 146 164 162
119. 141 162 164 165 163; 143 146 147 166 164; 144 164 166 167 165
120. 146 147 148 168 166; 147 166 168 169 167; 149 148 149 170 168
121. 150 168 170 171 169; 152 149 150 172 170; 153 170 172 173 171
122. 162 176 180 181 178; 163 178 181 17 14; 165 180 182 183 181; 166 181 183 19 17
123. 168 182 184 185 183; 169 183 185 20 19; 171 184 186 187 185; 172 185 187 22 20
124. 174 186 188 189 187; 175 187 189 24 22; 177 188 190 191 189; 178 189 191 26 24
125. 184 160 163 196 194; 185 194 196 153 152; 186 163 165 197 196
126. 187 196 197 154 153; 188 165 167 198 197; 189 197 198 7 154
127. 190 167 169 199 198; 191 198 199 155 7; 192 169 171 200 199
128. 193 199 200 156 155; 194 171 173 201 200; 195 200 201 157 156
129. 200 13 16 205 203; 201 203 205 180 176; 202 16 18 206 205; 203 205 206 182 180
130. 204 18 8 207 206; 205 206 207 184 182; 206 8 21 208 207; 207 207 208 186 184
131. 208 21 23 209 208; 209 208 209 188 186; 210 23 25 210 209; 211 209 210 190 188
132. 214 26 212 213 35; 215 212 214 215 213; 216 214 27 36 215; 217 35 213 216 44
133. 218 213 215 217 216; 219 215 36 45 217; 220 44 216 218 52; 221 216 217 219 218
134. 222 217 45 53 219; 223 52 218 220 61; 224 218 219 221 220; 225 219 53 62 221
135. 226 61 220 222 70; 227 220 221 223 222; 228 221 62 71 223; 229 70 222 224 79
136. 230 222 223 225 224; 231 223 71 80 225; 232 79 224 226 88; 233 224 225 227 226
137. 234 225 80 89 227; 235 88 226 228 97; 236 226 227 229 228; 237 227 89 98 229
138. 238 97 228 230 106; 239 228 229 231 230; 240 229 98 107 231
139. 241 106 230 232 115; 242 230 231 233 232; 243 231 107 116 233
140. 244 115 232 234 123; 245 232 233 235 234; 246 233 116 124 235
141. 247 123 234 236 132; 248 234 235 237 236; 249 235 124 133 237
142. 250 132 236 238 141; 251 236 237 239 238; 252 237 133 142 239
143. 253 141 238 240 150; 254 238 239 241 240; 255 239 142 151 241
144. 256 150 240 242 172; 257 240 241 243 242; 258 241 151 174 243
145. 259 172 242 244 173; 260 242 243 245 244; 261 243 174 175 245
146. 262 190 246 247 191; 263 246 248 249 247; 264 248 192 193 249
147. 265 191 247 212 26; 266 247 249 214 212; 267 249 193 27 214
148. 268 173 244 250 201; 269 244 245 251 250; 270 245 175 202 251
149. 271 201 250 252 157; 272 250 251 253 252; 273 251 202 9 253
150. 274 25 254 255 210; 275 254 256 257 255; 276 256 10 211 257

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151. 277 210 255 246 190; 278 255 257 248 246; 279 257 211 192 248
152. 280 15 258 259 29; 281 258 260 261 259; 282 260 14 28 261; 283 29 259 262 38
153. 284 259 261 263 262; 285 261 28 37 263; 286 38 262 264 47; 287 262 263 265 264
154. 288 263 37 46 265; 289 47 264 266 55; 290 264 265 267 266; 291 265 46 54 267
155. 292 55 266 268 64; 293 266 267 269 268; 294 267 54 63 269; 295 64 268 270 73
156. 296 268 269 271 270; 297 269 63 72 271; 298 73 270 272 82; 299 270 271 273 272
157. 300 271 72 81 273; 301 82 272 274 91; 302 272 273 275 274; 303 273 81 90 275
158. 304 91 274 276 100; 305 274 275 277 276; 306 275 90 99 277
159. 307 100 276 278 109; 308 276 277 279 278; 309 277 99 108 279
160. 310 109 278 280 118; 311 278 279 281 280; 312 279 108 117 281
161. 313 118 280 282 126; 314 280 281 283 282; 315 281 117 125 283
162. 316 126 282 284 135; 317 282 283 285 284; 318 283 125 134 285
163. 319 135 284 286 144; 320 284 285 287 286; 321 285 134 143 287
164. 322 144 286 288 159; 323 286 287 289 288; 324 287 143 158 289
165. 325 159 288 290 161; 326 288 289 291 290; 327 289 158 160 291
166. 328 177 292 293 179; 329 292 294 295 293; 330 294 176 178 295
167. 331 179 293 258 15; 332 293 295 260 258; 333 295 178 14 260
168. 334 161 290 296 195; 335 290 291 297 296; 336 291 160 194 297
169. 337 195 296 298 11; 338 296 297 299 298; 339 297 194 152 299
170. 340 12 300 301 204; 341 300 302 303 301; 342 302 13 203 303
171. 343 204 301 292 177; 344 301 303 294 292; 345 303 203 176 294
172. START GROUP DEFINITION
173. ELEMENT
174. _MX 15 16 19 20 23 24 27 28 31 32 35 36 39 40 43 44 47 48 51 52 55 56 59 60 -
175. 63 64 67 68 71 72 75 76 79 80 83 84 87 88 91 92 95 96 99 100 103 104 107 -
176. 108 111 112 115 116 119 120 123 124 137 138 140 141 149 150 152 153 162 163 -
177. 165 166 174 175 177 178 184 TO 187 192 TO 195 200 TO 203 208 TO 211 -
178. 214 TO 345
179. _MY 15 TO 20 23 TO 28 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 TO 84 87 TO 92 -
180. 111 TO 116 119 TO 124 137 138 140 141 143 144 146 147 149 150 152 153 162 -
181. 163 165 166 168 169 171 172 174 175 177 178 184 TO 195 200 TO 211 -
182. 214 TO 219 226 TO 243 250 TO 285 292 TO 309 316 TO 345
183. _SHEAR 63 TO 68 71 TO 76 137 138 140 141 143 144 146 147 149 150 152 153 162 -
184. 163 165 166 168 169 171 172 174 175 177 178 184 TO 195 200 TO 211 -
185. 214 TO 345
186. END GROUP DEFINITION
187. ELEMENT PROPERTY
188. 15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
189. 80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 138 140 141 -
190. 143 144 146 147 149 150 152 153 162 163 165 166 168 169 171 172 174 175 177 -
191. 178 184 TO 195 200 TO 211 214 TO 345 THICKNESS 0.55
192. DEFINE MATERIAL START
193. ISOTROPIC CONCRETE
194. E 2.17185E+007
195. POISSON 0.17
196. DENSITY 23.5616
197. ALPHA 1E-005
198. DAMP 0.05
199. TYPE CONCRETE
200. STRENGTH FCU 27579
201. END DEFINE MATERIAL
202. MEMBER PROPERTY AMERICAN
203. 1 TO 4 PRIS YD 0.7 ZD 0.5
204. 5 PRIS YD 0.5 ZD 0.3
205. CONSTANTS
206. MATERIAL CONCRETE ALL
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207. ****************************************
208. SUPPORTS
209. 1 2 7 TO 303 ELASTIC MAT DIRECT Y SUBGRADE 72000 PRINT
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JOINT	INFLUENCE AREA
1	0.062500
2	0.062500
7	0.028125
8	0.028125
9	0.015937
10	0.015937
11	0.015937
12	0.015937
13	0.030000
14	0.066666
15	0.035416
16	0.028125
17	0.062500
18	0.028125
19	0.062500
20	0.062500
21	0.028125
22	0.062500
23	0.028125
24	0.062500
25	0.030000
26	0.066666
27	0.035416
28	0.066666
29	0.035416
30	0.062500
31	0.062500
32	0.062500
33	0.062500
34	0.062500
35	0.066666
36	0.035416
37	0.066666
38	0.035416
39	0.062500
40	0.062500
41	0.062500
42	0.062500
43	0.062500
44	0.066666
45	0.035416
46	0.066666
47	0.035416
48	0.062500
49	0.062500
50	0.062500
51	0.062500
52	0.066666
53	0.035416

54	0.066666
55	0.035416
56	0.062500
57	0.062500
58	0.062500
59	0.062500
60	0.062500
61	0.066666
62	0.035416
63	0.066666
64	0.035416
65	0.062500
66	0.062500
67	0.062500
68	0.062500
69	0.062500
70	0.066666
71	0.035416
72	0.066666
73	0.035416
74	0.062500
75	0.062500
76	0.062500
77	0.062500
78	0.062500
79	0.066666
80	0.035416
81	0.066666
82	0.035416
83	0.062500
84	0.062500
85	0.062500
86	0.062500
87	0.062500
88	0.066666
89	0.035416
90	0.066666
91	0.035416
92	0.062500
93	0.062500
94	0.062500
95	0.062500
96	0.062500
97	0.066666
98	0.035416
99	0.066666
100	0.035416
101	0.062500
102	0.062500
103	0.062500
104	0.062500
105	0.062500
106	0.066666
107	0.035416
108	0.066666
109	0.035416

110	0.062500
111	0.062500
112	0.062500
113	0.062500
114	0.062500
115	0.066666
116	0.035416
117	0.066666
118	0.035416
119	0.062500
120	0.062500
121	0.062500
122	0.062500
123	0.066666
124	0.035416
125	0.066666
126	0.035416
127	0.062500
128	0.062500
129	0.062500
130	0.062500
131	0.062500
132	0.066666
133	0.035416
134	0.066666
135	0.035416
136	0.062500
137	0.062500
138	0.062500
139	0.062500
140	0.062500
141	0.066666
142	0.035416
143	0.066666
144	0.035416
145	0.062500
146	0.062500
147	0.062500
148	0.062500
149	0.062500
150	0.066666
151	0.035416
152	0.030000
153	0.028125
154	0.028125
155	0.028125
156	0.028125
157	0.030000
158	0.066666
159	0.035416
160	0.063333
161	0.033645
162	0.062500
163	0.059375
164	0.062500
165	0.059375

166	0.062500
167	0.059375
168	0.062500
169	0.059375
170	0.062500
171	0.059375
172	0.066666
173	0.063333
174	0.035416
175	0.033645
176	0.063333
177	0.033645
178	0.066666
179	0.035416
180	0.059375
181	0.062500
182	0.059375
183	0.062500
184	0.059375
185	0.062500
186	0.059375
187	0.062500
188	0.059375
189	0.062500
190	0.063333
191	0.066666
192	0.033645
193	0.035416
194	0.060000
195	0.031875
196	0.056250
197	0.056250
198	0.056250
199	0.056250
200	0.056250
201	0.060000
202	0.031875
203	0.060000
204	0.031875
205	0.056250
206	0.056250
207	0.056250
208	0.056250
209	0.056250
210	0.060000
211	0.031875
212	0.070834
213	0.070834
214	0.070834
215	0.070834
216	0.070834
217	0.070834
218	0.070834
219	0.070834
220	0.070834
221	0.070834

222	0.070834
223	0.070834
224	0.070834
225	0.070834
226	0.070834
227	0.070834
228	0.070834
229	0.070834
230	0.070834
231	0.070834
232	0.070834
233	0.070834
234	0.070834
235	0.070834
236	0.070834
237	0.070834
238	0.070834
239	0.070834
240	0.070834
241	0.070834
242	0.070834
243	0.070834
244	0.067292
245	0.067292
246	0.067292
247	0.070834
248	0.067292
249	0.070834
250	0.063750
251	0.063750
252	0.031875
253	0.031875
254	0.031875
255	0.063750
256	0.031875
257	0.063750
258	0.070834
259	0.070834
260	0.070834
261	0.070834
262	0.070834
263	0.070834
264	0.070834
265	0.070834
266	0.070834
267	0.070834
268	0.070834
269	0.070834
270	0.070834
271	0.070834
272	0.070834
273	0.070834
274	0.070834
275	0.070834
276	0.070834
277	0.070834

278	0.070834
279	0.070834
280	0.070834
281	0.070834
282	0.070834
283	0.070834
284	0.070834
285	0.070834
286	0.070834
287	0.070834
288	0.070834
289	0.070834
290	0.067292
291	0.067292
292	0.067292
293	0.070834
294	0.067292
295	0.070834
296	0.063750
297	0.063750
298	0.031875
299	0.031875
300	0.031875
301	0.063750
302	0.031875
303	0.063750
TOTAL	17.280014

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210. ****************************************
211. LOAD 1 DEAD LOAD DL
212. SELFWEIGHT Y -1
213. JOINT LOAD
214. 5 FX -0.297 FY -155.824 FZ -1.798
215. 6 FX -0.238 FY -145.789 FZ 13.698
216. ELEMENT LOAD
217. 15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
218. 80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 138 140 141 -
219. 143 144 146 147 149 150 152 153 162 163 165 166 168 169 171 172 174 175 177 -
220. 178 184 TO 195 200 TO 211 214 TO 345 PR GY -44.1
221. ****************************************
222. LOAD 2 LIVE LOAD LL
223. JOINT LOAD
224. 5 FX -0.164 FY -78 FZ -4.072
225. 6 FX -0.117 FY -78 FZ 4.072
226. ****************************************
227. LOAD 3 MATERIAL LOAD LL
228. JOINT LOAD
229. 5 FX -0.129 FY -70.3 FZ -3.715
230. 6 FX -0.11 FY -71.1 FZ 3.715
231. ****************************************
232. LOAD 4 WIND LOAD ALONG CONVEYOR WL (+Z)
233. JOINT LOAD
234. 5 FX 0.189 FY 265.049 FZ 39.623
235. 6 FX -0.189 FY -265.049 FZ 39.577
236. ****************************************
237. LOAD 5 WIND LOAD ALONG CONVEYOR WL (-Z)
238. JOINT LOAD
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239. 5 FX -0.185 FY -264.046 FZ -39.473
240. 6 FX 0.185 FY 264.046 FZ -39.427
241. ****************************************
242. LOAD 6 WIND LOAD ACROSS THE CONV.  WL (+X)
243. JOINT LOAD
244. 5 FX 3.21 FY 0 FZ 0
245. 6 FX 3.21 FY 0 FZ 0
246. ****************************************
247. LOAD 7 WIND LOAD ACROSS THE CONV.  WL (-X)
248. JOINT LOAD
249. 5 FX -3.21 FY 0 FZ 0
250. 6 FX -3.21 FY 0 FZ 0
251. ****************************************
252. LOAD 8 TEMP LOAD   TL1
253. JOINT LOAD
254. 5 FX 0 FY 0 FZ -15.912
255. 6 FX 0 FY 0 FZ 15.912
256. ****************************************
257. LOAD 9 TEMP LOAD   TL2
258. JOINT LOAD
259. 5 FX 0 FY 0 FZ 15.912
260. 6 FX 0 FY 0 FZ -15.912
261. ****************************************
262. LOAD 10 SEISMIC LOAD ALONG THE CONV. SL (+X)
263. JOINT LOAD
264. 5 FX 0.018 FY 0 FZ 0.003
265. 6 FX 0.021 FY 0.1 FZ -0.003
266. ****************************************
267. LOAD 11 SEISMIC LOAD ACROSS THE CONV. SL (+Z)
268. JOINT LOAD
269. 5 FX -0.002 FY 15.194 FZ 2.379
270. 6 FX 0.002 FY -15.194 FZ 2.375
271. ****************************************
272. *****LOAD COMBINATION FOR BEARING CAPACITY CHECK
273. LOAD 301 1.0 DL
274. REPEAT LOAD
275. 1 1.0
276. LOAD 302 1.0 (DL + TL1) + 1.0  (LL+ML)
277. REPEAT LOAD
278. 1 1.0 8 1.0 2 1.0 3 1.0
279. LOAD 303 1.0 (DL + TL1) + 1.0  (LL+ML)
280. REPEAT LOAD
281. 1 1.0 9 1.0 2 1.0 3 1.0
282. LOAD 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
283. REPEAT LOAD
284. 1 1.0 4 0.75 2 0.75 3 0.75
285. LOAD 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
286. REPEAT LOAD
287. 1 1.0 5 0.75 2 0.75 3 0.75
288. LOAD 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
289. REPEAT LOAD
290. 1 1.0 6 0.75 2 0.75 3 0.75
291. LOAD 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
292. REPEAT LOAD
293. 1 1.0 7 0.75 2 0.75 3 0.75
294. LOAD 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
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295. REPEAT LOAD
296. 1 1.0 10 0.525 2 0.75 3 0.75
297. LOAD 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
298. REPEAT LOAD
299. 1 1.0 10 -0.525 2 0.75 3 0.75
300. LOAD 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
301. REPEAT LOAD
302. 1 1.0 11 0.525 2 0.75 3 0.75
303. LOAD 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
304. REPEAT LOAD
305. 1 1.0 11 -0.525 2 0.75 3 0.75
306. LOAD 312 0.6 DL + 1.0 WL(+Z)
307. REPEAT LOAD
308. 1 0.6 4 1.0
309. LOAD 313 0.6 DL + 1.0 WL(-Z)
310. REPEAT LOAD
311. 1 0.6 5 1.0
312. LOAD 314 0.6 DL + 1.0 WL(+X)
313. REPEAT LOAD
314. 1 0.6 6 1.0
315. LOAD 315 0.6 DL + 1.0 WL(-X)
316. REPEAT LOAD
317. 1 0.6 7 1.0
318. LOAD 316 0.6 DL + 0.7 EL(+X)
319. REPEAT LOAD
320. 1 0.6 10 0.7
321. LOAD 317 0.6 DL + 0.7 EL(-X)
322. REPEAT LOAD
323. 1 0.6 10 -0.7
324. LOAD 318 0.6 DL + 0.7 EL(+Z)
325. REPEAT LOAD
326. 1 0.6 11 0.7
327. LOAD 319 0.6 DL + 0.7 EL(-Z)
328. REPEAT LOAD
329. 1 0.6 11 -0.7
330. ****************************************
331. *****LOAD COMBINATION FOR FOUNDATION DESIGN
332. LOAD 401 1.4 DL
333. REPEAT LOAD
334. 1 1.4
335. LOAD 402 1.4 (DL + TL1) + 1.7  (LL+ML)
336. REPEAT LOAD
337. 1 1.4 8 1.4 2 1.7 3 1.7
338. LOAD 403 1.4 (DL + TL1) + 1.7  (LL+ML)
339. REPEAT LOAD
340. 1 1.4 9 1.4 2 1.7 3 1.7
341. LOAD 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
342. REPEAT LOAD
343. 1 1.2 4 1.6 2 0.5 3 0.5
344. LOAD 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
345. REPEAT LOAD
346. 1 1.2 5 1.6 2 0.5 3 0.5
347. LOAD 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
348. REPEAT LOAD
349. 1 1.2 6 1.6 2 0.5 3 0.5
350. LOAD 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
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351. REPEAT LOAD
352. 1 1.2 7 1.6 2 0.5 3 0.5
353. LOAD 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
354. REPEAT LOAD
355. 1 1.2 10 1.0 2 0.5 3 0.5
356. LOAD 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
357. REPEAT LOAD
358. 1 1.2 10 -1.0 2 0.5 3 0.5
359. LOAD 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
360. REPEAT LOAD
361. 1 1.2 11 1.0 2 0.5 3 0.5
362. LOAD 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
363. REPEAT LOAD
364. 1 1.2 11 -1.0 2 0.5 3 0.5
365. LOAD 412 0.9 DL + 1.6 WL(+Z)
366. REPEAT LOAD
367. 1 0.9 4 1.6
368. LOAD 413 0.9 DL + 1.6 WL(-Z)
369. REPEAT LOAD
370. 1 0.9 5 1.6
371. LOAD 414 0.9 DL + 1.6 WL(+X)
372. REPEAT LOAD
373. 1 0.9 6 1.6
374. LOAD 415 0.9 DL + 1.6 WL(-X)
375. REPEAT LOAD
376. 1 0.9 7 1.6
377. LOAD 416 0.9 DL + 1.0 EL(+X)
378. REPEAT LOAD
379. 1 0.9 10 1.0
380. LOAD 417 0.9 DL + 1.0 EL(-X)
381. REPEAT LOAD
382. 1 0.9 10 -1.0
383. LOAD 418 0.9 DL + 1.0 EL(+Z)
384. REPEAT LOAD
385. 1 0.9 11 1.0
386. LOAD 419 0.9 DL + 1.0 EL(-Z)
387. REPEAT LOAD
388. 1 0.9 11 -1.0
389. ****************************************
390. PERFORM ANALYSIS
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P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**303**	**NUMBER OF MEMBERS**	**5**
NUMBER OF PLATES	**264**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**299**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 290/ 30/ 93 DOF
TOTAL PRIMARY LOAD CASES = 49, TOTAL DEGREES OF FREEDOM = 921
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 86 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 14.9/ 53205.1 MB

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391. ****************************************
392. LOAD LIST 401 TO 419
393. START CONCRETE DESIGN
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394. CODE ACI 2008
395. FC 35000 MEMB 1 TO 5
396. FYMAIN 420000 MEMB 1 TO 5
397. TRACK 1 ALL
398. CLB 0.05 MEMB 1 TO 5
399. CLT 0.05 MEMB 1 TO 5
400. CLS 0.05 ALL
401. MAXMAIN 25 ALL
402. MINMAIN 16 ALL
403. MINSEC 8 ALL
404. REINF 0 MEMB 1 TO 4
405. DESIGN COLUMN 1 TO 4
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COLUMN NO. 1 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	329.82	0.21	0.00021

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	329.82	15.58	0.01575

==========

COLUMN NO. 2 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	335.62	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	335.62	0.00	0.00000

==========

COLUMN NO. 3 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	347.47	0.19	0.00019

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	347.47	14.71	0.01412

==========

COLUMN NO. 4 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	314.01	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	314.01	0.00	0.00000

**********END OF COLUMN DESIGN RESULTS**********

406. DESIGN BEAM 5

BEAM NO. 5 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

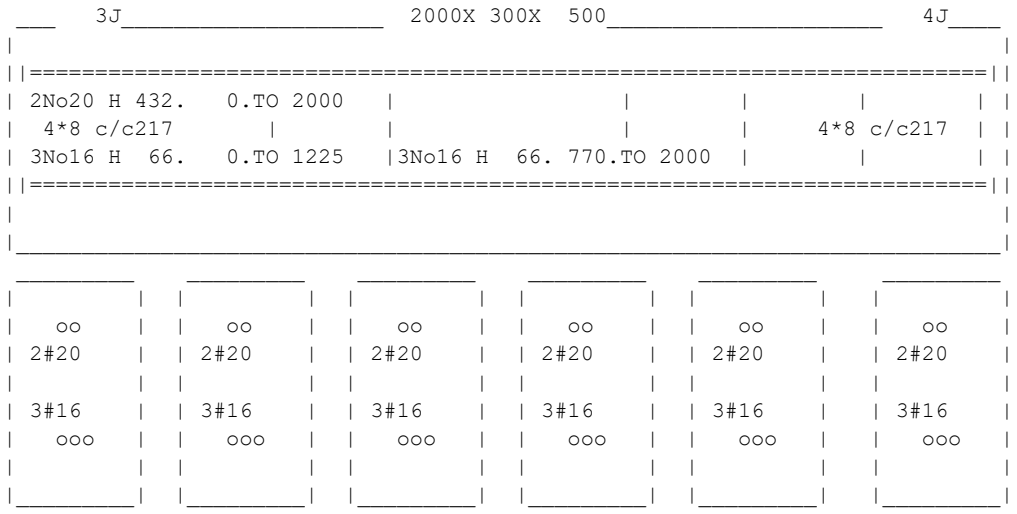
LEN - 2000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	3 - 16MM	0.	1225.	YES	NO
	CRITICAL POS MOMENT= 91.59 KN-MET AT 0.MM, LOAD 404					
	REQD STEEL= 576.MM2, RHO=0.0044, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 58050.0 cm^4					
2	66.	3 - 16MM	770.	2000.	NO	YES
	CRITICAL POS MOMENT= 73.63 KN-MET AT 2000.MM, LOAD 413					
	REQD STEEL= 460.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 58050.0 cm^4					
3	432.	2 - 20MM	0.	2000.	YES	YES
	CRITICAL NEG MOMENT= 99.10 KN-MET AT 2000.MM, LOAD 404					
	REQD STEEL= 625.MM2, RHO=0.0048, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 45./ 164. MMS					
	REQD. DEVELOPMENT LENGTH = 513. MMS					
	Cracked Moment of Inertia Iz at above location = 60046.8 cm^4					

B E A M N O. 5 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 92.95 KNS Vc= 125.36 KNS Vs= 0.00 KNS
Tu= 0.06 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 404
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 8 MM 2-LEGGED STIRRUPS AT 217. MM C/C FOR 566. MM

AT END SUPPORT - Vu= 97.75 KNS Vc= 125.36 KNS Vs= 4.97 KNS
Tu= 0.06 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 404
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT IS REQUIRED FOR SHEAR.
PROVIDE 8 MM 2-LEGGED STIRRUPS AT 217. MM C/C FOR 566. MM



**********END OF BEAM DESIGN**********

407. END CONCRETE DESIGN
408. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAR 19,2020 TIME= 9:47:55 ****

```
*************************************************************
*   For technical assistance on STAAD.Pro, please visit     *
*   http://selectservices.bentley.com/en-US/                *
*                                                           *
*   Details about additional assistance from                *
*   Bentley and Partners can be found at program menu       *
*   Help->Technical Support                                 *
*                                                           *
*      Copyright (c) 1997-2016 Bentley Systems, Inc.        *
*                http://www.bentley.com                     *
*************************************************************
```

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2			
Date	19-03-2020	30-04-2020	13/10/2023				Date	19-03-2020	30-04-2020	13/10/2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED C01	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED C01	AS BUILT			
Prepared By	APL	APL	APL				Prepared By	APL	APL	APL			
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

C01

ATTACHMENT 5 STAAD INPUT & OUTPUT FILE OF FOUNDATION FOR TR-4

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 30-Apr-20
JOB NAME MMGP-SAUDI ARABIA
JOB CLIENT MA'ADEN
JOB NO O19045/900707
JOB REV 1
JOB PART TRESTLE TR4 FOUNDATION
JOB REF FOUNDATIONS FOR CONVEYOR 3120-CNV-001
ENGINEER NAME APL
CHECKER NAME SBY
APPROVED NAME PRI
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 -3 0; 2 0 -3 6; 3 0 1.49012e-008 0; 4 0 1.49012e-008 6; 5 0 0.35 0;
6 0 0.35 6;
MEMBER INCIDENCES
1 1 3; 2 3 5; 3 2 4; 4 4 6; 5 3 4;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
1 TO 4 PRIS YD 0.7 ZD 0.5
5 PRIS YD 0.6 ZD 0.3
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
1 2 FIXED
*****************************************
LOAD 1 DEAD LOAD    DL
SELFWEIGHT Y -1
JOINT LOAD
5   FX  -0.063  FY  -98.466 FZ  -21.337
6   FX  0.058   FY  -203.433    FZ  32.537
*****************************************
LOAD 2 LIVE LOAD    LL
JOINT LOAD
5   FX  -0.015  FY  -71.967 FZ  -12.957
6   FX  0.015   FY  -72.033 FZ  12.957
*****************************************
LOAD 3 MATERIAL LOAD    LL
JOINT LOAD
5   FX  -0.01   FY  -57.467 FZ  -10.348
6   FX  0.01    FY  -57.533 FZ  10.348
*****************************************
```

```
LOAD 4 WIND LOAD ALONG CONVEYOR WL (+Z)
JOINT LOAD
5   FX  0.045   FY  226.499 FZ  47.075
6   FX  -0.045  FY  -226.499    FZ  47.074
*****************************************
LOAD 5 WIND LOAD ALONG CONVEYOR WL (-Z)
JOINT LOAD
5   FX  -0.045  FY  -226.499    FZ  -47.075
6   FX  0.045   FY  226.499 FZ  -47.074
*****************************************
LOAD 6 WIND LOAD ACROSS THE CONV.  WL (+X)
JOINT LOAD
5   FX  9.675   FY  0   FZ  0
6   FX  9.675   FY  0   FZ  0
*****************************************
LOAD 7 WIND LOAD ACROSS THE CONV.   WL (-X)
JOINT LOAD
5   FX  -9.675  FY  0   FZ  0
6   FX  -9.675  FY  0   FZ  0
*****************************************
LOAD 8 TEMP LOAD    TL1
JOINT LOAD
5   FX  0   FY  0   FZ  -55.184
6   FX  0   FY  0   FZ  55.184
*****************************************
LOAD 9 TEMP LOAD    TL2
JOINT LOAD
5   FX  0   FY  0   FZ  55.184
6   FX  0   FY  0   FZ  -55.184
*****************************************
LOAD 10 SEISMIC LOAD ALONG THE CONV.    SL (+X)
JOINT LOAD
5   FX  0.026   FY  -0.033  FZ  0
6   FX  0.03    FY  0.033   FZ  0
*****************************************
LOAD 11 SEISMIC LOAD ACROSS THE CONV.   SL (+Z)
JOINT LOAD
5   FX  -0.007  FY  12.024  FZ  2.135
6   FX  0.007   FY  -12.024 FZ  2.135
*****************************************
*****Load combination for bearing capacity check
LOAD COMB 301 1.0 DL
1 1.0
LOAD COMB 302 1.0 (DL + TL1) + 1.0  (LL+ML)
1 1.0 8 1.0 2 1.0 3 1.0
LOAD COMB 303 1.0 (DL + TL1) + 1.0  (LL+ML)
1 1.0 9 1.0 2 1.0 3 1.0
LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
1 1.0 4 0.75 2 0.75 3 0.75
LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
1 1.0 5 0.75 2 0.75 3 0.75
LOAD COMB 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
1 1.0 6 0.75 2 0.75 3 0.75
LOAD COMB 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
```

```
1 1.0 7 0.75 2 0.75 3 0.75
LOAD COMB 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
1 1.0 10 0.525 2 0.75 3 0.75
LOAD COMB 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
1 1.0 10 -0.525 2 0.75 3 0.75
LOAD COMB 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
1 1.0 11 0.525 2 0.75 3 0.75
LOAD COMB 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
1 1.0 11 -0.525 2 0.75 3 0.75
LOAD COMB 312 0.6 DL + 1.0 WL(+Z)
1 0.6 4 1.0
LOAD COMB 313 0.6 DL + 1.0 WL(-Z)
1 0.6 5 1.0
LOAD COMB 314 0.6 DL + 1.0 WL(+X)
1 0.6 6 1.0
LOAD COMB 315 0.6 DL + 1.0 WL(-X)
1 0.6 7 1.0
LOAD COMB 316 0.6 DL + 0.7 EL(+X)
1 0.6 10 0.7
LOAD COMB 317 0.6 DL + 0.7 EL(-X)
1 0.6 10 -0.7
LOAD COMB 318 0.6 DL + 0.7 EL(+Z)
1 0.6 11 0.7
LOAD COMB 319 0.6 DL + 0.7 EL(-Z)
1 0.6 11 -0.7
*****************************************
*****Load combination for foundation design
LOAD COMB 401 1.4 DL
1 1.4
LOAD COMB 402 1.4 (DL + TL1) + 1.7  (LL+ML)
1 1.4 8 1.4 2 1.7 3 1.7
LOAD COMB 403 1.4 (DL + TL1) + 1.7  (LL+ML)
1 1.4 9 1.4 2 1.7 3 1.7
LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
1 1.2 4 1.6 2 0.5 3 0.5
LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
1 1.2 5 1.6 2 0.5 3 0.5
LOAD COMB 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
1 1.2 6 1.6 2 0.5 3 0.5
LOAD COMB 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
1 1.2 7 1.6 2 0.5 3 0.5
LOAD COMB 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
1 1.2 10 1.0 2 0.5 3 0.5
LOAD COMB 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
1 1.2 10 -1.0 2 0.5 3 0.5
LOAD COMB 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
1 1.2 11 1.0 2 0.5 3 0.5
LOAD COMB 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
1 1.2 11 -1.0 2 0.5 3 0.5
LOAD COMB 412 0.9 DL + 1.6 WL(+Z)
1 0.9 4 1.6
LOAD COMB 413 0.9 DL + 1.6 WL(-Z)
1 0.9 5 1.6
LOAD COMB 414 0.9 DL + 1.6 WL(+X)
```

```
1 0.9 6 1.6
LOAD COMB 415 0.9 DL + 1.6 WL(-X)
1 0.9 7 1.6
LOAD COMB 416 0.9 DL + 1.0 EL(+X)
1 0.9 10 1.0
LOAD COMB 417 0.9 DL + 1.0 EL(-X)
1 0.9 10 -1.0
LOAD COMB 418 0.9 DL + 1.0 EL(+Z)
1 0.9 11 1.0
LOAD COMB 419 0.9 DL + 1.0 EL(-Z)
1 0.9 11 -1.0
*****************************************
PERFORM ANALYSIS
LOAD LIST 301 TO 319
PRINT SUPPORT REACTION ALL
*****************************************
LOAD LIST 401 TO 419
START CONCRETE DESIGN
CODE ACI 2008
FC 35000 ALL
FYMAIN 420000 ALL
TRACK 1 ALL
CLB 0.05 ALL
CLT 0.05 ALL
CLS 0.05 ALL
MAXMAIN 25 ALL
MINMAIN 16 ALL
MINSEC 8 ALL
REINF 0 MEMB 1 TO 4
DESIGN COLUMN 1 TO 4
DESIGN BEAM 5
END CONCRETE DESIGN
FINISH
```

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6           *
*           Version  20.07.11.70                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    APR 30, 2020                *
*           Time=    18:49:37                    *
*                                                *
*      USER ID: lnt                              *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: D:\APL\Assignments\13-Maaden-Crushing_n_Screening\Conv-3120-CNV-001\Staad\Civil\3120-CNV-0.. .STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 30-APR-20
    4. JOB NAME MMGP-SAUDI ARABIA
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045/900707
    7. JOB REV 1
    8. JOB PART TRESTLE TR4 FOUNDATION
    9. JOB REF FOUNDATIONS FOR CONVEYOR 3120-CNV-001
   10. ENGINEER NAME APL
   11. CHECKER NAME SBY
   12. APPROVED NAME PRI
   13. END JOB INFORMATION
   14. INPUT WIDTH 79
   15. UNIT METER KN
   16. JOINT COORDINATES
   17. 1 0 -3 0; 2 0 -3 6; 3 0 1.49012E-008 0; 4 0 1.49012E-008 6; 5 0 0.35 0
   18. 6 0 0.35 6
   19. MEMBER INCIDENCES
   20. 1 1 3; 2 3 5; 3 2 4; 4 4 6; 5 3 4
   21. DEFINE MATERIAL START
   22. ISOTROPIC CONCRETE
   23. E 2.17185E+007
   24. POISSON 0.17
   25. DENSITY 23.5616
   26. ALPHA 1E-005
   27. DAMP 0.05
   28. TYPE CONCRETE
   29. STRENGTH FCU 27579
   30. END DEFINE MATERIAL
   31. MEMBER PROPERTY AMERICAN
   32. 1 TO 4 PRIS YD 0.7 ZD 0.5
   33. 5 PRIS YD 0.6 ZD 0.3
   34. CONSTANTS
   35. MATERIAL CONCRETE ALL
   36. SUPPORTS
   37. 1 2 FIXED
   38. ****************************************
```

```
39. LOAD 1 DEAD LOAD DL
40. SELFWEIGHT Y -1
41. JOINT LOAD
42. 5 FX -0.063 FY -98.466 FZ -21.337
43. 6 FX 0.058 FY -203.433 FZ 32.537
44. ********************************************
45. LOAD 2 LIVE LOAD LL
46. JOINT LOAD
47. 5 FX -0.015 FY -71.967 FZ -12.957
48. 6 FX 0.015 FY -72.033 FZ 12.957
49. ********************************************
50. LOAD 3 MATERIAL LOAD LL
51. JOINT LOAD
52. 5 FX -0.01 FY -57.467 FZ -10.348
53. 6 FX 0.01 FY -57.533 FZ 10.348
54. ********************************************
55. LOAD 4 WIND LOAD ALONG CONVEYOR WL (+Z)
56. JOINT LOAD
57. 5 FX 0.045 FY 226.499 FZ 47.075
58. 6 FX -0.045 FY -226.499 FZ 47.074
59. ********************************************
60. LOAD 5 WIND LOAD ALONG CONVEYOR WL (-Z)
61. JOINT LOAD
62. 5 FX -0.045 FY -226.499 FZ -47.075
63. 6 FX 0.045 FY 226.499 FZ -47.074
64. ********************************************
65. LOAD 6 WIND LOAD ACROSS THE CONV.  WL (+X)
66. JOINT LOAD
67. 5 FX 9.675 FY 0 FZ 0
68. 6 FX 9.675 FY 0 FZ 0
69. ********************************************
70. LOAD 7 WIND LOAD ACROSS THE CONV.  WL (-X)
71. JOINT LOAD
72. 5 FX -9.675 FY 0 FZ 0
73. 6 FX -9.675 FY 0 FZ 0
74. ********************************************
75. LOAD 8 TEMP LOAD   TL1
76. JOINT LOAD
77. 5 FX 0 FY 0 FZ -55.184
78. 6 FX 0 FY 0 FZ 55.184
79. ********************************************
80. LOAD 9 TEMP LOAD   TL2
81. JOINT LOAD
82. 5 FX 0 FY 0 FZ 55.184
83. 6 FX 0 FY 0 FZ -55.184
84. ********************************************
85. LOAD 10 SEISMIC LOAD ALONG THE CONV. SL (+X)
86. JOINT LOAD
87. 5 FX 0.026 FY -0.033 FZ 0
88. 6 FX 0.03 FY 0.033 FZ 0
89. ********************************************
90. LOAD 11 SEISMIC LOAD ACROSS THE CONV. SL (+Z)
91. JOINT LOAD
92. 5 FX -0.007 FY 12.024 FZ 2.135
93. 6 FX 0.007 FY -12.024 FZ 2.135
94. ********************************************
```

```
 95. *****LOAD COMBINATION FOR BEARING CAPACITY CHECK
 96. LOAD COMB 301 1.0 DL
 97. 1 1.0
 98. LOAD COMB 302 1.0 (DL + TL1) + 1.0  (LL+ML)
 99. 1 1.0 8 1.0 2 1.0 3 1.0
100. LOAD COMB 303 1.0 (DL + TL1) + 1.0  (LL+ML)
101. 1 1.0 9 1.0 2 1.0 3 1.0
102. LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
103. 1 1.0 4 0.75 2 0.75 3 0.75
104. LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
105. 1 1.0 5 0.75 2 0.75 3 0.75
106. LOAD COMB 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
107. 1 1.0 6 0.75 2 0.75 3 0.75
108. LOAD COMB 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
109. 1 1.0 7 0.75 2 0.75 3 0.75
110. LOAD COMB 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
111. 1 1.0 10 0.525 2 0.75 3 0.75
112. LOAD COMB 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
113. 1 1.0 10 -0.525 2 0.75 3 0.75
114. LOAD COMB 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
115. 1 1.0 11 0.525 2 0.75 3 0.75
116. LOAD COMB 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
117. 1 1.0 11 -0.525 2 0.75 3 0.75
118. LOAD COMB 312 0.6 DL + 1.0 WL(+Z)
119. 1 0.6 4 1.0
120. LOAD COMB 313 0.6 DL + 1.0 WL(-Z)
121. 1 0.6 5 1.0
122. LOAD COMB 314 0.6 DL + 1.0 WL(+X)
123. 1 0.6 6 1.0
124. LOAD COMB 315 0.6 DL + 1.0 WL(-X)
125. 1 0.6 7 1.0
126. LOAD COMB 316 0.6 DL + 0.7 EL(+X)
127. 1 0.6 10 0.7
128. LOAD COMB 317 0.6 DL + 0.7 EL(-X)
129. 1 0.6 10 -0.7
130. LOAD COMB 318 0.6 DL + 0.7 EL(+Z)
131. 1 0.6 11 0.7
132. LOAD COMB 319 0.6 DL + 0.7 EL(-Z)
133. 1 0.6 11 -0.7
134. ****************************************
135. *****LOAD COMBINATION FOR FOUNDATION DESIGN
136. LOAD COMB 401 1.4 DL
137. 1 1.4
138. LOAD COMB 402 1.4 (DL + TL1) + 1.7  (LL+ML)
139. 1 1.4 8 1.4 2 1.7 3 1.7
140. LOAD COMB 403 1.4 (DL + TL1) + 1.7  (LL+ML)
141. 1 1.4 9 1.4 2 1.7 3 1.7
142. LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
143. 1 1.2 4 1.6 2 0.5 3 0.5
144. LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
145. 1 1.2 5 1.6 2 0.5 3 0.5
146. LOAD COMB 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
147. 1 1.2 6 1.6 2 0.5 3 0.5
148. LOAD COMB 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
149. 1 1.2 7 1.6 2 0.5 3 0.5
150. LOAD COMB 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
```

```
151. 1 1.2 10 1.0 2 0.5 3 0.5
152. LOAD COMB 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
153. 1 1.2 10 -1.0 2 0.5 3 0.5
154. LOAD COMB 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
155. 1 1.2 11 1.0 2 0.5 3 0.5
156. LOAD COMB 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
157. 1 1.2 11 -1.0 2 0.5 3 0.5
158. LOAD COMB 412 0.9 DL + 1.6 WL(+Z)
159. 1 0.9 4 1.6
160. LOAD COMB 413 0.9 DL + 1.6 WL(-Z)
161. 1 0.9 5 1.6
162. LOAD COMB 414 0.9 DL + 1.6 WL(+X)
163. 1 0.9 6 1.6
164. LOAD COMB 415 0.9 DL + 1.6 WL(-X)
165. 1 0.9 7 1.6
166. LOAD COMB 416 0.9 DL + 1.0 EL(+X)
167. 1 0.9 10 1.0
168. LOAD COMB 417 0.9 DL + 1.0 EL(-X)
169. 1 0.9 10 -1.0
170. LOAD COMB 418 0.9 DL + 1.0 EL(+Z)
171. 1 0.9 11 1.0
172. LOAD COMB 419 0.9 DL + 1.0 EL(-Z)
173. 1 0.9 11 -1.0
174. ****************************************
175. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**6**	**NUMBER OF MEMBERS**	**5**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**2**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

```
  ORIGINAL/FINAL BAND-WIDTH=     2/     2/     18 DOF
TOTAL      PRIMARY LOAD CASES =    11, TOTAL DEGREES OF FREEDOM =      24
TOTAL LOAD COMBINATION  CASES =    38  SO FAR.
  SIZE OF STIFFNESS MATRIX =        1 DOUBLE  KILO-WORDS
  REQRD/AVAIL. DISK SPACE  =     12.0/  49575.2 MB
```

```
176. LOAD LIST 301 TO 319
177. PRINT SUPPORT REACTION ALL
```

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
1	301	0.06	136.54	-3.76	-9.49	-0.01	-0.19
	302	0.08	265.97	-13.34	-16.32	-0.02	-0.26
	303	0.08	265.97	0.13	-6.72	-0.02	-0.26
	304	0.05	49.15	-41.20	-85.50	-0.01	-0.14
	305	0.11	418.09	29.41	63.48	-0.02	-0.35
	306	-7.18	233.62	-5.90	-11.01	-0.01	24.06
	307	7.33	233.62	-5.90	-11.01	-0.01	-24.55
	308	0.06	233.63	-5.90	-11.01	-0.02	-0.20
	309	0.09	233.60	-5.90	-11.01	-0.01	-0.29
	310	0.08	226.84	-7.02	-13.38	-0.02	-0.26
	311	0.07	240.39	-4.77	-8.64	-0.01	-0.23
	312	-0.01	-164.03	-49.33	-105.02	0.00	0.02
	313	0.08	327.88	44.82	93.62	-0.02	-0.25
	314	-9.64	81.92	-2.26	-5.70	-0.01	32.30
	315	9.71	81.92	-2.26	-5.70	-0.01	-32.52
	316	0.02	81.95	-2.26	-5.70	-0.01	-0.05
	317	0.05	81.90	-2.26	-5.70	-0.01	-0.17
	318	0.04	72.89	-3.75	-8.85	-0.01	-0.13
	319	0.03	90.96	-0.76	-2.54	-0.01	-0.10
2	301	-0.05	246.06	-7.44	-14.38	-0.01	0.17
	302	-0.08	375.62	2.14	-7.55	-0.02	0.25
	303	-0.08	375.62	-11.33	-17.15	-0.02	0.25
	304	-0.04	527.70	-40.61	-87.35	-0.01	0.13
	305	-0.10	158.76	30.00	61.63	-0.02	0.33
	306	-7.33	343.23	-5.30	-12.86	-0.01	24.54
	307	7.18	343.23	-5.30	-12.86	-0.01	-24.08
	308	-0.09	343.21	-5.30	-12.86	-0.02	0.28
	309	-0.06	343.25	-5.30	-12.86	-0.01	0.18
	310	-0.08	350.01	-6.43	-15.23	-0.02	0.24
	311	-0.07	336.46	-4.18	-10.49	-0.01	0.22
	312	0.01	393.59	-51.54	-107.95	0.00	-0.03
	313	-0.07	-98.32	42.61	90.69	-0.02	0.24
	314	-9.71	147.63	-4.46	-8.63	-0.01	32.51
	315	9.64	147.63	-4.46	-8.63	-0.01	-32.31
	316	-0.05	147.61	-4.46	-8.63	-0.01	0.17
	317	-0.01	147.66	-4.46	-8.63	-0.01	0.03
	318	-0.04	156.67	-5.96	-11.79	-0.01	0.12
	319	-0.03	138.60	-2.97	-5.47	-0.01	0.09

********** END OF LATEST ANALYSIS RESULT **********

```
178. ****************************************
179. LOAD LIST 401 TO 419
180. START CONCRETE DESIGN
181. CODE ACI 2008
182. FC 35000 ALL
183. FYMAIN 420000 ALL
184. TRACK 1 ALL
185. CLB 0.05 ALL
186. CLT 0.05 ALL
187. CLS 0.05 ALL
188. MAXMAIN 25 ALL
189. MINMAIN 16 ALL
190. MINSEC 8 ALL
191. REINF 0 MEMB 1 TO 4
192. DESIGN COLUMN 1 TO 4
```

==========

COLUMN NO. 1 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	240.80	0.02	0.00003

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	240.80	3.87	0.00536

==========

COLUMN NO. 2 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	212.08	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	212.08	0.00	0.00000

==========

COLUMN NO. 3 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	476.68	0.02	0.00001

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	476.68	17.08	0.01194

==========

COLUMN NO. 4 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	438.16	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	438.16	0.00	0.00000

**********END OF COLUMN DESIGN RESULTS**********

193. DESIGN BEAM 5

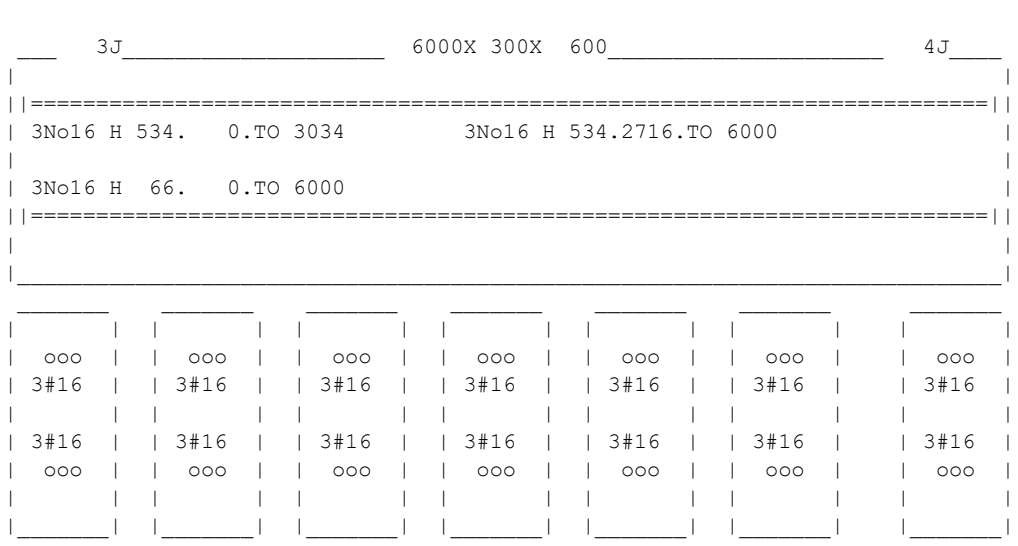
BEAM NO. 5 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 6000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	3 - 16MM	0.	6000.	YES	YES
	CRITICAL POS MOMENT= 88.30 KN-MET AT 0.MM, LOAD 412					
	REQD STEEL= 562.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 90761.5 cm^4					
2	534.	3 - 16MM	0.	3034.	YES	NO
	CRITICAL NEG MOMENT= 101.04 KN-MET AT 0.MM, LOAD 405					
	REQD STEEL= 562.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 90761.5 cm^4					
3	534.	3 - 16MM	2716.	6000.	NO	YES
	CRITICAL NEG MOMENT= 117.42 KN-MET AT 6000.MM, LOAD 404					
	REQD STEEL= 600.MM2, RHO=0.0038, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 90033.2 cm^4					

B E A M N O. 5 D E S I G N R E S U L T S - SHEAR

```
AT START SUPPORT - Vu=   11.46 KNS  Vc=  148.48 KNS  Vs=    0.00 KNS
Tu=    0.01 KN-MET  Tc=    6.3 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
AT END   SUPPORT - Vu=   17.83 KNS  Vc=  148.48 KNS  Vs=    0.00 KNS
Tu=    0.01 KN-MET  Tc=    6.3 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
```



**********END OF BEAM DESIGN**********

194. END CONCRETE DESIGN
195. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= APR 30,2020 TIME= 18:49:38 ****

```
*************************************************************
*   For technical assistance on STAAD.Pro, please visit     *
*   http://selectservices.bentley.com/en-US/                *
*                                                           *
*   Details about additional assistance from                *
*   Bentley and Partners can be found at program menu       *
*   Help->Technical Support                                 *
*                                                           *
*      Copyright (c) 1997-2016 Bentley Systems, Inc.        *
*                http://www.bentley.com                     *
*************************************************************
```

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2			
Date	19-03-2020	30-04-2020	13/10/2023				Date	19-03-2020	30-04-2020	13/10/2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED C01	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED C01	AS BUILT			
Prepared By	APL	APL	APL				Prepared By	APL	APL	APL			
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

C01

ATTACHMENT 6 STAAD INPUT & OUTPUT FILE OF FOUNDATION FOR VGTU

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 30-Apr-20
JOB NAME MMGP-SAUDI ARABIA
JOB CLIENT MA'ADEN
JOB NO O19045/900707
JOB REV 1
JOB PART VGTU FDN
JOB REF FOUNDATIONS FOR CONVEYOR 3120-CNV-001
ENGINEER NAME APL
CHECKER NAME SBY
APPROVED NAME PRI
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 -3 0; 2 0 -3 5; 3 4 -3 0; 4 4 -3 5; 5 8 -3 0; 6 8 -3 5; 7 0 0.35 0;
8 0 0.35 5; 9 4 0.35 0; 10 4 0.35 5; 11 8 0.35 0; 12 8 0.35 5; 13 0 1 0;
14 0 1 5; 15 4 1 0; 16 4 1 5; 17 8 1 0; 18 8 1 5;
MEMBER INCIDENCES
1 1 7; 2 7 13; 3 2 8; 4 8 14; 5 3 9; 6 9 15; 7 4 10; 8 10 16; 9 5 11; 10 11 17;
11 6 12; 12 12 18; 13 7 8; 14 9 10; 15 11 12; 16 7 9; 17 9 11; 18 8 10;
19 10 12;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
9 TO 12 PRIS YD 0.6 ZD 0.6
13 TO 15 PRIS YD 0.5 ZD 0.3
MEMBER PROPERTY AMERICAN
16 TO 19 PRIS YD 0.6 ZD 0.3
MEMBER PROPERTY AMERICAN
1 TO 8 PRIS YD 0.6 ZD 0.6
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
1 TO 6 FIXED
*****************************************
LOAD 1 DEAD LOAD    DL
SELFWEIGHT Y -1
JOINT LOAD
13  FX  -26.371 FY  -249.267    FZ  -11.129
14  FX  -25.779 FY  -312.574    FZ  17.043
15  FX  -24.34  FY  -19.859 FZ  0.217
16  FX  -23.594 FY  -91.235 FZ  4.168
17  FX  0.002   FY  -17.4   FZ  0.144
18  FX  0.082   FY  -20.833 FZ  0.957
```

```
****************************************
LOAD 2 LIVE LOAD    LL
JOINT LOAD
13  FX  -2.176  FY  -243.759    FZ  -12.866
14  FX  -3.119  FY  -204.114    FZ  12.947
15  FX  3.174   FY  -271.907    FZ  -8.1
16  FX  2.147   FY  -281.229    FZ  9.183
17  FX  -0.003  FY  -62.072 FZ  -0.597
18  FX  -0.023  FY  -51.935 FZ  -0.567

****************************************
LOAD 3 MATERIAL LOAD    LL
JOINT LOAD
13  FX  -0.019  FY  -29.065 FZ  -1.699
14  FX  -0.044  FY  -28.435 FZ  1.595
15  FX  0.044   FY  -28.401 FZ  0.069
16  FX  0.021   FY  -29.099 FZ  0.169
17  FX  0   FY  -0.234  FZ  -0.067
18  FX  -0.002  FY  0.234   FZ  -0.067

****************************************
LOAD 4 WIND LOAD ALONG CONVEYOR WL (+Z)
JOINT LOAD
13  FX  -1.249  FY  95.805  FZ  27.496
14  FX  1.246   FY  -95.805 FZ  27.721
15  FX  -1.647  FY  98.912  FZ  23.828
16  FX  1.641   FY  -98.912 FZ  23.822
17  FX  -0.001  FY  21.194  FZ  10.181
18  FX  0.009   FY  -21.194 FZ  10.177

****************************************
LOAD 5 WIND LOAD ALONG CONVEYOR WL (-Z)
JOINT LOAD
13  FX  1.249   FY  -95.805 FZ  -27.496
14  FX  -1.246  FY  95.805  FZ  -27.721
15  FX  1.647   FY  -98.912 FZ  -23.828
16  FX  -1.641  FY  98.912  FZ  -23.822
17  FX  0.001   FY  -21.194 FZ  -10.181
18  FX  -0.009  FY  21.194  FZ  -10.177

****************************************
LOAD 6 WIND LOAD ACROSS THE CONV.  WL (+X)
JOINT LOAD
13  FX  39.901  FY  161.852 FZ  6.16
14  FX  39.826  FY  163.052 FZ  -6.423
15  FX  39.907  FY  -161.296    FZ  0.099
16  FX  41.16   FY  -163.605    FZ  0.485
17  FX  1.188   FY  -0.556  FZ  -0.16
18  FX  -0.079  FY  0.553   FZ  -0.16

****************************************
LOAD 7 WIND LOAD ACROSS THE CONV.   WL (-X)
JOINT LOAD
```

```
13  FX  -39.901 FY  -161.852    FZ  -6.16
14  FX  -39.826 FY  -163.052    FZ  6.423
15  FX  -39.907 FY  161.296 FZ  -0.099
16  FX  -41.16  FY  163.605 FZ  -0.485
17  FX  -1.188  FY  0.556   FZ  0.16
18  FX  0.079   FY  -0.553  FZ  0.16

****************************************
LOAD 8 TEMP LOAD    TL1
JOINT LOAD
13  FX  3.038   FY  -0.075  FZ  -42.464
14  FX  -1.4    FY  19.304  FZ  32.335
15  FX  5.459   FY  -14.698 FZ  0.616
16  FX  0.516   FY  -4.531  FZ  1.663
17  FX  0.08    FY  -0.61   FZ  -0.625
18  FX  0.507   FY  0.61    FZ  0.275

****************************************
LOAD 9 TEMP LOAD    TL2
JOINT LOAD
13  FX  -3.038  FY  0.075   FZ  42.464
14  FX  1.4 FY  -19.304 FZ  -32.335
15  FX  -5.459  FY  14.698  FZ  -0.616
16  FX  -0.516  FY  4.531   FZ  -1.663
17  FX  -0.08   FY  0.61    FZ  0.625
18  FX  -0.507  FY  -0.61   FZ  -0.275

****************************************
LOAD 10 SEISMIC LOAD ALONG THE CONV.    SL (+X)
JOINT LOAD
13  FX  1.494   FY  7.426   FZ  0.325
14  FX  1.575   FY  7.646   FZ  -0.292
15  FX  1.488   FY  -7.368  FZ  -0.021
16  FX  1.579   FY  -7.704  FZ  -0.002
17  FX  0   FY  -0.018  FZ  -0.005
18  FX  -0.007  FY  0.018   FZ  -0.005

****************************************
LOAD 11 SEISMIC LOAD ACROSS THE CONV.   SL (+Z)
JOINT LOAD
13  FX  -0.045  FY  4.946   FZ  1.361
14  FX  0.045   FY  -4.946  FZ  1.372
15  FX  -0.066  FY  4.22    FZ  0.935
16  FX  0.066   FY  -4.22   FZ  0.935
17  FX  0   FY  1.836   FZ  0.763
18  FX  0.001   FY  -1.836  FZ  0.663

****************************************
*****Load combination for bearing capacity check
LOAD COMB 301 1.0 DL
1 1.0
LOAD COMB 302 1.0 (DL + TL1) + 1.0  (LL+ML)
1 1.0 8 1.0 2 1.0 3 1.0
LOAD COMB 303 1.0 (DL + TL1) + 1.0  (LL+ML)
```

```
1 1.0 9 1.0 2 1.0 3 1.0
LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
1 1.0 4 0.75 2 0.75 3 0.75
LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
1 1.0 5 0.75 2 0.75 3 0.75
LOAD COMB 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
1 1.0 6 0.75 2 0.75 3 0.75
LOAD COMB 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
1 1.0 7 0.75 2 0.75 3 0.75
LOAD COMB 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
1 1.0 10 0.525 2 0.75 3 0.75
LOAD COMB 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
1 1.0 10 -0.525 2 0.75 3 0.75
LOAD COMB 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
1 1.0 11 0.525 2 0.75 3 0.75
LOAD COMB 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
1 1.0 11 -0.525 2 0.75 3 0.75
LOAD COMB 312 0.6 DL + 1.0 WL(+Z)
1 0.6 4 1.0
LOAD COMB 313 0.6 DL + 1.0 WL(-Z)
1 0.6 5 1.0
LOAD COMB 314 0.6 DL + 1.0 WL(+X)
1 0.6 6 1.0
LOAD COMB 315 0.6 DL + 1.0 WL(-X)
1 0.6 7 1.0
LOAD COMB 316 0.6 DL + 0.7 EL(+X)
1 0.6 10 0.7
LOAD COMB 317 0.6 DL + 0.7 EL(-X)
1 0.6 10 -0.7
LOAD COMB 318 0.6 DL + 0.7 EL(+Z)
1 0.6 11 0.7
LOAD COMB 319 0.6 DL + 0.7 EL(-Z)
1 0.6 11 -0.7
*****************************************
*****Load combination for foundation design
LOAD COMB 401 1.4 DL
1 1.4
LOAD COMB 402 1.4 (DL + TL1) + 1.7  (LL+ML)
1 1.4 8 1.4 2 1.7 3 1.7
LOAD COMB 403 1.4 (DL + TL1) + 1.7  (LL+ML)
1 1.4 9 1.4 2 1.7 3 1.7
LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
1 1.2 4 1.6 2 0.5 3 0.5
LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
1 1.2 5 1.6 2 0.5 3 0.5
LOAD COMB 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
1 1.2 6 1.6 2 0.5 3 0.5
LOAD COMB 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
1 1.2 7 1.6 2 0.5 3 0.5
LOAD COMB 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
1 1.2 10 1.0 2 0.5 3 0.5
LOAD COMB 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
1 1.2 10 -1.0 2 0.5 3 0.5
LOAD COMB 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
```

```
1 1.2 11 1.0 2 0.5 3 0.5
LOAD COMB 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
1 1.2 11 -1.0 2 0.5 3 0.5
LOAD COMB 412 0.9 DL + 1.6 WL(+Z)
1 0.9 4 1.6
LOAD COMB 413 0.9 DL + 1.6 WL(-Z)
1 0.9 5 1.6
LOAD COMB 414 0.9 DL + 1.6 WL(+X)
1 0.9 6 1.6
LOAD COMB 415 0.9 DL + 1.6 WL(-X)
1 0.9 7 1.6
LOAD COMB 416 0.9 DL + 1.0 EL(+X)
1 0.9 10 1.0
LOAD COMB 417 0.9 DL + 1.0 EL(-X)
1 0.9 10 -1.0
LOAD COMB 418 0.9 DL + 1.0 EL(+Z)
1 0.9 11 1.0
LOAD COMB 419 0.9 DL + 1.0 EL(-Z)
1 0.9 11 -1.0
*****************************************
PERFORM ANALYSIS
LOAD LIST 301 TO 319
PRINT SUPPORT REACTION ALL
*******************************************************************************
LOAD LIST 401 TO 419
START CONCRETE DESIGN
CODE ACI 2008
FC 35000 ALL
FYMAIN 420000 ALL
TRACK 1 ALL
CLB 0.05 ALL
CLT 0.05 ALL
CLS 0.05 ALL
MAXMAIN 25 ALL
MINMAIN 16 ALL
MINSEC 8 ALL
REINF 0 MEMB 1 TO 12
DESIGN COLUMN 1 TO 12
DESIGN BEAM 13 TO 19
END CONCRETE DESIGN
FINISH
```

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6           *
*           Version  20.07.11.70                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    APR 30, 2020                *
*           Time=    16:46:55                    *
*                                                *
*      USER ID: lnt                              *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: D:\APL\Assignments\13-Maaden-Crushing_n_Screening\Conv-3120-CNV-001\Staad\Civil\3120-CNV-0.. .STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 30-APR-20
    4. JOB NAME MMGP-SAUDI ARABIA
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045/900707
    7. JOB REV 1
    8. JOB PART VGTU FDN
    9. JOB REF FOUNDATIONS FOR CONVEYOR 3120-CNV-001
   10. ENGINEER NAME APL
   11. CHECKER NAME SBY
   12. APPROVED NAME PRI
   13. END JOB INFORMATION
   14. INPUT WIDTH 79
   15. UNIT METER KN
   16. JOINT COORDINATES
   17. 1 0 -3 0; 2 0 -3 5; 3 4 -3 0; 4 4 -3 5; 5 8 -3 0; 6 8 -3 5; 7 0 0.35 0
   18. 8 0 0.35 5; 9 4 0.35 0; 10 4 0.35 5; 11 8 0.35 0; 12 8 0.35 5; 13 0 1 0
   19. 14 0 1 5; 15 4 1 0; 16 4 1 5; 17 8 1 0; 18 8 1 5
   20. MEMBER INCIDENCES
   21. 1 1 7; 2 7 13; 3 2 8; 4 8 14; 5 3 9; 6 9 15; 7 4 10; 8 10 16; 9 5 11; 10 11 17
   22. 11 6 12; 12 12 18; 13 7 8; 14 9 10; 15 11 12; 16 7 9; 17 9 11; 18 8 10
   23. 19 10 12
   24. DEFINE MATERIAL START
   25. ISOTROPIC CONCRETE
   26. E 2.17185E+007
   27. POISSON 0.17
   28. DENSITY 23.5616
   29. ALPHA 1E-005
   30. DAMP 0.05
   31. TYPE CONCRETE
   32. STRENGTH FCU 27579
   33. END DEFINE MATERIAL
   34. MEMBER PROPERTY AMERICAN
   35. 9 TO 12 PRIS YD 0.6 ZD 0.6
   36. 13 TO 15 PRIS YD 0.5 ZD 0.3
   37. MEMBER PROPERTY AMERICAN
   38. 16 TO 19 PRIS YD 0.6 ZD 0.3
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39. MEMBER PROPERTY AMERICAN
40. 1 TO 8 PRIS YD 0.6 ZD 0.6
41. CONSTANTS
42. MATERIAL CONCRETE ALL
43. SUPPORTS
44. 1 TO 6 FIXED
45. ******************************************
46. LOAD 1 DEAD LOAD DL
47. SELFWEIGHT Y -1
48. JOINT LOAD
49. 13 FX -26.371 FY -249.267 FZ -11.129
50. 14 FX -25.779 FY -312.574 FZ 17.043
51. 15 FX -24.34 FY -19.859 FZ 0.217
52. 16 FX -23.594 FY -91.235 FZ 4.168
53. 17 FX 0.002 FY -17.4 FZ 0.144
54. 18 FX 0.082 FY -20.833 FZ 0.957
56. ******************************************
57. LOAD 2 LIVE LOAD LL
58. JOINT LOAD
59. 13 FX -2.176 FY -243.759 FZ -12.866
60. 14 FX -3.119 FY -204.114 FZ 12.947
61. 15 FX 3.174 FY -271.907 FZ -8.1
62. 16 FX 2.147 FY -281.229 FZ 9.183
63. 17 FX -0.003 FY -62.072 FZ -0.597
64. 18 FX -0.023 FY -51.935 FZ -0.567
66. ******************************************
67. LOAD 3 MATERIAL LOAD LL
68. JOINT LOAD
69. 13 FX -0.019 FY -29.065 FZ -1.699
70. 14 FX -0.044 FY -28.435 FZ 1.595
71. 15 FX 0.044 FY -28.401 FZ 0.069
72. 16 FX 0.021 FY -29.099 FZ 0.169
73. 17 FX 0 FY -0.234 FZ -0.067
74. 18 FX -0.002 FY 0.234 FZ -0.067
76. ******************************************
77. LOAD 4 WIND LOAD ALONG CONVEYOR WL (+Z)
78. JOINT LOAD
79. 13 FX -1.249 FY 95.805 FZ 27.496
80. 14 FX 1.246 FY -95.805 FZ 27.721
81. 15 FX -1.647 FY 98.912 FZ 23.828
82. 16 FX 1.641 FY -98.912 FZ 23.822
83. 17 FX -0.001 FY 21.194 FZ 10.181
84. 18 FX 0.009 FY -21.194 FZ 10.177
86. ******************************************
87. LOAD 5 WIND LOAD ALONG CONVEYOR WL (-Z)
88. JOINT LOAD
89. 13 FX 1.249 FY -95.805 FZ -27.496
90. 14 FX -1.246 FY 95.805 FZ -27.721
91. 15 FX 1.647 FY -98.912 FZ -23.828
92. 16 FX -1.641 FY 98.912 FZ -23.822
93. 17 FX 0.001 FY -21.194 FZ -10.181
94. 18 FX -0.009 FY 21.194 FZ -10.177
96. ******************************************
97. LOAD 6 WIND LOAD ACROSS THE CONV.  WL (+X)
98. JOINT LOAD
99. 13 FX 39.901 FY 161.852 FZ 6.16
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100. 14 FX 39.826 FY 163.052 FZ -6.423
101. 15 FX 39.907 FY -161.296 FZ 0.099
102. 16 FX 41.16 FY -163.605 FZ 0.485
103. 17 FX 1.188 FY -0.556 FZ -0.16
104. 18 FX -0.079 FY 0.553 FZ -0.16
106. ****************************************
107. LOAD 7 WIND LOAD ACROSS THE CONV.  WL (-X)
108. JOINT LOAD
109. 13 FX -39.901 FY -161.852 FZ -6.16
110. 14 FX -39.826 FY -163.052 FZ 6.423
111. 15 FX -39.907 FY 161.296 FZ -0.099
112. 16 FX -41.16 FY 163.605 FZ -0.485
113. 17 FX -1.188 FY 0.556 FZ 0.16
114. 18 FX 0.079 FY -0.553 FZ 0.16
116. ****************************************
117. LOAD 8 TEMP LOAD   TL1
118. JOINT LOAD
119. 13 FX 3.038 FY -0.075 FZ -42.464
120. 14 FX -1.4 FY 19.304 FZ 32.335
121. 15 FX 5.459 FY -14.698 FZ 0.616
122. 16 FX 0.516 FY -4.531 FZ 1.663
123. 17 FX 0.08 FY -0.61 FZ -0.625
124. 18 FX 0.507 FY 0.61 FZ 0.275
126. ****************************************
127. LOAD 9 TEMP LOAD   TL2
128. JOINT LOAD
129. 13 FX -3.038 FY 0.075 FZ 42.464
130. 14 FX 1.4 FY -19.304 FZ -32.335
131. 15 FX -5.459 FY 14.698 FZ -0.616
132. 16 FX -0.516 FY 4.531 FZ -1.663
133. 17 FX -0.08 FY 0.61 FZ 0.625
134. 18 FX -0.507 FY -0.61 FZ -0.275
136. ****************************************
137. LOAD 10 SEISMIC LOAD ALONG THE CONV. SL (+X)
138. JOINT LOAD
139. 13 FX 1.494 FY 7.426 FZ 0.325
140. 14 FX 1.575 FY 7.646 FZ -0.292
141. 15 FX 1.488 FY -7.368 FZ -0.021
142. 16 FX 1.579 FY -7.704 FZ -0.002
143. 17 FX 0 FY -0.018 FZ -0.005
144. 18 FX -0.007 FY 0.018 FZ -0.005
146. ****************************************
147. LOAD 11 SEISMIC LOAD ACROSS THE CONV. SL (+Z)
148. JOINT LOAD
149. 13 FX -0.045 FY 4.946 FZ 1.361
150. 14 FX 0.045 FY -4.946 FZ 1.372
151. 15 FX -0.066 FY 4.22 FZ 0.935
152. 16 FX 0.066 FY -4.22 FZ 0.935
153. 17 FX 0 FY 1.836 FZ 0.763
154. 18 FX 0.001 FY -1.836 FZ 0.663
156. ****************************************
157. *****LOAD COMBINATION FOR BEARING CAPACITY CHECK
158. LOAD COMB 301 1.0 DL
159. 1 1.0
160. LOAD COMB 302 1.0 (DL + TL1) + 1.0  (LL+ML)
161. 1 1.0 8 1.0 2 1.0 3 1.0
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162. LOAD COMB 303 1.0 (DL + TL1) + 1.0  (LL+ML)
163. 1 1.0 9 1.0 2 1.0 3 1.0
164. LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
165. 1 1.0 4 0.75 2 0.75 3 0.75
166. LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
167. 1 1.0 5 0.75 2 0.75 3 0.75
168. LOAD COMB 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
169. 1 1.0 6 0.75 2 0.75 3 0.75
170. LOAD COMB 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
171. 1 1.0 7 0.75 2 0.75 3 0.75
172. LOAD COMB 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
173. 1 1.0 10 0.525 2 0.75 3 0.75
174. LOAD COMB 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
175. 1 1.0 10 -0.525 2 0.75 3 0.75
176. LOAD COMB 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
177. 1 1.0 11 0.525 2 0.75 3 0.75
178. LOAD COMB 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
179. 1 1.0 11 -0.525 2 0.75 3 0.75
180. LOAD COMB 312 0.6 DL + 1.0 WL(+Z)
181. 1 0.6 4 1.0
182. LOAD COMB 313 0.6 DL + 1.0 WL(-Z)
183. 1 0.6 5 1.0
184. LOAD COMB 314 0.6 DL + 1.0 WL(+X)
185. 1 0.6 6 1.0
186. LOAD COMB 315 0.6 DL + 1.0 WL(-X)
187. 1 0.6 7 1.0
188. LOAD COMB 316 0.6 DL + 0.7 EL(+X)
189. 1 0.6 10 0.7
190. LOAD COMB 317 0.6 DL + 0.7 EL(-X)
191. 1 0.6 10 -0.7
192. LOAD COMB 318 0.6 DL + 0.7 EL(+Z)
193. 1 0.6 11 0.7
194. LOAD COMB 319 0.6 DL + 0.7 EL(-Z)
195. 1 0.6 11 -0.7
196. ****************************************
197. *****LOAD COMBINATION FOR FOUNDATION DESIGN
198. LOAD COMB 401 1.4 DL
199. 1 1.4
200. LOAD COMB 402 1.4 (DL + TL1) + 1.7  (LL+ML)
201. 1 1.4 8 1.4 2 1.7 3 1.7
202. LOAD COMB 403 1.4 (DL + TL1) + 1.7  (LL+ML)
203. 1 1.4 9 1.4 2 1.7 3 1.7
204. LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
205. 1 1.2 4 1.6 2 0.5 3 0.5
206. LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
207. 1 1.2 5 1.6 2 0.5 3 0.5
208. LOAD COMB 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
209. 1 1.2 6 1.6 2 0.5 3 0.5
210. LOAD COMB 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
211. 1 1.2 7 1.6 2 0.5 3 0.5
212. LOAD COMB 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
213. 1 1.2 10 1.0 2 0.5 3 0.5
214. LOAD COMB 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
215. 1 1.2 10 -1.0 2 0.5 3 0.5
216. LOAD COMB 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
217. 1 1.2 11 1.0 2 0.5 3 0.5
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218. LOAD COMB 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
219. 1 1.2 11 -1.0 2 0.5 3 0.5
220. LOAD COMB 412 0.9 DL + 1.6 WL(+Z)
221. 1 0.9 4 1.6
222. LOAD COMB 413 0.9 DL + 1.6 WL(-Z)
223. 1 0.9 5 1.6
224. LOAD COMB 414 0.9 DL + 1.6 WL(+X)
225. 1 0.9 6 1.6
226. LOAD COMB 415 0.9 DL + 1.6 WL(-X)
227. 1 0.9 7 1.6
228. LOAD COMB 416 0.9 DL + 1.0 EL(+X)
229. 1 0.9 10 1.0
230. LOAD COMB 417 0.9 DL + 1.0 EL(-X)
231. 1 0.9 10 -1.0
232. LOAD COMB 418 0.9 DL + 1.0 EL(+Z)
233. 1 0.9 11 1.0
234. LOAD COMB 419 0.9 DL + 1.0 EL(-Z)
235. 1 0.9 11 -1.0
236. ****************************************
237. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**18**	**NUMBER OF MEMBERS**	**19**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**6**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 6/ 6/ 42 DOF
TOTAL PRIMARY LOAD CASES = 11, TOTAL DEGREES OF FREEDOM = 72
TOTAL LOAD COMBINATION CASES = 38 SO FAR.
SIZE OF STIFFNESS MATRIX = 4 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.1/ 49587.7 MB

```
238. LOAD LIST 301 TO 319
239. PRINT SUPPORT REACTION ALL
```

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
1	301	14.37	309.77	-3.24	-8.09	-0.09	-36.75
	302	11.99	583.37	-10.79	-6.59	1.34	-31.19
	303	16.77	582.71	-2.60	-15.79	-1.19	-43.02
	304	14.99	433.48	-26.31	-67.39	-0.55	-38.57
	305	13.77	595.97	14.65	46.57	0.62	-35.46
	306	-0.73	380.25	-4.70	-9.28	0.06	4.29
	307	29.48	649.19	-6.97	-11.55	0.02	-78.32
	308	13.99	510.49	-5.80	-10.40	0.03	-35.95
	309	14.76	518.96	-5.86	-10.42	0.04	-38.08
	310	14.40	511.81	-6.53	-12.34	-0.00	-37.06
	311	14.36	517.64	-5.14	-8.48	0.07	-36.97
	312	9.44	77.54	-29.25	-80.82	-0.83	-24.13
	313	7.80	294.18	25.36	71.12	0.72	-19.97
	314	-11.52	6.57	-0.43	-3.34	-0.03	33.03
	315	28.76	365.15	-3.46	-6.37	-0.08	-77.13
	316	8.11	180.21	-1.91	-4.83	-0.06	-20.63
	317	9.13	191.51	-1.98	-4.87	-0.05	-23.46
	318	8.65	181.97	-2.87	-7.43	-0.10	-22.11
	319	8.59	189.75	-1.02	-2.27	-0.00	-21.99
2	301	14.12	375.57	-2.53	-8.25	-0.21	-36.01
	302	14.74	588.26	13.76	14.60	0.79	-37.76
	303	14.83	630.64	-12.18	-25.15	-0.98	-37.68
	304	14.01	632.22	-20.48	-62.96	-0.71	-35.73
	305	15.23	469.74	20.40	50.93	0.46	-38.85
	306	-0.48	415.54	-1.05	-6.80	-0.05	3.98
	307	29.72	686.43	0.97	-5.23	-0.19	-78.56
	308	14.22	546.62	-0.09	-6.07	-0.13	-36.18
	309	15.02	555.34	0.00	-5.96	-0.12	-38.41
	310	14.60	553.90	-0.73	-7.95	-0.16	-37.25
	311	14.64	548.07	0.65	-4.09	-0.09	-37.34
	312	7.65	333.67	-28.77	-80.88	-0.90	-19.53
	313	9.29	117.02	25.73	70.98	0.65	-23.68
	314	-11.66	44.75	-2.87	-6.00	-0.03	33.42
	315	28.60	405.93	-0.17	-3.90	-0.22	-76.63
	316	7.93	219.53	-1.58	-5.02	-0.13	-20.12
	317	9.01	231.16	-1.46	-4.88	-0.12	-23.09
	318	8.45	229.23	-2.44	-7.53	-0.17	-21.55
	319	8.50	221.46	-0.60	-2.38	-0.08	-21.66
3	301	20.12	79.09	0.25	-3.50	-0.29	-42.37
	302	17.12	391.89	-2.91	-7.62	0.63	-36.09
	303	23.29	362.62	-1.51	-5.38	-1.19	-49.34
	304	21.27	220.72	-18.59	-53.23	-2.04	-44.68
	305	19.10	384.71	15.41	41.74	1.48	-40.58
	306	-3.79	423.66	-1.73	-6.13	-0.30	7.51
	307	44.16	181.77	-1.46	-5.36	-0.26	-92.77
	308	19.57	306.57	-1.59	-5.74	-0.28	-41.34

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	309	20.80	298.85	-1.60	-5.76	-0.28	-43.93
	310	20.21	300.26	-2.10	-7.16	-0.33	-42.68
	311	20.16	305.17	-1.09	-4.34	-0.23	-42.58
	312	13.53	-61.87	-22.52	-65.41	-2.52	-28.16
	313	10.62	156.79	22.81	61.21	2.18	-22.68
	314	-19.89	208.72	-0.04	-2.61	-0.20	41.43
	315	44.03	-113.80	0.33	-1.58	-0.15	-92.27
	316	11.25	52.60	0.15	-2.08	-0.18	-23.70
	317	12.90	42.31	0.14	-2.11	-0.17	-27.15
	318	12.11	44.18	-0.52	-3.97	-0.23	-25.50
	319	12.03	50.73	0.82	-0.22	-0.11	-25.35
4	301	19.47	151.85	-4.40	-8.43	-0.39	-41.22
	302	20.49	464.13	-2.66	-8.09	0.14	-43.23
	303	20.03	455.54	-2.92	-7.49	-1.12	-42.68
	304	18.97	464.84	-20.19	-55.43	-2.23	-40.47
	305	21.15	300.85	13.81	39.53	1.30	-44.58
	306	-3.70	505.69	-3.34	-8.36	-0.45	7.37
	307	43.83	260.00	-3.04	-7.54	-0.48	-92.41
	308	19.41	386.87	-3.19	-7.94	-0.47	-41.16
	309	20.71	378.81	-3.20	-7.96	-0.47	-43.88
	310	20.03	385.30	-3.69	-9.36	-0.51	-42.47
	311	20.09	380.39	-2.69	-6.54	-0.42	-42.58
	312	10.23	200.44	-25.30	-68.37	-2.59	-21.99
	313	13.13	-18.22	20.03	58.25	2.12	-27.47
	314	-20.01	254.90	-2.84	-5.60	-0.21	41.79
	315	43.37	-72.68	-2.44	-4.52	-0.26	-91.25
	316	10.82	96.49	-2.63	-5.05	-0.24	-22.92
	317	12.54	85.74	-2.64	-5.07	-0.23	-26.54
	318	11.64	94.38	-3.31	-6.94	-0.30	-24.66
	319	11.72	87.84	-1.97	-3.18	-0.17	-24.80
5	301	16.32	57.26	2.03	0.79	-0.28	-37.62
	302	12.34	123.61	2.27	1.89	-0.34	-30.47
	303	18.30	118.81	2.60	2.26	-0.42	-43.23
	304	16.65	84.18	-6.42	-22.77	-1.94	-39.05
	305	14.49	126.26	11.08	26.27	1.23	-35.03
	306	-6.09	118.74	2.41	1.98	-0.39	9.74
	307	37.24	91.69	2.25	1.53	-0.32	-83.83
	308	15.00	105.57	2.34	1.76	-0.36	-35.82
	309	16.14	104.87	2.33	1.74	-0.35	-38.26
	310	15.60	104.05	1.95	0.68	-0.38	-37.09
	311	15.54	106.39	2.71	2.83	-0.33	-36.99
	312	11.23	6.30	-10.45	-32.22	-2.28	-25.25
	313	8.35	62.41	12.88	33.17	1.95	-19.89
	314	-19.09	52.39	1.32	0.77	-0.22	39.81
	315	38.68	16.32	1.11	0.18	-0.12	-84.96
	316	9.03	34.83	1.22	0.48	-0.17	-20.95

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	317	10.55	33.88	1.21	0.46	-0.16	-24.20
	318	9.83	32.79	0.71	-0.96	-0.19	-22.64
	319	9.76	35.92	1.72	1.90	-0.14	-22.51
6	301	15.61	62.07	-3.50	-4.96	-0.28	-36.43
	302	15.12	114.37	-2.87	-3.48	-0.39	-36.81
	303	14.97	115.30	-3.00	-3.54	-0.54	-36.66
	304	14.11	122.69	-11.83	-28.39	-2.00	-34.65
	305	16.26	80.60	5.68	20.65	1.17	-38.67
	306	-6.64	114.24	-3.00	-3.65	-0.46	10.22
	307	37.01	89.05	-3.15	-4.09	-0.37	-83.54
	308	14.59	101.99	-3.07	-3.86	-0.42	-35.38
	309	15.78	101.30	-3.08	-3.88	-0.42	-37.94
	310	15.16	102.82	-3.47	-4.96	-0.44	-36.61
	311	15.21	100.47	-2.68	-2.79	-0.40	-36.71
	312	7.93	65.30	-13.77	-35.67	-2.28	-19.18
	313	10.80	9.19	9.57	29.72	1.95	-24.53
	314	-19.74	54.04	-1.99	-2.68	-0.23	40.66
	315	38.46	20.45	-2.20	-3.27	-0.11	-84.37
	316	8.57	37.71	-2.09	-2.96	-0.17	-20.15
	317	10.16	36.78	-2.10	-2.99	-0.17	-23.57
	318	9.33	38.81	-2.62	-4.42	-0.20	-21.79
	319	9.40	35.68	-1.57	-1.53	-0.14	-21.92

********** END OF LATEST ANALYSIS RESULT **********

```
240. ******************************************************************************
241. LOAD LIST 401 TO 419
242. START CONCRETE DESIGN
243. CODE ACI 2008
244. FC 35000 ALL
245. FYMAIN 420000 ALL
246. TRACK 1 ALL
247. CLB 0.05 ALL
248. CLT 0.05 ALL
249. CLS 0.05 ALL
250. MAXMAIN 25 ALL
251. MINMAIN 16 ALL
252. MINSEC 8 ALL
253. REINF 0 MEMB 1 TO 12
254. DESIGN COLUMN 1 TO 12
```

==========

COLUMN NO. 1 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	605.99	24.52	0.01208

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	605.99	5.98	0.00295

==========

COLUMN NO. 2 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

```
 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    536.88      0.00    0.00000
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    536.88      0.00    0.00000
 ---------------------------------------------------------
```

==========

COLUMN NO. 3 DESIGN PER ACI 318-08 - AXIAL + BENDING

```
FY - 420.0  FC -  35.0 MPA,  SQRE SIZE - 600.0 X 600.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3600.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
 ---------------------------------------------------------

 12 - 20 MM               1.047      401      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER    8 SPACING 320.00 MM

    COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    747.72     24.31    0.00970
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    747.72      0.50    0.00020
 ---------------------------------------------------------
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==========

COLUMN NO. 4 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	673.24	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	673.24	0.00	0.00000

==========

COLUMN NO. 5 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	109.15	53.91	0.14744

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	109.15	9.30	0.02544

==========

COLUMN NO. 6 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	42.77	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	42.77	0.00	0.00000

==========

COLUMN NO. 7 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

12 - 20 MM 1.047 401 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	265.86	51.68	0.05802

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	265.86	13.58	0.01524

==========

COLUMN NO. 8 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

12 - 20 MM 1.047 401 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	196.51	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	196.51	0.00	0.00000

==========

COLUMN NO. 9 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	62.12	36.74	0.17654

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	62.12	12.93	0.06212

==========

COLUMN NO. 10 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

```
 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
   M0        P-tens.   Des.Pn    Des.Mn     e/h
 411.95  -1583.37     37.48      0.00    0.00000
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
   M0        P-tens.   Des.Pn    Des.Mn     e/h
 411.95  -1583.37     37.48      0.00    0.00000
 ---------------------------------------------------------
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==========

COLUMN NO. 11 DESIGN PER ACI 318-08 - AXIAL + BENDING

```
FY - 420.0  FC -  35.0 MPA,  SQRE SIZE - 600.0 X 600.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3600.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
 ---------------------------------------------------------

 12 - 20 MM               1.047      401      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER    8 SPACING 320.00 MM

    COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
   M0        P-tens.   Des.Pn    Des.Mn     e/h
 411.95  -1583.37     72.49      34.16    0.14066
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
   M0        P-tens.   Des.Pn    Des.Mn     e/h
 411.95  -1583.37     72.49      14.55    0.05991
 ---------------------------------------------------------
```

==========

COLUMN NO. 12 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	44.87	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	44.87	0.00	0.00000

**********END OF COLUMN DESIGN RESULTS**********

255. DESIGN BEAM 13 TO 19

BEAM NO. 13 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 16MM	0.	5000.	YES	YES

CRITICAL POS MOMENT= 48.56 KN-MET AT 0.MM, LOAD 412
REQD STEEL= 400.MM2, RHO=0.0031, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 168. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

Cracked Moment of Inertia Iz at above location = 41100.8 cm^4

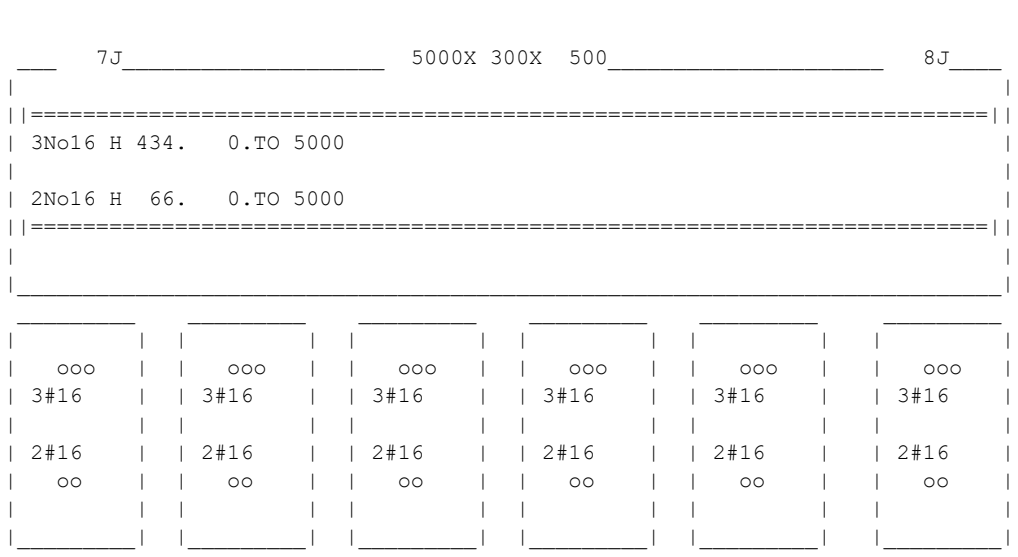
2	434.	3 - 16MM	0.	5000.	YES	YES

CRITICAL NEG MOMENT= 66.22 KN-MET AT 5000.MM, LOAD 404
REQD STEEL= 457.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

Cracked Moment of Inertia Iz at above location = 58050.0 cm^4

B E A M N O. 13 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 8.36 KNS Vc= 122.93 KNS Vs= 0.00 KNS
Tu= 0.03 KN-MET Tc= 5.0 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 12.08 KNS Vc= 122.93 KNS Vs= 0.00 KNS
Tu= 0.03 KN-MET Tc= 5.0 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 14 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

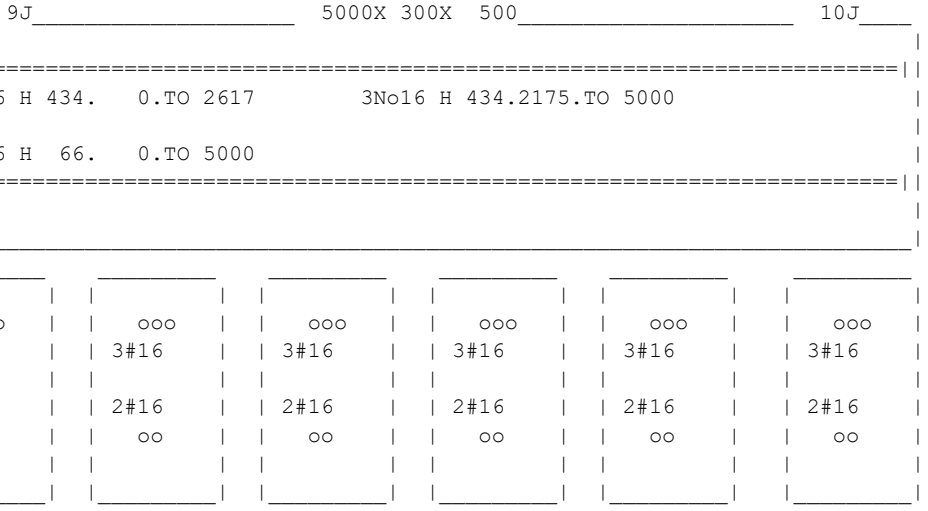
LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 16MM	0.	5000.	YES	YES
	CRITICAL POS MOMENT= 39.49 KN-MET AT 0.MM, LOAD 412					
	REQD STEEL= 324.MM2, RHO=0.0025, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 168. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 41100.8 cm^4					
2	434.	3 - 16MM	0.	2617.	YES	NO
	CRITICAL NEG MOMENT= 48.99 KN-MET AT 0.MM, LOAD 405					
	REQD STEEL= 404.MM2, RHO=0.0031, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 58050.0 cm^4					
3	434.	3 - 16MM	2175.	5000.	NO	YES

```
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     55.03 KN-MET  AT  5000.MM, LOAD  404|
|   REQD STEEL=   455.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./   84. MMS         |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                        |
|--------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    58050.0 cm^4
```

B E A M N O. 14 D E S I G N R E S U L T S - SHEAR

```
 AT START SUPPORT - Vu=    8.92 KNS  Vc=  133.03 KNS  Vs=    0.00 KNS
 Tu=    0.01 KN-MET  Tc=    5.2 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
 AT END   SUPPORT - Vu=   11.52 KNS  Vc=  125.38 KNS  Vs=    0.00 KNS
 Tu=    0.01 KN-MET  Tc=    5.2 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
```



BEAM NO. 15 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 16MM	0.	5000.	YES	YES

```
|--------------------------------------------------------------|
|   CRITICAL POS MOMENT=     18.04 KN-MET  AT     0.MM, LOAD  412|
|   REQD STEEL=   147.MM2, RHO=0.0011, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./  168. MMS         |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                       |
|--------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    41100.8 cm^4

  2        434.     2 - 16MM         0.         2617.        YES  NO
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     30.39 KN-MET  AT     0.MM, LOAD  405|
|   REQD STEEL=   249.MM2, RHO=0.0019, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./  168. MMS         |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                       |
|--------------------------------------------------------------|

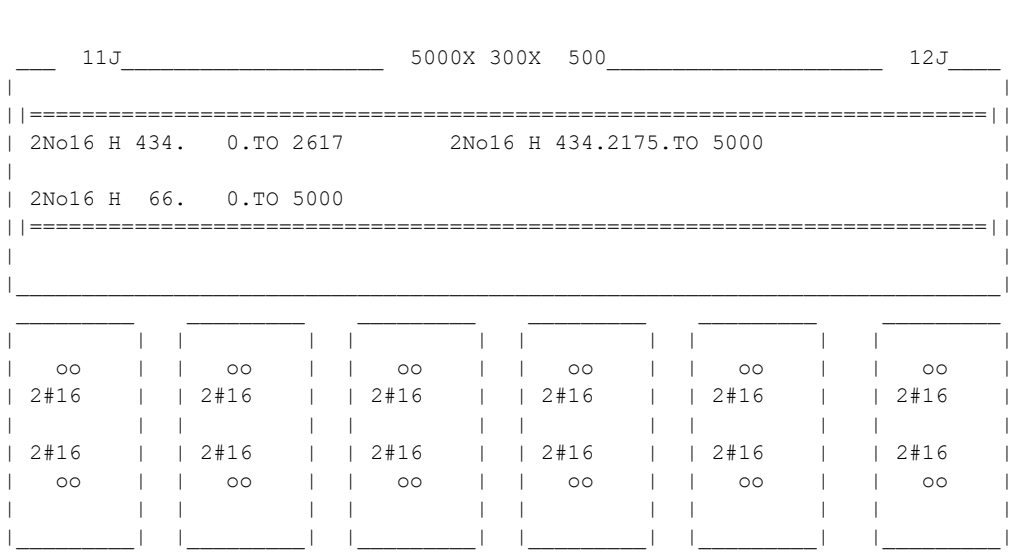
  Cracked Moment of Inertia Iz at above location =    41100.8 cm^4

  3        434.     2 - 16MM      2175.         5000.        NO   YES
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     32.09 KN-MET  AT  5000.MM, LOAD  404|
|   REQD STEEL=   263.MM2, RHO=0.0020, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./  168. MMS         |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                       |
|--------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    41100.8 cm^4
```

B E A M N O. 15 D E S I G N R E S U L T S - SHEAR

```
AT START SUPPORT - Vu=    9.70 KNS  Vc=  125.41 KNS  Vs=    0.00 KNS
Tu=    0.00 KN-MET  Tc=    5.2 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
AT END   SUPPORT - Vu=   10.75 KNS  Vc=  123.93 KNS  Vs=    0.00 KNS
Tu=    0.00 KN-MET  Tc=    5.2 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
```



BEAM NO. 16 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

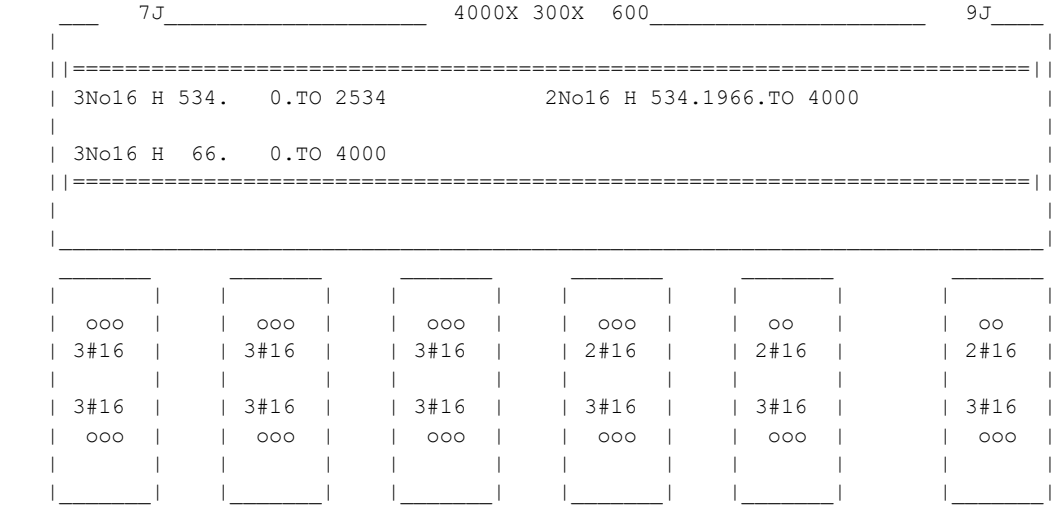
LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	3 - 16MM	0.	4000.	YES	YES
	CRITICAL POS MOMENT= 67.66 KN-MET AT 4000.MM, LOAD 407					
	REQD STEEL= 453.MM2, RHO=0.0028, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 90761.5 cm^4					
2	534.	3 - 16MM	0.	2534.	YES	NO
	CRITICAL NEG MOMENT= 96.08 KN-MET AT 0.MM, LOAD 407					
	REQD STEEL= 562.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 90761.5 cm^4					
3	534.	2 - 16MM	1966.	4000.	NO	YES

```
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     38.11 KN-MET  AT  4000.MM, LOAD  414|
|   REQD STEEL=   253.MM2, RHO=0.0016, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./  168. MMS         |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                       |
|-------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    63898.6 cm^4
```

B E A M N O. 16 D E S I G N R E S U L T S - SHEAR

```
 AT START SUPPORT - Vu=   23.52 KNS  Vc=  153.13 KNS  Vs=    0.00 KNS
 Tu=    0.22 KN-MET  Tc=    6.5 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
 AT END   SUPPORT - Vu=    6.11 KNS  Vc=  153.13 KNS  Vs=    0.00 KNS
 Tu=    0.22 KN-MET  Tc=    6.5 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
```



BEAM NO. 17 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	3 - 16MM	0.	4000.	YES	YES

```
|--------------------------------------------------------------|
|   CRITICAL POS MOMENT=     75.41 KN-MET  AT  4000.MM, LOAD  407|
|   REQD STEEL=   505.MM2, RHO=0.0032, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./   84. MMS         |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                        |
|--------------------------------------------------------------|
```

Cracked Moment of Inertia Iz at above location = 90761.5 cm^4

```
 2        534.     3 - 16MM         0.         2200.        YES  NO
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     86.15 KN-MET  AT     0.MM, LOAD  407|
|   REQD STEEL=   562.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./   84. MMS         |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                        |
|--------------------------------------------------------------|
```

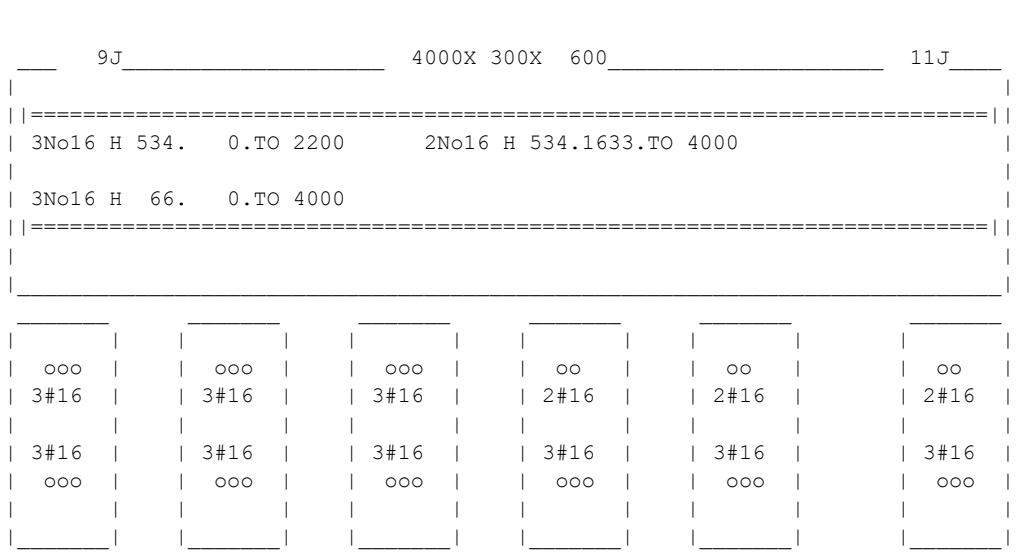
Cracked Moment of Inertia Iz at above location = 90761.5 cm^4

```
 3        534.     2 - 16MM      1633.         4000.        NO   YES
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     40.88 KN-MET  AT  4000.MM, LOAD  414|
|   REQD STEEL=   272.MM2, RHO=0.0017, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./  168. MMS         |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                        |
|--------------------------------------------------------------|
```

Cracked Moment of Inertia Iz at above location = 63898.6 cm^4

B E A M N O. 17 D E S I G N R E S U L T S - SHEAR

```
AT START SUPPORT - Vu=   24.13 KNS  Vc=  151.61 KNS  Vs=    0.00 KNS
Tu=    0.22 KN-MET  Tc=    6.4 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
AT END   SUPPORT - Vu=    6.72 KNS  Vc=  151.61 KNS  Vs=    0.00 KNS
Tu=    0.22 KN-MET  Tc=    6.4 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
```



BEAM NO. 18 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	3 - 16MM	0.	4000.	YES	YES

CRITICAL POS MOMENT= 68.28 KN-MET AT 4000.MM, LOAD 407
REQD STEEL= 457.MM2, RHO=0.0029, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

Cracked Moment of Inertia Iz at above location = 90761.5 cm^4

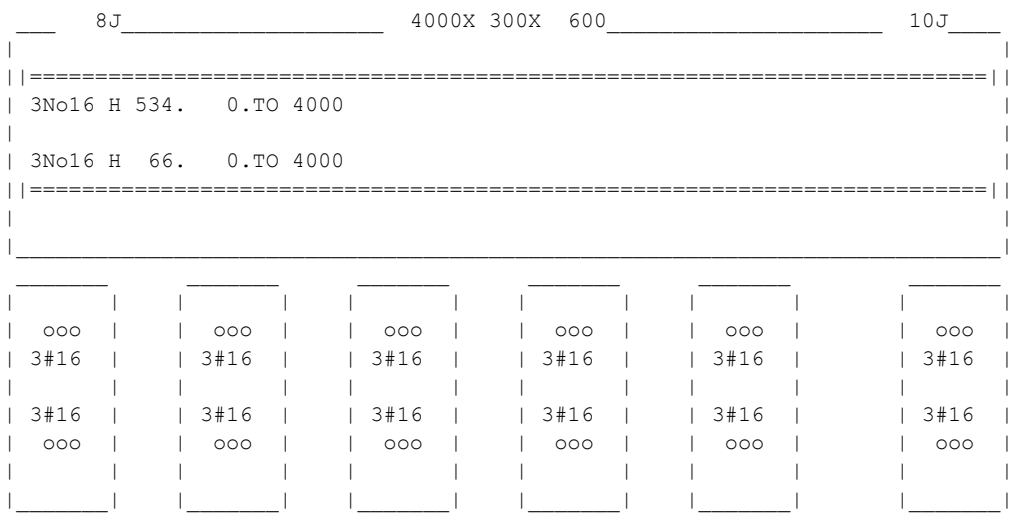
LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
2	534.	3 - 16MM	0.	4000.	YES	YES

CRITICAL NEG MOMENT= 96.24 KN-MET AT 0.MM, LOAD 407
REQD STEEL= 562.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

Cracked Moment of Inertia Iz at above location = 90761.5 cm^4

B E A M N O. 18 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 23.29 KNS Vc= 153.25 KNS Vs= 0.00 KNS
Tu= 0.51 KN-MET Tc= 6.5 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 5.88 KNS Vc= 153.25 KNS Vs= 0.00 KNS
Tu= 0.51 KN-MET Tc= 6.5 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 19 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	3 - 16MM	0.	4000.	YES	YES

CRITICAL POS MOMENT= 73.76 KN-MET AT 4000.MM, LOAD 407
REQD STEEL= 494.MM2, RHO=0.0031, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

Cracked Moment of Inertia Iz at above location = 90761.5 cm^4

2	534.	3 - 16MM	0.	2200.	YES	NO

```
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     84.87 KN-MET  AT     0.MM, LOAD  407|
|   REQD STEEL=   562.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./   84. MMS         |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                       |
|-------------------------------------------------------------|

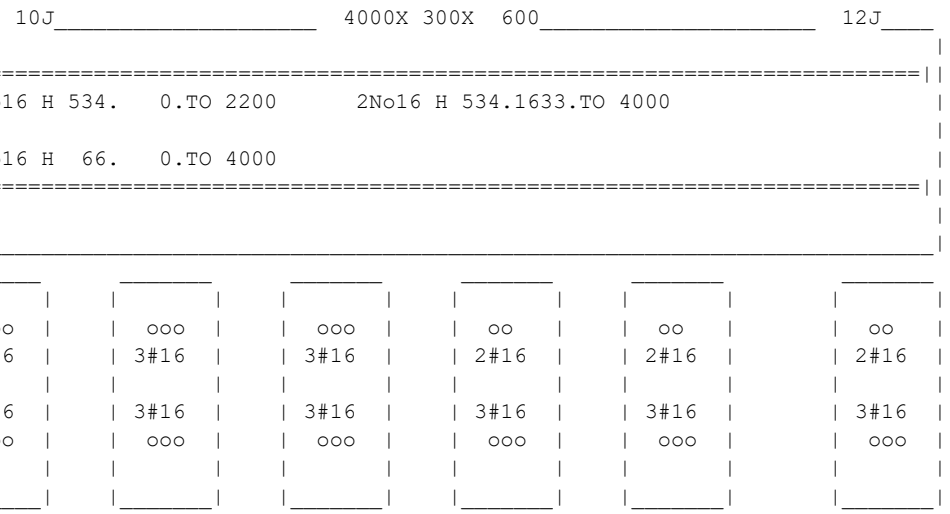
  Cracked Moment of Inertia Iz at above location =    90761.5 cm^4

  3        534.     2 - 16MM      1633.         4000.        NO   YES
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     41.68 KN-MET  AT  4000.MM, LOAD  414|
|   REQD STEEL=   277.MM2, RHO=0.0017, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./  168. MMS         |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                       |
|-------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    63898.6 cm^4
```

B E A M N O. 19 D E S I G N R E S U L T S - SHEAR

```
  AT START SUPPORT - Vu=   23.24 KNS  Vc=  151.81 KNS  Vs=    0.00 KNS
  Tu=    0.34 KN-MET  Tc=    6.4 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                   STIRRUPS ARE NOT REQUIRED.
  AT END   SUPPORT - Vu=    5.83 KNS  Vc=  151.81 KNS  Vs=    0.00 KNS
  Tu=    0.34 KN-MET  Tc=    6.4 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                   STIRRUPS ARE NOT REQUIRED.
```




```
  ********************END OF BEAM DESIGN**************************


  256. END CONCRETE DESIGN
  257. FINISH
```

********** END OF THE STAAD.Pro RUN **********

**** DATE= APR 30,2020 TIME= 16:46:56 ****

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* *
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